

M/Chip Requirements

for Contact and Contactless

29 April 2025



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M/Chip Requirements for Contact and Contactless overview

The M/Chip Requirements for Contact and Contactless communicates the Mastercard requirements that customers must meet and best practices that customers should follow to ensure secure and successful face-to-face contact and contactless transactions with their Mastercard products.

Audience

This document is intended for use by customers, service suppliers, and product vendors involved in chip implementation projects that have a general understanding of how chip works.

Product

All terminals and card-related products used in face-to-face contact and contactless transactions.

The following table is a summary of the metadata details that are used for this product.

Table 1: Details

Metadata	Value
Audience	Acquirers, Issuers, Processors, Service suppliers, Vendors
Product	Chip, Contact and Contactless
Region	Global
Type	Reference
Publish Date	29 April 2025

The following table is a summary of changes that have occurred since the previous publication of the manual.

Table 2: Summary of changes

Description of change	Where to look
Minor editorial, grammar, typographic fixes, or formatting changes, where appropriate.	Throughout
Chapter 3	

Description of change	Where to look
Add paragraph that starts with "Application Transaction Counter monitoring...." Update RI457.25 row	Application Transaction Counter monitoring
Update RI459.25 row	Authorization of fallback transactions
Add new last paragraph Add a new table BI248.25 after this paragraph	Clearing of contactless transactions
Update first sentence	Transaction types > ATM
Update paragraph starting with "Refund transactions..." Change first row table from BI119.22 to RI458.25 and update column 5	Refunds
Change RI082.19 to BI246.24 in table and update column 5	CVM requirements for Fleet Cards
Update sentence before first bullet list	PSD2
Change RI223.22 to RI223.25 and update column 5	Application Priority Indicator
Chapter 4	
Change RA184.13 to RA184.25 and update column 5	Terminal requirements > Chip and contactless acceptance
Add new section with new table rows.	Merchant Category Code '9F15'
Update paragraph that is before the final table	Currency conversion
Update row RA036.24 Change row RA561.24 to RA561.25 and update column 5 Change row RA556.24 to RA556.25 and update column 5 Replace RA077.14 to RA077.24 in paragraph that starts with "Requirement..."	Cardholder verification > Contact transactions
Add a new sentence after paragraph that starts with "All contactless-enabled..." Update paragraph that starts with "Purchase with cash back..." Add new RA567.25 table row.	Cardholder verification > Contactless transactions
Update first sentence Add new row (RA566.25) after BA050.13 table row.	Use of PIN

Description of change	Where to look
Update paragraph that starts with "The decision to decline..."	Subsequent contact transactions
Create a new paragraph starting with "If the authorization response ..." including sentence from prior paragraph.	
Update paragraph that starts with "If the payment device..."	
Change table row RA278.19 to RA278.25 and update column 5.	
Change table row RA445.19 to RA445.25	Terminal types > ATM > Technology selection
Add new paragraph and table row BA127.25	Data in the online request message
Add new row (BA128.25) below BA072.14 table row	Clearing requirements
Change RA545.24 to RA545.25 and update column 5	Asia/Pacific region > Chip and contactless acceptance
Add RA565.25 row	
Update paragraph that starts with "In China, all attended POS..."	Asia/Pacific region > CVM requirements
Change RA504.24 to RA504.25 and update column 5	
Change RA001.18 to RA001.25 and update column 5.	Europe region > Chip and contactless acceptance
Change RA491.24 to RA491.25 and update column 5.	
Remove table row RA402.20.	
Change RA447.24 and update column 5.	
Change RA299.24 to RA299.25 and update column 5.	Maestro acceptance at No CVM locations > CVM requirements
Remove table row RA371.24.	
Change RA372.24 to RA372.25 and update column 5.	
Change RA500.24 to RA500.25 and update column 5.	Latin America and the Caribbean region > CVM requirements
Update paragraph that starts with "In Peru..." and add new sentence after this paragraph.	
Change RA529.23 to RA529.25 and update column 5.	

Description of change	Where to look
Add new bullet point to the bullet list in this section and add new rows after BA1110.16.	United States region > PIN Entry Bypass
Update this section and move this section to be subsection under "M/Chip Fast"	United States region > Purchase with cash back
Chapter 5	
Update sentence that starts with "The first byte..."	Application Selection Registered Proprietary Data
Split section "Network data" into additional subsections as presented:	Network data Network data legend
Network data legend. New subsection which was originally in "Network data" section originally as table 8.	DE 55 data for messages from Acquirer DE 55 data for 0110 and 0210 response messages
Update row RA546.23 and RA547.23 to RA546.24 and RA547.24	Other data in messages from Acquirer IPM Clearing messages from Acquirer requirements
DE 55 data for messages from Acquirer. New subsection which was originally in Network data section originally as table 8.	
DE 55 data for 0110 and 0210 response messages. New subsection which was originally in Network data section originally as table 9.	
Other data in messages from Acquirer subsection which was originally in Network data section originally as table 10.	
IPM Clearing messages from Acquirer requirements. New subsection which was originally in Network data section was originally table 11.	
Appendix A	
Update multiple rows with new content or removed some content.	Data dictionary
Appendix B	
Change RA546.23 and RA547.23 to RA546.24 and RA547.24	Terminal Capabilities, '9F33'
Add new section	Kernel Identifier-Terminal, '96'

Chapter 1 Introduction

This chapter provides an overview of this document.

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Purpose

The *M/Chip Requirements for Contact and Contactless* communicates the Mastercard requirements that customers must meet and best practices that customers should follow to ensure secure and successful face-to-face contact and contactless transactions with their Mastercard products.

The purpose of this document is to:

- Define the chip requirements that Mastercard® has established in addition to EMV® specifications for use with Mastercard products
- Propose recommendations, which constitute best practices for chip implementations
- Define when and how the optional functions in EMV must be used
- Address what is not covered in the Standards, such as explanatory guidance concerning data items to be personalized in the chip or parameterized in terminals

NOTE: The Standards include the information contained in the *Mastercard Rules* manual and other manuals, along with guides, bulletins, and policies that may be updated from time to time.

This document does not provide an introduction or explanation of how chip works nor does it duplicate or reproduce existing standards, such as EMV or the existing requirements for Magnetic Stripe technology.

Scope

This document covers contact and contactless chip technology and Mastercard Consumer-Presented QR.

The use of contactless payment technology by both cards and access devices other than cards, for example, mobile phones, wearables, stickers, and key fobs, is also within scope of this manual.

This document contains the requirements relating to Mastercard, Debit Mastercard®, Maestro®, and Cirrus® card products and requirements for products that include the U.S. Maestro Application Identifier (AID). The requirements for performing transactions on Automated Teller Machines (ATMs), Bank Branch Terminals (BBTs), Point-Of-Sale (POS) terminals, and Cardholder Activated Terminals (CATs) are also part of this manual.

The following requirements do not cover general Magnetic Stripe processing. Requirements for in-store proximity payments are in the *Mastercard Alternate Proximity Payment Requirements* manual.

This document does not discuss general brand rules or requirements, except to explain how certain rules are implemented in chip cards. For full details of the rules and requirements for specific card brands, refer to the relevant brand-specific documentation on Mastercard Connect®.

Requirements and best practices in this document are card application independent unless specifically stated.

The following topics are not in the scope of this document because they are already addressed in other dedicated documents. They are mentioned in the manual to add clarity in the context and not to cover them in detail.

- Card application specifications, for example, M/Chip Advance, and Common Payment Application
- Digital specifications, for example, M/Chip Mobile, Mastercard TEE Based Payment (MTBP), and Mastercard Cloud-Based Payment (MCBP)
- Terminal or reader specifications
- Card personalization data
- Value-added programs, such as Chip Authentication Program
- Data storage functionality

This document contains requirements and best practices that apply globally and also those that are region-specific or market specific.

Audience

This document is intended for use by customers, service suppliers, and product vendors involved in chip implementation projects that have a general understanding of how chip works.

The target audience includes:

- Staff working on chip implementation projects
- Operations staff who need to understand the impact of chip on their activities
- Staff responsible for security and for business risk management
- Personalization bureaus, card vendors, terminal and reader vendors, and service suppliers

NOTE: Readers who want an explanation of how chip works for payments should begin by reading the *M/Chip Program Guide*, which gives a high-level view of the subject.

Support

For support and assistance with implementing the requirements in this document, contact the M/Chip Requirements team.

The email address for the M/Chip Requirements team is chip_requirements@mastercard.com.

Requirements and best practices

Requirements and best practices are provided throughout this manual, using different conventions, to assist the customer when determining if functional elements must be implemented or are Mastercard recommendations.

Requirements as identified in this document are functional elements that must be implemented to achieve the required level of acceptance for Mastercard chip cards on chip-enabled terminals. They are mandatory.

Requirements are always expressed using the word *must*. Requirements are contained in tables and are indicated by a capital R in the left column.

Best practices are Mastercard recommendations for the best ways to implement EMV options. If customers choose not to follow them, their chip implementation will still work but may not be as effective or efficient as it could be.

Best practices are written using the word *should*. Best practices are formatted in the same way as requirements but are preceded by the letters BP in the table.

Requirements and best practices include an indication of whether they apply to all products or just specific brands. They also indicate whether they apply to transactions using the contact, contactless, or both interfaces.

The reference numbers associated with the requirements and best practices are used to help communication during testing or other correspondence. The year extension indicates when a requirement or best practice has most recently been updated. The second character indicates whether the requirement or best practice applies to the issuer (I) or the acquirer (A).

Example table

In this example table:

- R is for requirement.
- All means that the requirement applies to all products.
- CT and CL mean that the requirement applies to contact and contactless interfaces, respectively.
- RI001.21 represents the reference number, where:
 - R is for Requirement.
 - I is for Issuer.
 - 001 is the reference number.
 - 21 stands for 2021, the year that the reference was last updated.
- The text describes the requirement.

R	All	CT CL	RI001.21	All chip cards and contactless payment devices must use one of the approved applications.
----------	------------	----------	----------	---

Table 3: Key to table

R	Requirement
BP	Best Practice, in the first column
B	Best Practice, in reference
All	Applies to all products
NOTE: In the United States region, "All" includes USM. For Mastercard Consumer-Presented QR transactions, "All" only includes Maestro and Mastercard.	
MC	Applies to both Mastercard and Debit Mastercard products, unless explicitly stated otherwise in the text
MS	Applies to Maestro product
CI	Applies to Cirrus product
USM	Applies to U.S. Maestro AID
QR	Applies to Mastercard Consumer-Presented QR
CT	Applies to the contact interface
CL	Applies to the contactless interface

Related information

The documents and resources listed in this section provide information related to the subjects discussed in this document.

Related documents on TRC

Documents available on Mastercard Connect > Technical Resource Center (TRC) and some documents may only be available to customers with appropriate entitlements for the document.

- *AN 4458 Mastercard Announces Global Availability of the Contactless Risk Manager Service*
- *Account Management System User Manual*
- *Card Personalization Validation Guide*
- *Contactless Risk Manager - Global Counter and Accumulator Limits*

- *Chargeback Guide*
- *Customer Interface Specification*
- *Digital Secure Remote Payments UCAF Formats*
- *Fleet Implementation Guide Global*
- *Fleet Implementation Guide for the US Region*
- *IPM Clearing Formats*
- *Mastercard Contactless Kernel Configuration*
- *Mastercard Contactless Reader Specification*
- *Mastercard Contactless Reader Release Note*
- *Mastercard Global Transit Implementation Guide*
- *Mastercard Network Processing - Dual Message Authorization System Guide*
- *Mastercard Network Processing - Single Message System Guide*
- *Mastercard Rules*
- *Mastercard Transit Terminal Requirements*
- *Mastercard Dictionary*
- *M/Chip Advance Card Application Specification - Payment*
- *M/Chip Advance Card Application Specification - Payment and Data Storage*
- *M/Chip Advance - Common Personalization Specifications*
- *M/Chip Advance Data Storage Operator Security Guidelines*
- *M/Chip Advance - Issuer Guide*
- *M/Chip Card Application Cryptographic Algorithms*
- *M/Chip Card Personalization Standard Profiles*
- *M/Chip Personalization Data Specifications For Contact and Contactless*
- *M/Chip Processing Services - Service Description Guide*
- *M-TIP Process Guide*
- *Payment Systems Public Keys for M/Chip*
- *Public Key Infrastructure (PKI) - Overview*
- *Public Key Infrastructure (PKI) - Policy*
- *Quick Reference Booklet*
- *Security Rules and Procedures*
- *Single Message System Specifications*
- *Transaction Processing Rules*

Mastercard Cloud Based Payments documentation

Mastercard Cloud Based Payments related documents, which require a license.

Contact mcbp@mastercard.com.

- *Mastercard Cloud-Based Payments Product Guide*
- *Mastercard Cloud-Based Payments Cryptographic Algorithms*
- *Mastercard Cloud-Based Payments MPA Specification*
- *Mastercard Cloud-Based Payments - Personalization Specification*
- *Mastercard Cloud-Based Payments Security Guidelines for MPA Development*

- *Mastercard Cloud-Based Payments Server Components Functional Description*
- *Mastercard Cloud-Based Payments Server Requirements*
- *Mastercard Cloud-Based Payments Wallet Implementation Guide*

Trusted Execution Environment Based Payment documents

Mastercard Trusted Execution Environment Based Payment related documents, which require a license.

Contact chip_product_management@mastercard.com.

- *Mastercard TEE Based Payments Product Guide*
- *Mastercard TEE Based Payment - Issuer Cryptographic Algorithms*
- *Mastercard TEE Based Payment - Personalization Specification and Standard Profiles*
- *Mastercard TEE Based Payments Functional Description V1.2.2*

M/Chip Mobile documents

Mastercard M/Chip Mobile related documents, which require a license.

Contact chip_product_management@mastercard.com.

- *M/Chip Mobile Product Guide*
- *M/Chip Mobile - Issuer Cryptographic Algorithms*
- *M/Chip Mobile Personalization Profiles*
- *M/Chip Mobile Specification Version 1.1.3*
- *M/Chip Mobile V1.1 - Implementation Guide*

Other documentation

Other documents available upon request from chip_requirements@mastercard.com.

- *M/Chip Fast Technical Requirements U.S. Market*
- *Fleet Implementation Guide - Mastercard Corporate Fleet Card Program* (for Canada and U.S. Region before 11 April 2025)

Fleet Card additional documentation

In addition to the Mastercard Fleet Implementation Guides listed above, the following documents are available from Conexus at <https://www.conexus.org/public-standards>:

- Conexus EMV Fleet Tag Specifications, which includes documents:
 - *Implementation Guide - EMV Fleet Data Prompting, (also known as IFSF Part 3-28 Additions for EMV Fuel Cards)*
 - *Implementation Guide - Retail Financial Transactions - EMV Fleet Tags*

Mastercard brand resources

Mastercard webpage resources, for downloading all branding information: <http://brand.mastercard.com>.

- Includes Mastercard Brand Mark Guidelines
- Includes Mastercard Contactless Branding Standards

EMV documentation

EMV Documentation available on <https://www.emvco.com>:

- *EMV Acquirer and Terminal Security Guidelines*
- *EMV Contactless Specifications for Payment Systems, Book A - Architecture and General Requirements*
- *EMV Contactless Specifications for Payment Systems, Book B - Entry Point Specification*
- *EMV Contactless Specifications for Payment Systems, Book C-2 - Kernel 2 Specification*
- *EMV Integrated Circuit Card Specifications for Payment Systems: Book 1-Application Independent ICC to Terminal Interface Requirements*
- *EMV Integrated Circuit Card Specifications for Payment Systems: Book 2-Security and Key Management*
- *EMV Integrated Circuit Card Specifications for Payment Systems: Book 3-Application Specification*
- *EMV Integrated Circuit Card Specification for Payment Systems Book 4-Cardholder, Attendant and Acquirer Interface Requirements*
- *EMV Issuer and Application Security Guidelines*
- *EMV Payment Tokenization Specification - Technical Framework*

Notational conventions

The notational conventions that are used throughout this manual are set out in this section.

Table 4: Notational conventions

Notation	Meaning	Example
'0' to '9' and 'A' to 'F'	Hexadecimal notation. Values expressed in hexadecimal form are enclosed in straight single quotation marks.	'6B75'
1001b	Binary notation. Values expressed in binary form are followed by the letter b and not enclosed in quotation marks.	00001000b
340	Decimal notation. Values expressed in decimal form are not enclosed in quotation marks.	12

Notation	Meaning	Example
"abcd"	This notation is used for the following values, which are always enclosed in straight double quotation marks: 1. An "an" or "ans" format text string (ASCII unless otherwise specified), where "an" is an alphanumeric character (a-z, A-Z, and 0-9) and "ans" is an alphanumeric special character (printable ASCII characters) 2. Name of a bit	1. "05", "MASTERCARD/CARD", " /" 2. "Application Not Yet Effective" bit
xx	Undefined value	"2xx", "05x"
[] []	For multibyte data objects, a byte index and a bit index are enclosed in brackets.	TVR [1] [8] refers to bit 8 of byte 1 of the Terminal Verification Results.

Additional notational conventions

EMV Card commands

EMV Card commands are written in all all-uppercase letters.

Example: GENERATE AC

Data elements

When the contents of data elements are shown in the text, spaces may be included for clarity of reading. The spaces should be omitted in the actual personalization.

Example in text: TAC Denial 'F8 50 80 A0 00'

Example in personalization: 'FA5080F800'

Chapter 2 Definitions

This chapter contains the definitions of key terms used throughout the document.

Definitions..... 29

Definitions

The common terms used in relation to chip technology are defined in this section.

Definitions of other terms used are available in the *Mastercard Dictionary*.

Access device

A device other than a plastic card that uses at least one EMV®-capable payment application to provide access to an account in compliance with the Standards. A contactless payment device is a type of access device. The access device may support transactions in other ways than by the contactless interface, for example, Mastercard® Consumer-Presented QR.

Application Cryptogram (AC)

The term *Application Cryptogram* is used for any of the three kinds of cryptogram the card can produce in response to a terminal request. ACs are used by the issuer host to verify that the card is genuine and to ensure that none of the critical data associated with the transaction has been changed. An AC can only be verified by the issuer, or an authority delegated by the issuer with knowledge of the secret key used to compute it.

An AC is a Message Authentication Code (MAC) calculated using symmetric cryptography, with transaction data specified by the issuer as input. Refer to *EMV Book 2, Security and Key Management*, for more details on AC generation.

The three types of AC are:

- Transaction Certificate (TC) - The card produces a TC when a transaction is approved, either offline or, for contact transactions, following an online approval from the issuer.
- Authorization Request Cryptogram (ARQC) - The card produces an ARQC when a transaction is to be sent online for authorization. The issuer uses the ARQC to perform online Card Authentication Method (CAM).
- Application Authentication Cryptogram (AAC) - The card produces an AAC when it declines a transaction, usually either offline or, for contact transactions, in response to an online decline from the issuer. A terminal may request an AAC for both contact and contactless refund transactions.

Application Identifier (AID)

Often referred to as an AID, the application identifier is used by the card and terminal to find all mutually supported applications. The AID is associated with a set of parameters and processing rules that are configured in the card and terminal and used to complete the transaction. A different AID is normally used for each different product. Also refer to section "Application Selection" in chapter "Issuer requirements."

Attended POS Terminal

A POS Terminal where a representative of the merchant or financial institution is in attendance to operate the terminal and perform the transaction.

- It excludes CAT1, CAT2, CAT3, CAT4, CAT7 terminals.
- It includes MPOS (CAT9) and iPOS terminals (unless they are CAT1, CAT2, CAT3, CAT4, CAT7), and bank-branch terminals.

Biometric Card

A Biometric Card is a card that allows the cardholder to be verified using their fingerprint. Biometric Cards feature an embedded fingerprint sensor that captures the cardholder's physical fingerprint and then matches it locally with a card-stored fingerprint template, without recourse to an external biometric sensor or process.

Card

This document uses the term *card* generically to refer to any plastic card or access device performing a transaction and the term *cardholder* to the person using that device and *card data* being the data personalized onto the device. Specific reference is made to the interface or the type of contactless payment device, where appropriate.

Card Authentication Method (CAM)

Card authentication is the process by which the terminal or the issuer host system checks that the card being used is genuine.

There are two main types of CAMs defined by EMV:

- Online CAM, which is performed remotely by the issuer host during an online authorization
- Offline CAM, which is performed locally by the terminal

Cardholder Verification Method (CVM)

A Cardholder Verification Method (CVM) is a technique used to confirm that the person presenting a card is the legitimate cardholder. Chip cards support several CVMs:

- No CVM, which corresponds to the absence of any actual check
- Signature, where the POS terminal attendant checks a handwritten signature against the signature present on the card signature panel
- Online Personal Identification Number (PIN), where a PIN entered on the terminal PIN pad is sent online to the issuer for validation against a reference value
- Offline plaintext PIN and Offline enciphered PIN, where the chip card validates a PIN entered on the terminal PIN pad against a reference value it holds in its secure memory. The two methods differ insofar that Offline enciphered PIN is transmitted enciphered to the card

In addition to these four methods, payment devices may also support the Consumer Device Cardholder Verification Method.

Chip card

A chip card is an ID-1 format plastic card, which includes a contact chip interface. A chip card performs contact chip transactions in accordance with EMV Books 1, 2, 3, and 4. A chip card may be a dual interface card.

Chip grade and Magnetic Stripe grade issuers

Magnetic Stripe grade was a term that was used to relate to issuers that did not process chip data during an authorization request or financial transaction request or use the data in DE 55 (Integrated Circuit Card [ICC] System Related Data) to improve authorization decisions.

Due to the risks involved, Mastercard does not permit issuers to operate in Magnetic Stripe grade anymore. Issuers that do not have the capability on their host system to process chip data must enroll for Mastercard Cryptogram Pre-Validation Service. All issuers must operate in chip grade.

Refer to the *M/Chip Processing Services - Service Description Guide* for full details of this service.

Consumer Device Cardholder Verification Method (CDCVM)

A Consumer Device Cardholder Verification Method (CDCVM) is a CVM whereby the cardholder is verified locally or remotely by the payment device instead of the payment terminal. CDCVM may be based on PIN, password, biometric, or behavioral patterns.

Contactless limits

For a given contactless transaction, three limits are used by the terminal. The same limit may have different values for different products, transaction types, and different CVM options.

The Terminal Contactless Transaction Limit is a maximum transaction amount above which a contactless transaction must not be performed. This limit consists of two separate limits: the Terminal Contactless Transaction Limit, No On-device CVM, and the Terminal Contactless Transaction Limit, On-device CVM. The former indicates the maximum transaction amount for transactions without CDCVM and the latter the maximum transaction amount for transactions with the use of CDCVM. In most markets, a terminal must be able to perform contactless transactions up to the same value as other technologies. Exceptionally, in markets that support Offline PIN for contact transactions but do not support Online PIN, the value of contactless transactions may be limited.

Terminals do not require cardholder verification for transaction amounts equal to or below the Terminal CVM Required Limit and, unless specifically requested by the cardholder, do not require a printed receipt. For contactless transactions above the Terminal CVM Required Limit, normal cardholder verification and receipt printing procedures apply. The *Chargeback Guide* lists the CVM Limit amounts that are applicable in each market.

NOTE: Some mobile payment device implementations may require CDCVM for transaction amounts equal to or below the appropriate CVM Limit, for example, in the case of lost and stolen protection. Also, the issuer or the card may specifically request a CVM for transaction amounts equal to or below the Terminal CVM Required Limit, for example, in the case of Strong Customer Authentication (SCA).

The Terminal Contactless Floor Limit is a transaction amount above which online issuer authorization is required.

EMV mode transactions equal to or below the Terminal Contactless Floor Limit may be authorized offline. To provide the acquirer protection against authorization-related chargebacks, transaction amounts above this limit should be authorized online by the issuer.

There are no specific floor limits applicable to EMV mode contactless transactions. The same limits apply as for contact transactions.

Magnetic Stripe mode transactions always require online issuer authorization.

Contactless operating modes

Mastercard supports two modes of operation for contactless transactions:

- Magnetic Stripe mode
- EMV mode

Magnetic Stripe mode is typically used with the term *transaction*, that is, Magnetic Stripe mode transaction, to indicate contactless payment based on Track 1, Track 2 data, or both obtained from the contactless interface. Magnetic Stripe mode is designed for contactless payments using authorization networks that currently support only Magnetic Stripe authorization.

EMV mode is typically used with the term *transaction*, that is, EMV mode transaction, to indicate contactless payment using a full chip infrastructure carrying EMV data. The transaction may require online authorization but may be approved offline by the card and terminal. EMV mode is designed for contactless payments in markets that have migrated to chip technology for EMV contact payments.

Contactless payment device

A contactless payment device is a type of access device, other than a dual interface card, that exchanges data with the terminal by means of radio frequency communications in accordance with EMV Books A, B, and C-2.

CVM Fallback

In certain markets, Offline PIN is the preferred CVM for chip cards and should be used whenever the card and terminal both support it for contact transactions. In some circumstances, Offline PIN cannot be used, such as when the cardholder has unsuccessfully entered their PIN too many times. In these cases, a different CVM, usually Signature or No CVM, can be used, subject to certain constraints. This process, whereby a less preferred CVM is used instead, is called CVM Fallback.

Data Element (DE) 55

The ISO 8583 standard Financial transaction card originated messages - Interchange message specifications define data element (DE) 55 (Integrated Circuit Card [ICC] System Related Data) as used by the Mastercard Authorization Platform and Clearing and Settlement Platform to transport chip data between the acquirer and the issuer.

DE 55 is fully supported by all Mastercard networks and interfaces except for contactless Magnetic Stripe mode transactions.

Data Object List (DOL)

To carry out a transaction, a chip card may need a terminal to send some specific information. The items that are needed are specified in a data structure called a Data Object List (DOL).

A DOL contains a list of the tags and lengths of the data items the card will receive. The terminal concatenates the required values into a single data field that is sent to the card with a command. Because the card application knows the position and lengths of the data the terminal sends, it can unpack the data it receives into the correct fields.

Examples of DOLs include CDOL (Card Data Object List) and PDOL (Processing Options Data Object List).

Data storage

M/Chip Advance applications include a feature known as Data Storage where data can be stored within the chip in different ways, with different read and write access controls. Integrated data storage enables the reading and writing of data to be integrated with the commands associated with a normal transaction. It might be used with functions such as loyalty or ticketing that are linked to the completion of the transaction.

For details of how data storage works and how it might be used, refer to section "Introduction to M/Chip Advance Card Application Specifications - Data Storage" in *M/Chip Advance Card Application Specification - Payment and Data Storage v1.2.3*.

Domestic

When referring to transactions, a transaction is defined as Domestic if both of the following are true:

- Issuer Country Code tag '5F28' is present on the card
- Issuer Country Code tag '5F28' is identical to Terminal Country Code (tag '9F1A')

Digital Secure Remote Payments (DSRP)

Digital Secure Remote Payment (DSRP) is a protocol whereby a secure cryptogram is generated by an access device and delivered to the issuer to protect non-face-to-face transactions. This includes e-commerce transactions or those associated with in-app purchases.

Dual interface card

A dual interface card is an ID-1 plastic card that supports both contactless and contact transactions. Contactless transactions are performed in accordance with EMV Books A, B, and C-2.

Dual interface terminal

A dual interface terminal supports transactions using both contact and contactless chip technology. Either technology can be used to access the payment application in the terminal. Note that the two payment applications may be entirely separate or integrated.

EMVCo LLC

EMVCo LLC is the organization that supports and maintains the global standard for chip transactions and provides certain testing and certification services for cards and terminals. The

specifications are often referred to as EMV and transactions that follow the specifications as EMV transactions. The EMV specifications cover both contact and contactless technologies.

The EMV specifications detail the card-to-terminal interface requirements for chip transactions and also define the minimum data that must be transmitted in authorization and clearing messages from the point of interaction to the issuer. This is typically the data that the card signs in the authentication cryptograms. This document does not repeat the EMV requirements.

Hybrid card

Hybrid cards have both a chip and a Magnetic Stripe. To be considered hybrid, the chip on the card must carry an EMV payment application that supports the same payment product that is encoded on the Magnetic Stripe.

Hybrid terminal

A hybrid terminal is a device that can accept transactions using both contact chip and Magnetic Stripe technologies. Except in exceptional situations, chip terminals are also hybrid terminals to ensure that the terminal can accept magnetic-stripe-only cards.

Mastercard considers a terminal to be hybrid only if the terminal has successfully passed EMV Type Approval and Mastercard certification. This means that a device that can accept chip cards, but that has not successfully completed the Mastercard Terminal Integration Process (M-TIP), is considered by Mastercard to be a magnetic-stripe-only terminal.

Contact terminal has the same meaning as hybrid terminal in this document, unless specifically indicated, and applies to all terminal types, for example, POS or ATM.

Integrated Point-Of-Sale (IPOS)

An Integrated Point-Of-Sale (IPOS) terminal refers to acceptance architectures where the merchant's cash register is integrated with the card-reading technology. They are typically merchant-owned and deployed by large retailer chains and stores. This definition may include fuel dispenser terminals that have integrated payment functionality, though it does not include any devices that can be deployed as standalone payment terminals.

Interface

An interface is the technology used by the card and terminal to communicate when completing a transaction. The interfaces within the scope of this document are:

- Contact - Communications in accordance with EMV Books 1, 2, 3, and 4
- Contactless - Communications using Near Field Communications (NFC) technology in accordance with EMV Books A, B, and C-2
- Mastercard Consumer-Presented QR Code

Liability shift

Mastercard has introduced the concept of liability shift:

- To encourage migration to chip technology by issuers and acquirers

- To encourage support of PIN at POS for contact transactions
- To encourage support of more secure cardholder verification methods (CVMs) of Online PIN and CDCVM, for contactless transaction amounts that exceed the CVM limit (high-value contactless transactions)
- To provide fraud protection to chip-capable issuers or acquirers when the other party in the transaction does not support the more secure technology

The Chip Liability shift makes the non-chip party in a transaction liable for contact counterfeit fraud losses. The Chip and PIN Liability shift makes the non-PIN party in a transaction liable for lost, stolen, and card not received fraud losses.

The geographical coverage of the Liability Shift programs is detailed in the *Chargeback Guide*.

Mastercard Cloud-Based Payments (MCBP)

Mastercard Cloud-Based Payment application is a solution used on mobile payment devices. The solution is implemented outside of the secure element and thus does not need access to this environment. MCBP transactions use one-time session keys, which are periodically replenished to the device. As with contactless NFC transactions, the application also supports In-app (Remote) payments using DSRP and Mastercard Consumer-Presented QR transactions.

Mastercard Consumer-Presented QR

Mastercard Consumer-Presented QR enables consumer mobile wallets to generate personal QR codes that can be scanned by merchants in physical locations to receive payments.

Mastercard Consumer-Presented QR is built on Mastercard Cloud-Based Payment (MCBP) technology and conforms to the *EMV QR Code Specification for Payment Systems - Consumer-Presented Mode* specification released in July 2017.

Mastercard Digital Enablement Service (MDES)

The Mastercard Digital Enablement Service is an on-behalf service that supports issuers in provisioning data onto access devices. The service supports data preparation and tokenization. The service also de-tokenizes transactions forwarding the authorization to the issuer for approval containing the underlying funding Primary Account Number (PAN) of the account.

Mastercard TEE Based Payment (MTBP)

Mastercard TEE Based Payment (MTBP) enables consumers to digitize their cards into a TEE (Trusted Execution Environment), a secure area of a mobile phone. The TEE protects the confidentiality and integrity of the payment application code and data to secure contactless mobile payments. MTBP is currently deployed by Original Equipment Manufacturers (OEMs), as it requires the device to support a Trusted Execution Environment (TEE). Along with contactless NFC transactions, the application also supports In-app (Remote) payments using DSRP.

M/Chip

M/Chip is a Mastercard chip implementation. M/Chip is a generic term applicable to cards, access devices, terminals, acquirers, or issuers that use a chip implementation endorsed by

Mastercard. M/Chip Advance refers to a specific version of the Mastercard card application specifications.

M/Chip Mobile

M/Chip Mobile is the Mastercard platform for enabling contactless payments using a Secure Element. The platform allows Mastercard credit, debit, and prepaid payment products to be digitized into consumer devices equipped with a Secure Element. Along with contactless transactions, the application also supports In-app (Remote) payments using DSRP and it can enable separate authentication transactions that might be used to secure e-banking login.

Mobile payment device

A mobile payment device is a cardholder-controlled device containing a payment application compliant with the Standards and uses an integrated or nonintegrated keyboard or input device and screen or other output device to access an account. The term is used for any consumer payment device that includes a battery power source, a keypad or data entry capability, and a display or other output capability.

In this document, it is assumed that most mobile payment devices are contactless payment devices, noting that they may also support transactions in other ways than by the contactless interface. Alternatively, they might not support contactless NFC transactions but instead support other transaction types, such as DSRP and Mastercard Consumer-Presented QR.

Newly Deployed (including replacement)

Newly deployed (including replacement) terminals include new additional terminals deployed in the market and terminals deployed as a replacement for old existing terminals.

Terminals that change ownership or move location are not considered newly deployed.

Newly Deployed (excluding replacement)

Newly deployed (excluding replacement) terminals include new additional terminals deployed in the market, but the following exclusions apply:

- An individual terminal replacement is required due to an existing fault. In these cases, existing terminals may be replaced like-for-like.
- An existing terminal requires a software upgrade
- The addition of one or more new terminals in an existing store that is already equipped with Mastercard-accepting terminals

For retailer environments that do not consist of fixed physical stores (for example, travel or transport merchants) consult Mastercard to determine if these exclusions apply.

Terminals that change ownership or move location are not considered newly deployed

Nonmobile payment device

A nonmobile payment device is any contactless payment device that is not a mobile payment device, such as stickers or key fobs.

Payment application

The payment application is the set of code within the chip that processes the digital transaction. The underlying payment application is transparent to the terminal, although the issuer will need to interpret the data generated by the application and generate the data required by the application for personalization, authorization processing, and parameter updates. The payment application may also support other payment methods.

PIN Entry Bypass

PIN Entry Bypass is an optional function that may be supported at attended terminals. It allows the merchant or cardholder to force the terminal to stop prompting for PIN and to continue CVM processing with the next entry in the CVM List. In practice, this usually means using Signature instead of Online or Offline PIN.

PIN Entry Bypass allows a cardholder who has forgotten their PIN to complete a transaction. It may be useful during the early stages of a migration to PIN for particular products but should not be supported in mature implementations.

PIN-preferring cards

A PIN-preferring card is one where the CVM List has any PIN option, Online PIN or Offline PIN, either enciphered or plaintext, before Signature for POS terminal use, indicating that PIN is preferred to Signature at any terminal that supports it.

Provisioning

Provisioning is the preparation and personalization of a digitized card and payment device into an access device or wallet.

Public-key cryptography

A public-key cryptographic system uses two related keys called public and private keys.

Messages encrypted with the public key can only be decrypted using the private key and messages signed with the private key can be authenticated using the corresponding public key. Two entities that do not share a secret key can therefore use this cryptography to ensure the integrity or secrecy of data.

Public-key cryptography is used in EMV to encrypt Offline PINs and to perform offline data authentication.

Relay Resistance Protocol (RRP)

The Relay Resistance Protocol (RRP) is a security mechanism that was introduced in v2.5 of *EMVCo Book C-2* and protects contactless transactions from relay attacks.

RRP is designed to detect when transactions may be the subject of relay fraud, whereby a genuine contactless payment device is tricked into interacting with a fraudulent terminal device that transmits essential transaction data to an accomplice device being used to perform a transaction at a genuine terminal.

RRP works by detecting unexpected timing delays between the generation of the response by the genuine device and the receipt of the response by the genuine terminal. Unexpected delays result in the transaction being forced online to the issuer.

On terminals, the Mastercard *Contactless Reader Specifications* v3.1 and later supports RRP.

On cardholder devices, RRP is supported by the latest versions of M/Chip Advance, MTBP, and M/Chip Mobile.

Submitted for M-TIP testing

When this is used in a requirement, it applies to terminals submitted for M-TIP testing for any reason.

Symmetric key cryptography

A symmetric key cryptographic system uses the same key to perform encryption and decryption and uses the same key for both generating and verifying Message Authentication Codes (MACs). Keys used in a symmetric cryptographic system must be kept secret by the parties involved. Symmetric key cryptography is used to generate the application cryptograms in an EMV transaction and is often used to secure secret data, for example, PIN, in network messages.

Tag, Length, Value (TLV) format

Data in EMV transactions is often transmitted in strings of data using a format called Tag, Length, Value, or TLV. In this format, the first 1, 2, or 3 bytes contain the tag that identifies the data item, the following one or more bytes contain the length L of the data, and the following L bytes contain the data value itself.

Refer to *EMV Book 3, Application Specification* for a full definition of TLV encoding.

Technical fallback

Technical fallback may occur when the contact interface of a hybrid card is used at a hybrid terminal. When chip technology should be used, but a technical failure at the card or terminal prevents the chip transaction from processing, technical fallback occurs. In such cases, subject to certain constraints, the terminal can proceed with a Magnetic Stripe transaction instead or may use other technology, such as PAN Key Entry. Technical fallback is not supported in mature chip implementations.

Tokenization

Tokenization is the mapping of Primary Account Numbers (PANs) to Payment Tokens, which replace the PAN in the payment's ecosystem. A Payment Token is a 13-digit to 19-digit numeric value that must pass basic validation rules of an account number, including the Luhn check digit. Payment Tokens are generated within a Bank Identification Number (BIN) range that has been designated as a Token BIN Range and flagged accordingly in all appropriate BIN tables. Payment Tokens must not have the same value as or conflict with a real PAN. Payment tokens can help prevent fraud by making it easy to identify when data stolen from one technology or interface

are fraudulently presented using a different technology and can give cardholders confidence by not needing the genuine PAN to be disclosed to different merchants.

User interface

A mobile payment device will have a user interface that enables messages to be displayed to the cardholder and for instructions to be given to the device by the cardholder. This might support the cardholder in choosing which payment product they wish to use from a number available within a wallet or prompting the cardholder for CDCVM.

Wallet

A mobile payment device may contain multiple instances of separate payment products. This might be data representing several different products from the same issuer or products from different issuers, or payment schemes, or both. A wallet refers to an app that assists with the management of these different products, especially enabling the cardholder to choose which product to use for a particular transaction.

Chapter 3 Issuer requirements

This chapter contains the requirements that issuers must comply with to issue chip cards and contactless payment devices that support EMV technology. It also contains best practices to help ensure issuers receive the maximum benefit from those implementations.

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Introduction

Requirements apply to all Mastercard programs unless it is explicitly stated that they are program-specific or brand-specific.

Issuers should refer to chapter "Data requirements" for specific data requirements on chip cards and contactless payment devices.

Issuer device applications

Mastercard supports the following applications for chip cards and contactless payment devices:

- M/Chip Advance v1.1 or v1.2
- EMVCo CPA (Common Payment Application)
- M/Chip Mobile v1.1.3
- Mastercard TEE Based Payments v1.1.3
- Mastercard Cloud-Based Payments Mobile Payment Application (MPA) v1.0, v1.0+, V2.0, V2.1, V2.2

M/Chip Advance is the current card application supporting both contact and contactless interfaces. The same application may be used on nonmobile payment devices to drive the contactless interface. M/Chip Advance features highly configurable card risk management options and supports integrated data storage. The latest versions support Relay Resistance Protocol to guard against relay fraud on the contactless interface.

M/Chip Mobile is the current application for secure element-based mobile payment device implementations. Along with contactless payments, M/Chip Mobile supports Digital Secure Remote Payment (DSRP) for e-commerce and in-app purchases and a separate configuration to perform authentication transactions.

Mastercard® TEE Based Payments (MTBP) is part of the Mastercard M/Chip Digital suite of products and consists of specifications, requirements, and processes to simplify the licensing, development, and deployment of mobile payment services using a Trusted Execution Environment (TEE). MTBP is currently deployed by Original Equipment Manufacturers (OEMs) to perform Contactless Near Field Communications (NFC) and In-app (Remote) payments. It requires the device to support a TEE to protect the sensitive data.

Mastercard Cloud-Based Payments (MCBP) is available for mobile payment devices where access to the secure element is not available. This application also supports DSRP and Mastercard Consumer-Presented QR transactions.

Mastercard has announced the end of life for products supporting the legacy specifications, M/Chip 2. Issuers must not issue EMV® cards based on M/Chip 2 specifications and must migrate to M/Chip Advance.

More specifically, all issuance programs of chip cards and nonmobile payment devices must be implemented using an M/Chip Advance product.

Mastercard no longer certifies new implementations of MCBP 1.0/1.0+. Already approved implementations of MCBP 1.0/1.0+ can continue to be deployed and submitted for recertification.

Issuers should use the most recent version of the specification available.

R	All	CT CL	RI001.14	All chip cards and contactless payment devices must use one of the approved applications.
R	All	CT CL	RI370.22	Issuers are required to issue all chip cards and nonmobile payment devices using an M/Chip Advance product.
BP	All	CT CL	BI096.14	Issuers should use the most recent version of the application specification available.
R	MC MS	QR	RI388.22	Issuers that support Mastercard Consumer-Presented QR must use the Mastercard Cloud-Based Payments Mobile Payment Application (MPA) v2.1 or later specifications.

Issuer device requirements

Chip cards and contactless payment devices issued with a Mastercard brand must conform to requirements specified by Mastercard.

Hybrid cards

Hybrid cards have both a chip and a Magnetic Stripe. All contact chip cards must have a Magnetic Stripe and, therefore, are hybrid cards.

The product that is supported on the Magnetic Stripe must also be supported by the chip.

R	All	CT	RI003.14	All chip cards used for Mastercard payment products must be hybrid. This means that the cards must have both a chip and a Magnetic Stripe.
R	All	CT	RI005.11	Cards that have a chip must support the same product on the chip that is encoded on the Magnetic Stripe.

An exception to requirement RI003.14 is in chapter "Issuer requirements," section "Regional variations," subsection "Europe region," sub-subsection "Maestro chip-only cards."

Mastercard has put in place a timetable to remove the physical Magnetic Stripe from the back of the card globally, excluding Switzerland, with the following exemptions for which it is optional:

- Reloadable and non-reloadable prepaid card programs in the United States and Canada regions
- Non-reloadable prepaid card programs in the Asia/Pacific, Europe, Middle East/Africa, and the Latin America and the Caribbean regions

R	All	CT	RI427.22	Effective 1 April 2029, newly issued chip cards, except cards issued in Switzerland, must not have a physical Magnetic Stripe.
R	All	CT	RI428.22	Effective 1 April 2033, all cards in circulation, except in Switzerland, must support EMV chip technology and must not have a physical Magnetic Stripe.

Issuers in the following regions may optionally remove the physical Magnetic Stripe on newly issued chip cards:

- Canada
- Europe, except Switzerland
- Latin America and the Caribbean
- Middle East/Africa
- Asia Pacific

For the United States region, the physical Magnetic Stripe may be removed from cards issued after 1 April 2027.

Dual interface cards

ID-1 plastic cards that support both contactless and contact transactions are known as dual interface cards.

If contactless EMV mode is implemented on an ID-1 plastic card, the card must support both Magnetic Stripe and an EMV contact chip, that is, it must be a hybrid card.

Cards should be issued as hybrid dual interface and support both EMV contact and EMV mode contactless transactions.

R	All	CT CL	RI104.19	Contactless cards supporting EMV mode that are of ID-1 format must be dual interface cards supporting EMV contact chip transactions.
BP	All	CT CL	BI069.18	Cards should be issued as hybrid dual interface and support EMV mode on the contactless interface.

Newly issued Mastercard and Maestro cards must be dual interface cards except for the following cards, for which it is optional to support the contactless interface:

- Anonymous non-reloadable prepaid cards
- Cards with full metal body design
- Cards issued under the Mastercard Emergency Card Replacement (ECR) program

Issuers that issue multi-account cards with more than one application of a Mastercard brand on the contact interface can decide whether to include the same applications or a subset of them on the contactless interface. At least one Mastercard brand must be supported on the contactless interface.

R	MC	CT	RI381.22	Newly issued cards, except anonymous non-reloadable prepaid cards, cards with full metal body design, and cards issued under the ECR program, must be dual interface enabled with both EMV contact and EMV mode contactless functionality.
	MS	CL		

Requirement RI381.22 does not apply to Japan, the Canada region, and the United States region.

Refer to the regional variations in chapter "Issuer requirements," section "Regional variations," subsections "Asia/Pacific region" and "Europe region."

Mobile payment devices

Devices that have an input capability, for example, a keypad, output capability, for example, a user display, and independent power supply are referred to in this document as mobile payment devices. Examples include mobile phones, tablets, smart watches, and certain display cards.

Mobile payment devices perform contactless transactions in accordance with EMV Book C-2. The availability of a keypad and user display, and maybe other ways for the user to interact with the device, support opportunities not available on plastic cards. For example, mobile payment devices may support Consumer Device Cardholder Verification Method (CDCVM).

A single mobile may support multiple cards representing different products and issued by different issuers. Before a contactless transaction can be performed, the user must ensure that the required card is selected from the options. This may be a default card configured within the wallet to minimize the user interaction. This will be the Application Identifier (AID) selected by the terminal.

Mobile payment device solutions may be constructed whereby the application, parameters, and cryptographic keys to complete the transaction are securely provisioned onto the mobile in a manner similar to how all of the required data is personalized onto a plastic card. The M/Chip Mobile and Mastercard TEE Based Payment (MTBP) specification supports this kind of implementation. Alternatively, the mobile is empowered to perform a limited number of transactions independently. The Mastercard Cloud-Based Payments (MCBP) solution operates in this way. The requirements in this document for mobile apply to these types of implementations. More information and documentation about M/Chip Mobile, MCBP, and MTBP are available when licensed at digitallicensing@mastercard.com.

Along with supporting the NFC interface for contactless transactions, the mobile payment device can communicate over-the-air (OTA), which can enable new applications to be loaded, new products to be added, and offline risk controls to be managed. OTA communication is central to Mastercard Cloud-Based Payments to allow new session keys to be replenished as required.

Mobile payment devices may support alternative options for conducting payments such as In-app (Remote) payments using Digital Secure Remote Payment Transactions (DSRP) or Mastercard Consumer-Presented QR transactions.

More information on DSRP is available on Mastercard Connect®.

More information on Mastercard Consumer-Presented QR is available to MCBP licensees.

Nonmobile payment devices

Contactless functionality can be present in form factors other than traditional payment cards.

Examples of different form factors are key fobs and wristbands.

These nonmobile payment devices perform contactless transactions in accordance with *EMV Book C-2* exactly as contactless cards.

When cards use offline risk management features, an interaction with the card is required to manage the offline risk management counters. This cannot be performed in a contactless transaction because response data from the issuer is not returned to the card. Therefore, nonmobile payment devices are configured as online-only.

Support for EMV mode and Magnetic Stripe mode

A contactless card or contactless payment device using the Cirrus® or the Maestro® brand must support EMV mode transactions and must not support Magnetic Stripe mode transactions.

R	MS CI	CL	RI113.17	A contactless card or contactless payment device that supports the Cirrus or the Maestro brand must not support Magnetic Stripe mode for these AIDs.
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The mandate to remove the support of Magnetic Stripe mode contactless functionality is detailed in chapter "Issuer requirements," section "Issuer requirements," subsection "Regional variations," sub-subsections "Asia/Pacific region," "Europe region," "Latin America and the Caribbean region," and "Middle East/Africa region."

Also refer to the regional requirements in sub-subsections "Canada region" and the "United States region."

Branding, appearance, and physical requirements

Mastercard branding requirements can be found on the [Mastercard Brand Center](#).

Contactless card and contactless payment device delivery

Contactless data can be read by any reader that can power the contactless card or nonmobile payment device and send the correct commands.

Therefore, it is feasible that data could be captured while a contactless card or nonmobile payment device is in transit to the cardholder if the contactless interface is currently active. Issuers should consider appropriate control methods to reduce the risks and impact of card or data interception. This might be by disabling the contactless interface until the card has been activated by the cardholder.

M/Chip Advance has a feature that allows the card to be issued with the contactless interface disabled and only activated after a contact transaction has been initiated.

When a card that includes a contactless interface is issued to a cardholder for the first time, the cardholder may be unaware that a transaction can be completed without the contact chip being used. Issuers must alert cardholders to the presence of contactless functionality on their card.

BP	All	CL	BI094.14	Issuers should ensure that contactless transactions are not possible until the card has been activated by the cardholder.
R	All	CL	RI152.14	Issuers must alert cardholders that contactless functionality exists when issuing new payment devices.

Card expiration date

As security deteriorates over time, when using long life cards, issuers need to be prepared for security breaches that may result in financial loss, brand damage, and the immediate rollout of card replacement in the field.

To minimize this risk and to ensure a good level of security in the field, Mastercard recommends having an expiration date that does not exceed 3 years from the date of issuance.

BP	All	CT CL	BI237.22	Issuers should not issue cards with an expiration date that is more than 3 years after the card's issuance date.
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Issuer device approval processes

Launching a chip card or a contactless payment device requires the successful completion of certain vendor product approval processes, which focus on the product features, and issuer certification processes, which focus on the customization of the configuration of the product.

The goal of these processes is to ensure global interoperability and optimal brand acceptance.

All card programs must successfully complete the Vendor Product Approval Process and the Card Personalization Validation.

Mastercard vendor product approval processes

Mastercard has established a system of tests to maintain interoperability between chip cards, contactless payment devices, and terminals to help ensure that Mastercard products are accepted at Mastercard payment terminals worldwide.

Vendors are responsible for completing the testing requirements.

Approval processes of M/Chip card products

All M/Chip application cards issued are required to have Mastercard vendor product approval. Issuers are responsible for confirming that all products have received this approval.

A full card Letter of Approval (LoA) is only granted to a card when it has successfully completed all of the following cardholder device approval processes:

- Interface and application testing consists of functional testing that is performed by laboratories that have been accredited by Mastercard.
- Compliance Assessment and Security Testing (CAST) performs a security evaluation of the chip, considering the Integrated Circuit (IC), the operating system, and the payment application.
- Card Quality Management (CQM) ensures that chip cards are reliable, are manufactured to a consistent standard, and meet hardware manufacturing requirements.

When ordering cards from a card manufacturer, the issuer must ensure that the card manufacturer has a current LoA for the product being purchased. All products must have a valid LoA at the time the product is issued. A list of approved M/Chip products is on Mastercard Connect > Technical Resource Center > References.

Approval processes of NFC mobile payment products

There are three architectures for a Near Field Communications (NFC) mobile payment device in a Mastercard contactless mobile implementation:

- An NFC mobile payment device with an embedded Secure Element (SE)
- An NFC mobile payment device with a removable SE
- An NFC mobile payment device with Host Card Emulation (HCE) support, with or without Trusted Execution Environment (TEE)

The certification of an NFC mobile payment product is divided into two processes:

- Certification of the SE or Mobile Payment Application (MPA) in the case of HCE wallets
- Certification of the NFC mobile device

Approval processes of the SE payment products

The SE must be certified as a standalone component for use with any approved NFC mobile device. The SE solution must undergo functional testing (Mastercard Test Assessment) and CAST and EMVCo security evaluation. The results of the functional testing are summarized in a formal statement called the Test Assessment Summary (TAS), which is issued to the vendor.

If the SE security evaluation meets the requirements of the CAST Program, a CAST certificate with a certificate number (Card Certification Number [CCN] or Mobile Payment Certification Number [MPCN]) is issued with a validity period of 3 years.

Once the M/Chip implementation has obtained a CAST certificate and the TAS document, the corresponding Component Conformity Statement (CCS) can be requested from Mastercard.

Vendors incorporating the SE in a mobile device, including wearable devices, will therefore, partially or fully, inherit results when submitting their own product for approval by Mastercard.

Issuers using an SE solution are required to ensure that, when provisioning cards, the SE has passed the applicable Mastercard testing and the CCS is within the validity period.

For more details on the SE certification process and certificate life cycle, refer to the following documents:

- *M/Chip Mobile Secure Element Approval Guide*
- *Compliance Assessment and Security Testing Program Manual*

Alternatively, contact chip_certification_chd@mastercard.com or CAST@mastercard.com.

Restricted CCN or MPCN

When an SE-based product receives a CAST certificate (CCN or MPCN), it is normally valid for 3 years. This period may be extended by completing the CAST renewal or refresh security evaluation process. In case the renewal security evaluation is not performed, the vendor may submit a request for a restricted CAST certificate to extend product provisioning for up to 1 year. If the request is approved, a corresponding Restricted CCN (RCCN) or Restricted MPCN (RMPCN) will be granted.

While on a restricted certificate, the product can continue to be supplied to existing and planned customers. However, it must not be supplied to any new customers. The CAST-approved products list will also be updated to indicate that the SE product is restricted. Requests for a restricted certificate should be sent to CAST@mastercard.com, which will provide further information about the process.

The CCS linked to the previously issued CCN or MPCN will be extended for a period strictly the same as that of the restricted CAST certificate (RCCN or RMPCN).

Approval processes for Mastercard TEE Based Payment products

The Mastercard TEE Based Payment (MTBP) solution consists of an MPA communicating with a Trusted Payment Application (TA) implemented in a TEE. The complete solution must undergo functional testing, Mastercard Test assessment, User Interface (UI) testing, and a CAST and EMVCo Software-Based Mobile Payments (SBMP) security evaluation. The results of the functional testing are summarized in a formal statement called the Test Assessment Summary (TAS), which is issued to the vendor. If the security evaluation meets the requirements of the CAST Program, a CAST Security Test Assessment Summary (STAS) is issued with a validity period of 1 year.

Once the MTBP implementation has obtained a CAST STAS and functional TAS document, the corresponding CCS can be requested from Mastercard.

Vendors incorporating the TEE in a mobile device, including wearable devices, will therefore, partially or fully, inherit results when submitting their own product for approval by Mastercard.

Issuers using an MTBP service are required to ensure that, when provisioning cards to the MPA, the MPA has passed the applicable Mastercard testing and the CCS is within its validity period.

For more details on the certification process for MTBP, refer to the *MTBP Mobile Payment Application Approval Guide* or contact Software_Evaluations@Mastercard.com or CAST@mastercard.com.

Approval processes for Mastercard Cloud-Based Payments products

The Mastercard Cloud-Based Payment (MCBP) solution includes a device component referred to as the Mobile Payment Application (MPA), which is typically developed using a Software Development Kit (SDK). The complete solution must undergo:

- Functional testing
- Mastercard test assessment
- UI testing
- A CAST and EMVCo SBMP security evaluation

The results of the functional testing are summarized in a formal statement called the Test Assessment Summary (TAS), which is issued to the vendor. If the security evaluation meets the requirements of the CAST Program, a CAST STAS is issued with a validity period of 1 year.

Once the MCBP implementation has obtained a CAST STAS and functional TAS document, the corresponding CCS can be requested from Mastercard.

Issuers using an MCBP service are required to ensure that, when provisioning cards to the MPA, the MPA has passed the applicable Mastercard testing and the CCS is within its validity period.

For more details on the certification process for MCBP, refer to the *MCBP Mobile Payment Application Approval Guide* or contact Software_Evaluations@Mastercard.com or CAST@mastercard.com.

Restricted STAS/CCS in case annual security audit is not completed

When an MPA or SDK receives an approval in the form of a CCS, it is normally valid for 2 years. This period is subject to the successful completion of an annual CAST security renewal evaluation. Where the annual CAST security renewal evaluation is not completed by the renewal due date, on request, the STAS and CCS can be updated to a restricted state (Restricted STAS [RSTAS] and Restricted CCS [RCCS]).

While on an RSTAS and RCCS, the MPA or SDK can continue to be supplied to existing and planned customers. However, it must not be supplied to any new customers. A restricted STAS and CCS must be requested by sending an email to CAST@mastercard.com and copy Software_Evaluations@mastercard.com.

Approval processes of the NFC mobile device

The successful completion of EMV Mobile Level 1 Approval Testing is the prerequisite for EMVCo to issue an LoA to the product vendor for the NFC mobile device. They must list the technical details of the product and relevant components and any restrictions that may apply. Mastercard recognizes the LoA issued by EMVCo and does not operate its own testing program.

For questions related to the approval process for a mobile device, contact chip_certification_chd@mastercard.com.

R	All	CT CL	RI006.14	Issuers must ensure that all chip cards and contactless payment devices are covered by a valid LoA at the time they are issued.
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Approval processes for UI

The UI application evaluation process determines whether brand identifiers have been used in the relevant functions. This will also determine if the UI being evaluated is compliant with the requirements defined in the *Mastercard Brand Mark Guidelines*.

For any questions related to the approval process for a UI, contact Software_Evaluations@mastercard.com.

CPA vendor product approval process

CPA-based applications must be approved by EMVCo.

R	All	CT	RI007.14	All EMVCo CPA-compliant payment products must be approved by EMVCo.
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Issuer device personalization data

Guidance is given on the values required for certain data elements. For full details, refer to the *M/Chip Personalization Data Specifications For Contact and Contactless* manual and *Mastercard Profile Advisor* tool.

On dual interface cards, data may be interface-specific or shared by both interfaces.

Data objects must be personalized on a chip card or a contactless payment device in compliance with appendix A, "Data dictionary." If a data object is retrieved from the chip card or the contactless payment device in a template, it must be in the template as indicated. Data objects that are not to be retrieved in a template are indicated as such.

Customers that are interested in MCBP implementations can refer to the *Mastercard Cloud-Based Payments - Personalization Specification*.

R	All	CT CL	RI164.14	Data objects must be personalized on the chip card or the contactless payment device in accordance with appendix A, "Data dictionary."
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Token service providers, trusted service managers, device manufacturers, and issuers that are interested in M/Chip Mobile or MTBP implementations can refer to the *M/Chip Mobile Personalization Profiles* and *Mastercard TEE Based Payment - Personalization Specification and Standard Profiles*, respectively, which are available under license at digitallicensing@mastercard.com.

Card Personalization Validation

The personalization of each card and contactless payment device must be approved using the Card Personalization Validation (CPV) service before production.

CPV validates that the settings chosen by the issuer comply with the Mastercard requirements for a particular AID. A valid LoA is a prerequisite to initiate the Card Personalization Validation process. For more details, refer to the *Card Personalization Validation Guide*.

For products based on M/Chip Advance and MCBP 2.x, the Mastercard Profile Advisor tool must be used to define personalization profiles that meet both customer needs and Mastercard product rules. The tool defines the data elements required for the personalization profile and suggests values for these data elements. It produces profiles in line with the *M/Chip Personalization Data Specifications for Contact and Contactless* manual and may produce profiles compliant with M/Chip Card Personalization Standard Profiles for issuers willing to keep such configuration.

R	All	CT CL QR	RI011.22	CPV must be successfully completed for all issued cards and contactless payment devices, except for Mastercard Digital Enablement Service (MDES) implementations.
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Provisioning

Provisioning is the term used for the preparation and personalization of data onto a mobile or nonmobile payment device over-the-air (OTA).

Mastercard provides a provisioning service known as Mastercard Digital Enablement Service (MDES) that also tokenizes the funding Primary Account Number (PAN) to generate a unique digital PAN for each request. Provisioning includes:

- Personalized data preparation - MDES performs aggregation of the personalization data required to enable tokens. Data preparation is performed in line with the requirements of M/Chip Advance, M/Chip Mobile, MCBP, and MTBP Issuer Cryptographic Algorithms. This includes (EMV) key generation, which is necessary for transaction validation.
- Script generation - MDES encapsulates personalization data inside GlobalPlatform scripts for secure distribution. For devices with a Secure Element (SE), GlobalPlatform scripts are prepared using confidential keys to satisfy the secure access conditions.
- Package distribution - MDES prepares and delivers a package to perform digitization. This is provided by MDES to the Secure Element Issuer Trusted Service Manager (SEI TSM), Credentials Management System (CMS) Dedicated, or Wallet Provider for further transmission to the device.

More details on provisioning and digital services provided by Mastercard are available on Mastercard Connect.

Zero length tags

Data exchanged between a chip card or a contactless payment device and the terminal during a transaction uses TLV (Tag, Length, Value) encoding.

Some terminals may have problems with cards personalized with zero length tags, where the Length subfield is set to '00' and no value is present. For this reason, issuers are advised not to personalize zero length tags on their cards.

BP	All	CT CL	BI003.12	Chip cards and contactless payment devices should not be personalized with any tag that is read by the terminal during a transaction that has a length of zero and no associated data.
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Proprietary tags

Proprietary data objects included in the card personalization that are not documented in the Mastercard specifications may require dedicated functionality on the terminal. Issuers should be aware of potential conflicts when using such tags.

BP	All	CT CL	BI083.14	Issuers should be aware of the risks and limitations associated with using proprietary tags in their card personalization.
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Track 2 Equivalent Data

The Track 2 Equivalent Data of the primary application contains the same data objects that are present in the Track 2 of the Magnetic Stripe.

Components of this data object will also appear in individual data objects on the chip, for example, PAN and Expiration Date. The values in these individual data objects must be consistent with the value in the Track 2 Equivalent Data, which in turn must be consistent with the values on the Magnetic Stripe of the card if the application on the chip corresponds to the Magnetic Stripe. There may be more applications supporting different data.

An exception to this requirement for consistency is the Track 2 Discretionary Data. The CVC 1 value must not be present in the chip data to ensure that an authentic Magnetic Stripe cannot be produced from data read from the chip interface. The chip data should instead carry a Chip Card Validation Code (CVC). Refer to chapter "Issuer requirements," section "Issuer device personalization data," subsection "CVC values in chip data."

The Track 2 Discretionary Data may differ in other ways.

A second exception is that issuers may choose to use an Application PAN on the contactless interface, which is different from the PAN present on the contact interface or the Magnetic Stripe or that is displayed on the front of the card.

In cases where the contact interface supports multiple applications and these applications have different Track 2 Data, including PAN, the application whose Track 2 data contains the same PAN as that encoded on the Magnetic Stripe is called the primary application.

R	All	CT CL	RI016.14	The Track 2 Equivalent Data of the primary application must replicate the contents of the Magnetic Stripe, except for the Track 2 Discretionary Data and the Application PAN on the contactless interface that may be different.
R	All	CT CL	RI017.11	The individual data elements corresponding to the subfields within the Track 2 Equivalent Data must contain the same values as the actual subfields within the Track 2 Equivalent Data.

R	MS	CT	RI077.12	On Maestro and Cirrus cards that do not support CVC 1, some aspect of the discretionary data field within the Magnetic Stripe track must be unique to the Magnetic Stripe and unpredictable from the data in the Track 2 Equivalent Data on the chip. (Varying all or part of the PAN Sequence Number or Expiration date does not qualify as unpredictable.)
	CI	CL		

The padding of the Application PAN may be different for the Track 2 Equivalent Data and the data element Application PAN (tag '5A'). If the PAN on the card is composed of an odd number of digits, the Application PAN in tag '5A' will have one and only one trailing hexadecimal 'F' to ensure that it consists of a whole number of bytes, but the PAN in the Track 2 Equivalent Data must not be padded. The Track 2 Equivalent Data as a whole may be padded with one trailing hexadecimal 'F' to ensure that it consists of a whole number of bytes in length.

Application PAN

The Application Primary Account Number (PAN) identifies the main account linked to the application.

The PAN personalized on a chip card or contactless payment device might be the actual funding PAN of the underlying account, known as the Funding Primary Account Number (FPAN), or might be a digitized token of the FPAN, known as a Device Primary Account Number (DPAN).

Issuers may choose to use a value of the application PAN that is unique to a specific interface to prevent cross-contamination where data obtained from one interface is used to effect fraud through another. The issuer or an agent will map the application PAN used on the interface concerned onto the account concerned. In this document, *application PAN* refers to the PAN that is personalized in the chip data.

The appropriate Bank Identification Number (BIN) range for the product must be used, and this must be compatible with the AID of the card.

R	All	CT	RI070.11	The BIN range must be compatible with the product identified by the AID.
		CL		

If the PAN on the card is composed of an odd number of digits, it must be padded with a single trailing hexadecimal 'F' to ensure that it consists of a whole number of bytes.

If the PAN has an odd number of digits, the padding 'F' is included in the string of static data to be authenticated.

R	All	CT CL	RI019.11	PANs that have an odd length must be padded with a single trailing 'F'.
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Payment Account Reference

The Payment Account Reference (PAR) (tag '9F24') is a data object associated with an Application PAN.

The PAR allows acquirers and merchants to link transactions, whether tokenized or not, that are associated to the same underlying Application PAN. The PAR, which does not contain financially sensitive data, cannot be used for any financial purpose.

For details on how to get PAR, refer to the [Mastercard Developers](https://developer.mastercard.com/product/payment-account-management/) website at <https://developer.mastercard.com/product/payment-account-management/>.

BP	All	CT CL	BI127.20	All chip cards and access devices should be personalized with the Payment Account Reference (PAR) in tag '9F24'.
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NOTE: BI127.20 is not applicable to MCBP 2.x profiles. MCBP 2.x Personalization Specification mandates the use of PAR for all MCBP 2.x wallets.

PAN Sequence Number

The PAN Sequence Number enables the issuer to differentiate between cards or applications that have the same PAN.

Examples of are:

- Cards that have been renewed or replaced
- Multiple cards corresponding to the same account
- Different products on a multi-account card that use the same PAN

BP	All	CT CL	BI087.14	Issuers should not use the same combination of PAN and PAN Sequence Number on separate chip cards and contactless payment devices, even if linked.
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The PAN Sequence Number data element in EMV (tag '5F34') is two digits long. When a PAN Sequence Number is used in a Magnetic Stripe environment, it typically contains only one digit.

The Application PAN Sequence Number is normally used to calculate the Application Cryptogram Integrated Circuit Card (ICC) keys and Secure Messaging ICC keys. This is why it is essential that terminals and acquirers accurately forward the PAN sequence number in online authorization and clearing messages, to prevent unnecessary declines and disputes.

Application Effective Date

The Application Effective Date is the date from which the application may be used.

It is expressed in the format YYMMDD. Terminals check whether the Application Effective Date is later than the current date and, if so, must set the "Application Not Yet Effective" bit in the Terminal Verification Results (TVR).

R	All	CT CL	RI021.22	If present, the Application Effective Date of the primary application must match the "valid date" embossed on the physical card.
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NOTE: For MCBP and MTBP implementations, the Application Effective Date must have a date that is later than the Application Expiration Date. Issuers are required to use the date of 31 December 2049.

Application Expiration Date

The Application Expiration Date is the date after which the application may no longer be used. The expiration date of the card should be consistent across all technologies supported.

It is expressed in the format YYMMDD. DD must be set to the last day in the month indicated by MM. Terminals check whether the Application Expiration Date is earlier than the current date and if so, set the "Application Expired" bit in the TVR.

R	All	CT CL	RI023.11	The Application Expiration Date of the primary application must match the corresponding date (Track 1 or Track 2) encoded on the Magnetic Stripe or present on the physical card.
R	All	CT CL	RI024.11	The day in the Application Expiration Date must be the last day of the month indicated.

Cardholder name

The presence of Cardholder Name (tag '5F20') in the chip data is conditional on the presence in Track 1 of the Magnetic Stripe of contact cards.

When Track 1 is present on the Magnetic Stripe, the Cardholder Name (tag '5F20') is normally identical to the value of the cardholder name subfield in Track 1.

For card programs where the Cardholder Name is not known, such as instant issuance and anonymous prepaid cards, a dummy value can be used on the Magnetic Stripe and in the chip, for example, "{space}"/" in Track 1 and "MASTERCARD/CARD" in the chip.

Mastercard prohibits accessing the cardholder name in the data read through the contactless interface to prevent unauthorized disclosure. Mastercard recommends using a space character followed by the surname separator " /" in the Track 1 Data.

To avoid the risk that files intended to be used for contact transactions are read through the contactless interface, M/Chip Advance has a feature that certain files, for example, the file that contains the cardholder name or the Track 1 discretionary data, may be filtered such that they cannot be read through a specific interface.

R	All	CL	RI137.14	The name of the cardholder must not be readable over the contactless interface.
R	All	CL	RI229.22	Issuers must use either a space followed by a slash (" /") or coding that complies with the Name field as defined for Track 1 data in ISO/IEC 7813 for the cardholder name in the data read through the contactless interface.
R	MC MS	QR	RI389.19	For Mastercard Consumer-Presented QR transactions, the actual cardholder's name must not be encoded in tag '5F20'.

CVC values in chip data

The CVC is a three-digit value that the issuer algorithmically derives to make it more difficult for counterfeiters to create cards or alter data and use them for fraudulent purposes.

Mastercard defines three types of CVCs for chip cards:

- CVC 1 - Value encoded on the Magnetic Stripe of the card
- CVC 2 - Value printed on the back of the card
- Chip CVC - Value encoded in the Track 2 Equivalent Data (EMV tag '57'), Track 1 Discretionary Data (EMV tag '9F1F'), or Track 2 Discretionary Data (EMV tag '9F20') field of either interface on the chip

NOTE: Track 1 Discretionary Data and Track 2 Discretionary Data are optional data elements on the chip.

Issuers must derive the value for Chip CVC in a different way than the CVC 1 encoded on the Magnetic Stripe. This prevents compromised chip data from being used to fraudulently create valid counterfeit Magnetic Stripe cards. However, due to the pseudo-random method by which the three-digit values for CVC 1, CVC 2, and Chip CVC are generated, there is a 1 in 1,000 chance that the CVC 1 and Chip CVC values for a new or reissued card will be identical. If by chance the CVC 1 and Chip CVC values are the same, then the use of these values for a new or reissued card is acceptable.

To generate the Chip CVC, issuers may either use their own proprietary method or a method described by Mastercard. For more information on implementing the Chip CVC, refer to chapter "Issuer requirements," section "Issue host requirements," subsection "Chip CVC validation" and the *Security Rules and Procedures* manual.

R	All	CT CL	RI020.14	If CVC 1 is present on the Magnetic Stripe, chip cards must use a Chip CVC derived in a different way than the CVC 1 in every data object where it is present. CVC 1 must not appear in the chip data.
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Maestro and Cirrus cards that do not contain a CVC 1 do not need to contain a Chip CVC. However, refer to the requirement in chapter "Issuer requirements," section "Issuer device personalization data," subsection "Track 2 Equivalent Data" and the information about unpredictable data on the Magnetic Stripe.

Service codes

Particular values for the service code on the Magnetic Stripe are used to indicate that a card is hybrid. This allows a hybrid terminal to recognize a chip card if the Magnetic Stripe is swiped so it can prompt the merchant or cardholder to use the chip reader.

Multi-account chip cards containing at least one international application must ensure that the service code on the Magnetic Stripe indicates that the card contains a chip capable of supporting international transactions.

A value for the service code may be found several times on a card, for example:

- On the Magnetic Stripe of the card in both Track 1 and Track 2
- In Track 1 Data (tag '56') and Track 2 Data (tag '9F6B') accessed through the contactless interface
- In Track 2 Equivalent Data (tag '57') accessed through the contactless interface
- In Track 2 Equivalent Data (tag '57') accessed through the contact interface

Cards should be personalized with a service code appropriate for the product.

The service code values used in the contactless application should be consistent in each data object where the service code appears. Although not recommended, contactless issuers may choose to use service code values in the contactless application that differ from those used on the Magnetic Stripe of the same card. If the issuer does use a different service code value on the contactless interface, the value may be acted on by some terminals. In particular, terminals that process the service code may reject international cards that have a service code value starting with 5 (national use only).

R	All	CT	RI004.11	Hybrid cards must have a service code on the Magnetic Stripe containing either: • "2xx", indicating an international card with a chip, or • "6xx", indicating a domestic-only card with a chip
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R	All	CT	RI073.13	Multi-application cards that support at least one international application must contain a service code value "2xx" on the Track 2 of the Magnetic Stripe.
BP	All	CT CL	BI074.14	Issuers should use a value of the service code appropriate for the product.
BP	All	CT CL	BI075.14	Issuers should use the same value of the service code each time the service code appears on the card.

Third Party Data

Third Party Data may be used by a terminal for proprietary processing.

Issuers that intend to participate in a scheme using this data object must request a Unique Identifier from Mastercard. Contact chip_requirements@mastercard.com. A subfield of this data object is also used to carry the Device Type. For contactless cards and payment devices, the presence of Device Type in the Third Party Data object is mandatory. It is not necessary to request a Unique Identifier from Mastercard if the Third Party Data is only used to support the Device Type. Refer to chapter "Data requirements" for more information.

The Third Party Data must be added to the File Control Information (FCI) Issuer Discretionary Data that is returned during application selection.

R	All	CL	RI139.24	Contactless cards and contactless payment devices must be personalized with the Device Type present in the Third Party Data (tag '9F6E') on the contactless interface.
R	All	CT CL	RI385.19	If present, the Third Party Data must be included in the FCI Issuer Discretionary Data that is returned when the application is selected.
R	MC MS	QR	RI390.19	Access devices supporting Mastercard Consumer-Presented QR transactions must include the Device Type in the Third Party Data (tag '9F6E').

Issuers should be aware that the contents of Third Party Data can be freely read and, therefore, should not contain sensitive cardholder information.

BP	All	CT CL	BI086.14	Issuers should respect relevant local data privacy laws when personalizing the Proprietary Data subfield of Third Party Data on the card.
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Contactless EMV mode personalization requirements

Some older versions of the contactless reader specifications used a predefined file structure in the contactless card or contactless payment device. If the predefined file structure is not followed, the Application File Locator (AFL) specified for the predefined file structure must not be used.

R	All	CL	RI141.14	If data objects are not organized according to the rules specified for the predefined file structure, the predefined values for the AFL must not be used.
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Contactless Magnetic Stripe mode personalization requirements

This section applies to contactless issuers that support Magnetic Stripe mode transactions.

In Magnetic Stripe mode transactions, the terminal reads a single record from the contactless interface on the card. That record contains all required data objects to perform the transaction.

The first record of the file Short File Identifier (SFI) 1 must include the data objects necessary to perform the contactless Magnetic Stripe mode transactions.

Track 1 Data and Track 2 Data must be personalized with the same values of PAN and Expiration Date. The last digit of both Track 1 and Track 2 must not be used by the issuer, as this is used by the terminal to indicate the number of digits of the Unpredictable Number (nUN). The length of the nUN must not be fewer than three digits and Mastercard strongly recommends that it be four or five digits.

The following table shows recommended uses of the available positions in Track 2 Data.

Available positions	Positions used for			
	ATC	UN	CVC 3	nUN
10	2	4	3	1
11	3	4	3	1
12	3	4	4	1
13	3	5	4	1

Issuers that support dynamic CVC 3 validation in stand-in should use three digits for the value of Application Transaction Counter (ATC) to maximize acceptance. Refer to requirement RI335.16 in chapter "Issuer requirements," section "Issuer host requirements," subsection "Stand-In Processing." However, that should not result in reducing the number of the digits used for the Unpredictable Number (UN).

Issuers that use contactless on-behalf CVC 3 validation services can refer to the *Contactless On-behalf Services Guide* for more details on the values and the positions of Track Data.

R	MC	CL	RI147.16	Record 1 of SFI 1 must contain the data to perform a Magnetic Stripe mode transaction.
R	MC	CL	RI168.14	Track 1 Data and Track 2 Data must be personalized with the same values of PAN and Expiration Date.
R	MC	CL	RI148.14	The last digit of both Track 1 Data and Track 2 Data must not be used by the issuer.
R	MC	CL	RI150.17	The Unpredictable Number used for the dynamic CVC 3 must be at least three digits in length.
BP	MC	CL	BI110.17	The Unpredictable Number used for the dynamic CVC 3 should be four or five digits.

Online processing

Cards may be configured to manage risk in different ways. This may be through offline risk controls in the application, requiring online authorization by the issuer host, or both.

The exact parameters and data configuration required to achieve the different options are payment application dependent. Refer to the appropriate application Issuer Guide for more information.

Offline-capable configuration

Cards that are personalized to be able to approve transactions without always requesting online authorization are referred to as offline-capable.

Mastercard cards and Maestro cards may be personalized as offline-capable.

Refer to the regional requirements for the United States region in chapter "Issuer requirements," section "Regional variations," subsection "United States region."

Online-preferring configuration

Online-preferring chip cards always request an online authorization but may accept transactions offline when an online connection is not available. The cards are therefore offline-capable but prefer to seek online authorization.

Mastercard cards and Maestro cards may be personalized as online-preferring.

Mastercard recommends this configuration for contact cards where the issuer wants to achieve an optimal balance between tight credit control and good cardholder acceptance. It is also recommended for the contactless interface.

BP	MC	CT	BI235.22	Mastercard and Maestro chip cards should be configured as online-preferring.
	MS	CL		

Online-only configuration

An online-only card never approves a transaction offline. If the terminal indicates that it is not online-capable, the card declines offline.

This may be achieved by personalizing the card with the support for Combined DDA/AC Generation (CDA)-only, configured to always go online or no support for offline data authentication on the card.

The support of offline Card Authentication Method (CAM) on an online-only card may have greater acceptance than a card configured without any support for offline CAM because it may enable transactions to be completed offline at the acquirer's risk. This can occur in certain acceptance situations, for example, in environments that do not have an online connection, such as vending machines. In addition, the support for CDA on online-only cards protects the issuer against the fraudulent manipulation of data on the card-to-terminal interface, known as a wedge device attack. This is especially important if the card supports Offline Personal Identification Number (PIN).

Maestro and Mastercard contact cards may be personalized as online-only. However, Mastercard does not recommend this because this configuration may limit the acceptance of the card. Specifically, online-only cards may not be accepted in the following environments:

- Limited amount unattended terminals (CAT Level 3)
- At any merchant type during temporary communication outages

Therefore, Maestro cards and Mastercard contact cards should be configured to have some offline capability in order to enable systematic acceptance at terminals that are not able to obtain an online authorization.

BP	MC	CT	BI019.20	The contact interface of Mastercard and Maestro chip cards should not be configured as online-only.
	MS			

Cirrus cards must be configured as online-only.

R	CI	CT	RI045.22	Cirrus cards must be configured as online-only.
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Online-only configuration is also required for access devices and contactless payment devices that have technical restrictions to reset the offline risk management counters.

R	All	CL	RI371.22	Access devices and contactless payment devices that do not provide a mechanism to reset the offline risk management counters must be configured as online-only.
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Also, MCBP, MTBP, and M/Chip Mobile must be configured as online-only.

R	All	CL QR	RI430.22	MCBP, MTBP, and M/Chip Mobile applications must be configured as online-only.
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Contactless Magnetic Stripe mode transactions are always authorized online, either in real time or deferred. The card has no input into the decision to seek authorization.

Offline-only configuration

Dual interface Mastercard and Maestro cards must not be configured as offline-only.

As some contactless-enabled terminals operate exclusively online, EMV mode contactless cards and contactless payment devices should be configured to support online transaction approval.

R	MC MS	CT CL	RI431.22	Dual interface Mastercard and Maestro cards must not be configured as offline-only.
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Application selection

EMV defines a mechanism for a terminal to determine a list, the Candidate List, of applications that are supported by both the card and the terminal.

The application selection mechanism is described in more detail in the *M/Chip Program Guide*.

The different applications are identified by the Application Identifier (AID). A standard Mastercard AID is 7 bytes long. An extended AID is longer than 7 bytes. Terminals normally find applications to include in the Candidate List using the standard AID and will find all extended AIDs based on the same standard AID.

Contact terminals may perform application selection using the Payment System Environment (PSE) and must support application selection using the list of AIDs method. Therefore, it is optional whether cards contain a PSE.

If any of the following data objects are present in both the Application Definition File (ADF) of the application and data referenced by the PSE, they should have the same value:

- Application Label (tag '50')

- Application Priority Indicator (tag '87')
- Application Preferred Name (tag '9F12')
- Issuer Code Table Index (tag '9F11')
- Language Preference (tag '5F2D')

BP	All	CT	BI111.15	If any of the following data objects are present, it should have the same value in both ADF and PSE, if present: • Application Label (tag '50') • Application Priority Indicator (tag '87') • Application Preferred Name (tag '9F12') • Issuer Code Table Index (tag '9F11') • Language Preference (tag '5F2D')
BP	All	CT	BI101.14	Issuers should not pad the PSE.

Contactless-enabled terminals perform application selection using the Proximity Payment System Environment (PPSE) on the card. All contactless cards and contactless payment devices must contain a PPSE.

R	All	CL	RI119.22	All contactless cards and contactless payment devices must contain a PPSE.
BP	All	CL	BI106.14	Issuers should not personalize the Kernel Identifier (tag '9F2A') in the PPSE directory entry on the card.

The application selection process can be different for cards and terminals processing transactions in the Europe region, the Latin America and Caribbean region, and the United States region. For more details, refer to the regional sections.

There is a distinction between the EMV application selection process and the user process for selecting applications from within a wallet. Where available, this must be initiated by the cardholder before the contactless transaction starts and cannot be performed during a transaction. The AID used within the wallet may have an extended form to enable the user to differentiate between different cards while the single AID included in the PPSE following user selection may use just a 7-byte AID appropriate to the product selected.

Application Identifiers

The Application Identifiers (AIDs) supported for Mastercard products are shown in chapter "Data requirements," section "Application identification," subsection "Application identifiers." The same AIDs are used for both contact and contactless transactions.

Identification of the product in contactless transactions uses the 7-byte product AID without any extension. An extended AID may be used by issuers and is considered as a successful match by the terminal when partial AID matching is supported. However, Mastercard recommends not

to use any AID extension, as some legacy contactless-enabled terminals do not support partial AID matching.

BP	All	CL	BI076.22	AID extension should not be used on the contactless interface.
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If the same account is accessed through the contact and contactless interfaces, the AID used on each interface might be different. The contact AID might contain an AID extension, but the contactless AID excludes this extension.

Mobile payment devices that support multiple product instances should use the process described in chapter "Data requirements," section "Application identification," subsection "Application identifiers" to ensure that each AID is unique even if products for multiple banks are contained in the same wallet. When a specific product is chosen over the user interface, the mobile payment application creates a Proximity Payment System Environment (PPSE) that contains the appropriate AID or AIDs without any extension.

Multiple application support

Multi-account cards support more than one instance of a Mastercard product that might be used for a contact or contactless transaction.

For example, the card may support both a personal Mastercard application and a business Mastercard application or both a Maestro application and a Mastercard application when allowed by the product rules. In some situations, two different applications access the same underlying funding account.

This document does not discuss cards or contactless payment devices that support applications for nonpayment purposes.

Sometimes there may be two or more applications on a card that use the standard AID but have different extended AIDs. All applications corresponding to the same brand must use an extended AID. Cards must not contain one application identified by the standard AID only and other applications with extended AIDs.

When using explicit application selection for contact transactions, terminals select these applications by selecting the standard AID and then using the SELECT command again with the Next Occurrence option.

R	All	CT	RI013.11	Cards that carry more than one instance of a Mastercard brand on the chip must use the corresponding brand AID with different extended AIDs present in each of them.
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Cardholders using mobile payment devices that contain multiple product instances must select the product they wish to use for a transaction before the mobile payment device is tapped. If they do not make the selection, then the default product will be used.

For contactless transactions, the selected applications will be the only ones to appear in the PPSE for that transaction.

For a Mastercard Consumer-Presented QR transaction, the generated QR code can only contain data corresponding to a single application.

R	MC MS	QR	RI391.24	The generated QR Code must only contain data corresponding to a single application.
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Issuers of cards with multiple applications that access different underlying funding accounts should be able to identify the application selected from data within the mandatory data set in network messages. This means that different applications should use different PANs or PAN Sequence Numbers. The issuer may not receive the AID selected at the time of the transaction in authorization and clearing messages.

BP	All	CT CL	BI004.22	Issuers of multi-account cards that access different underlying funding accounts should use different PANs or PAN Sequence Numbers for the different applications.
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Issuers should ensure that the cardholder knows that cardholder selection may not be available on some terminals. Terminals that do not support chip or chip terminals that do not support cardholder selection may not enable the use of all applications.

Application Priority Indicator

If there is more than one application in the Candidate List, the terminal normally sorts the applications according to the issuer-defined Application Priority Indicator (API).

For contact transactions, applications are presented to the cardholder in the order the issuer has defined in the API when a choice needs to be made.

For contactless transactions, the terminal automatically selects the application with the highest API in the Candidate List. Issuers must set a unique value for the API for each application. For mobile wallet contactless transactions, either the default card or the card selected by the cardholder is used.

The primary application should have the highest application priority to ensure this is the product automatically used in both Magnetic Stripe environments and chip terminals that do not support cardholder selection.

R	All	CT	RI015.15	The Application Priority Indicator must be present in the File Control Information (FCI) of each application in a multi-application card and in each entry in the record referenced by the PSE.
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R	All	CL	RI120.14	Issuers must set a unique value for the Application Priority Indicator in each directory entry in the FCI of the PPSE.
BP	All	CT CL	BI005.15	Cards that support multiple applications should attach the highest application priority to the primary application.

Refer to the regional requirements for the Europe region, the United States region, the Middle East/Africa region, and the Asia/Pacific region in chapter "Issuer requirements," section "Regional variations."

Cardholder Confirmation

EMV allows an issuer to request that before a terminal automatically selects an application, the cardholder must confirm the application to be used for a contact transaction. This is indicated in a bit in the Application Priority Indicator.

The Cardholder Confirmation setting contained in the Application Priority Indicator is only relevant when the terminal either has only one application in the Candidate List or when the terminal does not support cardholder selection. Cardholder confirmation will always slow down the transaction flow by introducing a manual step.

For a single application card, Cardholder Confirmation serves no useful purpose. The cardholder has decided to use the application when they use the card. For a multi-application card, Cardholder Confirmation would only be performed when the card is used at a terminal that supports only one of the applications supported by the card. If the "Cardholder Confirmation" bit is set, rather than automatically selecting the application, the terminal will request confirmation from the cardholder. This means that the cardholder is being alerted as to which application will actually be used. For example, if a card that supports both a Mastercard application and a Maestro application is used at a terminal that supports Maestro but not Mastercard, the cardholder is alerted that Maestro will be used. If the cardholder intended to spend from their Mastercard account rather than from their Maestro account, they can choose not to proceed with the transaction because the Mastercard option is not available.

Cardholder Confirmation is never used in the following scenarios:

- When the card and terminal mutually support multiple applications and the terminal supports Cardholder Selection. If there is more than one choice in the Candidate List, the cardholder will be asked to choose the application whether or not the "Cardholder Confirmation" bit is set on any of the available applications.
- On a multi-account card that supports multiple occurrences of the same AID, for example, Mastercard credit and Debit Mastercard, at terminals supporting cardholder selection. In this scenario, the terminal can never have only one Mastercard application in the Candidate List.

If an application requires Cardholder Confirmation but the terminal does not support it, then that application cannot be selected. There may be other applications available. Otherwise, the transaction will be terminated.

For this reason, Mastercard recommends that issuers do not set the "Cardholder Confirmation" bit on the contact interface.

Cardholder Confirmation must not be set in the API on the contactless interface.

R	All	CL	RI121.14	Issuers must not set the "Cardholder Confirmation" bit in the Application Priority Indicator in the File Control Information (FCI) of the PPSE.
BP	All	CT	BI006.11	Issuers should not set the "Cardholder Confirmation" bit (bit 8) in the Application Priority Indicator.

Approval processes of the NFC mobile device

The successful completion of EMV Mobile Level 1 Approval Testing is the prerequisite for EMVCo to issue an LoA to the product vendor for the NFC mobile device. They must list the technical details of the product and relevant components and any restrictions that may apply. Mastercard recognizes the LoA issued by EMVCo and does not operate its own testing program.

For questions related to the approval process for a mobile device, contact chip_certification_chd@mastercard.com.

R	All	CT CL	RI006.14	Issuers must ensure that all chip cards and contactless payment devices are covered by a valid LoA at the time they are issued.
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Approval processes for UI

The UI application evaluation process determines whether brand identifiers have been used in the relevant functions. This will also determine if the UI being evaluated is compliant with the requirements defined the *Mastercard Brand Mark Guidelines*.

For any questions related to the approval process for a UI, contact Software_Evaluations@mastercard.com.

Application Label

Each AID must have an associated Application Label. An Application Label is a text field that can be displayed on the terminal to make it easier for cardholders to select a payment application during a transaction. It may appear on any receipts.

The Application Label is mandatory in Mastercard applications.

Issuers must use one of the defined values for the Application Label and must not use an Application Label for a different product, for example, Mastercard on a Maestro card. For the mandated values, refer to chapter "Data requirements," section "Application identification," subsection "Application identifiers."

R	All	CT	RI122.19	Issuers must use one of the available values as defined by Mastercard for the Application Label.
		CL		
		QR		
R	All	CT	RI420.20	Issuers must not use an Application Label for a different product.
		CL		
		QR		

Refer to the regional requirements in chapter "Issuer requirements," section "Regional variations," subsection "United States region."

Application Preferred Name

The Application Preferred Name is an optional field defining an alternative label for an application that the issuer would like to have displayed during application selection. The Application Preferred Name is encoded using the character set indicated in the Issuer Code Table Index.

If the Application Preferred Name is present and if the terminal supports the character set indicated by the Issuer Code Table Index, the terminal displays the Application Preferred Name during application selection. Otherwise, the terminal displays the Application Label.

There are no requirements specifying the values or character set to be used for the Application Preferred Name, and issuers may choose their preferred value for this field. However, the Application Preferred Name, if present, should contain a value that is appropriate for the application.

Refer to the regional requirements in chapter "Issuer requirements," section "Regional variations," subsection "United States region."

Card authentication

Card authentication is the process by which either the terminal or the issuer checks the genuineness of the card and the integrity of data on the card and terminal interface.

Online card authentication requirements

Online CAM is the process that enables issuers to authenticate their card during an online chip transaction. All cards must support online CAM by generating a dynamic cryptogram that is delivered to the issuer in the authorization.

Contactless Magnetic Stripe mode transactions generate a dynamic CVC 3 that is delivered to the issuer in the authorization.

In MCBP implementations, for both contactless and Mastercard Consumer-Presented QR transactions, there are two card cryptograms to validate: a User and Mobile Device (UMD) authentication cryptogram and a Mobile Device (MD) authentication cryptogram. Issuers must ensure that they always verify MD and also that they verify UMD, where appropriate.

The results of the two cryptogram validations are forwarded to the issuer's authorization system.

Issuers must always validate these cryptograms before approving transactions. The validation may be performed by the issuer or by a Mastercard pre-validation service.

Table 5:

R	MC	CL	RI128.14	All contactless cards and contactless payment devices that support contactless Magnetic Stripe mode transactions must support dynamic CVC 3.
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NOTE: Online card authentication requirements performed by the issuer host are in Chapter 3 (Issuer) - > "Issuer host requirements" -> "Online request" -> "Card authentication" section.

Offline card authentication requirements

Dynamic authentication provides the strongest counterfeit protection.

Although Dynamic Data Authentication (DDA) ensures that the payment device being used is genuine, only Combined DDA/AC Generation (CDA) provides full security and protects against wedge device attacks. All new cards that support offline CAM must be personalized to support CDA, meaning cards may support CDA only or both DDA and CDA. However, any card supporting DDA but not CDA must be out of circulation.

Static Data Authentication (SDA) is vulnerable to fraud attacks. That is why any card containing SDA must no longer be in circulation.

BP	All	CT	BI118.22	Chip cards that support offline CAM should support DDA.
R	All	CT	RI226.20	All chip cards containing SDA must be out of circulation.
R	All	CT	RI227.22	Any card containing DDA but not CDA must be out of circulation.
R	All	CT	RI228.22	Newly issued and reissued chip cards that support offline CAM must be personalized to support CDA.

For contactless transactions, CDA is mandated for all contactless cards and contactless payment devices, even for online-only profiles, to ensure acceptance at transit terminals, which must require CDA.

SDA and DDA are not supported on the contactless interface.

Local Dynamic Authentication (LDA) is a form of CDA that is used on some online-only mobile payment devices to provide local authentication at the terminal. LDA enables offline authentication without the need to store private keys in a secure element within the device. Mobile payment devices that support LDA are personalized as online-only in data that is signed by the card certificate (Issuer Action Code).

R	All	CL	RI123.19	Contactless cards and contactless payment devices must support CDA on the contactless interface.
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Public key requirements for offline CAM support

To support the use of offline CAM, Offline enciphered PIN, or both, issuers need to be able to produce issuer keys and have these keys certified by the Mastercard Certification Authority.

Issuers that support DDA, CDA, Offline enciphered PIN, or a combination of these need to produce a public/private key pair for each card. The issuer uses its private key to produce public key certificates for the card public keys.

The choice by the chip issuer of issuer-level and card-level key lengths and of related parameters should consider a balance between security risks, both fraud losses and adverse publicity, and functionality, including transaction time.

RSA algorithm keys, including both ICC Keys and Issuer Keys, must be chosen to be sufficiently long to withstand factorization attacks during the lifetime of both the key and the data protected by the key. Therefore, minimum RSA key lengths increase over time as the capabilities to factorize the public key improve. Such improvements are inevitable, even in the absence of any development in factorization algorithms, simply due to the ever-increasing power and availability of computers.

Compromise of a single card key would allow fraudsters to create clones of the compromised card to perform fraudulent offline transactions against the account of the compromised card until the expiration of the public key certificate associated with the compromised card key or until the card was blocked.

In contrast, compromise of an issuer key would allow fraudsters to create clones that could perform fraudulent offline transactions against any account under the issuer BIN that the issuer key is certified with and until the issuer public key certificate expires.

As the compromise of a single issuer key would impact the issuer's operations at a higher level, issuer key lengths should be longer than card key lengths.

Although longer keys are more secure against factorization attacks, they have worse performance characteristics such as marginally slower transaction times. Mastercard recommends a key length of 144 bytes (1,152 bits) but 176 bytes (1,408 bits) is preferred.

For performance reasons, issuers may use a card public exponent of 3.

Local considerations of the terminal base, national cryptographic policy, and local requirements may condition an issuer's choice of key lengths and public exponents.

When certifying an issuer public key, issuers should select the shortest Payment System Public Key (PSPK) with an expiration date that accommodates the cards to be issued. The key expiration date must not be earlier than the expiration date of the card.

The length of the issuer key cannot exceed the length of the PSPK that is used to certify it. Issuers should use an issuer key with the same length as the certifying PSPK, except when the PSPK is 1,984 bits. In this case, issuers should use the maximum issuer key length of 1,976 bits, although 1,920 bits may be preferred to align with 8-byte boundaries. For performance reasons, issuers may again use a public exponent of 3.

BP	All	CT CL	BI061.12	Issuers should select the shortest PSPK with an expiration date that accommodates the expiration dates of the cards to be issued.
BP	All	CT CL	BI058.18	Issuers should use an issuer key with the same length as the certifying PSPK, except when the PSPK is 1,984 bits. In this case, issuers should use the maximum issuer key length of 1,976 bits, although 1,920 bits may be preferred to align with 8-byte boundaries.
BP	All	CT CL	BI055.24	Card key lengths should be at least 144 bytes (1152 bits.)

Mastercard recommends using the same ICC Key pair for transactions on the contact and contactless interface. The ICC Public Key Certificate cannot be shared between the contact and contactless interface even if the same keys are used because some of the data elements signed in the certificate are different.

BP	All	CT CL	BI078.14	Issuers should use the same Issuer and ICC Public Keys across both the contact and contactless interface.
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For other requirements for public keys supporting offline CAM, refer to the latest version of the annual announcement "Updated Payment System Public Keys for M/Chip," the *Public Key Infrastructure (PKI) - Overview* manual, and the *Security Guidelines for Chip, Contactless, and Tokenized Transactions* manual.

Certificate expiration

Public key certificates contain an expiration date, which terminals check as part of offline CAM.

If the current date is later than a certificate's expiration date, the terminal considers the certificate invalid and offline CAM fails.

The expiration date of each public key certificate used to produce the card is checked for:

- The issuer's public key certificate
- The ICC public key certificate, for DDA and CDA cards
- The ICC PIN encipherment public key certificate, if present

R	All	CT CL	RI032.14	Cards must not have an expiration date later than the expiration date of any of the public key certificates they contain.
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Non-expiring Maestro and Cirrus cards

Maestro and Cirrus rules allow issuers to produce non-expiring cards that have an expiration date of December 2049.

If non-expiring cards were to contain public key certificates, those certificates would expire before the card, resulting in a failure of offline CAM. Therefore, non-expiring cards must not support offline CAM. In addition, these cards must be configured to send all transactions online and decline if online authorization is not possible.

R	MS CI	CT	RI031.13	Non-expiring cards must not support offline CAM and must be configured to be online-only.
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Cardholder verification

All cards must follow specific Cardholder Verification Method (CVM) requirements according to the brand on the card and the interface being used.

For more information, refer to chapter "Data requirements." For full details about CVM Lists for each card brand and interface, refer to the *M/Chip Personalization Data Specifications for Contact and Contactless* manual.

For contact and contactless EMV mode transactions, the CVM used for a transaction is determined by the terminal based on the terminal capabilities and the issuer preferences expressed in the Application Interchange Profile (AIP) and the CVM List as well as dedicated functionality in the cardholder device.

Contactless transaction amounts equal to or below the CVM Limit

For contactless transaction amounts equal to or below the appropriate CVM Limit, no cardholder verification is required regardless of the setting of the Service Code for Magnetic Stripe mode or CVM List for EMV mode transactions. For certain transaction types, for example, cash withdrawal and purchase with cash back, this limit may be set to zero.

Some mobile payment or biometric device implementations may require CDCVM for transaction amounts equal to or below the appropriate CVM Limit, for example, in the case of lost and stolen protection. On terminals supporting Single Tap and PIN Request, a PIN may be requested for transaction amounts below the CVM Required Limit.

Magnetic Stripe mode transactions

For Magnetic Stripe mode transactions, the cardholder verification method is determined by the terminal. The terminal is not required to refer to the Service Code, which is in multiple data elements. For a mobile payment device and Biometric Cards, support for and performance of CDCVM is communicated to the terminal as part of the transaction data.

EMV mode transactions

For EMV mode transactions, the CVM is determined by the reader application in the terminal based on the terminal capabilities and the issuer preferences expressed in the AIP and the CVM List as well as dedicated functionality in the cardholder device.

DSRP and Mastercard Consumer-Presented QR transactions

For DSRP and Mastercard Consumer-Presented QR transactions, CDCVM is the only cardholder verification method that is permitted.

International transactions

For international transactions, CVM List entries should not rely on the use of the X and Y values to influence the availability of a particular CVM, as the terminal may not understand the card application currency. This means that condition codes '06', '07', '08', or '09' should not be used.

BP	All	CT CL	BI081.14	CVM List entries should not make use of the X and Y values to influence the availability of a particular CVM.
R	All	CT	RI043.11	If the CVM List contains an amount-based condition, that is, condition code values '06', '07', '08', or '09', the card must contain the Application Currency Code.
R	All	CL	RI177.14	Contactless cards and contactless payment devices must contain the Application Currency Code.

Offline PIN is not supported for contactless transactions. Issuers must not include Offline PIN options in the CVM List read through the contactless interface.

R	All	CL	RI130.14	Contactless cards and contactless payments devices must not support Offline plaintext PIN or Offline enciphered PIN over the contactless interface.
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CDCVM enables cardholder verification to be completed on a mobile payment device using the input and output options of the mobile payment device. The CDCVM can be specific to an individual card profile. In this case, where users have more than one card digitized in the Mobile Payment Application, they may potentially have different verification methods and values for each one.

Issuers or wallet providers can also use a wallet-level CDCVM, where the CDCVM is specific to an individual wallet and therefore all cards stored in the wallet share the same verification method. A device-level CDCVM is a CVM method that is shared between many applications on the Mobile Device. For example, the use of device unlock can be used as CDCVM. The wallet-level or device-level CDCVM is called a shared CVM.

Support for CDCVM is recommended for all mobile payment devices. A mobile payment device that supports CDCVM must also support CDA.

The current mobile implementations of M/Chip Mobile, MCBP, and MTBP are capable of supporting CDCVM ALWAYS, FLEXIBLE CDCVM, and CARD-LIKE CVM Models.

While MCBP used to be a CDCVM ALWAYS product, the latest specifications of MCBP provide the feature of CARD-LIKE CVM profiles without the support of CDCVM. Profiles with CDCVM enabled are still supported.

BP	All	CL	BI080.14	CDCVM should be supported on all contactless-enabled mobile payment devices.
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For Mastercard Consumer-Presented QR, all CVM Models will be offered as options. But in regions where there are specific government or regulatory requirements, support of certain CVM Models may be mandated or prohibited. For more information on allowed CVM Models and QR display times, refer to the *Mastercard Consumer-Presented QR sections in the Mastercard Cloud-Based Payment (MCBP) Product Guide*.

R	MC MS	QR	RI394.22	Contactless payment devices must support CDCVM for all Mastercard Consumer-Presented QR transactions. For non-wearable Mobile Devices, Prolonged authentication must not be supported and CDCVM must be explicitly performed for each QR code that is generated. For wearable Mobile Devices, Persistent authentication may be supported.
BP	MC MS	QR	BI121.19	Issuers that support Mastercard Consumer-Presented QR Code payments should set the contactless CVM Model to CDCVM_ALWAYS to have a consistent CVM Model for all supported transaction types, Contactless, DSRP, and QR Code.

Mastercard requires the support of Online PIN for tokenized contactless ATM transactions in the CVM List for newly enabled account ranges on mobile devices. In addition, Mastercard strongly

recommends the support of Online PIN in the CVM List for all newly provisioned mobile wallets to enable transactions at contactless ATMs.

R	All	CL	RI401.19	All newly enabled account ranges must support Online PIN ('4201') in the CVM List for tokenized contactless ATM transactions.
BP	All	CL	BI122.19	All newly provisioned mobile devices supporting contactless should support Online PIN ('4201') in the CVM List for tokenized contactless ATM transactions.

CVM requirements for Cirrus

Cirrus is an ATM brand. Therefore, all Cirrus cards must support Online PIN.

R	CI	CT CL	RI033.16	Cirrus cards must support Online PIN.
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CVM requirements for Maestro

Maestro chip cards must not support Signature.

Maestro chip cards must support Online PIN and Offline PIN on the contact interface.

No CVM must be supported on the contactless interface, except for Biometric Cards, where it is optional.

Maestro contactless cards and nonmobile payment devices must support Online PIN.

Maestro mobile payment devices must support either Online PIN, CDCVM, or both. Support for CDCVM is recommended. Support for Online PIN is recommended for all Maestro enabled mobile payment devices to enable ATM withdrawals.

R	MS	CT	RI034.14	Maestro chip cards must support Online PIN.
R	MS	CT	RI098.14	Maestro chip cards must support Offline PIN.
R	MS	CT CL	RI035.19	Maestro cards and contactless payment devices must not support Signature.
R	MS	CT	RI407.20	Maestro cards must not support No CVM over the contact interface.
R	MS	CL	RI129.22	Maestro contactless cards, except Biometric Cards, and contactless payment devices must support No CVM over the contactless interface for transaction amounts equal to or below the CVM Limit.

R	MS	CL	RI134.18	Maestro contactless cards and nonmobile payment devices must support Online PIN for contactless transaction amounts above the CVM Limit.
R	MS	CL	RI135.14	Maestro mobile payment devices must support Online PIN, CDCVM, or both for Maestro contactless transaction amounts above the CVM Limit.
BP	MS	CL	BI079.14	Support for Online PIN is recommended for all Maestro mobile payment devices for contactless transaction amounts above the CVM Limit.
BP	MS	CL	BI109.14	Support for CDCVM is recommended for all Maestro mobile payment devices for contactless transaction amounts above the CVM Limit.

NOTE: Requirement RI098.14 does not apply to issuers in the United States region as the support of Offline PIN is optional.

Refer to the regional requirements for the Europe region in chapter "Issuer requirements," section "Regional variations."

With prior approval, Maestro cards that are part of a prepaid program may restrict card acceptance to certain environments or merchants. The requirements listed in the table are relaxed if a CVM is not required in the environments where the card is accepted.

Maestro No CVM authorization processing

Refer to the regional requirements for the Europe region in chapter "Issuer requirements," section "Regional variations."

CVM requirements for Mastercard and Debit Mastercard

All Mastercard chip cards, on both interfaces, and nonmobile payment devices must support Online PIN for use at ATM and CAT Level 1 terminals. Mastercard chip cards must also support Signature on contact interface for use at POS terminals.

Mastercard cards may support Offline PIN on the contact interface. Support for Offline PIN is recommended.

Mastercard chip cards, on both interfaces, and contactless payment devices must support No CVM, except on the contactless interface of a Biometric Card, where it is optional.

All Mastercard mobile payment devices must support Online PIN, CDCVM, or both. Support for CDCVM is recommended for all Mastercard mobile payment devices. Support for Online PIN is recommended for all Mastercard mobile payment devices to enable ATM withdrawals.

R	MC	CT	RI036.24	Mastercard and Debit Mastercard chip cards must support Online PIN on the contact interface. This does not apply to Mastercard signature-only cards issued in the US region.
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R	MC	CL	RI442.24	Mastercard and Debit Mastercard chip cards and nonmobile payment devices must support Online PIN for contactless transaction amounts above the CVM Limit. This does not apply to Mastercard signature-only cards issued in the US region.
R	MC	CT	RI037.15	Mastercard and Debit Mastercard chip cards must support Signature on the contact interface.
BP	MC	CT	BI013.14	Mastercard and Debit Mastercard chip cards should support Offline PIN on the contact interface.
R	MC	CT	RI038.24	Mastercard and Debit Mastercard chip cards must support No CVM.
R	MC	CL	RI443.23	Mastercard and Debit Mastercard chip cards and contactless payment devices must support No CVM for contactless transaction amounts equal to or below the CVM Limit, except on the contactless interface of a Biometric Card.
R	MC	CL	RI133.14	Mastercard and Debit Mastercard mobile payment devices must support Online PIN, CDCVM, or both for contactless transaction amounts above the CVM Limit.
BP	MC	CL	BI067.14	Support for Online PIN is recommended for all Mastercard and Debit Mastercard mobile payment devices for contactless transaction amounts above the CVM Limit.
BP	MC	CL	BI114.15	Support for CDCVM is recommended for all Mastercard and Debit Mastercard mobile payment devices for contactless transaction amounts above the CVM Limit.
BP	MC	CT	BI053.12	Mastercard and Debit Mastercard chip cards that support Offline plaintext PIN and not Offline enciphered PIN should list Online PIN before Signature in the CVM List for use at POS.

Refer to the regional requirements for the United States region in chapter 3, "Issuer requirements," section "Regional variations," subsection "United States region."

With prior approval, Mastercard cards that are part of a prepaid program may restrict card acceptance to certain environments or merchants. The requirements listed in the table are relaxed if a CVM is not required in the environments where the card is accepted.

CVM requirements for U.S. Maestro AID

For details of the CVM requirements for U.S. Maestro AID, refer to chapter "Issuer requirements," section "Regional variations," subsection "United States region."

General requirements for Offline PIN support

Chip cards that support Offline PIN may support Offline enciphered PIN, Offline plaintext PIN, or both.

Cards that support dynamic offline CAM and Offline PIN should support offline enciphered PIN. To ensure the maximum security, all new chip cards undergoing Mastercard certification that are Offline PIN enabled must support Offline enciphered PIN and CDA.

To further enhance PIN security, Offline Plain-text PIN verification is being phased out from 1 January 2025 onwards. The only Offline PIN method going forwards will be Offline enciphered PIN.

BP	All	CT	BI054.12	Chip cards that support dynamic offline CAM and support Offline PIN should support both Offline enciphered PIN and Offline plaintext PIN.
R	All	CT	RI334.16	Offline enciphered PIN is mandatory for all new chip cards undergoing Mastercard certification that are Offline PIN-enabled and support any form of dynamic offline CAM.
R	All	CT	RI451.24	Effective 1 January 2025, newly issued and reissued cards that support Offline PIN over the contact interface must support CDA and Offline Enciphered PIN.

Issuers of Mastercard chip cards that support offline transactions and Offline PIN should request an online authorization for transactions attempted where Offline PIN is not successfully performed. However, transactions may still be approved in Unable to Go Online mode or at offline terminals when Offline PIN has not been attempted.

BP	All	CT	BI052.12	Online authorization should be requested whenever Offline PIN is not successfully performed on an Offline PIN-supporting Mastercard chip card.
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Cardholders do not know that there is a difference between Online and Offline PIN verification nor do they know which is being prompted for, so when they are prompted to enter a PIN, they will always use the same value. Therefore, issuers must ensure that the values of Offline PIN and Online PIN are the same.

R	All	CT	RI042.11	Issuers must ensure that the values of Offline PIN and Online PIN are the same, even if PIN change is allowed, to avoid cardholder confusion.
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Some old terminals incorrectly attempt to read the Offline PIN Try Counter when Offline PIN is not being performed, even for cards that do not support Offline PIN, and abort the transaction

if the data element is not present. Even in cases where it is present with a value set to '00', these terminals may consider that the Offline PIN Try Counter is exceeded and display a PIN Exceeded message that can be misleading for the merchants. Therefore, issuers should set the Offline PIN Try Counter to a value of at least '01', even on cards where Offline PIN is not present in the CVM List.

BP	All	CT	BI016.11	Issuers should set the Offline PIN Try Counter to a value of at least '01', even on cards that do not support Offline PIN.
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CVM requirements for Biometric Cards

Issuers must configure the Biometric Card to fall back to the CVM List when CDCVM is not attempted or fails during a contact transaction, for example, when the fingerprint sensor of a Biometric Card is not accessible once inserted into a POS terminal.

R	MC MS	CT	RI432.22	When CDCVM fails or is not attempted, the Biometric Card must fall back to the CVM List.
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When CDCVM fails or is not attempted during a contactless transaction, the issuer may configure the Biometric Card to either decline the transaction or fall back to the CVM List.

CVM fallback

CVM fallback refers to situations when the preferred CVM, normally Offline PIN, cannot be used and a different CVM, usually Signature, is used instead.

CVM fallback normally occurs either when the PIN Try Counter of the card is exceeded or when the cardholder or merchant uses the PIN Entry Bypass function that is available on some terminals.

Refer to the *M/Chip Program Guide* for a full description of CVM fallback. CVM fallback is only allowed at POS terminals and CATs and not ATMs or BBTs.

If the issuer's preferred CVM is Offline PIN, CVM fallback is allowed:

- To Signature or Online PIN at POS terminals
- To No CVM at CAT, only if the CAT supports No CVM

If the issuer's preferred CVM is Online PIN, CVM fallback to Signature is possible only as a result of PIN Entry Bypass.

Support for PIN Entry Bypass is not available at all terminals, so holders of cards whose primary CVM for POS use is either Online PIN or Offline PIN should be made aware that they may need to use their PIN at POS terminals or their transactions may fail.

PIN Entry Bypass is not allowed on CAT terminals (refer to requirement RA103.11) and must not be supported for contactless transactions (except for US Maestro AID transactions - refer to requirement RA264.14).

The issuer controls how and when CVM fallback proceeds through the personalization of data in the card. This impacts the CVM List and the Issuer Action Codes (IACs).

Magnetic Stripe-based PVV

Changing some of the data on a chip card may not be possible or easy. Any existing Magnetic Stripe processes that rely on rewriting data to the Magnetic Stripe after the card has been issued need to be evaluated.

In particular, this may affect Magnetic Stripe-based PIN Verification Value (PVV) solutions for Online PIN verification if PIN change is supported.

BP	All	CT CL	BI082.14	Magnetic Stripe-based PVV methods should not be used for Online PIN verification if PIN change is supported.
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Single Tap and PIN Request

For more information, refer to chapter 4 "Acquirer Requirements," section "Cardholder Verification," subsection, "Single Tap and PIN Request," or the *Authorization Manual*.

Important additional notes for the issuer:

- When a contactless transaction with no cardholder verification is received, if the terminal supports Single Tap and PIN Request, the issuer can request a PIN-based follow-up online authorization request by setting:
 - DE 39 (Response Code) to "65"
 - DE 48, SE 22, SF 4 (Issuer PIN Request in Single Tap Mode) to "1"
- When the follow-up online authorization request arrives from the merchant with the online PIN (DE 48, SE 22, SF 3 = "1"), the DE 55 chip data is identical to the chip data in the original authorization request. Specifically:
 - All DE 55 tags are the same, including Cryptogram (tag '9F26'), ATC (tag '9F36'), Unpredictable Number (tag '9F37') and (if present) Transaction Sequence Counter (tag '9F41').
 - CVM results (tag '9F34') and TVR (tag '95') incorrectly show Online PIN CVM as not being used, since they are from the original transaction.
 - If M/Chip Cryptogram Pre-validation Service (DE 48, SE 71, SF 1 = "02") or M/Chip Cryptogram Validation in Stand-In Processing (DE 48, SE 71, SF 1 = "03") are used, the issuer should expect DE 48, SE 71, SF 2 (on-behalf Result 1) to be "E" (Valid Application Cryptogram, ATC Replay).

- Transaction should not be declined due to the DE 55 data being a duplicate of the original, as this is expected.
- See subsection "Application Transaction Counter monitoring" in the "Issuer host requirements" section.
- If the optional DE 48, SE 63 is provided, it indicates the original trace ID.
- Issuers must set their card profiles as online-only if they use Mastercard Contactless Risk Manager to trigger Single Tap and PIN Requests.
- See subsection "Contactless Risk Manager Service" in the "Mastercard on-behalf services" section.

R	All	CL	RI453.24	When a contactless online authorization request with no cardholder verification is received, where Single Tap and PIN Request is supported by the terminal and acquirer: if the issuer wants to decline and request a follow-up transaction with PIN in Single Tap mode they must respond with: • Value "65" in DE 39 (Response Code) • Value "1" in DE 48, SE 22, SF 4 (Issuer PIN Request in Single Tap mode)
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Card Risk Management

Card Risk Management (CRM) is performed when the terminal first issues a GENERATE AC command. The card application determines whether the transaction should be declined, require online authorization, or can be approved offline.

For contact transactions, Card Risk Management will also be performed when the terminal issues a second GENERATE AC command.

For contactless transactions, interaction with the card or the payment device is terminated after the response to the first GENERATE AC command has been received by the terminal.

During CRM, the card or payment device carries out its own checks on the risk associated with the transaction and any other checks that the issuer chooses.

The actual checks performed depend on the underlying payment application being used. Refer to the appropriate application Product or Issuer Guide for a full explanation of Card Risk Management. Some applications, such as MCBP and MTBP, always request online authorization and cannot approve transactions offline. Other applications, such as M/Chip Mobile or M/Chip Advance, can be configured to approve transactions offline with appropriate security and credit control or can be configured to always require online authorization.

For requirements relating to online processing, refer to section "Online Processing" in chapter "Issuer requirements."

If the card application supports the configuration of a maximum transaction amount, it must only be used to influence the decision to authorize the transaction online or offline. A transaction must not be declined based solely on this parameter.

R	All	CT CL	RI142.14	If a maximum transaction amount is used in CRM, it must not lead to transactions being declined offline based solely on this parameter.
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Currency conversion in CRM

CRM is application-dependent, but most payment applications include a mechanism to track offline spending by maintaining a cumulative count of the value of offline transactions.

The cumulative offline amount is compared with issuer-determined limits. The results of these tests are used to decide if a transaction should be sent for online authorization, if it can be approved, or if it should be declined offline. Refer to the *M/Chip Program Guide* for a full explanation of the offline counters used in CRM.

When a transaction is carried out in a currency other than the application currency of the card, it can only be included in the cumulative offline amount if the amount is converted to this currency. EMV allows this conversion to be performed by the terminal using the Reference Currency method, if the card requests it, although very few terminals support this function. Mastercard recommends that cards do not request terminals to perform currency conversion using the Reference Currency method. This recommendation can be implemented by ensuring that tag '9F3A' (Amount, Reference Currency) is not included in the CDOL1.

BP	All	CT CL	BI018.14	Issuers should not request terminals to perform currency conversion using the Reference Currency method.
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Some payment applications, such as M/Chip Advance, and M/Chip Mobile, can perform the currency conversion internally using exchange rates personalized onto the card by the issuer.

After performing CRM, the card responds with the appropriate AC depending on the result of CRM and the terminal's request. The rules governing this are defined by EMV. For more details, refer to *EMV Book 3 - Application Specification*.

Card action after online authorization

For contact transactions, a second Card Risk Management (CRM) may be performed when the terminal issues a second GENERATE AC command.

The card carries out its own checks on the risk associated with the transaction and any other checks that the issuer chooses. The actual checks performed depend on the underlying card application being used.

Often, when a contact transaction has been authorized online, the card does not perform a second CRM because the issuer host has analyzed the risk. The chip may authenticate the issuer response by checking the Issuer Authentication Data, if present. If Issuer Authentication Data is not present, the card may approve or decline the transaction depending on the card settings.

There may be some situations where the card is expecting Issuer Authentication Data in the response message but this data is not present. These situations may arise due to an error in the acquirer's processing. For this reason, Mastercard recommends that issuers configure cards so that they do not decline transactions because the Issuer Authentication Data is not present in the response message. This card setting is known as semi-grade.

After processing the online response, the chip produces a TC or Application Authentication Cryptogram (AAC) depending on the issuer's decision, the presence and accuracy of the Issuer Authentication Data, and the card configuration.

BP	All	CT	BI021.11	Issuers should personalize their cards as semi-grade.
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In addition to using the online response to decide for the current transaction, the card application may use the online response to decide whether to reset or adjust the offline counters used in Card Risk Management. Cards that are set up as chip grade and semi-grade only reset the counters when the Authorization Response Cryptogram (ARPC) is present in the online response and validated by the card.

Card action in Unable to Go Online mode

For contact transactions, the second GENERATE AC command may indicate that although online authorization has been requested, a response has not been received from the issuer. The terminal will have performed a second Terminal Risk Management and is requesting a final decision from the card.

The card carries out its own checks on the risk associated with the transaction and any other checks that the issuer chooses. The actual checks performed depend on the underlying card application being used.

Managing the contactless controls

On a dual interface card, the issuer can manage the offline counters and parameters for the contactless interface during the online response to a contact chip transaction. They cannot be managed during a contactless transaction, as no online response is ever delivered to the card.

The contactless EMV mode transaction may trigger an online request at the next contact transaction to enable management of the offline counters.

Other contactless payment devices may have alternative ways to manage the offline counters and parameters. For example, the issuer can manage the offline counters and parameters on a mobile phone by instructions sent to the phone over the air.

Issuer host requirements

This section describes the impact of chip transactions on network interfaces and the processing of authorization and clearing messages. Both contact and contactless transactions are considered.

Online request

Online requests are approved against the account balance or open to buy position in the usual way. Also, issuers will receive additional information in online requests originating from a chip transaction, which may help in the risk assessment of the transaction.

Issuers must ensure authorization systems are capable of correctly receiving and processing online request messages containing specific values for Data Element (DE) 22 (POS Entry Mode) and DE 61 (POS Data).

DE 22 (POS Entry Mode), subfield 1 may have the following values.

For contact transactions:

- "05" is used for a contact chip transaction
- "79" is used for a voice authorization associated with a technical fallback transaction.
- "80" is used for a technical fallback transaction.

For contactless transactions:

- "07" is used for a contactless EMV mode transaction.
- "82" is used for transactions routed through a PAN auto-entry server, issuer, acquirer, or third-party vendor system.
- "91" is used for a contactless Magnetic Stripe mode transaction, even if performed at an EMV mode capable terminal.

For DSRP transactions:

- "09" is used in case full EMV data is carried in DE 55.
- "81" is used for Universal Cardholder Authentication Field (UCAF)-based transactions.

For Mastercard Consumer-Presented QR:

- "03" is used for Mastercard Consumer-Presented QR transactions.

For Mastercard Consumer-Presented QR transactions, issuers must also ensure that the following are validated:

- Length of the Issuer Application Data must be 18 bytes.
- CVR byte 6, bits 2 and 1 must be set to 11b.
- The transaction time as encoded within the UN data to ensure that it is within the permitted window.

For additional information on Issuer Host Requirements for Mastercard Consumer-Presented QR transactions, refer to the *MCBP - Requirements for Server Components* document.

For values in DE 61 (POS Card Data Terminal Input Capability Indicator), subfield 11, refer to the *Customer Interface Specification* or the *Single Message System Specifications* manual.

Issuers must support online request messages where DE 22 has a value of "05x" or "07x" and DE 55 is present or not and DE 23 is present or not.

R	All	CT	RI153.22	Issuers must support on their authorization network interface and host system transactions as described in this section.
		CL		
		QR		

However, issuers should treat transactions with DE 22 of "05x" or "07x" and without DE 55 carefully and not approve them without further consideration.

Issuers should expect Refund (DE 3.1 value "20") and Account Status Inquiry (DE 61.7 value "8") transactions to arrive without DE 55, and not decline either just because of the lack of DE 55.

When DE 22 contains a value of "03x", "05x", or "07x", DE 35 contains the Track 2 Equivalent Data (tag '57').

Card authentication

An Application Cryptogram (AC) allows issuers to confirm that the card is genuine by verifying the cryptogram received in an online request message (online CAM).

Issuers perform online CAM in one of the following ways:

- Validating that the AC received from a contact, contactless EMV mode, DSRP, or Mastercard Consumer-Presented QR transaction
- Validating the dynamic CVC 3 received from a contactless Magnetic Stripe mode transaction

If the AC is correct, the issuer knows that the genuine card has generated the cryptogram and that the associated data is consistent with the data used by the card and has not been altered. Other checks may be necessary, such as checking the Relay Resistance Protocol data, to check that the genuine cryptogram has not been fraudulently obtained. If the dynamic CVC 3 is correct, the card is likely to be genuine, although this is a less secure cryptogram than the AC due to the less variable input used in its generation. For MCBP implementations, the MD must always be verified. UMD may also be verified depending on the issuer's configuration.

If the AC or dynamic CVC 3 is not correct, the card may be counterfeit. However, the cryptogram from a genuine transaction may be incorrect because of a processing error in the acquirer or issuer systems.

For transactions where the AC is invalid or the key for validating the AC cannot be derived, for example, due to an incorrect Key Derivation Index (KDI) received in the Issuer Application Data, issuers may use the outcome of these tests to inform their authorization decision. A failure to validate the AC, combined with at least one other failed test, should be a sufficient indication of fraud to warrant an issuer decline.

- Perform the ATC best practices described in Chapter 3 "Issuer requirements," section "Issuer host requirements," subsection "Online request", "Application Transaction Counter Monitoring."
- Validate the Chip CVC.
- Ensure that the PAN and Expiration Date are consistent and correspond to a legitimately issued and currently valid chip card.
- Ensure that standard fraud and risk management processes are followed, for example, velocity checking and geographic bounding.

Once the issuer has verified the cryptogram, they can perform any risk management or other checks that form part of their normal authorization process. Issuers should not routinely validate the chip CVC except when the AC validation has failed.

Some data items are present in DE 55 and also in other data elements in the online request message. The AC can only be verified using the exact data contained in DE 55, as this is the data used by the chip application to generate the cryptogram.

The cryptogram type (ARQC, TC, or AAC) is determined using the Card Verification Results (CVR, part of tag '9F10') and issuers should check its consistency with the Cryptogram Information Data (CID, tag '9F27'). If the data is inconsistent, then this may indicate fraudulent manipulation of the transaction data and the issuer should factor this into their authorization decision.

R	All	CT CL	RI217.15	Issuers must always perform online CAM in one of these ways: • Validating the application cryptogram received in online request messages • Supporting the M/Chip Cryptogram Pre-Validation service
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R	MC MS	QR	RI392.19	Issuers must always perform online CAM for Mastercard Consumer-Presented QR transactions in one of these ways: - Validating the UMD authentication cryptogram and an MD cryptogram received in online request messages - Using MDES or third party TSP for cryptogram validation
R	MC	CL	RI155.15	For contactless Magnetic Stripe mode, issuers must always perform online CAM in one of these ways: - Validating the dynamic CVC 3 received in online request messages - Supporting the dynamic CVC 3 Pre-validation Service
BP	All	CT CL	BI240.23	Issuers should check that the CVR (part of tag '9F10') and CID (tag '9F27') are consistent and consider declining if inconsistent.

For more information, refer to chapter 3, "Issuer requirements," section "Card authentication," subsection "Online card authentication requirements."

Cardholder verification

The CVM used for a transaction will depend on the terminal capability and the card preference and should adhere to the product rules for the type of transaction.

The appropriate CVM may be contingent on the transaction amount. The issuer is responsible to ensure that transactions are approved only when an appropriate CVM has been used.

The TVR indicates that the terminal considers cardholder verification not successful when bit [3] [8] is set.

The verification status of Offline PIN or CDCVM is determined using the Card Verification Results (CVR, part of tag '9F10') and issuers should check its consistency with the CVM Results (tag '9F34'). If the data is inconsistent, for example, if CVR indicates that Offline PIN or CDCVM was not successfully performed by the card but CVM Results indicates that the terminal considers it was successfully performed, then this may indicate fraudulent manipulation of the transaction data and the issuer should factor this into their authorization decision.

NOTE: For Mobile Payment Devices, these may indicate that CDCVM was performed and successful, but a different CVM may have been chosen by the terminal, for example Online PIN or NoCVM. This is to be expected and is not a reason to decline.

BP	All	CT CL	BI116.23	Issuers should check that the CVR (part of tag '9F10') and CVM Results (tag '9F34') are consistent and consider declining if inconsistent.
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Certain low value transactions may be completed with No CVM. Consecutive transactions with No CVM should be monitored to guard against lost and stolen fraud. As part of the authorization decision process, issuers should consider the number of consecutive transactions done with No CVM.

BP	All	CT CL	BI089.14	Issuers should manage the risk of consecutive transactions performed with No CVM.
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Terminal Verification Results

The Terminal Verification Results (TVR, tag '95') is mandatory in the online request message for EMV transactions and contains the result of the tests performed by the terminal during the transaction.

The issuer should consider that bits that are not set in the TVR included in the authorization request of a contactless EMV mode transaction may not always reflect the final outcome of the terminal tests performed. An example of this is when card authentication may have been completed after the GENERATE AC command was issued to the card or after the TVR was signed.

Tests may fail for many reasons. For example, if the offline CAM performed by the terminal is unsuccessful, this might be due to attempted fraud with a counterfeit card, or in the case of CDA, fraud with a legitimate card, but the data on the card or terminal interface has been fraudulently altered. However, it may also indicate that the correct Mastercard public key was not present in the terminal. The validation of an AC is a more reliable card authentication method and should be the primary method used when processing online requests.

BP	All	CT CL	BI031.11	Issuers should inspect the TVR and use the information to improve their authorization decision making or note it for subsequent action.
BP	All	CT CL	BI032.14	Issuers should not decline a chip transaction where the online application cryptogram has been validated if the TVR shows that offline CAM was unsuccessful or not performed unless there are other indications of possible fraud.
BP	All	CT CL	BI033.11	Issuers should monitor offline CAM failures and transactions where offline data authentication is not performed to detect systemic failures or possible fraud attacks.

ICC Data Missing (TVR [1][6])

If this bit is set and the "Cardholder Verification Not Successful" bit is not set, CVM fallback has occurred and the terminal recognizes that another CVM was successfully completed.

Cardholder Verification Not Successful (TVR [3][8])

This bit is set if none of the CVM options has been successfully completed.

BP	All	CT CL	BI035.11	If the TVR in an online request indicates that cardholder verification was not successful, the issuer should decline the transaction.
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Refer to the regional exception for the Europe region in chapter "Regional variations," section "Europe region."

PIN Try Limit Exceeded (TVR [3][6])

This bit is set when the terminal identifies that the PIN Try Counter on a contact card has reduced to zero. If this bit is set and the "Cardholder Verification Not Successful" bit is not set, then CVM fallback has occurred and another CVM has been completed successfully.

BP	All	CT	BI036.11	If the TVR indicates that the PIN Try Limit in the card is exceeded and the issuer does not know if the card has been lost or stolen, the issuer should contact the cardholder.
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PIN Entry Required and PIN Pad not Present or Not Working (TVR [3][5])

This bit may be set when the terminal identifies that a PIN CVM is required and supported but the PIN pad is not functioning or when certain CVM Condition Codes are set on the card.

If this bit is set and the "Cardholder Verification Not Successful" bit is not set, then CVM fallback has occurred and another CVM has been completed successfully.

PIN Entry Required, PIN Pad Present, but PIN Was Not Entered (TVR [3][4])

This bit is set when PIN Entry Bypass has been performed by the merchant or the cardholder for a contact transaction.

If this bit is set and the "Cardholder Verification Not Successful" bit is not set, CVM fallback has occurred and another CVM has been completed successfully.

Online PIN Entered (TVR [3][3])

If the TVR indicates that Online PIN was entered, the issuer normally receives the encrypted PIN block in the online request and performs PIN verification.

In certain environments, when the final amount of the transaction may not be known at the time of the EMV process, a placeholder amount may be used. For such transactions, the "Online PIN entered" TVR bit [3][3] may be set, but the PIN block may not be present in the online request message if the final transaction amount in DE 4 is low value. For these transactions, issuer systems should ignore the TVR bit and handle them in the same way as any other No CVM transaction.

BP	All	CT CL	BI117.17	Issuers should not routinely decline low value transactions where the TVR indicates that Online PIN was entered but there is no PIN block present.
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Relay Resistance Threshold Exceeded (TVR [5][4])

This bit is set when the terminal has performed the Relay Resistance Protocol and found that the time taken to perform the exchange was long enough, compared to a limit configured in the terminal, that there could potentially have been a relay attack performed. This technology requires fine calibration and some terminals may set this bit when no relay attack occurred. It should not be used to routinely decline transactions. This bit is only used for contactless transactions.

BP	All	CL	BI120.18	Issuers should not routinely decline transactions where the TVR indicates that Relay Resistance Threshold is exceeded.
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Relay Resistance Time Limits Exceeded (TVR [5][3])

This bit is set when the terminal has performed the Relay Resistance Protocol and found that the time taken to perform the exchange was longer than that expected by the issuer, and, therefore, there is a possibility that a relay attack could have been performed. It should not be used to routinely decline transactions. This bit is only used for contactless transactions.

BP	All	CL	BI244.24	Issuers should not routinely decline transactions where the TVR indicates that Relay Resistance Time Limits is exceeded.
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Amount, Other (Cash Back Amount)

When a purchase with cash back is made, Amount, Other contains the cash back amount. The Amount Authorized field is then the sum of the purchase amount and the cash back amount.

For a normal purchase transaction with no cash back, Amount, Other (tag '9F03') is either filled with binary zeros in DE 55 or is not present in DE 55.

R	All	CT CL	RI433.22	If Amount, Other is not present in DE 55, issuers must assume that a value of binary zeros was used in the cryptogram generation and use the same value in the AC verification.
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Application Transaction Counter monitoring

The Application Transaction Counter (ATC) is incremented by the card for each transaction and is an input to the calculation of the card cryptogram, either the Application Cryptogram or the magstripe contactless dynamic CVC 3.

This acts as an identifier for each chip transaction, helps in the detection of replayed cryptograms, and, in conjunction with the terminal Unpredictable Number, it ensures freshness of the cryptogram.

For multi-application cards, issuers must establish how the ATC is managed on the card. This is normally either:

- 1. ATC is shared across all applications on the chip
- 2. ATC is unique per application

When monitoring the ATC, issuers must take this into account. If the Dedicated File Name (tag '84') is not received from the acquirer, issuers should consider if this impacts their ATC tracking.

Although ATC values are generated sequentially, they may not be presented to the issuer in this way. Transactions may sometimes be completed offline or with deferred authorization, or not completed at all. In these situations, ATC values could be missing in the sequence received by the issuer or they could be received out of sequence. Issuers might also receive duplicate ATC values for valid reasons. Issuers should not routinely decline transactions with duplicate ATC values or if the ATC values arrive out of sequence.

For transactions where the cryptogram has been successfully validated, issuers should keep a record of the highest received ATC. Issuers should set a range outside of which the receipt of an ATC value is considered suspicious and possibly indicative of fraud, however issuers should not decline the transaction for this reason alone. The range depends on the market environment where the card is used. For example, if deferred authorizations, offline chip transactions, and contactless aggregated transit transactions are frequently performed, a wider range may be required. The range does not have to be centered on the highest received ATC. The acceptable window could be wider for numbers less than this value than for numbers greater than this value.

R	All	CT CL	RI457.25	For multi-application cards, ATC tracking at the issuer host must take into account if the ATC is shared between applications on the card
BP	All	CT CL	BI059.23	Issuers should raise a post-event alert for out-of-range ATC values and conduct further investigation.

Issuers should be able to detect when they receive a transaction that has the same ATC value as a previously approved transaction for the same card, as this may indicate an attempted replay fraud, and such transactions should be treated as suspicious. However, issuers should note that duplicate ATC values may also occur for legitimate reasons in transactions where an authorization request was not initially approved and is repeated. Examples of such transactions

are Single Tap and PIN contactless transactions or repeated resubmissions of deferred authorizations pending funds or credit availability.

BP	All	CT CL	BI060.23	Issuers should consider declining authorization requests with an ATC value that has been previously submitted in another approved transaction for the same card.
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Receiving authorization requests for the same card with the same ATC value and with valid but different cryptograms may indicate that the secret keys of the card have been compromised.

BP	All	CT CL	BI241.23	Issuers should consider blocking a card and investigate further if the card appears in two or more transactions with the same ATC value and with valid but different cryptograms.
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Issuer Application Data

The Issuer Application Data is defined by the card application and is sent in the online chip online request message.

The contents of the Issuer Application Data are set entirely by the chip, rather than the terminal, and are especially useful to the issuer because it does not have to rely on the terminal to set up the data correctly. Part of the content of the Issuer Application Data is included in the Authorization Request Cryptogram (ARQC) calculation, so it is protected against corruption and alteration.

The exact contents of the Issuer Application Data are application-dependent, but most implementations include:

- Information about the AC calculation, for example, key index, type of cryptogram
- The Card Verification Result (CVR)
The contents of the CVR are also application-dependent but typically include information from the chip about the transaction, such as whether Offline PIN was performed successfully, and certain status information about the card, for example, the result of script processing.

Some acquirers may incorrectly pad or truncate the Issuer Application Data resulting in the value being sent in the online request message with a different length than expected by the issuer.

BP	All	CT CL	BI038.11	Issuers should not decline a chip transaction where the application cryptogram has been validated if the Issuer Application Data received is longer or shorter than expected.
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Chip CVC validation

CVC provides a simple, usually static form of card authentication that can reduce counterfeit fraud. The validation of an online chip application cryptogram provides a more reliable card authentication method.

For chip transactions where chip data is present and the ARQC can be validated, the Chip CVC does not need to be checked.

In the unlikely event that a chip transaction is received but the chip data is not present, the Chip CVC should be validated.

The Mastercard *Security Rules and Procedures* manual specifies how to cryptographically derive CVC values from the card PAN, expiration date, and service code. For Chip CVC, the derivation requires a service code of "999" and CVC 2 requires a service code of "000".

For Magnetic Stripe-read transactions, the CVC 1 should be validated. To guard against counterfeit fraud, the issuer should ensure that the service code received in the online request message corresponds to the service code that was personalized on the card Magnetic Stripe and is not "000" or "999".

BP	All	CT CL	BI039.11	When the online application cryptogram has been validated, the Chip CVC should not be checked or the result of its validation used.
BP	All	CT CL	BI099.14	For chip-read transactions where the chip data is not present, the Chip CVC should be validated.
BP	All	CT CL	BI100.20	For Magnetic Stripe-read transactions when validating the CVC 1, the issuer should ensure that the service code received in the online request message corresponds to the service code that was personalized on the card Magnetic Stripe and is not "000" or "999".

Consistency of data

Although the information in DE 55 is usually consistent with other fields, there may be some differences for certain data elements.

- DE 3 (Processing Code) and tag '9C' Transaction Type
- DE 4 (Transaction Amount) and tag '9F02' Amount Authorized
- DE 13 (Transaction Date) and tag '9A' Transaction Date
- DE 43 (Card Acceptor Name and Location) and tag '9F1A' Terminal Country Code
- DE 49 (Transaction Currency Code) and tag '5F2A' Transaction Currency Code
- DE 54 (Additional Amounts) and tag '9F03' Amount, Other

If the AC is correct, the decision to approve or decline the transaction should use the information in these data elements rather than the corresponding information in DE 55, where present.

This is particularly important in transit implementations where a zero value contactless transaction may be used to enable entrance to the transit system but an online authorization for a different amount is made after entrance is granted.

Refer to Appendix B (Analysis of DE55 tags), section "Mandatory subelements in online request", subsection "Amount Authorized, '9F02' " for a list of situations where the '9F02' Amount Authorized may differ from DE 4 (Transaction Amount.)

BP	All	CT CL	BI026.11	If the application cryptogram is correct, the decision to approve or decline the transaction should use the information in the data elements listed in this section rather than the corresponding information in DE 55, where present.
BP	All	CT CL	BI027.11	If the application cryptogram is correct, the issuer should not decline a transaction simply because the data in DE 55 is different from the values in the data elements listed in this section.

Unexpected tags

Mastercard publishes a list of mandatory and recommended tags to be transmitted in DE 55 in the network messages. However, acquirers may add tags that are not included in the mandatory and recommended list. The presence of unexpected tags is not a reason to decline a transaction.

BP	All	CT CL	BI030.11	Issuers should not decline an online request because of the presence of tags in DE 55 that are not listed as mandatory or recommended.
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POS terminal capability data

Data in the online request message in DE 22 (Point-Of-Service [POS] Entry Mode), DE55 tag '9F33' (EMV Terminal Capabilities), and DE 61 (Point-Of-Service [POS] Data) indicate the actual circumstances of the transactions and the capability of the terminal, for example, if the terminal is chip-capable or whether PIN was used to validate the transaction.

Issuers should not routinely decline transactions simply because the terminal capabilities information is inconsistent with other data in the message. For example:

- If the terminal is not identified as a chip terminal but a valid ARQC is received
- If the terminal is not identified as PIN-capable but a valid PIN is received

BP	All	CT CL	BI024.11	Issuers should not routinely decline transactions simply because the terminal capabilities information is inconsistent with other data in the message.
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Transaction Certificate received in online request

In certain acceptance environments, a Transaction Certificate (TC) may be received from the card and used in a subsequent online request.

An example of this is certain transit environments. Issuers must not decline an authorization for the sole reason that the cryptogram received in an online request is a TC.

R	All	CT CL	RI050.11	Issuers must not decline an online request for the sole reason that the cryptogram received is a TC.
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Handling merchant-approved online requests

Issuers may receive online requests for transactions with a timestamp, DE 7 (Transmission Date and Time), that is later than the date and time of the transaction DE 12 (Time, Local Transaction) and DE 13 (Date, Local Transaction).

In addition, these transactions will contain an AAC instead of an ARQC and the ATC received may also be out of sequence.

Issuers must be prepared to receive online requests with these characteristics and must not routinely decline them. Such transactions have been accepted by the acquirer at its own risk.

R	MS MC	CT	RI055.20	Issuers must be prepared to receive Authorization Request/0100 or Financial Transaction Request/0200 messages for transactions with any of the following characteristics and must not routinely decline them: • The cryptogram included in the message is an AAC. • The ATC received is out of sequence. • The timestamp is later than the date and time of the transaction.
R	MS MC	CL	RI421.20	Issuers must be prepared to receive Authorization Request/0100 or Financial Transaction Request/0200 messages for contactless transactions with any of the following characteristics and must not routinely decline them: • The ATC received is out of sequence. • The timestamp is later than the date and time of the transaction.

Contactless kernel verification

For contactless transactions, if the Contactless Kernel Identifier-Terminal (tag '96') is provided, the issuer should verify it is as expected for the card or access device used, to prevent fraud.

BP	All	CL	BI247.25	If the Contactless Kernel Identifier-Terminal (tag '96') is present, issuers should verify it matches the expected Mastercard compatible kernel for the card or access device. If different, issuers should monitor, and take this into account in their risk management decision.
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Stand-In Processing

EMV issuers are mandated to use the Mastercard Cryptogram Validation for Stand-In Processing service to maximize acceptance and mitigate potential fraud patterns.

R	All	CT CL	RI361.17	EMV issuers must use the Mastercard Cryptogram Validation for Stand-In Processing service.
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Contactless issuers that support Magnetic Stripe mode transactions must participate in the Dynamic CVC 3 Validation in Stand-In Service for all contactless-enabled Mastercard credit card programs.

R	MC	CL	RI335.16	Issuers of Mastercard credit contactless-enabled programs that support Magnetic Stripe mode transactions must enroll for the Dynamic CVC 3 Validation in Stand-In Service.
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Refer to the regional requirements for the United States region in chapter "Issuer requirements," section "Regional variations," subsection "United States region."

Online response

Chip data should not be included in the online response for contactless transactions or Mastercard Consumer-Presented QR transactions.

BP	All	CL QR	BI107.19	Chip data should not be included in the online response for contactless or Mastercard Consumer-Presented QR transactions.
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Issuer Authentication Data

When the issuer host responds to an online request, it constructs a response message that includes Issuer Authentication Data (DE 55, subelement '91').

The Issuer Authentication Data must be present in the online response message when the following conditions are met. Any transactions originated from Mastercard Stand-In are excluded.

- Subfield 1 of DE 22 (POS Entry Mode) contains a value of "05".
- DE 55 is present in the authorization request message, including all mandatory tags.
- The ARQC has been validated successfully.
- The Issuer's Response Code indicates an approval.

Issuer Authentication Data is application-dependent but, subject to successful ARQC validation, contains an Authorization Response Cryptogram (ARPC) and often some data that allows the issuer to control the card's processing of the response, for example, whether the card offline counters should be reset.

For online transactions where the ARQC validation is not successful, an ARPC should not be submitted in the online response message. The response should, therefore, either omit that data item or replace it by a fixed value of zeroes or by a random value. Semi-grade cards should omit the ARPC.

In such cases, issuers should also not reply with a Response Code of "91" (Authorization System or issuer system inoperative), "92" (Unable to route transaction), and "96" (System error).

To avoid certain fraud cases, issuers must not echo the chip data present in the online request to the online response message.

R	All	CT	RI178.15	The Authorization Request Response/0110 or Financial Transaction Request Response/0210 message must contain DE 55, subelement 91 when the following conditions are met. Any transactions originated from Mastercard Stand-In are excluded. • Subfield 1 of DE 22 (POS Entry Mode) contains a value of "05". • DE 55 is present, including all mandatory tags. • The ARQC has been validated successfully. • The Issuer's Response Code indicates an approval.
R	All	CT	RI209.15	Issuers must not submit a valid ARPC in the online response message based on an ARQC from the online request that cannot be validated or is invalid.
BP	All	CT CL	BI112.15	Issuers should not reply with a Response Code of "91" (Authorization System or issuer system inoperative), "92" (Unable to route transaction), and "96" (System error) if the ARQC from the online request cannot be validated or is invalid.

R	All	CT CL	RI210.22	Issuers must not echo the chip data from the online request to the online response message.
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Issuer script requirements

Issuer scripts are commands sent by the issuer to their cards to change the state of the card or update the data on the card. They may only be sent in the response to contact transactions.

BP	All	CT	BI044.11	Mastercard recommends that issuers support at least the APPLICATION BLOCK script.
R	All	CT	RI051.13	Issuers must not send more than one template '71' script or one template '72' script with each Authorization Request Response/0110 or Financial Transaction Request Response/0210 message, although each script template may contain multiple script commands.
R	All	CT	RI052.16	The value field length of all scripts sent, excluding the tag and length identifiers, must not be more than 127 bytes.

Issuer script results

There is no Mastercard requirement for acquirers to carry this data element in authorization or clearing messages. Issuers should use the Issuer Application Data in the transaction to determine whether a script has been successfully delivered to the card during the preceding transaction.

Additional Processing Information

Data Element (DE) 48, subelement 74 (Additional Processing Information) provides additional information about chip transaction processing and may be used by Mastercard in the research and resolution of chip interoperability issues.

BP	All	CT CL	BI028.22	Issuers should include DE 48 (Additional Data), subelement 74 in the Authorization Request Response/0110 or the Financial Transaction Request Response/0210 message when there is an issue with chip cryptogram validation.
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Partial approval and approval of purchase only

The following two values for the Response Code DE 39 indicate that an issuer has approved part of a transaction but not the whole amount:

- DE 39 containing a value of "10" allows issuers to approve a portion of the requested purchase transaction amount.
- DE 39 containing a value of "87" allows issuers to approve the purchase portion of a purchase with cash back transaction.

Acquirers indicate in the authorization request message whether they support the new values. If so, the issuer may use these values.

Issuers that want to use these Response Codes for chip transactions must ensure that their card applications correctly accept and interpret these values in the online response. Some payment applications may decline online-authorized transactions if this is not set up.

R	All	CT	RI058.11	Issuers that support partial approval or approval of purchase only in cash back transactions must ensure that the ARPC Response Code indicated in the Issuer Authentication Data has the same value as for authorizations where DE 39 contains a value of "00" (Approved or completed successfully).
R	All	CT	RI059.11	Issuers that support partial approval or approval of purchase only in cash back transactions must ensure that the card application treats values of "10" (Partial Approval) and "87" (Purchase Amount Only, No Cash Back Allowed) contained in the Authorization Response Code (tag '8A') as approvals.

Issuers of cards that request the Amount Authorized or the Amount, Other in the CDOL2 must be aware that the value of these data elements in the clearing DE 55 will be the new final transaction values approved following the online response and not the values sent in the ARQC data.

Issuers of cards that do not request the Amount Authorized or the Amount, Other in the CDOL2 must be aware that the value of these data elements in the clearing DE 55 will be the same value as in the ARQC data and not the values that were approved by the issuer and submitted in the non-chip clearing data.

Contactless transactions

Because the consumer remains in control of the contactless card or contactless payment device throughout the transaction, the opportunity for merchants to pick up these cards is limited. Issuers should not use a capture card authorization response for contactless transactions.

For EMV mode transaction authorization responses, the issuer should not generate Issuer Authentication Data or issuer script messages, because the contactless-enabled terminal is not able to pass Issuer Authentication Data or issuer script messages to the contactless card or contactless payment device.

As a result of contactless-specific risk management, the issuer may wish to decline and prompt the cardholder to perform a contact card transaction with CVM, where possible. In this case, the

issuer should use the authorization Response Code "65" (Exceeds withdrawal count limit). The issuer should be aware that if a decline Response Code other than "65 Exceeds Withdrawal Count Limit" is used, the terminal might not prompt for a subsequent contact transaction to be performed. Except for US and Canada regions, Response Code "65 Exceeds Withdrawal Count Limit" along with DE 48, SE 22, SF 4 value "1" may be used to indicate a Single Tap and PIN Request. Refer to the Europe region requirements in chapter 3 "Issuer requirements," section "Regional variations," subsection "Europe region."

An authorization Response Code "01" (Refer to card issuer) must not be used for a contactless online request.

R	All	CL	RI157.14	Issuers must not use an authorization Response Code "01" (Refer to card issuer) for a contactless transaction.
BP	All	CL	BI091.14	Issuers should not use an authorization Response Code "04" (Capture card) for a contactless transaction.
BP	All	CL	BI093.14	Issuers should not send scripts with online responses to EMV mode transactions.
R	All	CL	RI158.14	Issuers that use a PAN mapping service must return the genuine PAN in the authorization response message, even if an alternative PAN was used in the online request.
R	All	CL	RI454.24	If the issuer declines a contactless transaction because they require a subsequent contact transaction, or a subsequent cardholder verified transaction, they must respond with Response Code "65" (Exceeds withdrawal count limit).

Online advice

Issuers may receive an online advice message generated by one of the Mastercard Stand-In services. The authorization advice informs the issuer of the decision taken by the Stand-In service.

If the original online request message contained DE 55, it is copied to the advice message sent to the issuer.

Authorization of fallback transactions

Technical fallback occurs when the Magnetic Stripe of a hybrid card is used instead of the contact chip.

This may be caused by a problem with the chip technology, card, or terminal, or an incorrect processing by merchants. Technical fallback can also occur when a fraudster deliberately disables the contact interface on a stolen card to avoid chip security.

All issuers in the Canada region, the Europe region, the Latin America and the Caribbean region, and the Middle East/Africa region must decline technical fallback transactions originating from any of those regions.

Effective 1 September 2025, all issuers in the Asia/Pacific region must decline technical fallback transactions, unless the merchant is located in the United States Region.

If there is no decline mandate in place, issuers can choose to approve or decline as appropriate.

R	All	CT	RI434.22	Issuers in the Canada region, the Europe region, the Latin America and the Caribbean region, and the Middle East/Africa region must decline technical fallback transactions acquired in those regions.
R	All	CT	RI459.25	Effective 1 September 2025, issuers in the Asia/Pacific region must decline all chip to magnetic stripe technical fallback transactions, except if the merchant is located in the United States region

Issuer host grade

Magnetic Stripe grade for issuers host were introduced in the early years of EMV migration as a transitional solution. Magnetic Stripe grade issuers do not validate the ARQC or generate an ARPC, and, therefore, they do not achieve the same benefits as chip grade issuers.

To maximize security and mitigate potential fraud attacks, Magnetic Stripe grade is not allowed and issuers hosts must either operate as Chip grade or enroll for Mastercard M/Chip Cryptogram Pre-Validation service.

R	All	CT CL	RI172.14	All EMV issuers must either support Chip grade host processing or enroll for Mastercard M/Chip Cryptogram Pre-Validation service.
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Magnetic Stripe mode transactions

Contactless Magnetic Stripe mode transactions appear like regular Magnetic Stripe transactions, but the dynamic CVC 3 will have been inserted into the track data.

Magnetic Stripe mode transactions may be completed using either Track 1 or Track 2 data according to the acquirer preference.

R	MC	CL	RI156.14	Issuers who support contactless Magnetic Stripe mode must be able to process transactions if either Track 1 Data or Track 2 Data is present in the online request message.
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Clearing requirements

Chip transactions are presented in clearing records for posting to cardholder accounts as for other transaction technologies.

Chip transactions received by the issuer may or may not contain the chip data in DE 55.

R	All	CT CL	RI063.11	Issuers must be able to receive clearing messages that contain DE 55 unless they use the Chip to Magnetic Stripe Conversion Service for clearing messages.
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Transaction certificates in clearing messages

A verified Transaction Certificate (TC) in a clearing message proves to the issuer the results listed in this section.

- The genuine card was present when the transaction took place.
- The associated data in DE 55 have the same values they had when presented to the card.
- If the TC is associated with a transaction that was approved offline, this verifies that the card really did approve the transaction.
- If the TC is associated with a contact transaction that was approved online, this confirms that the card received an online authorization.

For offline-approved transactions, the data in the other fields of the clearing message should be the same as the equivalent subelements in DE 55 because this is the information the chip used to make its decision to approve. Issuers may have more chargeback rights if an acquirer submits different data. For details, refer to the *Chargeback Guide*.

Validation of DE 55 chip data in clearing messages

If the clearing message relates to an online authorized transaction, validation of the chip data should have occurred at authorization time and no further validation is required.

However, Mastercard recommends to validate the clearing cryptogram and other chip data in other cases, including:

- If the clearing message does not relate to an online authorized transaction
- If the transaction is disputed
- If the issuer considers the transaction as suspicious
- If the clearing message relates to an online authorization, but contains different chip data, except for differences due only in the case where online authorization provided the 1st GENERATE AC response, but clearing provided the 2nd GENERATE AC response.

If anomalies are found, the issuer should consider what action to take on a case-by-case basis.

BP	All	CL	BI243.24	Issuers are recommended to validate clearing chip data including the cryptogram, unless identical data was received in a matching online authorization which has already been verified.
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ARQC in clearing messages

In some circumstances, acquirers may have technical reasons why they populate the clearing message with an ARQC instead of a TC. This situation can also occur in single message systems where the terminal does not return the TC to the acquirer host system.

For contactless payments, the cryptogram in the clearing is the cryptogram generated in the first GENERATE AC Response.

Chip-declined Mastercard transactions in clearing

Sometimes a merchant may legitimately accept transactions that the card application declined by producing an AAC. This may happen as a result of Merchant Approved or at on-board terminals.

In this case, if DE 55 is present in the clearing message, it contains an AAC, and a value of "F" (Offline Chip) in DE 22 (Point of Service Entry Mode), subfield 7 (Card Data Input Mode) indicates that the terminal processed an offline chip transaction. Issuers can charge back transactions that were declined offline by the chip only if they suffer a loss.

R	All	CT CL	RI065.11	Issuers must be able to accept clearing messages for transactions that were declined offline by the chip.
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Clearing of contactless transactions

Contactless issuers must ensure that host systems are capable of correctly receiving and processing existing subfields within the clearing message containing specific values of the data input capability and the data input profile, DE 22 (POS Entry Code).

DE 22, subfield 1 identifies the terminal capabilities and must contain the following values:

- The value "M" for a transaction at an EMV mode capable terminal
- The value "A" for a transaction at a Magnetic Stripe mode only terminal

DE 22, subfield 7 identifies the card data input profile for this transaction and must contain the following values:

- The value "M" for an EMV mode transaction
- The value "A" for a Magnetic Stripe mode transaction

R	All	CL	RI160.14	Issuers must support contactless transactions as described in this section, "Clearing of contactless transactions" on their clearing interface and host system.
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As there is only one GENERATE AC command in a contactless transaction, the cryptogram and related data included in the clearing message always relates to the first GENERATE AC. The cryptogram may be a TC or an ARQC.

To monitor fraud, if Kernel Identifier-Terminal (tag '96') is present, issuers should check it matches the expected kernel id for the card or access-device. Any differences are likely to be a sign of fraud.

BP	All	CL	BI248.25	Issuers should monitor cleared contactless transactions where Kernel Identifier-Terminal (tag '96') differs to the expected kernel for the card or access-device.
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Mastercard on-behalf services

Mastercard provides various network-based chip processing services for both chip and digital environments. These services allow issuers to perform secure chip or digital transactions without significant implementation at their host systems.

Chip contact and contactless on-behalf services

A brief description of the chip on-behalf services is given in this section. For full details, refer to *M/Chip Processing Services - Service Description Guide* and *Contactless On-behalf Services Guide*.

Chip to Magnetic Stripe Conversion Service

This service converts online requests from chip format into Magnetic Stripe format, allowing issuers to issue chip cards while leaving their host systems unchanged.

The service removes all chip-related data from the message. The Chip to Magnetic Stripe Conversion Service is available on both the Mastercard Dual Message Authorization System and the Single Message System. The service is also available to issuers that receive their dual message system Clearing Presentment messages from the Global Clearing Management System (GCMS).

This service cannot be used as a stand-alone service. It is only available with M/Chip Cryptogram Pre-Validation Service.

Chip CVC to CVC 1 conversion service

The Chip CVC to CVC 1 Conversion Service provides a solution for issuers that are deploying chip CVC-compliant chip cards but have not upgraded their processing system to support chip CVC validation.

The service validates the Chip CVC in the online request message and, if the Chip CVC is valid, replaces it with the card's CVC 1 value.

M/Chip Cryptogram Pre-Validation Service

The M/Chip Cryptogram Pre-Validation Service performs dynamic online CAM on behalf of the issuer. This allows the issuer to begin issuing chip cards without implementing all the cryptographic functions associated with online CAM.

For each online request, the M/Chip Cryptogram Pre-Validation Service validates the ARQC and optionally performs other tests specified by the issuer before sending the request to the issuer host.

The issuer host produces its response, and the M/Chip Cryptogram Pre-Validation Service adds the Issuer Authentication Data and sends the response back to the card.

When the issuer is not available, the Stand-In system considers the results of the cryptogram validation and the decision-making parameters provided by the issuer when processing the transaction.

M/Chip Cryptogram Validation in Stand-In Processing Service

This service is similar to the M/Chip Cryptogram Pre-Validation Service in that the ARQC is validated using issuer-provided keys. In addition, issuer-defined tests are performed on the chip data in the online request message.

There are two differences between the services:

- Stand-In Processing only takes place when the issuer host is not available.
- The Stand-In system also decides whether to approve the transaction because the issuer host is not available.

As with the M/Chip Cryptogram Pre-Validation Service, the Stand-In system generates the Issuer Authentication Data, including the ARPC.

Dynamic CVC 3 Pre-validation Service

This service is available only to contactless issuers that do not have the resources to implement CVC 3 validation in-house or whose processors have not upgraded their systems to support it.

With the Dynamic CVC 3 Pre-validation Service, Mastercard validates the dynamic CVC 3 included in the contactless online request before it reaches the participating issuer.

CVC 3 Validation in Stand-In Service

The CVC 3 Validation in Stand-In Service is an enhancement of the existing Mastercard Stand-In Processing service.

If the issuer host is not available, the service:

- Validates the dynamic CVC 3
- Authorizes or declines transactions based on issuer parameters, including CVC 3 validation results
- Sends an issuer advice, including the CVC 3 validation result

Contactless Risk Manager Service

Contactless Risk Manager Service is a network-level counter and accumulator management service for issuers that prompts for cardholder authentication at pre-established limits.

The limits can be set up as the cumulative amount (Accumulator), total number of transactions (Counters), or both depending on issuer preference. Predefined limits for Accumulators and Counters for each country are published in the "*Contactless Risk Manager – Global Counter and Accumulator Limits*" document, available on Mastercard Connect.

To ensure that the service works correctly for a given card, all the transactions for that card must be routed through Mastercard. This requires cards to be configured as online-only.

For additional information, refer to the Contactless Risk Manager Playbook (available on request) and Announcement AN4458 *Mastercard Announces Global Availability of the Contactless Risk Manager Service*, available on Mastercard Connect.

R	All	CL	RI452.24	Issuers using the Mastercard Contactless Risk Manager service must configure their cards as online-only.
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Digital on-behalf services

Mastercard also provides a set of on-behalf services through the Mastercard Digital Enablement Service (MDES) for issuers that support digital payments. A list of the available services supported by MDES are contained in this section.

For more details on these services, refer to the *Mastercard Digital Enablement Service Issuer Implementation Guide*.

Pre-digitization Service

This configurable service lets issuers determine whether a card is available and eligible for digitization. Eligibility can be determined using pre-digitization network messages or predefined eligibility rules, default decisions, or both.

Pre-digitization may also determine that authentication of the cardholder is required and an issuer specified Card Eligibility process must be completed before digitization.

Tokenization Service

This service allocates a token Account PAN to be included in the digitized card credentials that are generated as part of the digitization process.

Digitization (Provisioning) Service

The service packages the digital account credentials and generates the relevant cryptographic keys and personalization scripts for provisioning to the Secure Element (SE), Trusted Execution Environment (TEE), or Host Card Emulation (HCE) solution of the cardholder's mobile device.

Transaction Validation and Token to Account PAN Mapping Service

The Mastercard Transaction Processing Service validates the EMV cryptogram or dynamic CVC3, applies token controls, and maps the token to the Account PAN.

Transaction Detail Service

Mastercard Transaction Detail Service (TDS) manages the interactions between Mastercard, the Wallet Provider, and the cardholder's payment application for the registration, notification, and delivery of transaction details, such as the merchant's name and the transaction amount.

Customer service tools

The Mastercard Customer Service application and Customer Service Application Programming Interface (API) are available for issuers and Wallet Providers to manage cardholder queries.

Reporting tools

Several reports specific to the Mastercard Digital Enablement Service (MDES) are available for issuers.

Chip transactions involving multiple requests

Chip transactions involving multiple requests may occur when a nonfinancial request (any transaction that does not cause a transfer of funds) follows a financial or nonfinancial request. In this scenario, Mastercard recommends a new transaction with fresh chip data (refer to requirement BA016.16).

However, issuers must be able to receive the same chip data (same ATC and ARQC) if the acquirer does not generate a new one.

R	All	CT CL	RI445.23	When a nonfinancial request follows a financial or nonfinancial request, issuers must be able to receive duplicate chip data.
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Transaction types

Special considerations may apply when processing different transaction types using chip technology.

ATM

The CVM used for ATM transactions is Online PIN.

Issuers should support ATM transactions on the contactless interface.

As not all ATMs validate the settings of the card, issuers of contactless cards and contactless payment devices should be aware that they may receive transactions from ATMs even if support for ATM is not indicated in the Application Usage Control or if support for Online PIN is not included in the CVM List.

BP	All	CL	BI071.14	The Application Usage Control should indicate support for ATM transactions.
BP	All	CL	BI088.14	Issuers should adapt their host systems to receive contactless transactions from ATMs even if the card configuration indicates that the card is not valid for use on ATMs.

Refer to the regional requirements for the Europe region in chapter "Issuer requirements," section "Regional variations," subsection "Europe region."

Balance inquiry

Balance inquiries normally involve an EMV transaction for a zero amount, which is sent online to allow the issuer to respond with the account balance. They may occur at ATMs or POS terminals.

The transaction uses normal EMV CVM processing.

Issuers should consider whether they wish to reset or update the offline counters on the card when responding positively to balance inquiry transactions by using an appropriate ARPC Response Code.

BP	All	CT	BI097.14	Issuers that want to update or reset the offline counters when responding positively to balance inquiry transactions should use an appropriate ARPC Response Code.
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PIN Management transactions

PIN Management is the term used to describe cardholder initiated transactions to either change their allocated PIN or to reset the Offline PIN try counter if this has become blocked.

PIN Management transactions take place at ATMs or other specialist devices that enable cardholder interaction and are used to manage the PIN used with a chip card. There are two types of transactions:

- PIN Unblock (Transaction Type 91)
- PIN Change (Transaction Type 92)

Issuers usually respond to a PIN Management request with a script or other instruction to the card. For PIN change requests, the issuer will also need to update the Online PIN in the issuer host system.

The cryptogram received in a PIN Unblock request will usually be an ARQC but may be an AAC in some situations.

PIN Management requests involve an ATM-based EMV transaction for a zero amount verified by Online PIN.

Issuers may use the following Response Codes if required:

- "57" (Transaction not permitted to issuer/cardholder)
- "58" (Transaction not permitted to acquirer/terminal)
- "70" (Contact Card Issuer)
- "71" (PIN Not Changed)
- "85" (Not declined)
- "89" (Unacceptable PIN - Transaction Declined - Retry)

Refer to the regional requirements for the Europe region in chapter "Issuer requirements," section "Regional variations," subsection "Europe region."

Mastercard offers a service to issuers to help them support PIN change and PIN unblock for their chip cards.

Refer to the *Authorization Manual* for more details.

Some products enable changes to Offline PIN or PIN-based CDCVMs without special transactions and issuer intervention. M/Chip Advance permits a cardholder to change their Offline PIN through a personal cardholder device such as a balance reader. The old PIN must be entered to enable a new PIN to be defined.

Various mobile implementations enable users to change their PIN-based CDCVM on the device without any need for online communication to the issuer.

Purchase with cash back transactions

Debit Mastercard and Maestro cards should support purchase with cash back on the contact and the contactless interfaces.

BP	MC	CT	BI022.11	Maestro and Debit Mastercard cards should indicate support for purchase with cash back transactions in the Application Usage Control.
	MS	CL		

Issuers must enable support for purchase with cash back on all newly issued and reissued Debit Mastercard, including prepaid, and Maestro, including prepaid, account ranges. This requirement is not applicable to card programs on which cash access is prohibited, such as anonymous prepaid programs.

Issuers must configure the Application Usage Control (AUC) accordingly to indicate support for domestic and international purchase with cash back (PWCB) transactions on both interfaces.

Purchase with cash back transactions always require cardholder verification, regardless of the amount. Refer to requirement RA267.14.

All Mastercard purchase with cash back transactions are authorized online.

Issuers have the option to support the use of Authorization Response Code "87" (Purchase Amount Only, No Cash Back Allowed). Refer to chapter "Issuer requirements," section "Online response," subsection "Partial approval and approval of purchase only."

R	MC	CT	RI422.24	Newly issued and reissued Debit Mastercard cards including prepaid, Maestro cards including prepaid, and access devices must indicate support for domestic and international PWCB transactions in the Application Usage Control (AUC) unless this is a card program on which cash access is prohibited.
	MS	CL		

NOTE: Requirement RI422.24 does not apply to the Asia/Pacific region, Colombia, or Venezuela.

Refer to the regional specifics for the Asia/Pacific region, Canada region, Europe region, Latin America and the Caribbean region, and the Middle East/Africa region in chapters "Issuer requirements" and "Acquirer requirements."

Refunds

Mastercard contactless issuers must be able to support the processing of refund transactions initiated through the contactless interface.

R	MC	CL	RI159.22	Issuers must be able to process refunds initiated through the contactless interface.
	MS			

Refund transactions may be sent online for the issuers to be aware in real time that a refund has been given to the cardholder and not to wait until they receive the clearing. However, refund transactions might not always contain an ARQC in the online message. These transactions might contain an AAC or a TC or no chip data at all depending on the card's or device's configuration and how refund has been implemented at terminal level. Therefore, issuers must not decline refund transactions based on the type of the cryptogram or the presence or omission of chip data in the online message.

R	MC	CT	RI458.25	Issuers must not decline refund transactions based on the type of the cryptogram or the presence or omission of chip data in the online message.
	MS	CL		
R	MC	CT	RI418.20	All issuers must support the online authorization of a refund transaction for all programs except for non-reloadable prepaid card programs.
	MS	CL		
BP	MC	CT	BI128.20	Issuers should only use the authorization Response Code value of "00" (Approved or completed successfully) in DE 39 (Response Code), subfield 1 for approvals. The value of "85" (Not declined) should not be used.
	MS	CL		

Transit transactions

Issuers should consider the presence of transit indicators in DE 48 (Additional Data), subelement 64 (Transit Program) of the online request message during the authorization decision process. This may indicate a new period of aggregation, and hence, the issuer liability is greater than the Transaction Amount in DE 4.

Issuers may choose to use an Application PAN on the contactless interface that is different to the PAN present on the Magnetic Stripe or that appears on the face of the card. If this option is chosen, the issuer must be aware of the requirements to return the value of the embossed PAN in the response message for contactless transit transactions.

For more details, refer to the *Authorization Manual*.

For more information on implementation requirements, guidelines, and standards, refer to the *Mastercard Global Transit Implementation Guide*.

BP	All	CL	BI090.14	Issuers should adapt the authorization decision process when appropriate for transit-based transactions.
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Account Status Inquiry

Issuers should expect DE 55 to be optional in Account Status Inquiry (ASI) transactions, and not to decline them just because of this.

BP	All	CT	BI245.24	Issuers should not decline ASI transactions just because DE 55 is omitted.
		CL		
		QR		

Product variations

This section describes variations related to specific Mastercard products or programs.

Fleet Cards

This section applies to Mastercard Corporate Fleet Card, MultiCard, and Government Commercial Card programs.

The term Fleet Card is used here to refer to all of the programs. For information on the Mastercard Corporate Fleet Card Program, refer to the appropriate document for the region as follows:

- *Fleet Implementation Guide For the U.S. Region* (for U.S. region fleet programs from 11 April 2025 onwards).
- *Fleet Implementation Guide - Mastercard Corporate Fleet card Program* (for Canada region and U.S. region before 11 April 2025).
- *Fleet Implementation Guide Global* (for all regions except U.S. and Canada).
- Or an alternative Mastercard agreed specification.

BIN Range

Certain Fleet Implementation Guides require specific BIN ranges, depending on the type of Fleet Card being issued. This ensures that terminals can correctly detect fleet cards by BIN range, and act accordingly.

R	All	CT	RI455.24	If a specific BIN range is required by the appropriate Fleet Implementation Guide, this must be used
		CL		

Application identifiers

Fleet Cards use the appropriate AID for the underlying brand.

For more information, refer to chapter 3 "Issuer requirements," section "Application selection," subsection "Application identifiers."

An AID extension may be used where terminals adopt special processing for particular programs within the overall Fleet program, for example, collecting different data or extending special

service to a subset of Fleet Cards. Where the special processing relates to a given fleet scheme, the 'FC uu uu' AID extension should be used.

Refer to chapter "Data requirements," section "Application identification," subsection "Extended AIDs with special acceptance functionality," sub-subsection "Fleet Cards." Alternatively, programs can apply to Mastercard for the allocation of a club identifier. Refer to chapter "Data requirements," section "Application identification," subsection "Extended AIDs with special acceptance functionality," sub-subsection "Club AID extension."

Fleet specific data on the card

An important benefit of Fleet Cards is the flexibility to control what can be purchased and what information is prompted for at the fleet merchant point-of-sale locations.

This is achieved by encoding fleet specific data on the card, to instruct the terminal what information to request from the cardholder. Certain Fleet Implementation Guides may also support encoding of fixed fleet specific information, such as driver or vehicle number, on the chip itself.

The appropriate Fleet Implementation Guide document makes clear which fleet-specific data must be encoded on the card, both on the chip and magnetic stripe.

If the Fleet Implementation Guide adopts the Conexus IFSF standards, then the Conexus documents including the *Implementation Guide - EMV Fleet Data Prompting, (also known as IFSF Part 3-28 Additions for EMV Fuel Cards)* provide additional information on fleet specific chip data.

R	All	CT CL	RI175.24	Fleet chip cards must personalize fleet data as appropriate for the country or region of issue, as defined by the appropriate Fleet Implementation Guide.
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For Fleet Cards that support contactless, refer to section "Contactless technology and Fleet Cards."

Online Processing

Fleet cards must be configured as online-only, so the issuer can ensure that only appropriate transactions for the card are approved.

This enables the issuer to verify the merchant, transaction type, and any required information entered by the cardholder at the fleet terminal.

R	All	CT CL	RI456.24	Fleet cards must be configured as online-only.
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CVM requirements for Fleet Cards

Fleet Cards follow the CVM Requirements for the underlying payment brand.

Mastercard Fleet Cards that are issued as vehicle cards with no specified single cardholder are accepted without verifying the cardholder. The CVM List of such cards should include Signature as POS terminals may not support No CVM. Although CVM processing results in Signature being identified as the cardholder verification method, the card will be accepted without Signature being completed according to existing Fleet Card processes.

BP	MC	CT	BI246.24	Mastercard cards issued as vehicle cards should support both Signature and No CVM. They should not support PIN for purchase transactions.
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Contactless technology and Fleet Cards

Contactless technology can be used on Fleet Cards.

Only EMV mode transactions are allowed through the contactless interface for fleet cards following these specifications:

- *Fleet Implementation Guide For the U.S. Region* (for US region fleet programs from 11 April 2025 onwards)
- *Fleet Implementation Guide - Mastercard Corporate Fleet card Program* (for Canada region and US region before 11 April 2025)
- *Fleet Implementation Guide Global* (for all regions except US and Canada), only EMV mode transactions are allowed through the contactless interface. Magnetic Stripe mode transactions must not be supported by the contactless interface, as data positions required by contactless Magnetic Stripe mode transactions are already used in the product personalization requirements of these products.

When an alternative specification is used, requirement RI211.15 may not apply.

Magnetic Stripe mode transactions must not be supported by the contactless interface, as data positions required by contactless Magnetic Stripe mode transactions are already used in the product personalization requirements of these products.

R	All	CL	RI211.15	Magnetic Stripe mode transactions must not be supported over the contactless interface of Fleet Cards.
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Regional variations

This section describes regional requirements and best practices that vary from the global requirements.

Asia/Pacific region

The additional requirements described in this section apply to issuers in Asia/Pacific region.

For cards issued in American Samoa, Guam, and North Mariana Islands, also refer to requirements relating to the Durbin Regulations in section "United States region."

Chip mandates

This EMV mandate applies to all cards in Australia and New Zealand, with the exception of non-reloadable prepaid cards.

R	All	CT CL	RI090.22	All cards in Australia and New Zealand, except for non-reloadable prepaid cards, must be EMV.
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Dual interface cards

All cards, except for non-reloadable prepaid cards, prepaid travel cards, and commercial cards, issued in Australia and New Zealand must be contactless-enabled.

R	MC MS	CT CL	RI176.14	All cards in Australia and New Zealand, except for non-reloadable prepaid cards, prepaid travel cards, and commercial cards, must be enhanced with contactless payment functionality.
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For new cards issued in Japan from 1 January 2026, the following requirement applies.

R	MC MS	CT CL	RI444.24	Effective 1 January 2026, newly issued cards (excluding non-reloadable prepaid cards) in Japan must be dual interface chip cards enabled with both EMV contact and EMV mode contactless functionality.
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Support for EMV mode and Magnetic Stripe mode

This mandate applies to issuers in Asia/Pacific.

R	All	CL	RI400.20	Any new contactless-enabled chip cards in the Asia/Pacific region submitted for Card Personalization Validation (CPV) testing must only support EMV mode contactless functionality and must not support Magnetic Stripe mode contactless functionality.
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R	All	CL	RI402.24	Effective 1 April 2025, newly issued and reissued contactless-enabled access devices in the Asia/Pacific region must only support EMV mode contactless functionality and must not support Magnetic Stripe mode contactless functionality.
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Single Tap and PIN Request

Mastercard has introduced the Single Tap and PIN Request functionality.

Single Tap and PIN Request provides a seamless cardholder experience when low value contactless transactions require an SCA. Cardholders can enter their PIN for the same contactless transaction instead of having to dip their card.

R	All	CL	RI437.23	Issuers in the Asia/Pacific region must be prepared to receive DE 48, subelement 22, subfield 02 (Single Tap Indicator) and 03 (Response to PIN Request).
BP	All	CL	BI238.23	Issuers in the Asia/Pacific region should support Single Tap and PIN Request functionality.

Application priority indicator

The Mastercard brand must have a higher priority than other payment applications on the cards.

R	All	CT CL	RI224.22	Issuers must attach an application priority indicator to Mastercard, Maestro, or Cirrus payment application on cards and access devices that is higher than the value of the application priority indicator of any other payment chip application co-residing on the card.
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Purchase with cash back

The support for purchase with cash back is mandatory on all Debit Mastercard cards issued in Australia.

R	MC	CT CL	RI416.22	Debit Mastercard cards issued in Australia must indicate support for purchase with cash back transactions for domestic usage in the Application Usage Control unless this is a card program on which cash access is prohibited, such as anonymous prepaid program.
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CVM requirements

Another CVM requirement applies to Mastercard and Debit Mastercard cards issued in Australia.

R	MC	CT CL	RI166.24	Issuers in Australia must personalize newly issued and reissued Mastercard and Debit Mastercard cards as PIN-preferring cards unless an individual cardholder has a genuine need for a Signature-preferring card.
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Canada region

The additional requirements described in this section apply to issuers in the Canada region.

Support for EMV mode and Magnetic Stripe mode

The mandate described in this section applies to issuers in Canada.

R	MC	CL	RI360.24	Newly issued and reissued Mastercard contactless cards and contactless payment devices in the Canada region must support EMV mode.
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Purchase with cash back

The support for purchase with cash back is mandatory on all Debit Mastercard cards, Prepaid Mastercard cards, and access devices issued in the Canada region.

R	MC	CT CL	RI336.20	Debit Mastercard, Prepaid Mastercard cards, and access devices in the Canada region must indicate support for purchase with cash back transactions for both domestic and international usage in the Application Usage Control.
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Europe region

The requirements described in this section apply to issuers in the Europe region.

Chip mandates

The requirements listed in this section apply to all issuers residing in any of the Single European Payments Area (SEPA) countries and in a list of selective non-SEPA countries.

The list of countries and territories that are part of the SEPA region are in *Mastercard Rules*.

R	All	CT	RI002.18	All contact cards in the SEPA region and in Albania, Bosnia and Herzegovina, Kosovo, Macedonia, Moldova, Montenegro, and Serbia, except for non-reloadable prepaid cards and Maestro chip-only cards, must support both Magnetic Stripe and EMV chip technology.
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Maestro chip-only cards

Maestro issuers in the Europe region may issue chip-only cards.

Mastercard recommends the use of a dummy Magnetic Stripe on a chip-only card to ensure acceptance at EMV ATMs and to support other Magnetic Stripe activated functions. The dummy Magnetic Stripe should contain at least a Start Sentinel and End Sentinel and need not contain valid Track data.

For more details on this option, refer to the *Mastercard Rules*.

Support for EMV mode and Magnetic Stripe mode

The mandates described in this section apply to contactless cards and contactless payment devices issued in the Europe region.

R	All	CL	RI358.20	All contactless cards and contactless payment devices issued in the Europe region must be EMV mode capable.
R	All	CL	RI359.24	Newly issued and reissued contactless cards in the Europe region must be EMV mode-capable only and must not support Magnetic Stripe mode contactless functionality.
R	All	CL	RI406.24	Effective 1 April 2025, newly issued and reissued contactless-enabled access devices in the Europe region must only support EMV mode contactless functionality and must not support Magnetic Stripe mode contactless functionality.

Dual interface cards

The country mandates for dual interface cards that are described in this section apply to issuers in Italy and the United Kingdom.

R	MC MS	CT CL	RI204.22	Except for non-reloadable prepaid cards, all Mastercard and Maestro cards in circulation in Italy must be dual interface cards.
R	MC	CT CL	RI356.17	Except for commercial, prepaid, and ATM-only Mastercard cards, all Mastercard consumer credit cards in circulation in the United Kingdom must be dual interface.

PSD2

Payment Service Directive 2 (PSD2) and the related Regulatory Technical Standards (RTS) define how Strong Customer Authentication (SCA) must be applied for payments occurring within the European Economic Area (EEA), the United Kingdom, and Gibraltar.

These requirements maintain high security standards that protect cardholders and their accounts.

Under the RTS Articles 11 and 12, there are exemptions defined for which issuers are not required to apply SCA for payment at the point-of-interaction (POI).

The following exemptions apply to contactless transactions that are under either one of these articles:

- Article 11 - Individual contactless transactions that do not exceed EUR 50 and either the cumulative amount since the last SCA does not exceed EUR 150 or the consecutive number of contactless transactions since the last SCA does not exceed 5
- Article 12 - Transactions at an unattended payment terminal for paying a transport fare or a parking fee

The valid transport and parking card acceptor business codes (MCCs) are:

- 4111 (Local Commuter Transport)
- 4112 (Passenger Railways)
- 4131 (Bus Lines/Charters Tours)
- 4784 (Tolls/Bridges Fees)
- 4789 (Transportation Services Not Elsewhere Classified)
- 7523 (Parking Lots/Garages)

Issuers must implement one of the following controls by managing the requirement for card verification:

- In the host or on-behalf services and responding with an appropriate Authorization Response Code to force a Single Tap and PIN request or decline the contactless transaction and force a contact transaction with cardholder verification
- In the card risk management, declining the contactless transaction and force a contact transaction with cardholder verification

If using card risk management parameters, issuers should not use both the accumulator and counter for the same transaction to ensure that excessive cardholder verification is not requested.

R	All	CL	RI397.22	Issuers in the EEA countries, the United Kingdom, and Gibraltar must be able to comply with articles 11 and 12 of the RTS by using one of the following: • Host system functionality • Card risk management settings • Mastercard's Contactless Risk Manager
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BP	All	CL	BI126.22	Issuers in the EEA countries, the United Kingdom, and Gibraltar should either monitor the cumulative amount spent in contactless transactions without CVM or the number of contactless transactions without CVM, but not both at the same time.
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To meet the SCA requirements, all Mastercard cards should be configured as PIN-preferring.

BP	MC	CL	BI124.22	All contactless cards issued in EEA countries, the United Kingdom, and Gibraltar should be Online PIN-preferring.
BP	MC	CT	BI125.22	All contact cards issued in EEA countries, the United Kingdom, and Gibraltar should be PIN-preferring, Offline or Online.

Single Tap and PIN Request

Once the Article 11 exemption counter or accumulator reaches its limit during a contactless low-value transactions (LVT), Europe region issuers can use a Single Tap and PIN Request mechanism, also known as single tap processing, that allows them to request an Online PIN in a single tap mode.

The cardholder only needs to subsequently enter an Online PIN at the terminal PIN pad without the need to tap again or perform a Chip and PIN transaction, traditional usage of Response Code "65" (Exceeds withdrawal count limit).

R	All	CL	RI396.20	Issuers in the Europe region supporting Single Tap and PIN Request processing must support the indicators received in DE 48, subelement 22, subfield 02 (Single Tap Indicator) and 03 (Response to PIN Request). Also, issuers must be able to send back DE 48, subelement 22, subfield 04 (Issuer PIN Request in a Single Tap Mode) to request a PIN in Single Tap mode.
R	All	CL	RI404.19	Issuers in the Europe region must be able to support the indicator received in DE 48, subelement 22, subfield 02 (Single Tap Indicator) and ignore it if Single Tap mode is not supported by the issuer host.

Support of Single Tap and PIN Request is currently mandated for issuers in Andorra, Austria, Belgium, Bulgaria, Croatia, Cyprus, Czech Republic, Denmark, Estonia, Finland, France, Georgia, Germany, Greece, Hungary, Iceland, Italy, Kazakhstan, Latvia, Lithuania, Luxembourg, Malta, Monaco, Netherlands, Norway, Poland, Portugal, Romania, San Marino, Slovakia, Slovenia, Spain, Sweden, and Vatican City.

R	All	CL	RI413.24	Issuers in Andorra, Austria, Belgium, Bulgaria, Croatia, Cyprus, Czech Republic, Denmark, Estonia, Finland, France, Georgia, Germany, Greece, Hungary, Iceland, Italy, Kazakhstan, Latvia, Lithuania, Luxembourg, Malta, Monaco, Netherlands, Norway, Poland, Portugal, Romania, San Marino, Slovakia, Slovenia, Spain, Sweden, and Vatican City must support Single Tap and PIN Request functionality.
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Support of Single Tap and PIN Request is recommended in all other markets in the Europe region that support Online PIN at the Point-Of-Sale (POS). Any issuer in the Europe region that wants to support Single Tap and PIN Request functionality must comply with requirement RI396.20.

BP	All	CL	BI233.22	Any countries in the Europe region supporting Online PIN should support Single Tap and PIN Request functionality.
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Contactless Risk Manager

Mastercard has developed a Contactless Risk Manager service, previously called On-behalf Contactless Counters Service, that monitors low-value contactless transactions acquired in the EEA countries, the United Kingdom, and Gibraltar and prompts for SCA at the required threshold.

It enables issuers to comply with the SCA requirements described in articles 11 and 12. This service can be adopted optionally by issuers in EEA countries, the United Kingdom, and Gibraltar and has recently been updated to offer a similar counter mechanism to non-EEA issuers.

Application selection registered proprietary data

The requirement described in this section applies to issuers residing in the European Union (EU) and in Norway, Iceland and Liechtenstein. It no longer applies to the United Kingdom, and Gibraltar.

R	MC MS	CT	RI338.16	All new or reissued chip cards must contain the Application Selection Registered Proprietary Data (tag '9F0A') in the Directory Discretionary data (tag '73') within any Application Definition File (ADF) Directory Entry within the Payment System Environment (PSE), and in the FCI Issuer Discretionary data (tag 'BFOC') within the File Control Information (FCI) of each payment application ADF.
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R	MC MS	CL	RI339.16	All new or reissued contactless cards and contactless payment devices must contain the Application Selection Registered Proprietary Data (tag '9F0A') in each Directory Entry (tag '61') within the FCI of the Proximity Payment System Environment, and in the FCI Issuer Discretionary Data (tag 'BF0C') within the FCI of each payment application ADF.
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Application Priority Indicator

The requirement described in this section applies to cards issued in the European Union (EU) and in Norway, Iceland, Liechtenstein, the United Kingdom, and Gibraltar.

R	All	CT CL	RI223.25	For cards issued in the European Union (EU), Iceland, Liechtenstein, Norway, United Kingdom and Gibraltar an application priority indicator must be attached to the Mastercard, Maestro, or Cirrus payment application on cards and access devices. This application priority indicator must not be lower than the value of the application priority indicator of any other payment chip application co-residing on the card.
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CVM requirements for Maestro cards

Mastercard permits Maestro No CVM card acceptance at Europe transit vending machines, parking, garages, and tollways.

To promote acceptance in these environments, No CVM may be included in the CVM List on the contact interface of Maestro chip cards issued in the Europe region. If No CVM is included, it must be the last entry in the CVM List.

R	MS	CT	RI163.20	Maestro cards in the Europe region that support No CVM on the contact interface must include it as the last entry in the CVM List.
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Maestro No CVM authorization processing

Maestro contact cards are permitted to be accepted in the Europe region without PIN entry up to the maximum transaction amount indicated and in the acceptance environments listed in this section.

If the transaction amount exceeds the applicable limits, the PIN must be used according to standard Maestro processing or else the terminal must abort the transaction.

Acceptance environment maximum	Transaction amount
Tollways (MCC 4784)	EUR 100 or local currency equivalent

Acceptance environment maximum	Transaction amount
Parking Lots and Garages (MCC 7523)	EUR 50 or local currency equivalent
Unattended Transit Vending Machines (MCCs 4111, 4112, 4131 and 4789)	EUR 25 or local currency equivalent

If the Maestro card does not include No CVM in the CVM List on the contact interface, the transaction should still be processed by the terminal with no cardholder verification. In this case, the TVR will indicate that "Cardholder Verification was not Successful" and may indicate "PIN Entry Required and PIN Pad Not Present or Not Working".

Maestro contactless transaction amounts above the CVM Limit without CVM may also be performed in the MCCs listed in the table. Contactless issuers must not systematically decline such Maestro contactless transactions when completed without CVM.

R	MS	CT	RI054.13	Issuers must not routinely decline transactions on Maestro cards that do not support No CVM when the following conditions are satisfied: • The transaction occurs in the Europe region and it is identified with one of the permitted MCCs. • The transaction amount does not exceed the applicable limit for each MCC. • TVR Byte 3 bit 8 indicates that "Cardholder Verification was not Successful", and where applicable. • TVR Byte 3 bit 5 indicates "PIN Entry Required and PIN Pad Not Present or Not Working". All other authorization processing must proceed as normal for the transaction.
R	MS	CL	RI340.16	Issuers must not routinely decline Maestro contactless transaction amounts above the CVM limit with No CVM transactions if they have been performed in one of the permitted MCCs.

This requirement does not apply to Maestro issuers in the Netherlands for transactions acquired at Transit Vending Machines.

PIN management

The support of PIN Management Services is currently recommended for the Europe region issuers.

BP	All	CT	BI098.14	Issuers in the Europe region should support PIN Management Services.
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Purchase with cash back

Purchase with cash back must be supported by Maestro and Debit Mastercard cards issued in the Europe region for both domestic and international usage.

The mandate also applies to Mastercard credit cards issued in specific European countries, as described in these requirements. For all other European countries, support of purchase with cash back on Mastercard credit cards remains optional.

Mastercard credit cards issued in the Europe region that support purchase with cash back for international usage must also support purchase with cash back for domestic usage.

R	MS MC	CT CL	RI083.12	Maestro and Debit Mastercard cards issued in the Europe region must indicate support for purchase with cash back transactions for both domestic and international usage in the Application Usage Control.
R	MC	CT CL	RI368.17	Mastercard credit cards issued in the Europe region that support purchase with cash back for international transactions must also support purchase with cash back for domestic transactions and indicate that in the Application Usage Control.
R	MC	CT CL	RI352.24	Newly issued and reissued Mastercard credit cards in Germany, except for prepaid Mastercard cards, must indicate support for purchase with cash back transactions for both domestic and international usage in the Application Usage Control.
R	MC	CT CL	RI353.24	Newly issued and reissued Mastercard credit cards in Romania must indicate support for purchase with cash back transactions for domestic usage in the Application Usage Control.
R	MC	CT CL	RI393.24	Newly issued and reissued Mastercard credit cards in Russia must indicate support for purchase with cash back transactions for domestic and international usage in the Application Usage Control.
R	MC	CT CL	RI395.22	All Mastercard credit cards in Ukraine must indicate support for purchase with cash back transactions for domestic and international usage in the Application Usage Control.
R	MC	CT CL	RI441.24	Newly issued and reissued Mastercard prepaid cards and access devices in Italy must indicate support for purchase with cash back transactions for domestic and international usage in the Application Usage Control.

R	MC	CT CL	RI425.24	Newly issued and reissued Mastercard credit cards and access devices in Armenia, Belarus, Georgia, Kazakhstan, Kyrgyzstan, Tajikistan, Turkmenistan, and Uzbekistan must indicate support for purchase with cash back transactions for domestic and international usage in the Application Usage Control.
R	MC	CT CL	RI426.22	Effective 1 December 2025, all Mastercard credit cards and access devices in Armenia, Belarus, Georgia, Kazakhstan, Kyrgyzstan, Tajikistan, Turkmenistan, and Uzbekistan must indicate support for purchase with cash back transactions for domestic and international usage in the Application Usage Control.
R	MC	CT CL	RI448.23	Newly issued and reissued Mastercard cards and access devices in Moldova must indicate support for purchase with cash back transactions for both domestic and international usage in the Application Usage Control.
R	MC	CT CL	RI449.23	Effective 1 January 2025, all Mastercard cards and access devices in Moldova must indicate support for purchase with cash back transactions for both domestic and international usage in the Application Usage Control.

ATM

Cards issued in these countries must support ATM transactions on the contactless interface: Albania, Austria, Bosnia, Bulgaria, Croatia, the Czech Republic, Hungary, Israel, Macedonia, Montenegro, Poland, Romania, Serbia, Slovakia, and Slovenia.

In addition, issuers in the listed countries must accept transactions performed on ATMs.

R	MS MC	CL	RI201.22	Maestro and Mastercard contactless cards and contactless payment devices issued in the listed countries must indicate support for cash withdrawal transactions, domestic and international, at ATM in the Application Usage Control if it is also supported on the contact interface.
R	MS MC	CL	RI202.14	Issuers of Maestro and Mastercard contactless cards and contactless payment devices in the listed countries must not routinely decline transactions performed on ATMs.

Latin America and the Caribbean region

The additional requirements described in this section apply to issuers in the Latin America and the Caribbean region.

For cards issued in Puerto Rico and American Virgin Islands, also refer to requirements relating to the Durbin Regulations in section "United States region."

Support for EMV mode and Magnetic Stripe mode

The mandate in this section applies to the Latin America and Caribbean region.

R	All	CL	RI399.20	Any new contactless-enabled cards in the Latin America and Caribbean region submitted for Card Personalization Validation (CPV) testing must only support EMV mode contactless functionality and must not support Magnetic Stripe mode contactless functionality.
R	All	CL	RI403.24	Effective 1 April 2025, newly issued and reissued contactless-enabled access devices in the Latin America and Caribbean region must only support EMV mode contactless functionality and must not support Magnetic Stripe mode contactless functionality.

Single Tap and PIN Request

Mastercard has introduced the Single Tap and PIN Request functionality.

Single Tap and PIN Request provides a seamless cardholder experience when low value contactless transactions require an SCA. Cardholders can enter their PIN for the same contactless transaction instead of having to dip their card.

R	All	CL	RI438.23	Issuers in the Latin America and the Caribbean region must be prepared to receive DE 48, subelement 22, subfield 02 (Single Tap Indicator) and 03 (Response to PIN Request).
BP	All	CL	BI239.23	Issuers in the Latin America and the Caribbean region should support Single Tap and PIN Request functionality.

Middle East/Africa region

The additional requirements described in this section apply to issuers in the Middle East/Africa region.

Application Priority Indicator

The requirement in this section applies to payment applications used by the Middle East/Africa market.

R	All	CT CL	RI331.18	No payment application resident on the chip of a card issued in the Middle East/Africa region must have a higher application priority than the card's primary application.
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Support for EMV mode and Magnetic Stripe mode

The mandate in this section applies to the Middle East/Africa region.

R	All	CL	RI398.24	Newly issued and reissued contactless-enabled chip cards in the Middle East/Africa region must be enabled with only EMV mode contactless functionality and must not support Magnetic Stripe mode contactless functionality.
R	All	CL	RI405.24	Effective 1 April 2025, newly issued and reissued contactless-enabled access devices in the Middle East/Africa region must only support EMV mode contactless functionality and must not support Magnetic Stripe mode contactless functionality.

Single Tap and PIN Request

Mastercard has introduced the Single Tap and PIN Request functionality.

Single Tap and PIN Request provides a seamless cardholder experience when low value contactless transactions require an SCA. Cardholders can enter their PIN for the same contactless transaction instead of having to dip their card.

R	All	CL	RI439.23	Issuers in the Middle East/Africa region must support Single Tap and PIN Request functionality.
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Purchase with cash back

The requirement described in this section applies to issuers in Kenya.

R	MC	CT	RI435.22	Maestro, Mastercard, and Debit Mastercard cards and access devices issued in Kenya submitted to Card Personalization Validation (CPV) or Change Notification Statement (CNS) must indicate support for purchase with cash back and cash back without purchase transactions for domestic usage in the Application Usage Control.
	MS	CL		

United States region

The requirements described in this section apply to issuers in the United States region.

Introduction to Mastercard Debit Programs

The following programs, which are listed in this section, generate transactions that are routed across Mastercard networks and are subject to Mastercard Rules:

- Debit Mastercard

- PIN Only Debit
- Other debit products where Mastercard is a routing option available to merchants or acquirers

Debit Mastercard

A Debit Mastercard card carries the Mastercard brand and is accepted both domestically and internationally.

The card must carry the Mastercard AID for chip transactions. Mastercard chip cards must be hybrid and may also support contactless transactions.

Debit Mastercard cards must use an M/Chip card application.

Transactions initiated using the Mastercard AID must be routed to a Mastercard network for processing. Along with supporting the Mastercard AID, these cards must also support the U.S. Maestro AID. This AID may be used by merchants to originate transactions that may be routed through unaffiliated debit networks in accordance with the Durbin amendment to the Dodd-Frank Wall Street Reform and Consumer Protection Act (Durbin Regulations). Transactions originated from this AID may also be routed to Mastercard networks for processing.

Durbin Regulations also apply to U.S. Territories. Therefore, all requirements applicable to debit products also apply to cards issued in U.S. overseas territories.

PIN Only debit

A PIN Only debit card may carry the Maestro brand and is accepted both domestically and internationally.

The card must carry the Maestro AID for chip transactions. Maestro chip cards must be hybrid and may also support contactless transactions.

Maestro cards must use an M/Chip card application.

Transactions initiated using the Maestro AID must be routed to a Mastercard network for processing. Along with supporting the Maestro AID, these cards must also support the U.S. Maestro AID. This AID may be used by merchants to originate transactions that may be routed through unaffiliated debit networks in accordance with Durbin Regulations. Transactions that originate from this AID may also be routed to Mastercard networks for processing.

Other U.S. debit branded cards

Issuers may choose Maestro as an unaffiliated PIN Debit network on debit cards supporting another brand. The requirements for these cards are outside the scope of this document.

These cards may support the U.S. Maestro AID supporting an M/Chip card application. Alternatively, these cards may support an AID and application sponsored by the front-of-card brand. Use of such an AID and application is subject to the licensing agreement between Mastercard and the front-of-card brand. Chip transactions originating from these cards may be routed over Mastercard networks but may, at the acquirer's option, be routed over other networks.

U.S. Maestro AID

The U.S. Maestro AID is to be used for regulated Debit cards in the U.S. market and U.S. Territories only and will not be supported by any cross-border acceptance devices. It may be used for both contact and EMV mode contactless transactions.

Unaffiliated networks processing transactions from the U.S. Maestro AID must enter into a licensing agreement with Mastercard and are subject to the terms in the agreement. The AID must appear on U.S.-issued Debit Mastercard cards, prepaid Mastercard cards, and Maestro cards. It may also appear on other cards that are not a Mastercard product. Such cards are beyond the scope of this document.

Cards using the U.S. Maestro AID must use an M/Chip card application for transactions using this AID. When the card is a Debit Mastercard card, a prepaid Mastercard card, or a Maestro card, Mastercard Network must be a routing option.

Card application

This requirement applies to card applications used by the United States market.

R	All	CT	RI086.12	In the United States region, payment applications used on Mastercard payment chip cards must comply with the M/Chip specifications.
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Support for EMV mode and Magnetic Stripe mode

A contactless card or contactless payment device for the U.S. Maestro AID must support EMV mode transactions and not support Magnetic Stripe mode transactions.

R	USM	CL	RI341.16	Magnetic Stripe mode must not be supported for the U.S. Maestro AID.
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This mandate applies to contactless cards and contactless payment devices issued in the United States.

R	MC	CL	RI387.24	Newly issued and reissued Mastercard contactless cards and access devices issued in the U.S. must be enabled with EMV mode functionality. Contactless Magnetic Stripe mode functionality is optional.
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NOTE: The support of Magnetic Stripe mode contactless is optional for newly issued and reissued contactless cards in the United States region.

Online processing

EMV-capable cards issued in the United States region must be configured as online-only or online-preferring.

R	All	CT CL	RI115.16	EMV-capable cards issued in the United States region must be configured as online-only or online-preferring.
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Application Priority Indicator

No payment application may have a higher application priority than the card's primary application for cards issued in the United States region.

R	MC MS	CT	RI087.12	For cards issued in the United States region, no payment application may have a higher application priority than the card's primary application.
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Application label

These requirements apply to Debit Mastercard applications issued in the United States region.

R	MC	CT CL	RI378.18	In the United States region, the application label for the Debit Mastercard application must be "Mastercard Debit".
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This requirement applies to Mastercard Credit applications present on multi-account cards issued in the United States region.

R	MC	CT CL	RI379.18	In the United States region, when a Mastercard credit application is on the same card as a debit application, the application label for the credit Mastercard application must be "MastercardCredit".
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This requirement applies to U.S. Maestro AID.

R	USM	CT CL	RI342.16	The application label associated with the U.S. Maestro AID must not infer that the resulting transaction will be a Mastercard transaction.
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Application preferred name

The requirement described in this section applies to multi-account cards in the United States region.

R	All	CT CL	RI367.17	When a card supports multiple products that are linked to the same application label, the application preferred name must be present for each of these products to reduce potential cardholder confusion.
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Issuer Country Code

All United States debit and prepaid applications must include the two-character alpha Issuer Country Code of "US" (ASCII), or ('5553') in the data accessed at application selection.

The Issuer Country Code must not be present for all other card types issued in the United States region or a United States Territory if the Issuer Identification Number is also present.

R	All	CT CL	RI099.24	All debit applications and prepaid applications on cards issued in the United States region or in a United States Territory must include the two-character alpha Issuer Country Code of "US".
R	All	CT CL	RI351.16	Other than for debit and prepaid cards, the two-character alpha Issuer Country Code and the Issuer Identification Number must not both be present for cards issued in the United States region or in a United States Territory.

Issuer Identification Number

All United States debit and prepaid applications must include the Issuer Identification Number (IIN) in the data accessed at application selection.

The Issuer Identification Number must not be present for all other card types issued in the United States region or in a United States Territory if the two-character alpha Issuer Country Code is also present.

R	All	CT CL	RI180.22	All debit and prepaid applications on cards issued in the United States region must include the Issuer Identification Number in the data accessed at application selection.
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CVM requirements

CVM requirements for U.S. Maestro AID

The U.S. Maestro AID contact application must support Online PIN and No CVM. U.S. Maestro AID contact applications may also support Offline PIN. Either Offline plaintext PIN, Offline enciphered PIN, or both may be supported.

Signature is not permitted in the CVM List. However, local processing practices may lead to Signature being requested for higher value transactions where No CVM has been used.

Online PIN entry may be bypassed for U.S. Maestro AID contactless transactions. In this case issuers should be aware that the TVR indicates "Online PIN entered," even though it was bypassed. This happens because the TVR was sent to the card before it was known that the PIN was bypassed. Leaving the TVR unchanged ensures that the verification of the ARQC by the issuer is not affected.

R	USM	CT	RI343.16	U.S. Maestro AID applications must support Online PIN and No CVM in the CVM List read through the contact interface.
R	USM	CT	RI344.16	U.S. Maestro AID applications must not support Signature in the CVM List read through the contact interface.

U.S. Maestro AID contactless cards:

- Must support Online PIN
- Must support No CVM

U.S. Maestro AID contactless payment devices:

- Must support Online PIN
- Must support No CVM
- Must not support CDCVM

U.S. Maestro AID contactless cards and contactless payment devices must not support Signature.

R	USM	CL	RI345.16	U.S. Maestro AID contactless cards and contactless payment devices must support Online PIN and No CVM in the CVM List read through the contactless interface.
R	USM	CL	RI347.16	U.S. Maestro AID contactless payment devices must not support CDCVM.
R	USM	CL	RI348.16	U.S. Maestro AID applications must not support Signature in the CVM List read through the contactless interface.

BP	USM	CL	BI242.24	If Online PIN is bypassed, the issuer receives a TVR indicating Online PIN entered, but no PIN will be present in the message. This should be expected as normal in a PIN bypass scenario, and should not be routinely declined if the issuer permits PIN-bypass.
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CVM requirements for Mastercard

The requirement for "Online PIN (for all terminals)" to be present in the CVM list on Mastercard, for both contact and contactless interfaces, is optional as long as both:

- "Online PIN (if unattended Cash)" is present as the first CVM list entry
- Signature is present in the CVM list

R	MC	CT CL	RI450.24	Mastercard cards issued in the US region may omit "Online PIN (for all terminals)" from the CVM list, on the condition that "Online PIN (if unattended cash)" is the first CVM list entry, and "Signature" is in the CVM list.
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Biometric Cards

Issuers in the United States region must not issue Biometric Cards under Mastercard debit card programs.

R	MC MS	CT CL	RI436.22	Biometric Cards must not be issued under Mastercard debit card programs in the United States region.
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Stand-In Processing

Issuers in the United States region must follow the requirement described in this section in Stand-In Processing.

R	MC	CL	RI337.16	Issuers of contactless-enabled Mastercard and Debit Mastercard card programs that support Magnetic Stripe mode transactions must enroll for the Dynamic CVC 3 Validation in Stand-In service.
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Chapter 4 Acquirer requirements

This chapter contains the requirements acquirers must comply with to accept chip cards. These general requirements apply to both contact and contactless terminals unless specified. Requirements apply to all Mastercard card programs unless it is explicitly stated that they are program-specific or brand-specific.

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Terminal applications

Mastercard does not publish a terminal specification for contact transactions.

Terminals that support contact transactions must respect the requirements in this document and should be compliant with the most recent version of the EMV® specifications.

BP	All	CT	BA001.11	All contact terminals submitted for M-TIP testing should conform to the latest version of the EMV specifications.
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All contactless readers must be compliant with *Mastercard® Contactless Reader Specification v3.0* or later or *EMVCo Book C-2*.

R	All	CL	RA336.19	All contactless readers must be compliant with <i>Mastercard Contactless Reader Specification v3.0</i> or later or <i>EMVCo Book C-2</i> .
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Refer to the regional requirements for the United States region in chapter 4, "Acquirer requirements" section, "Regional variations."

In addition to the latest reader specifications, acquirers must ensure that the contactless readers are compliant with the Proximity Coupling Device (PCD) requirements. The Proximity Coupling Device is one of the key peripheral components of a contactless reader. This device handles the communication channel between the cardholder's contactless card or payment device and the terminal contactless reader.

Terminal requirements

To ensure interoperability and a good cardholder experience, Mastercard specifies various terminal requirements.

Chip and contactless acceptance

To ensure that Magnetic Stripe-only cards are still accepted, all merchants accepting contact chip must also be able to process Magnetic Stripe cards.

Terminals that can accept both contact chip and Magnetic Stripe technology and that have completed the Mastercard Terminal Integration Process (M-TIP) are called hybrid terminals.

All contact chip terminals must be hybrid terminals supporting both chip and Magnetic Stripe transactions. However, chip-only terminals may be deployed provided Magnetic Stripe cards are still accepted at the same merchant location. An exception to this requirement is the use of Chip-only Mobile Point-Of-Sale (MPOS) terminals. Refer to chapter 4, "Acquirer requirements" section "Terminal types," subsection "Mobile POS terminals."

Whenever a contact chip capable terminal that accepts Mastercard brands is deployed by an acquirer, the terminal must support all the Mastercard brands that it supports over the Magnetic Stripe interface over the chip interface. A terminal that is capable of supporting contact EMV transactions for other brands over the chip interface and Mastercard brands only over the Magnetic Stripe interface must not be deployed.

R	All	CT	RA184.25	All chip terminals, with the exception of Chip-only MPOS terminals and offline-only CATs, used to accept Mastercard chip cards must be hybrid unless there is another terminal in the same merchant location where Magnetic Stripe cards are accepted.
R	All	CT	RA004.13	A terminal that is capable of supporting EMV transactions for competing brands over the contact interface must support all the Mastercard brands on the contact interface that it supports on the Magnetic Stripe interface.

Apart from Magnetic Stripe and EMV contact transactions, terminals should also support contactless transactions. More specifically, all contactless cards and contactless payment devices should be valid for acceptance at all terminals, not just contactless cards. This includes, for example, mobile phones containing a contactless application.

BP	All	CT CL	BA113.18	All terminals should be hybrid and contactless-enabled.
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Refer to the regional requirements for the Asia/Pacific region, Canada region, Europe region, Latin American and Caribbean region, Middle East/Africa region, and the United States region in chapter "Acquirer requirements," section "Regional variations."

All contactless-enabled terminals that accept Mastercard should support EMV mode transactions as a minimum.

All contactless-enabled terminals that accept Cirrus® or Maestro® must support EMV mode transactions and must not support Magnetic Stripe mode transactions for these applications.

The EMV mode will be used if supported by the card and terminal. Attempted EMV mode transactions must not fall back to Magnetic Stripe mode. Terminals cannot change processing mode during a transaction once it is determined.

R	MS CI	CL	RA204.14	All contactless-enabled terminals accepting Cirrus or Maestro must not support Magnetic Stripe mode transactions for these AIDs.
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R	MS CI	CL	RA396.16	All contactless-enabled terminals accepting Cirrus or Maestro must support EMV mode transactions.
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A terminal that supports Magnetic Stripe and EMV chip contact transactions, hybrid terminal, that also supports contactless must support EMV mode.

For dual interface hybrid terminals, the same Mastercard applications must be supported over the contact and contactless interfaces.

R	All	CT CL	RA205.18	A dual interface hybrid terminal must support the same Mastercard applications on both contact and contactless interfaces.
R	All	CL	RA201.14	A contactless-enabled terminal that supports contact EMV transactions must also support EMV mode contactless transactions.

Contactless only terminals are not permitted except in:

- Unattended POS terminals identified as CATs
- Attended POS terminals deployed in specific countries and card acceptor business codes (MCCs) after a case-by-case approval by Mastercard
- Contactless-only MPOS solutions

For a full list of exceptions where contactless-only acceptance is allowed, refer to the *Transaction Processing Rules*.

R	All	CT CL	RA296.20	Contactless-enabled terminals, except for contactless-only MPOS, CATs, and attended POS deployed in specific countries and MCCs, must also support contact transactions.
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Terminal branding

Contactless-enabled terminals use common interfaces to provide a consistent consumer and merchant experience. Contactless-enabled terminals must meet the Mastercard branding requirements as they are described in the *Mastercard Contactless Branding Standards* manual.

Terminal design and ergonomics

Terminals that accept Mastercard contactless transactions must give the cardholder clear information where to tap the contactless device, ensure ease of use, and avoid certain fraud risks. The design requirements that contactless-enabled terminals must meet are described in the *Mastercard Contactless Terminal Implementation Requirements* manual.

Hybrid terminal card reader requirements

Hybrid terminals need a reader for the chip and another reader for the Magnetic Stripe.

The readers may be completely separate if there is a slot to swipe the Magnetic Stripe and a different slot to insert a chip card. Terminals may instead contain a combined reader that supports both technologies. The combined reader has a single slot used to insert both chip and Magnetic Stripe-only cards. Combined readers can select chip or Magnetic Stripe technology without the cardholder or merchant taking any action.

The kind of reader deployed affects how the terminal is used and, in particular, how technical fallback is implemented. For details, refer to chapter 4 "Acquirer requirements" section "Technology selection," subsection "Technical fallback."

POS terminals can support either combined or separate readers, but the cardholder or merchant may have to take action to ensure the correct technology is used.

ATMs must use a combined reader, meaning there is only one place for the cardholder to insert their card.

Wherever possible, CATs should use a combined reader to avoid cardholder confusion.

R	All	CT	RA010.11	Hybrid ATMs must use a combined chip and Magnetic Stripe reader.
BP	All	CT	BA002.11	CATs should use a combined chip and Magnetic Stripe reader.

Together with the issuing mandate that imposes the progressive removal of the physical Magnetic Stripe from the back of the cards (refer to requirements RI427.22 and RI428.22 in chapter "Issuer requirements"), terminals must be able to process cards that do not support the physical Magnetic Stripe in the Canada, Europe (except Switzerland), Latin America and Caribbean, Middle East /Africa, and Asia/Pacific regions.

R	All	CT	RA505.24	All chip-capable ATMs and POS terminals in deployment in the Canada region, Europe region, Latin America and Caribbean region, Middle East/Africa region and the Asia/ Pacific region must be able to correctly process chip cards with no physical Magnetic Stripe. Switzerland is excluded from this mandate.
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Refer to requirement RA512.22 in chapter "Acquirer requirements," section "Regional variations," subsection "United States region."

Contactless limits

In the contactless technical specifications, three limits are used by terminals in processing a contactless transaction. The same limit may have different values for different products and different CVM options.

The limits must be configurable for each AID accepted at the terminal. Acquirers should be aware that the transaction limits described here are managed and supported differently in the different versions of the contactless reader specifications.

The Terminal Contactless Transaction Limit is a maximum transaction amount above which a contactless transaction must not be performed. The limit may be different whether the transaction is being performed with CDCVM or not.

Terminals do not require cardholder verification for transaction amounts equal to or below the Terminal CVM Required Limit and, unless specifically requested by the cardholder, do not require a transaction receipt. For contactless transaction amounts above the Terminal CVM Required Limit, normal cardholder verification and receipt printing procedures apply.

The Terminal Contactless Floor Limit is a transaction amount above which online issuer authorization is required.

Terminal Contactless Transaction Limit

There is no maximum transaction amount published for contactless Mastercard transactions.

A terminal must be able to perform contactless transaction amounts up to the same value as other technologies whether the transaction is completed with CDCVM or another form of cardholder verification. An exception to this applies in markets that do not support Online PIN. In these markets, contactless cards and devices that do not support CDCVM do not need to be supported for contactless transaction amounts above the CVM Limit. However, mobile transaction amounts above the CVM Limit must be supported in all readers compliant with Mastercard Contactless Reader Specification v3.0 or later. Therefore, the Terminal Contactless Transaction Limit, On-device CVM, in these readers must be set to the same value as for the maximum transaction amount for contact.

For more details on processing of contactless transactions, refer to chapter 4 "Acquirer requirements" section "Cardholder Verification," subsection "Contactless transactions."

R	All	CL	RA353.15	A terminal must be able to perform contactless transaction amounts up to the same value as other technologies.
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For Mastercard Consumer-Presented QR transactions, there is no maximum transaction amount.

A terminal must be able to perform Mastercard Consumer-Presented QR transactions up to the same value as other technologies when the transaction is completed with CDCVM.

R	MC MS	QR	RA437.19	A terminal must be able to perform Mastercard Consumer-Presented QR transactions up to the same value as other technologies.
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Terminal CVM Required Limit

Transaction amounts equal to or below the Terminal CVM Required Limit do not require cardholder verification. Transaction amounts above the Terminal CVM Required Limit require cardholder verification.

Appendix C in the *Chargeback Guide* lists the amount to be used to configure the Terminal CVM Required Limit. Liability and chargeback rights, as defined in the *Chargeback Guide*, apply.

R	All	CL	RA486.20	The CVM Limit published by Mastercard that is relevant to the market must be used to set the Terminal CVM Required Limit.
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Requirement RA486.20 does not apply to ATM. Refer to requirement RA318.16 in chapter "Acquirer requirements," section "Terminal types," subsection "ATMs," sub-subsection "Contactless limits."

Terminal Contactless Floor Limit

EMV mode transactions less than or at the Terminal Contactless Floor Limit may be authorized offline. However, online issuer authorization may still be required if this is the outcome of the terminal and card risk management.

Transactions above this limit must be authorized online by the issuer to provide the acquirer protection against authorization-related chargebacks. For Magnetic Stripe mode transactions, online issuer authorization is normally obtained for all transactions.

Generally, there are no special floor limits that apply to contactless transactions. The same limit applies as for contact transactions. However, terminal implementations should be able to maintain and use this limit independently for each interface.

R	All	CL	RA235.14	Transactions above the Terminal Contactless Floor Limit must be authorized online by the issuer.
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Currency conversion

Mastercard does not recommend support for EMV currency conversion.

BP	All	CT CL	BA012.11	Terminals should not support EMV currency conversion.
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Some acquirers and merchants offer cardholders the option of converting the transaction amount from the local currency into the cardholder's billing currency before the transaction is authorized through the Mastercard system. This process is known as POI Currency Conversion (PCC).

R	All	CT CL	RA455.23	Terminals must not offer currency conversion if the Application Currency Code is present and matches the terminal's local currency.
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To perform PCC, the terminal or merchant gives the cardholder the choice of currency they want to be billed in, either the currency of the card or the local currency of the terminal. The transaction amount is then converted into the chosen currency using the exchange rate defined by the acquirer.

For contact transactions, the terminal then uses the converted values to perform a normal chip transaction.

BP	All	CT	BA078.14	For contact transactions, terminals that support POI Currency Conversion should set the Transaction Currency Code (tag '5F2A') to the selected billing currency before starting the chip transaction.
BP	All	CT	BA079.14	For contact transactions, terminals that support POI Currency Conversion should convert the Terminal Floor Limit (tag '9F1B') to the corresponding value in the selected billing currency before starting the chip transaction.
BP	All	CT	BA080.14	For contact transactions, terminals that support POI Currency Conversion should convert the Amount Authorized (tag '9F02'), Amount Authorized Binary (tag '81'), Amount, Other (tag '9F03'), and Amount, Other Binary (tag '9F04') to the corresponding values in the selected billing currency before starting the chip transaction.

For contactless transactions, the choice is offered to the cardholder after the contactless interaction between the card and terminal has been completed. POI Currency Conversion performed through the contactless interface will, therefore, result in inconsistency between

transaction amounts and currencies contained in DE 4, DE 49, and DE 55 of the online request message.

POI Currency Conversion is not supported for contactless transactions for amounts equal or below the relevant CVM Limit. Refer to Appendix C of the *Chargeback Guide* or Appendix E of the *Transaction Processing Rules* for the CVM limits.

R	All	CL	RA224.14	POI Currency Conversion must not be offered for transaction amounts that are equal to or below the CVM Limit.
R	All	CL	RA225.14	Terminals that support POI Currency Conversion must support cardholder selection of the POI Currency Conversion option.
R	All	CL	RA226.14	As POI Currency Conversion can only be offered based on data read from the card, it must not be offered before the card has been presented for the transaction.
R	All	CL	RA227.14	The implementation of POI Currency Conversion must not require a subsequent tap of the card on the terminal.
R	All	CL	RA228.14	The values of the CVM Required Limit, Contactless Transaction Limit, and Contactless Floor Limit data objects must not be converted. The terminal must compare the transaction amount to the transaction limits in the local currency, even if POI Currency Conversion is subsequently performed.

Data usage

Contactless acquirers and merchants must only use data read from the contactless interface for contactless transactions. This restriction does not apply to refunds and transit debt recovery transactions.

R	All	CL	RA237.14	Data read from the contactless interface must only be used for contactless transactions.
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Acquirers must manage cardholder data retrieved during a transaction in a secure manner. This means respecting industry practices such as Payment Card Industry (PCI) and any applicable local privacy laws.

R	All	CT CL	RA238.14	Acquirers must handle cardholder data in a secure manner. This includes compliance with Payment Card Industry Data Security Rules (PCI DSS) and local privacy laws.
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R	All	CT CL	RA239.14	Acquirers must respect local privacy laws when storing data read from the card.
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Track data consistency

For Magnetic Stripe mode contactless transactions, some POS systems collect Track 1 data, truncate it, and process it as Track 2.

Contactless Track 1 and Track 2 data may be different. For this reason, merchants and acquirers must make sure that Track 1 data is processed as Track 1 and that Track 2 Data is processed as Track 2. If data from one track is presented as the other, this may cause the transaction to be rejected by the card issuer as the dynamic CVC 3 cannot be verified correctly.

Track 2 Equivalent Data is mandatory for contact and EMV mode contactless transactions. It must be transmitted to the issuer in every online request message.

R	All	CT CL	RA240.15	Merchants and acquirers must make sure that Track 1 data is processed as Track 1 and that Track 2 data is processed as Track 2.
R	All	CT CL	RA241.14	Track 2 Equivalent Data must be used in the online request for contactless EMV mode transactions and all EMV contact transactions.

Cardholder name

Contactless cards and contactless payment devices must not include the cardholder name in the data read through the contactless interface. POS systems that normally obtain and use the cardholder name from Track 1 data obtained from a Magnetic Stripe must be able to accommodate this difference.

R	All	CL	RA245.14	Terminals that process Track 1 data must be able to handle the data without a fully populated cardholder name.
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Transaction receipts

EMV requires transaction receipts, where provided for chip transactions, to contain the AID of the application used in addition to the other data that normally appears for Magnetic Stripe transactions. The Application Label or Application Preferred Name must also be included.

R	All	CT CL	RA043.12	The transaction receipts must include the Application Preferred Name if present on the card and the terminal supports the character set indicated by the Issuer Code Table Index or otherwise the Application Label.
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The transaction receipt may also include the cryptogram and the data elements used to calculate the cryptogram.

The expiration date of the card must not be included on either the merchant's copy or the cardholder's copy of the transaction receipt. Also, the PAN of the application used for the transaction must be indicated in truncated form on the transaction receipt.

The *Transaction Processing Rules* detail the requirements and exceptions (including alternate methods of providing a receipt) for a paper transaction receipt to be offered to the cardholder at a POS terminal (including any MPOS or CAT device unless otherwise stated), regardless of if the outcome is an approval or decline.

Any receipt should specifically identify that contactless technology was used to complete the transaction. The input method should be shown as Contactless or CONTACTLESS for contactless transactions.

When a Signature is not required for cardholder verification as part of the transaction, the receipt must not include any text or placeholder for provision of a signature.

R	All	CL	RA274.23	A transaction receipt must be available for contactless transactions equal to or below the CVM Limit on cardholder request.
R	All	CT CL	RA275.24	At a POS Terminal (including any MPOS or CAT device unless otherwise stated), a paper Transaction Receipt must be offered to the Cardholder, regardless of if the outcome is an approval or decline. This does not apply to contactless transactions equal to or below the CVM limit. NOTE: See the <i>Transaction Processing Rules</i> for the detailed requirements and exceptions (including alternate methods of providing a receipt.)
R	All	CT CL	RA276.22	If Signature is not the chosen cardholder verification method for the transaction, then the receipt must not include text or placeholder for the provision of a signature.
BP	All	CL	BA065.23	Terminals should not routinely produce receipts for transactions below the CVM Limit.
BP	All	CL	BA066.14	Transaction receipts should identify contactless transactions by indicating "Contactless" or "CONTACTLESS".

Exceptions to requirements RA274.18 and RA275.18 are in chapter "Acquirer requirements," section "Regional variations," subsection "United States region."

Transaction details and terminal configuration

Acceptance problems that lead to transactions being declined offline can be very difficult to analyze and resolve because the data associated with the transaction is not usually kept.

It is easier to resolve these problems if the terminal can produce a record of the most important transaction details after the decline.

The command that produces this record is proprietary to the terminal vendor. Vendors might choose to automatically produce a record for all transactions declined offline. Any transaction data stored must conform to any applicable PCI requirements.

BP	All	CT CL	BA036.12	Hybrid terminals should be able to print, display, or record in an electronic file the terminal parameter and transaction data after an offline decline, including the data that would be included in DE 55, the Terminal Action Code (TAC) and IAC, PAN, and Application Pan Sequence Number, if present.
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Unpredictable Number

During a chip transaction, the chip cryptographically signs transaction-related data for the terminal and the issuer to check that the chip used is genuine.

The data signed by the chip includes a 4-byte Unpredictable Number (UN) generated by the terminal. The unpredictable nature of the UN enables the terminal and the issuer to be confident that the cryptogram is freshly generated and not a replay of a previously generated cryptogram.

EMV requires that the UN must be unpredictable.

R	All	CT CL	RA305.15	All chip-capable terminals in deployment must generate UNs that are unpredictable. Each possible 32-bit UN must be equally likely to be generated.
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Contactless transaction flow

To ensure a fast and effortless experience, terminals must avoid any superfluous prompts that require unnecessary merchant or cardholder action.

R	All	CL	RA489.22	Once the total transaction amount is displayed, the terminal must not prompt for cardholder confirmation before the card is presented except for the processing of special programs, such as preauthorization or M/Chip Fast.
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For contactless transactions, the aim, wherever possible, should be a single tap experience.

Terminal approval process

Acquirers and merchants must only use approved terminals.

All chip-capable terminals and readers must undergo type approval testing to help ensure that they comply with EMV and Mastercard standards.

The Vendor Terminal Approval Processes are initiated and performed by vendors to certify the terminals and readers.

- EMVCo Terminal Approval for contact terminals - The testing included in this process is performed by laboratories that have been accredited by EMVCo, and it is split into two levels of testing:
 - EMVCo Level 1 testing that focuses on the electrical behavior and the protocol
 - EMVCo Level 2 testing that focuses on the EMV kernel software

Successful completion of each of these levels is covered by a separate Letter of Approval. For more details, refer to the [EMVCo website](#).

- Contactless Reader Approval - The testing in this certification can follow either of the following independent processes:
 - Mastercard Contactless Reader Approval Process that tests whether the reader is compliant with the Mastercard contactless reader specifications
 - EMVCo Contactless Product Type Approval that tests whether the reader is compliant with EMV Contactless specifications

Both processes require the EMVCo Type Approval Contactless Terminal Level 1 to be completed as a prerequisite.

Successful completion of either Mastercard Contactless Reader Approval Process or EMVCo Contactless Product Type Approval is covered by a Letter of Approval.

- Terminal Quality Management (TQM) - This program is a Mastercard process that helps ensure the overall quality of terminal and reader production, including physical manufacturing and the vendor's control and customer service processes.
- Payment Card Industry-PIN Transaction Security (PCI-PTS) - This program is owned by the Payment Card Industry Security Standards Council and checks whether a submitted PIN Entry Device is compliant with PCI security standards, as set forth in the PIN Transaction Security (PTS) requirements.

Customer certification includes Mastercard certification processes that are initiated and performed by customers to ensure that their terminals are compliant with Mastercard Rules.

- Mastercard Terminal Integration Process (M-TIP) - M-TIP tests the behavior of chip terminal and reader once it has been integrated with the acquirer host in a test environment. M-TIP focuses on the conformance against important requirements that directly affect acceptance. Terminals submitted to M-TIP testing must have completed all the preceding Vendor Terminal Approval processes, as applicable.

Details on the prerequisites to initiate M-TIP are listed in the *M-TIP Process Guide*.

R	All	CT CL	RA007.14	All terminals that accept Mastercard chip cards must successfully complete M-TIP testing.
R	MC MS	QR	RA442.19	All terminals that accept Mastercard Consumer-Presented QR must successfully complete M-TIP testing.
R	All	CT	RA009.20	All contact terminals, except MPOS, submitted for M-TIP testing must have a valid: • EMVCo Level 1 Approval • EMVCo Level 2 Approval • Terminal Quality Management (TQM) Label or a TQM Action Plan • PCI-PTS approval, only applicable when the terminal supports a PIN Entry Device (PED)
R	All	CL	RA209.20	All contactless-enabled terminals, except MPOS, submitted for M-TIP testing must have a valid: • Mastercard Letter of Approval for Contactless Product or EMVCo Letter of Approval for Terminal Contactless Product • Terminal Quality Management (TQM) Label or a TQM Action Plan • PCI-PTS Approval, only applicable when the terminal supports a PIN Entry Device (PED)

NOTE: Requirements RA009.20 and RA209.20 do not apply to MPOS. For details on the certification of MPOS, refer to chapter "Acquirer requirements," section "Terminal types," subsection "Mobile POS Terminals," requirements RA494.22 and RA495.22.

Terminal types that have not completed the preceding testing will not qualify as chip terminals when considering liability for fraudulent transactions.

Terminals must also fulfill any other non-chip certification requirements, such as security requirements. Details of such requirements are outside the scope of this document.

Any exceptions regarding the prerequisites that terminals need to fulfill before they are submitted to M-TIP are listed in the *M-TIP Process Guide*.

Split terminal functions

Some terminals, particularly in large retail locations, may be split across different system components.

For example, the card-terminal interface functions may be performed by a combined card reader and PIN pad, which connects to the POS terminal. Several POS terminals may be

connected to a merchant host system that collects transactions and that may perform parts of the EMV processing, for example, checking warning bulletins.

Where the payment functionality is split between different devices, the overall system is considered to be the EMV terminal and must perform the functions required by EMV and be tested in the same way as any other POS terminal.

There may be several different variations of the components of a split system, such as different makes of card readers or different in-store servers and software. Therefore, several possible combinations of components exist for performing EMV functions.

R	All	CT CL	RA011.11	Split systems formed from different combinations of hardware and software must have a valid EMVCo approval and pass M-TIP in the same way as stand-alone terminals. In other words, the complete integrated system must undergo M-TIP testing.
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Terminal configuration

The Terminal Action Codes (TACs) settings for different types of contact terminals are provided in section "Terminal action codes" in chapter "Data requirements." For contactless-enabled terminals, the applicable TAC settings are in the *Mastercard Contactless Kernel Configuration* manual.

Terminal Capabilities

The Terminal Capabilities data element is used to indicate the basic capabilities of the terminal for the transaction in progress.

For the contact interface, refer to *EMV Book 4 - Cardholder, Attendant and Acquirer Interface Requirements* for the coding of the Terminal Capabilities.

For contactless interface, the applicable Terminal Capabilities settings are in the *Mastercard Contactless Kernel Configuration* manual.

Additional Terminal Capabilities

The Additional Terminal Capabilities data element is used to indicate the capabilities of the terminal above the basic capabilities expressed in the Terminal Capabilities data element.

Refer to *EMV Book 4 - Cardholder, Attendant and Acquirer Interface Requirements* for the coding of the Additional Terminal Capabilities.

Merchant Category Code '9F15'

Some card applications request tag '9F15' (Merchant Category Code) from the terminal, to control card behaviour.

As a result, it is important that '9F15' is configured correctly on the terminal, to ensure transactions are processed successfully.

BP	All	CT CL	BA129.25	All terminals should configure '9F15' (Merchant Category Code) to a value specified in the Quick Reference Booklet. If '9F15' is configured, it must match the value of DE 18 in the Authorization message.
R	All	CT CL	RA574.25	Effective 1 January 2026, all terminals submitted for M-TIP testing must configure '9F15' (Merchant Category Code) to a value specified in the Quick Reference Booklet. The value of DE 18 in the Authorization message must match the value used in the tag '9F15'.
R	All	CT CL	RA575.25	Effective 1 January 2026, newly deployed (including replacement) transit gate terminals must configure '9F15' (Merchant Category Code) to a value specified in the Quick Reference Booklet. The value of DE 18 in the Authorization message must match the value used in the tag '9F15'.
R	All	CT CL	RA576.25	Effective 1 July 2026, all transit gate terminals must configure '9F15' (Merchant Category Code) to a value specified in the Quick Reference Booklet. The value of DE 18 in the Authorization message must match the value used in the tag '9F15'.

Online processing

Terminals that accept Mastercard products must normally be online-capable. In some acceptance environments, online capability may not be possible or appropriate, such as a CAT Level 3 terminal, which is offline-only.

R	All	CT	RA005.11	All hybrid terminals must be online-capable, except: • Offline-only CATs • Terminals that accept Mastercard credit cards only, provided they support Voice Authorization • On-board terminals
BP	All	CL	BA056.14	Terminals should be online-capable unless they are CAT Level 3 terminals or other exceptions approved by Mastercard.

A service code read during a transaction indicating that the transaction must be processed online does not mean that the terminal needs to seek online authorization even for transactions below the appropriate floor limit. Also refer to the requirement RA362.16.

Refer to the regional requirements in chapter "Acquirer requirements," section "Regional variations," subsection "United States region."

Online-only terminals

Online-only terminals normally only approve transactions in real time by transmission of an online request message.

For contact terminals, online-only terminals differ from standard terminals in the following ways:

- Offline CAM does not need to be supported but must be performed if it is supported.
- Terminal Risk Management (TRM) tests do not need to be performed.
- If TRM tests are not supported, the Terminal Verification Results (TVR) bits relating to TRM are not set. If TRM is supported, the floor limit is always zero so the "Floor Limit Exceeded" bit in the TVR is always set.

Although the support of offline CAM at online-only terminals is optional, the support of CDA is recommended at online-only POS terminals. CDA protects the transaction against wedge device and replay attacks.

BP	All	CT	BA093.14	Online-only POS terminals should support CDA.
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Because some contactless cards and contactless payment devices may be configured to work as offline-only, Mastercard recommends that terminals not be configured to work online-only for contactless transactions. To maximize the benefits of chip processing, terminals should be able to complete transactions offline when required.

BP	All	CL	BA057.14	Terminals should not be online-only for contactless EMV mode transactions.
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Online-preferring terminals

Online-preferring contact terminals seek an online authorization for all transactions but may seek an offline card approval in the Unable to Go Online mode if an online response is not received.

The following terminals must perform offline CAM and Terminal Risk Management and may use one of these methods to seek an online authorization:

- Routinely request an ARQC.
- Set the "Transaction selected randomly for online processing" bit in the TVR and use Terminal Action Analysis to trigger the authorization.

The floor limit at these terminals should be set to the published Terminal offline chip authorization limit up to which transactions may be approved in Unable to Go Online mode.

BP	All	CT	BA074.22	The Terminal Floor Limits at online-preferring terminals should be set to the published terminal offline chip authorization limit.
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Technology selection

Technology selection is the process by which it is decided whether a contact or contactless transaction is to be performed and, for contact transactions, whether it can be completed using the chip or the Magnetic Stripe.

The goal of technology selection is to ensure that the most secure chip technology is used whenever possible.

If the transaction is eligible to be completed using contactless technology, the cardholder decides whether to use the contactless or contact interface on the card. The terminal does not drive this decision. If the cardholder chooses to use the contactless interface and both the card and terminal support EMV mode, this mode will be used to complete the transaction.

If a contact transaction is to be performed, chip technology will normally be used. POS terminals that have separate readers for chip and Magnetic Stripe should instruct the merchant to use the chip reader if the Magnetic Stripe of a chip card is read. POS terminals with a combined reader for chip and Magnetic Stripe may read the service code from the Magnetic Stripe and automatically start a chip transaction if it indicates that the card is hybrid or attempt to read the chip without reading the Magnetic Stripe.

BP	All	CT	BA003.11	Acquirers should train merchants to recognize chip cards and to prompt the cardholder to use the chip reader first when a chip card is presented.
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Terminals with separate readers and session manager software

Some terminals implement the transaction using a software session manager to operate the cardholder and merchant interface.

A session is usually represented by a complete transaction that includes the EMV transaction. A session could start with the selection of the amount and could end with the final acceptance or rejection of the transaction, including any referral processing or printing of receipts.

Using sessions has some implications for the way technical fallback transactions are processed, as explained in the sections "Magnetic Stripe is used first" and "Chip is used first."

Magnetic Stripe is used first

If the Magnetic Stripe is swiped, the brand and service code are retrieved.

The following requirements must be followed with a terminal that has separate readers using session management software when it first reads the Magnetic Stripe of a Mastercard card:

- If the service code is "1XX" or "5XX", the transaction must be processed as a normal Magnetic Stripe transaction.
- If the service code is "2XX" or "6XX" and there was no previous attempt to read the chip in the current session, the cardholder or merchant must be prompted to use the chip.
- If the service code is "2XX" or "6XX" and there was a previous attempt to read the chip in the current session and the terminal does not support technical fallback, the cardholder must be informed that the card is not accepted and the session must be closed.
- If the service code is "2XX" or "6XX" and there was a previous attempt to read the chip in the current session and the terminal is a POS terminal or CAT that supports technical fallback, the transaction proceeds as technical fallback.

R	All	CT	RA038.14	The requirements in this section must be followed with a terminal that has separate readers using session management software when it first reads the Magnetic Stripe of a Mastercard card.
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Chip is used first

If a contact chip transaction is attempted first in a terminal with session management software, the terminal attempts to read the chip.

If the chip cannot be read, the terminal may prompt the use of the Magnetic Stripe. If the Magnetic Stripe is read and service code is "1XX" or "5XX", the transaction must be processed as a normal Magnetic Stripe transaction. If the Magnetic Stripe is read and service code is "2XX" or "6XX", the transaction must be processed as a technical fallback if this is allowed.

Service code

The service code used for Magnetic Stripe transactions delivers basic instructions to the terminal regarding the performance of the transaction.

In contact and contactless EMV transactions, the way the terminal determines requirements for online processing and CVM requirements is driven by the EMV data and processes and the service code has no function. Accordingly, the service code must be ignored when performing contact or contactless transactions.

A service code read during the contactless transaction that indicates the presence of a chip card does not mean that the terminal needs to prompt for a contact transaction.

R	All	CL CT	RA362.16	A service code read during a chip transaction must be ignored by the terminal.
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The service code on the Magnetic Stripe is used to indicate the presence of a chip containing an EMV payment application. If the service code on the Magnetic Stripe is "2XX" or "6XX", the card has a chip that implements the same payment product as the one encoded on the Magnetic Stripe. If the service code of the Magnetic Stripe is read and indicates the presence of a chip, a chip transaction must be attempted.

R	All	CT CL	RA292.20	A chip transaction must be attempted if the Magnetic Stripe is read and indicates the presence of a chip.
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During a Mastercard Consumer-Presented QR transaction, the terminal must not interpret the service code and must not act based on the value of the service code.

R	MC MS	QR	RA438.22	A service code read during a Mastercard Consumer-Presented QR must be ignored by the terminal.
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Technical fallback

Technical fallback is the term used to describe when a Mastercard chip card is presented at a hybrid terminal but it is not possible to complete a contact chip transaction. Therefore, the Magnetic Stripe is used instead.

There is no technical fallback to Magnetic Stripe mode for contactless transactions. The contactless reader automatically selects EMV mode if this is available on both the card and terminal, and this mode is used to complete the transaction.

As a Mastercard contact payment application always resides on a hybrid card whose Magnetic Stripe contains "2XX" or "6XX", it follows that technical fallback may only take place when the Magnetic Stripe of the card contains "2XX" or "6XX".

R	All	CT	RA013.11	Transactions may only be processed as technical fallback on cards whose service code is "2XX" or "6XX".
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To ensure interoperability and to reduce the opportunity for fraud when technical fallback occurs, Mastercard has established a set of requirements and best practices concerning how terminals support technical fallback. Transactions that are declined by the issuer or the card or where all available AIDs on the chip are blocked must not be attempted in technical fallback.

If the Magnetic Stripe of a hybrid card cannot be read, transactions may be completed at a hybrid terminal using PAN Key Entry. These are also considered to be technical fallback transactions.

Acquirers have the option to withdraw support for technical fallback as the market matures and fallback rates are at a low level.

The requirements and the best practices in this section apply to all types of terminals. Refer to the section on each terminal type for specific technical fallback requirements and best practices.

R	All	CT	RA014.13	Magnetic Stripe-only terminals must not process transactions made using cards with a service code of "2XX" or "6XX" as technical fallback transactions.
R	All	CT CL	RA306.17	If the service code on the Magnetic Stripe is "2XX" or "6XX" and the transaction cannot proceed using chip technology, contact or contactless, any subsequent transaction using the Magnetic Stripe must be processed as technical fallback with a POS Entry Mode of "80".
R	All	CT	RA015.11	All technical fallback transactions must be online-authorized. If an online connection is not available, voice authorization may be used if permitted for the product.
R	All	CT	RA016.12	Technical fallback transactions, except those that use voice authorization, must have DE 22 (POS Entry Mode) set to "80".
R	All	CT	RA168.12	Technical fallback to PAN Key Entry transactions must have DE 22 (POS Entry Mode) set to "01".
R	All	CT	RA018.11	When a chip transaction fails and technical fallback to Magnetic Stripe is allowed, a terminal with separate chip and Magnetic Stripe readers must display a message instructing the use of the Magnetic Stripe reader.
R	All	CT	RA019.11	When a chip transaction fails and technical fallback is not allowed, a terminal with separate chip and Magnetic Stripe readers must display a message to the cardholder or merchant telling them that the card is not accepted.
R	All	CT	RA020.11	Terminals that support session management must not allow technical fallback if the chip transaction fails because the card is withdrawn before the transaction is complete.
R	All	CT	RA021.11	If a chip transaction is cancelled, either implicitly due to PIN entry time-out or explicitly by the merchant, for example, if the merchant or cardholder uses a "Cancel" button, the chip transaction may be restarted and technical fallback must not be attempted.

BP	All	CT	BA005.11	Acquirers should monitor rates of fallback at different locations to identify merchants who incorrectly use technical fallback and retrain the merchant to follow the correct procedure.
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This requirement applies if the chip fails for any reason during application selection.

R	All	CT	RA022.11	Terminals that support session management must not allow technical fallback if the card being used is blocked.
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Refer to requirement RA306.17 for cases where the candidate list is empty after application selection has been performed and session management is supported.

Application selection

The Mastercard core payment brands are represented by the following AIDs:

- Mastercard 'A0000000041010'
- Maestro 'A0000000043060'
- Cirrus 'A0000000046000'

A standard Mastercard AID is 7 bytes. An extended AID is longer than 7 bytes.

The terminal contains an Application Selection Indicator for each AID it supports. This indicator advises whether the AID in the card must match the AID in the terminal exactly or whether an AID that begins with the one known to the terminal can be selected. Valid Mastercard product AIDs may contain extended identifiers and these must be accepted at all terminals accepting the corresponding brand.

NOTE: Older versions of the contactless reader specification do not support partial name matching.

For contact transactions, EMV supports two methods for performing application selection. Terminals must support the List of AIDs method but may first attempt the PSE method. Not all cards support a PSE. If the terminal reveals any of these data objects in both the PSE and the ADF, the terminal must continue the transaction using the last value that it read. This should not be considered as duplicate data.

- Application Label (tag '50')
- Application Priority Indicator (tag '87')
- Application Preferred Name (tag '9F12')
- Issuer Code Table Index (tag '9F11')
- Language Preference (tag '5F2D')

R	All	CT	RA039.14	If a card has any of the listed fields in both the PSE and the Application Definition File, the terminal must continue the transaction using the last value for the field that it read. This should not be considered duplicate data.
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To allow the cardholder to select the preferred product from the Candidate List, all contact POS terminals, ATMs, Bank Branch Terminals, and On-board Terminals must support cardholder selection.

R	All	CT	RA078.19	POS terminals, ATMs, Bank Branch Terminals, and On-board Terminals must support cardholder selection as defined in EMV Book 1, section 12.4, step 4, to allow cardholders to choose their preferred application on a multi-application card.
BP	All	CT	BA027.11	Wherever possible, CATs should support cardholder selection as defined in EMV Book 1, section 12.4, step 4, to allow cardholders to choose their preferred application on a multi-application card.

For contactless transactions, terminals must support application selection without cardholder selection. And when supported by the terminal, clear indications on how to perform application selection must be provided to the cardholder.

R	All	CL	RA470.19	Terminals must allow the cardholder to perform a single tap contactless transaction with no cardholder interaction or selection using the application with the highest priority that is supported by the terminal. If supported by the terminal, the possibility to select any mutually supported application must be clearly indicated to the cardholder and instructions must be provided to guide the cardholder through the selection process.
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The Cardholder selection may be performed through but not limited to:

- A "Corr"/yellow button with function keys
- Other keys like softkeys or touchpad keys next to the POI screen
- A virtual button on the touchscreen of the POI

Refer to the regional requirements for the Europe region, the Asia/Pacific region, the Latin America and Caribbean region, and the United States region in chapter "Acquirer requirements," section "Regional variations."

Application Label and Application Preferred Name

A terminal allowing cardholder selection or confirmation must read from the ICC and display:

- The Application Preferred Name or Application Preferred Names, if present, and if the Issuer Code Table Index indicating the part of ISO/IEC 8859 to use is present and supported by the terminal as indicated in Additional Terminal Capabilities
- Otherwise, the Application Label or Application Labels by using the common character set of ISO/IEC 8859

BP	All	CT	BA011.12	Although EMV does not mandate the presence of an Application Label on the terminal, Mastercard recommends that terminals support a default Application Label for each AID supported by the terminal, for example, to be used in receipt printing, in case the card does not deliver an Application Label or an Application Preferred Name to the terminal.
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Terminal interface requirements for cardholder selection

Terminals that support cardholder selection need to be designed and configured to provide a simple user experience for merchant and cardholder.

Terminals should be designed to enable the cardholder to interact directly with the device when possible. This is typically achieved by using the PIN pad as the cardholder selection interface. The PIN pad may be integrated with the POS device and need to be physically handed to the cardholder to enable interaction or may be a separate customer facing screen or keypad.

BP	All	CT CL	BA043.19	Terminals should request amount entry before the application selection stage. This is particularly relevant for POS with an integrated PIN pad where both cardholder and merchant interact with the same screen or keypad.
BP	All	CT CL	BA045.19	If cardholder selection is required, terminals should prompt with an appropriate screen message for both cardholder and merchant.
BP	All	CT CL	BA047.19	Once an application has been selected by the cardholder, it should be identified to the merchant and cardholder while transaction processing completes.

These requirements are for terminals with a separate customer facing screen or keypad, for example, PIN pad.

R	All	CT CL	RA194.19	For devices with a separate customer screen or keypad application selection, choices must be displayed and actioned on the cardholder screen or keypad.
R	All	CT CL	RA195.19	For devices with a separate customer screen or keypad, the merchant interface must not permit the sales assistant to choose the application on behalf of the cardholder.
R	All	CT CL	RA196.19	For devices with a separate customer screen or keypad, only "transaction cancellation" must be available on merchant interface during cardholder application selection.
R	All	CT CL	RA197.19	Regardless of the location of the active card reader, facing merchant or cardholder, the cardholder must be in control of selecting the application for payment through a PIN pad or other interface facing the cardholder.
BP	All	CT CL	BA046.19	While the cardholder is choosing an application for payment, the merchant display should inform the merchant that this is taking place.

This best practice is for terminals where the PIN pad is integrated with the POS device and needs to be physically handed to the cardholder to enable interaction.

BP	All	CT CL	BA044.19	The screen where applications available for selection are displayed should clearly indicate to both merchant and cardholder that it is the latter that performs the cardholder selection.
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Screen size and an appropriate set of navigation keys are also essential to provide a consistent and clear user experience.

R	All	CT CL	RA198.19	The minimum requirements for the display on POS terminals and PIN pads are 2 lines with 16 characters in each line.
BP	All	CT CL	BA048.19	To allow clear instructions to the cardholder throughout the transaction process, the display on POS terminals and PIN pads should have 4 lines with 20 characters in each line.

R	All	CT CL	RA199.19	Terminals must present all mutually supported applications available for cardholder selection in accordance with display capacity: • If a terminal has sufficient display capacity to present all mutually supported applications, the terminal must do so. • If the applications in the candidate list cannot be displayed on a single screen, the cardholder must be able to navigate directly to the next screens to view the remaining applications.
R	All	CT CL	RA200.19	The "Cancel" key must not be used as part of the process of selecting an application but only for cancellation of the transaction.

Card authentication

All contact terminals, except online-only terminals, must support offline CAM.

Support of offline CAM is optional for online-only terminals, except for contactless online-only terminals that support CDCVM or Relay Resistance Protocol (RRP), which must support offline CAM.

For more details on how offline CAM is performed, refer to the *Public Key Infrastructure (PKI) - Overview* manual, which includes references to the procedures and tools used to support the Mastercard PKI.

At offline-capable terminals for contact transactions:

- DDA must be supported on all terminals.
- CDA must be supported on new terminals and should be supported on all terminals.

For contactless transactions, only CDA must be supported. SDA or DDA are not supported for contactless transactions.

If offline CAM fails or is not performed, the transaction is sent online for the issuer to approve or it is declined at an offline-only terminal. An exception to this may be if CDA fails, in which case the transaction may be declined offline if the AC cannot be recovered from the CDA envelope so online CAM cannot be performed.

R	All	CT	RA332.15	All offline-capable contact terminals must support DDA.
BP	All	CT	BA017.11	All offline-capable contact terminals should support CDA.
R	All	CT	RA074.14	All new offline-capable contact terminals undergoing Mastercard certification must support CDA.

R	All	CL	RA251.14	EMV mode terminals that support CDCVM must support CDA.
R	All	CL	RA398.16	EMV mode terminals that support RRP must support CDA.
R	All	CL	RA252.14	All offline-capable EMV mode contactless-enabled terminals must support CDA.
R	All	CL	RA253.22	EMV mode contactless-enabled terminals must not support SDA or DDA.

CDA

Version 4.2 and subsequent versions of *EMV Book 2 - Security and Key Management* features improvements to the CDA method over previous versions, and all offline-capable POS terminals and CATs must support it.

BP	All	CT CL	BA009.15	All terminals that support CDA should support CDA, as specified in EMV 4.2 or later.
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Different operating modes are defined for performing CDA. Terminals operating in Mode 1 have the most secure configuration and ensure that CDA is always performed whenever an offline approval or online authorization is requested. Terminals operating in Modes 2, 3, and 4 provide a lower level of security.

Currently, all CDA-capable terminals should support Mode 1. To mitigate against certain fraud attacks, all new terminals undergoing Mastercard certification that support CDA for contact transactions must support CDA in Mode 1.

BP	All	CT CL	BA010.14	If CDA is being implemented according to EMV 4.2 or later versions, the implementation should operate as Mode 1.
R	All	CT	RA338.15	All new terminals undergoing Mastercard certification that support CDA for contact transactions must support CDA in Mode 1.

Requirements for payment system public keys

The payment system public keys for contact and contactless transactions are the same values and may be shared. Terminals must contain all current keys and must be able to store up to six CA Public Keys for each Registered Application Provider Identifier (RID).

The terminal must associate each key with this key-related information that is used with the key:

- Certification Authority Public Key Check Sum, if required

- Certification Authority Public Key Exponent
- Certification Authority Public Key Index
- Certification Authority Public Key Modulus

Mastercard test public keys must not be held in operational terminals.

R	All	CT CL	RA032.13	All terminals that support Offline Data Authentication or Offline enciphered PIN must be loaded with all current Mastercard public keys.
R	All	CT CL	RA033.11	Terminals must only accept keys that the terminal can authenticate as originating from the genuine acquirer.
R	All	CT CL	RA034.11	Acquirers must verify that all appropriate keys are loaded into all terminals that generate transactions that they acquire.
R	All	CT CL	RA035.11	Live terminals must not contain test Mastercard public keys.

There are two Payment System Public Keys that are currently in use: keys with index 05 and 06. The lengths and the expiration dates of the Payment System Public Keys are reviewed annually, and their latest values are in the *Updated Payment System Public Keys for M/Chip* announcement on Mastercard Connect®.

Acquirers should monitor terminals for persistent failures of offline card authentication, as this may identify terminals either without the full operational set of public keys or with some cryptographic functional problem.

Mastercard has observed that groups of terminals may fail offline card authentication at the beginning of a new year because old expiration dates are stored in the terminal. Mastercard recommends that expiration dates of keys should not be stored in the terminal, but if they are, it is imperative that these expiration dates are maintained to be the same as the published expiration dates, as amended from time to time. Mastercard requires that expired keys must be removed from terminals within 6 months.

BP	All	CT CL	BA097.15	Acquirers should monitor terminals for persistent failures of offline card authentication.
R	All	CT CL	RA412.18	Acquirers must ensure that they remove the Payment System Public Keys from their operational set of public keys within 6 months after their expiration date.
BP	All	CT CL	BA051.14	The expiration date of keys should not be held in the terminal.

Processing restrictions

Warning Bulletin and BIN checking, for product and routing validity, by the terminal is recommended for terminals supporting offline transactions, including terminals supporting Merchant Approved and Deferred Authorization.

For contactless transactions, it may be done after the communication with the card is finished.

BP	All	CT CL	BA102.20	Offline-capable terminals, including terminals supporting Merchant Approved and Deferred Authorization, should support exception files and BIN checking.
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Cardholder verification

The terminal performs Cardholder verification as defined by EMV. For a detailed explanation of Cardholder verification, refer to the *M/Chip Program Guide*.

If a Signature is required for cardholder verification, this may be captured on a receipt or electronically.

Contact transactions

All ATM transactions must be verified by Online PIN. Bank Branch Terminals that accept Cirrus or Maestro cards must support Online PIN. POS terminals that accept Maestro cards must support Online PIN and Offline enciphered PIN.

To further enhance PIN security, Offline Plaintext PIN verification is being phased out from 1 January 2025 onwards. The only Offline PIN method going forward will be Offline enciphered PIN.

Attended terminals that accept Mastercard cards, except for MPOS terminals, must support Signature. The support of Signature is optional for MPOS terminals. Attended terminals may support PIN. Where PIN is supported, the terminal may support Online PIN, Offline PIN, or both. Support for Offline PIN is mandated on new POS terminals. In certain merchant categories, terminals may accept transactions with no cardholder verification and thus support No CVM.

Where Offline PIN is supported, Offline enciphered PIN must be supported.

R	All	CT	RA054.23	At ATMs the contact interface must only support Online PIN.
R	MS CI	CT	RA069.11	Bank Branch Terminals that support cash advance transactions and that accept Maestro or Cirrus cards must support Online PIN on the contact interface.

R	MS	CT	RA075.11	Attended POS terminals that accept Maestro cards must support Offline PIN and Online PIN on the contact interface.
R	MC	CT	RA070.20	Bank Branch Terminals that support cash advance transactions and that accept Mastercard cards must support Signature. Support of Online PIN and Offline PIN is optional on the contact interface.
R	MC	CT	RA076.20	Attended POS terminals, except MPOS terminals, that accept Mastercard cards must support Signature on the contact interface.

NOTE: For the deviation on requirement RA076.20, refer to chapter 4 "Acquirer requirements," section "Regional variations," subsection "Latin America and the Caribbean region."

R	All	CT	RA077.24	Newly deployed (including replacement) attended POS terminals, except chip-only MPOS terminals, chip-capable MPOS terminals, and terminals only accepting Maestro No CVM transactions, must support Offline enciphered PIN, on the contact interface.
R	All	CT	RA036.24	Terminals that support Offline PIN must support Offline enciphered PIN.
R	All	CT	RA561.25	Until 31 May 2027, attended Terminals supporting Offline PIN must support Offline Plaintext PIN and Offline enciphered PIN.
R	All	CT	RA555.24	Effective 1 January 2025, newly deployed (including replacement) unattended POS terminals must not support Offline Plaintext PIN.
R	All	CT	RA556.25	Effective 1 June 2027, newly deployed (including replacement) attended POS terminals must not support Offline Plaintext PIN.

Acquirers and terminals in certain countries in the Europe region do not support Online PIN. Therefore, they do not comply with requirements RA069.11 and RA075.11. The list of countries that Mastercard has given such consent to are in the *Transaction Processing Rules* manual.

Requirement RA077.24 does not apply to the United States region. Refer to the regional requirements for the United States region in chapter 4 "Acquirer requirements," section "Regional variations."

Cardholder Activated Terminals support CVM options according to their type. In CAT Level 2 and Level 3 terminals, transactions may be completed with no cardholder verification.

PIN Entry Bypass is not permitted at CATs.

R	MC	CT	RA103.11	CATs must not support PIN Entry Bypass.
R	MC	CT	RA104.11	CAT Level 1 terminals must support one of these combinations of CVM: Online PIN or Online PIN and Offline PIN.
R	MC	CT	RA105.11	CAT Level 2 terminals must support No CVM only.
R	MC	CT	RA172.12	CAT Level 3 terminals must support one of these combinations of CVM: Offline PIN and No CVM or No CVM.
R	MC	CT	RA189.13	CAT Level 4 terminals must support No CVM only.
R	MC	CT	RA524.22	CAT terminals that have dual capability as CAT Level 1 and CAT Level 2 (CAT 1/2) must support No CVM and either Online PIN, Offline PIN, or both.

Requirement RA103.11 does not apply to Automated Fuel Dispensers (AFDs) in the United States region.

For more details, refer to chapter 4 "Acquirer requirements," section "Terminal types," subsection "Cardholder activated terminals."

Contactless transactions

A CVM is not required for a contactless transaction amount equal to or below the CVM Limit unless specifically requested by the card or the issuer or at an ATM.

Acquirers must not request cardholder verification equal to or below this value.

For transaction amounts above the CVM Limit, a CVM is required.

There are some MCCs where no CVM has been part of the product proposition above the CVM limit up to a specific transaction limit, for example, tollways and parking. No CVM transactions may continue to be accepted in these environments up to the established limits.

Contactless terminals must not support Offline PIN on the contactless interface. Offline PIN may be supported at the same terminal but only for contact chip transactions. Terminals must ensure that Offline PIN is never selected as the CVM for a contactless transaction.

Contactless-enabled terminals that are compliant with the Mastercard Contactless Reader Specification v3.0 or later or EMVCo Book C-2 or were deployed after 1 January 2016 must follow the configurations described in the following table for transaction amounts above the CVM Limit.

If the contact interface of the POS terminal...	Then, for transactions above the CVM limit, high-value transactions, the contactless interface of the POS terminal...
Supports Online PIN	• Must support Online PIN • If Mastercard is accepted, must support Signature • Must support CDCVM
Does not support Online PIN	Must be configured to support one of the following: • Must support CDCVM For this configuration, a high-value transaction can only occur when a mobile payment device is used and CDCVM was successful. • Must support CDCVM and Signature For this configuration, a high-value transaction can occur with Signature as the CVM when Mastercard is accepted and may also be able to occur when a mobile payment device is used and CDCVM was successful.

Regarding the mandate to support Online PIN on both interfaces, refer to chapter "Acquirer requirements," section "Regional variations," subsection "CVM requirements" for the Asia/ Pacific, Europe, Middle East/ Africa, and Latin America and the Caribbean regions.

All contactless-enabled POS terminals must be configured to support CDCVM for transactions above the CVM Limit. Additionally, if the contactless-enabled POS terminal supports CDCVM and this was successfully performed on the access device, the POS terminal must not request another CVM (such as Online PIN) for the same transaction.

Bank Branch Terminals that accept Mastercard contactless transactions follow the same CVM requirements as contactless-enabled POS terminals. Bank Branch Terminals that accept Maestro and Cirrus contactless transactions must support Online PIN only.

For EMV mode transactions, the reader must complete CVM Processing for all transactions, both above and below the Terminal CVM Required Limit. The CVM is determined by:

- The issuer preferences expressed in the Application Interchange Profile (AIP) (tag '82') and the CVM List (tag '8E') as well as dedicated functionality in the cardholder device
- The CVM capabilities indicated by the reader application of the terminal

The CVM capabilities will be different above and below the Terminal CVM Required Limit.

Only No CVM must be supported in the Terminal Capabilities for transactions equal to or below the Terminal CVM Required Limit. Terminals must not support No CVM above the Terminal CVM Required Limit.

The use of No CVM must be positively identified by the EMV process. It does not mean skip CVM processing.

Purchase with cash back transactions must be completed with cardholder verification regardless of the amount. ATM transactions must be verified by Online PIN.

For Magnetic Stripe mode transactions, the CVM is determined by the terminal.

R	All	CL	RA258.22	Contactless-enabled terminals must not request cardholder verification for purchase transaction amounts equal to or below the CVM Limit unless specifically requested by the issuer, the card, or the access device.
R	All	CL	RA259.14	Contactless-enabled terminals must not support No CVM for transaction amounts above the CVM Limit.

For transactions above the CVM Limit, these requirements apply.

R	MC	CL	RA260.20	Attended Mastercard contactless-enabled terminals, except MPOS terminals, must support Signature for transaction amounts above the CVM Limit.
R	All	CL	RA339.24	Contactless-enabled terminals, except terminals in Israel, Ireland, and the United Kingdom, must support Online PIN for transaction amounts above the CVM Limit if Online PIN is also supported on the contact interface.
BP	All	CL	BA126.24	Contactless-enabled terminals in Israel, Ireland, and the United Kingdom, should support Online PIN for transaction amounts above the CVM Limit if Online PIN is also supported on the contact interface.
R	MC MS	CL	RA342.22	Contactless-enabled POS terminals must support CDCVM for transaction amounts above the CVM Limit.
R	All	CL	RA266.14	Contactless-enabled terminals must not support Offline PIN for contactless transactions.
R	All	CL	RA267.14	Contactless purchase with cash back transactions require cardholder verification regardless of the amount.
R	All	CL	RA268.20	ATM contactless transactions must be verified by Online PIN.
R	MC	CL	RA230.14	Signature verification must be supported on the contactless interface by terminals supporting manual cash advance for Mastercard.
R	MC	CL	RA430.18	Online PIN must be supported on the contactless interface by terminals supporting manual cash advance for Mastercard if it is also supported on the contact interface.
R	MS CI	CL	RA231.17	Online PIN verification must be the only supported CVM on the contactless interface for manual cash advance transactions for Maestro and Cirrus brand.

R	All	CL	RA567.25	If CDCVM is supported by the POS terminal and successfully performed on the access device, the POS terminal must not select another CVM for the same transaction.
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Refer to the regional variation of the requirement RA339.22 in chapter 4 "Acquirer requirements," section "Regional variations," subsection "Europe region," sub-subsection "CVM requirements."

For contactless transaction amounts below the CVM Limit, the terminal must not request cardholder verification. Terminals must not operate purely as CAT Level 1 terminals. Dual capability terminals must be used that support No CVM below the CVM Limit (CAT Level 2) and request cardholder verification above (CAT Level 1). Above the CVM Limit, CAT Level 1 terminals must support Online PIN and CDCVM.

Acquirers can choose to accept contactless transaction amounts above the CVM Limit without cardholder verification at CAT Level 2 terminals. The acquirer is liable for lost and stolen fraud on such transactions. CAT Level 2 and CAT Level 3 terminals must support No CVM. CAT Level 3 terminals must support CDCVM if they support contactless transaction amounts above the CVM Limit.

R	MC	CL	RA269.23	Terminals that operate as CAT Level 1 must have their CVM Limit set to 0. The CVMs supported for transaction amounts above the CVM Limit must be either Online PIN and CDCVM or, in markets where Online PIN is not supported, only CDCVM.
R	MC	CL	RA343.23	CAT terminals that support transaction amounts above the CVM Limit must support No CVM for transaction amounts equal to or below the CVM Limit and must support either Online PIN and CDCVM or only CDCVM for transaction amounts above the CVM Limit.
R	MC	CL	RA270.15	CAT Level 2 terminals must support only No CVM.
R	MC	CL	RA351.15	CAT Level 3 terminals must support No CVM.
R	MC	CL	RA354.15	CAT Level 3 terminals that support contactless transactions above the CVM Limit must support CDCVM.

A service code read during the contactless transaction indicating that PIN is required does not mean that PIN is required for a contactless transaction below the CVM Limit. Refer to requirement RA362.16.

Use of PIN

PINs can be 4 to 12 digits long. This means that the customer will need to press the "Enter" key after completing their PIN entry.

The display should present relevant information when prompting for PIN entry.

BP	All	CT CL	BA050.13	If the display has the capacity, terminals should present this information on the same screen while the cardholder is being prompted for PIN entry: • Application Preferred Name or Application Label of the application chosen for payment • Total amount of transaction • PIN Entry prompt
R	All	CT CL	RA566.25	Terminals which support PIN must support PINs from 4 to 12 digits in length. This applies for both Online PIN and Offline PIN validation.

When Online PIN is the chosen cardholder verification method for the transaction, the PIN must be entered or the transaction will be terminated.

R	All	CT CL	RA037.11	If PIN is requested but not entered after re-prompting and PIN entry is not bypassed, the terminal must terminate the chip transaction. Terminals must not then allow technical fallback.
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When Online PIN is used to verify the cardholder during a contact transaction, if the authorization is declined by the issuer because the PIN is incorrect, the transaction should be restarted and the cardholder prompted to reenter their PIN. For contactless transactions, a new transaction with new chip data is required in such cases.

BP	All	CT	BA064.19	If an online authorization is declined by the issuer because of an incorrect PIN, a new transaction should be reattempted and in that case a new PIN entry will be required.
R	All	CL	RA363.18	If a contactless transaction is declined by the issuer because of an incorrect PIN, then a new transaction must be started and the PIN must be reentered.

Refer to the regional requirements for the United States region in chapter "Acquirer requirements," section "Regional variations."

PIN Entry Bypass

Support by the terminal of PIN Entry Bypass is optional. If the terminal supports PIN Entry Bypass, the merchant should have appropriate procedures and training in place to ensure that PIN Entry Bypass is not used inappropriately.

For example, the use of PIN Entry Bypass may be restricted to a supervisor function. Usage of PIN Entry Bypass should be monitored and excessive use should be investigated by the merchant or acquirer.

PIN Entry Bypass is recommended during the early stages of a migration to PIN from Signature, but once cardholders become familiar with PIN usage, PIN Entry Bypass should be withdrawn.

PIN Entry Bypass must not be supported for contactless transactions, except for transactions on the US-Maestro AID.

BP	All	CT	BA018.11	Use of PIN Entry Bypass by the merchant should be monitored and controlled.
BP	All	CT CL	BA019.24	PIN Entry Bypass should be withdrawn once cardholders in the market are familiar with PIN usage.
R	MC MS CI	CL	RA264.14	PIN Entry Bypass must not be supported for contactless transactions.
R	All	CT CL	RA554.24	If Online PIN entry is bypassed, acquirers must send the same TVR in the online request as was sent to the card.

Refer to the regional requirements for the Europe and United States regions in chapter "Acquirer requirements," section "Regional variations."

Single Tap and PIN request

Single Tap and PIN Request is a transaction flow that allows issuers to request PIN verification on contactless transactions without the cardholder needing to re-tap the card.

Overview

Contactless transactions for amounts less than or equal to the CVM Required Limit do not require cardholder verification.

However, to manage cumulative risk or due to regulations like PSD2 in Europe, issuers may want to request online PIN for low-value contactless transactions whenever required by their risk management policies.

Single Tap and PIN Request is a transaction flow that, when supported by the terminal, permits the issuer, on receipt of an online contactless transaction without cardholder verification, to

request the terminal to prompt the cardholder for PIN entry (without the need for a second tap) and to send a follow-up online authorization request with that Online PIN included.

For the follow-up online authorization request, the card is not re-read by the terminal. Instead, the terminal stores the EMV data from the previous transaction, asks the cardholder to enter their PIN, and sends a new online authorization request, with EMV data from the original transaction, and the Online PIN just entered.

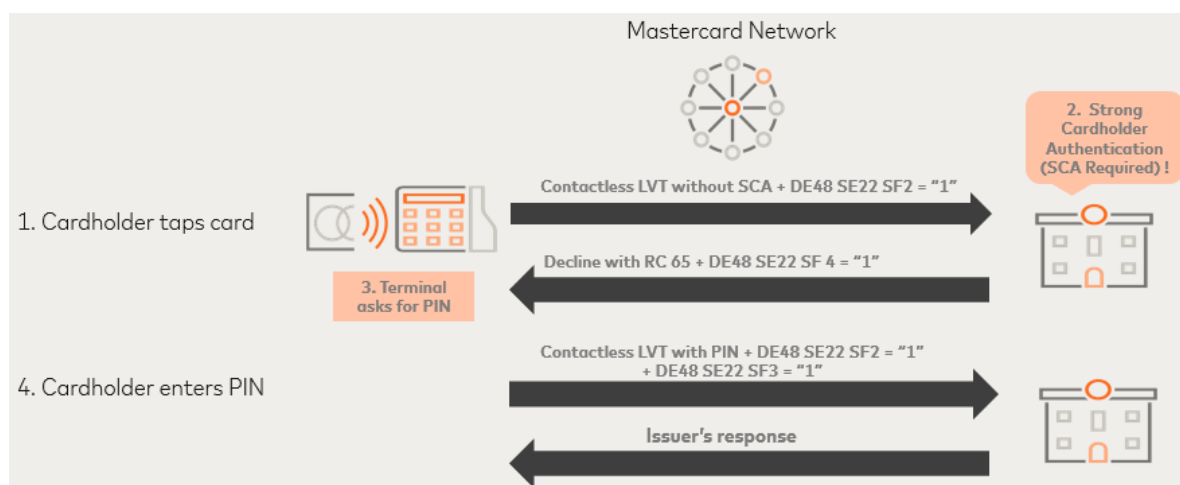
From a cardholder perspective, this provides a seamless experience like a high-value contactless transaction, because the card is only read once by the terminal, and the PIN only entered once.

Basic flow

1. In the initial low-value online authorization request from the acquirer, the terminal indicates support of Single Tap and PIN Request by setting DE 48, SE 22, SF 2 to value "1" (Merchant POI capable of single tap processing).
2. The issuer declines the online authorization request with response code value "65" in DE 39 and value "1" in DE 48, SE 22, SF 4 (issuer request PIN in Single Tap mode) to request a follow-up transaction with Online PIN.
3. In the follow-up online authorization request from the acquirer, the terminal sets DE 48, SE 22, SF 3 value "1" (Transaction contains an intentionally duplicated (replayed) ATC value) to indicate this is the follow-up online authorization request with Online PIN. This authorization request contains chip data from the original transaction.

The following figure demonstrates the Single Tap and PIN Request transaction flow.

Figure 1: Single Tap and PIN Request Transaction Flow



For a full explanation, refer to the *Authorization Manual*.

Single Tap and PIN Request functionality is currently available in the following regions: Europe, Asia/Pacific, Latin America and the Caribbean, and Middle East/Africa.

For regional specific requirements, refer to section, "Regional Variations."

R	All	CL	RA559.24	DE 48 SE 22 SF2 (Single Tap Indicator) must have value "1" (Merchant POI capable of single tap processing) in all contactless transactions from terminals supporting Single Tap and PIN Request.
R	All	CL	RA560.24	The Single Tap and PIN Request follow-up online authorization request must feature: • DE 2, DE 14, DE 23, DE 35, DE 55 identical to the original transaction • The Online PIN as entered by the cardholder • Value "1" in DE 48 SE 22 SF 2 (Single Tap Indicator) • Value "1" in DE 48 SE 22 SF 3 (Response to PIN Request) • All other fields set correctly to reflect this new transaction request and not intentionally copied from the original transaction. The cardholder must not be requested to re-tap their card.

CVM Fallback

CVM Fallback means that Signature at POS or No CVM at CAT is used in cases where PIN is the selected CVM but cannot be used.

CVM Fallback does not require special terminal processing. It is merely the logical outcome of the EMV-defined process according to the capabilities of the terminal and the instructions in the card.

If the selected CVM is offline or Online PIN, fallback to Signature at POS or No CVM at CAT may be allowed for the Mastercard product. It is not allowed for Maestro.

CVM Fallback transactions will normally be sent online by the Terminal Action Codes.

Skip CVM processing

Whenever the terminal skips the CVM processing, for instance, when biometric verification has been used, it must not prompt for Signature nor must it print any receipt including a signature line.

R	MC	CT CL	RA453.22	For transactions where the AIP indicates that no cardholder verification is supported, for example, Biometric Cards, attended terminals that support Signature must not prompt for signature nor print any signature line on the receipt.
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Mastercard Consumer-Presented QR transactions

For Mastercard Consumer-Presented QR transactions, CDCVM is the only valid CVM.

R	MC MS	QR	RA441.19	No form of terminal CVM processing must be supported for Mastercard Consumer-Presented QR transactions.
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Terminal Risk Management

Terminal Risk Management (TRM) is the term used to describe the checks performed on the transaction by the terminal, including floor limit checks, random selection of transactions to go online, and velocity checking.

EMV and Mastercard require that all offline-capable terminals always perform TRM, even if this is not requested by the card, to help prevent possible fraud.

R	All	CT CL	RA041.11	Offline-capable terminals must always perform TRM, even if the card does not request it.
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Random transaction selection does not need to be supported for contact transactions. Neither velocity checking by the terminal nor random transaction selection is performed for contactless transactions.

The floor limit maximum amount is a global maximum amount established by Mastercard, which applies to all Mastercard brands and all MCCs. The Mastercard published floor limits for contact and contactless transactions is in the *Quick Reference Booklet* under the sections "Terminal Offline Chip Authorization Limits" and "EMV Mode Contactless Transaction Terminal Offline Chip Authorization Limits."

Terminals may also perform a cumulative floor limit check by adding the last transaction in the terminal log file, if present and if performed by the same card, to the current transaction amount and comparing the total with the terminal floor limit.

R	All	CT	RA042.11	Hybrid terminals must have the capability to implement different floor limits for Magnetic Stripe and chip transactions.
BP	All	CT CL	BA013.14	The Terminal Floor Limit for chip transactions should not exceed the floor limit published by Mastercard.

R	All	CT	RA163.11	Terminals must only set the "Merchant Forced Transaction Online" bit in the TVR, if the merchant indicates that they are suspicious, and must not set it routinely as a way to force all transactions online.
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Terminal action analysis

The mandatory Terminal Action Codes (TACs) used for contact transactions are given in the section "Terminal Action Codes." The TACs may be different depending on the terminal capabilities.

The TACs for contactless readers can be found in the *Mastercard Contactless Kernel Configuration* manual.

The terminal maintains the TACs for contact and contactless transactions independently.

Authorization

For contact transactions, chip data in the issuer authorization request response, including Issuer Scripts, must be returned to the terminal if present in the authorization request response. For details, refer to chapter "Data requirements," section "Network data."

Chip data (DE 55, including all mandatory subelements) must be sent to the issuer. For details, refer to chapter "Data requirements," section "Network data."

For contactless EMV mode transactions, it is not required to deliver chip data from the issuer authorization request response, including Issuer Scripts, to the terminal in the online response. If the data is returned to the terminal, the terminal does not process the data. The terminal is not required to retain the data.

Unable to Go Online mode

For contact transactions, during the Unable to Go Online mode, the terminal performs the Second Terminal Action Analysis, as defined by EMV, by comparing the Terminal Verification Results with the default values of the Terminal Action Codes and of the Issuer Action Codes.

Depending on the result of this comparison, the terminal sets the Authorization Response Code to one of the following values:

- "Y3" if the result of the second Terminal Action Analysis is approved
- "Z3" if the result of the second Terminal Action Analysis is decline

The terminal then asks the card for its final decision using the preceding Authorization Response Code. The EMV transaction flow then completes with the card either approving or declining the transaction by generating the appropriate cryptogram.

Unable to Go Online mode is not supported for contactless transactions. If the card has generated an ARQC and issuer authorization is not possible, the transaction is normally declined, or, exceptionally, acquirers may choose to accept the current transaction and use Deferred Authorization at their own risk.

Mastercard Consumer-Presented QR transactions do not support Unable to Go Online mode. If the terminal is unable to go online, the transaction will be declined.

Script processing

Terminals do not need to support scripts for contactless transactions. Any script received from the issuer may be discarded.

For contact transactions, EMV states that terminals must be able to support scripts of 128 bytes. This includes the 1-byte tag and the 1-byte length, so in practice, the maximum length of the script that must be processed is 126 bytes. Acquirers must be able to process script messages sent before and after the second GENERATE AC.

R	All	CT	RA120.11	Acquirers must be able to process an Authorization Request Response/0110 message or Financial Transaction Request Response/0210 that contains either one Issuer Script Template 1 or one Issuer Script Template 2.
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Relay Resistance Protocol

RRP is mandatory on all readers compliant with Mastercard *Contactless Reader Specification* v3.1.1 or later (or *EMVCo Book C-2* v2.6 or later).

However, transit terminals that have high throughput constraints (MCCs 4111, 4131, and 4784), and Software MPOS Solutions are not required to support RRP. RRP may be supported in low throughput transit environments.

NOTE:

New implementations based on Mastercard *Contactless Reader Specifications* v3.1 must not have RRP enabled, as several vendors have not implemented RRP correctly with this version.

Regarding already deployed v3.1 terminals, RRP can optionally be deactivated without requiring a new M-TIP certification.

R	All	CL	RA389.19	Readers compliant with Mastercard <i>Contactless Reader Specification</i> v3.1.1 or later or EMVCo Book C-2 v2.6 or later must have the RRP activated, with the exception of Software MPOS Solutions and transit terminals with high throughput at entry and exit gates for MCCs 4111, 4131, and 4784.
R	All	CL	RA446.19	RRP must not be enabled in new implementations of readers based on Mastercard <i>Contactless Reader Specifications</i> v3.1.

Terminated transactions

A terminal may allow a merchant to cancel a contactless transaction for (a) EMV mode, before the GENERATE AC command is issued, or (b) Magnetic Stripe mode, before the terminal has requested an online authorization.

The terminal should monitor the number of aborted transactions. If the frequency is high, it is possible that a fraudster is trying to get a specific value of the Unpredictable Number (UN). The terminal should take appropriate measures to reduce the risks of an attack, such as introducing wait times after three aborted transactions.

BP	All	CL	BA067.14	The terminal should take appropriate measures to reduce the risk of an attack using aborted transactions.
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Subsequent contact transactions

If a contactless transaction is processed to completion and results in a decline, then the terminal should normally prompt for a contact transaction.

It should not be assumed that the transaction will be declined when the same card is used in a contact transaction. The decline of a contactless transaction with No CVM may well be because several consecutive transactions have been performed without cardholder verification and risk control requires that cardholder verification be performed this time, even for a low value transaction.

The decision to decline the transaction may be taken by the card, terminal, or issuer. An Authorization Response Code "65" (Exceeds withdrawal count limit) indicates that the issuer requires cardholder verification for this transaction.

If the Authorization Response additionally includes DE 48 SE 22 SF 4 set to value "1", this indicates a Single Tap and PIN request follow-up transaction is required - refer to Chapter 4,

section "Cardholder verification", subsection "Single Tap and PIN request". The remainder of this section describes the situation when Single Tap and PIN request is not indicated.

If the authorization response does not include a Single Tap and PIN request, a contact transaction should be performed. To determine if a contact interface is available, the terminal should use the Device Type if recovered from the contactless card or payment device.

If the payment device does not have a contact interface, a second contactless transaction may be performed with CDCVM having been completed, or Online PIN if supported by both card and terminal.

For the subsequent transaction, the terminal must be configured to support normal CVM options for the product and must be configured to require cardholder verification for this transaction and must not be configured to support only No CVM. There are no changes to network messages to identify these transactions as having previously been attempted using the contactless interface.

When an Authorization Response Code "65" is received, terminals should display an appropriate message to inform the cardholder that the transaction is declined and to prompt the cardholder to perform a contact transaction.

The following are recommended examples of the cardholder message that can be used:

- Transaction Declined - dip the card
- Declined - please insert your card
- Declined - use chip reader

To ensure an optimal cardholder experience, there should be no merchant intervention, that is, initiate a new contact transaction for the same amount after the declined contactless transaction with Response Code "65". Terminals should perform the fallback to the contact interface from the contactless interface during the same transaction session.

R	All	CL	RA277.19	If a contact interface with PIN pad is available on the terminal and the card has a contact interface, contactless terminals must display an appropriate message to prompt for a contact transaction to be attempted when a contactless transaction, other than a refund, is declined offline by either the card or the terminal.
R	All	CL	RA278.25	If a contact interface with PIN pad is available, real-time authorization is supported and the card has a contact interface, contactless terminals must display an appropriate message to prompt for a contact transaction to be attempted when a contactless transaction is declined by the issuer using an Authorization Response Code "65" (Exceeds withdrawal count limit) and DE 48 SE 22 SF 4 is not present (Single Tap and PIN not requested.).

R	All	CT	RA331.23	Contact transactions that are performed because a preceding contactless transaction has been declined using Response Code "65" (Exceeds withdrawal count limit) must be performed with cardholder verification. The terminal must support all the CVMs that are available for high value transactions.
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Refer to the regional variation of the requirement RA278.19 for the United States region in chapter 4 "Acquirer requirements," section "Regional variations."

Also refer to section "Single Tap and PIN Request" for the Europe region in chapter "Acquirer requirements," section "Regional variations."

Chip transactions involving multiple requests

Chip transactions involving multiple requests may occur when a nonfinancial request (any transaction that does not cause a transfer of funds) is followed by a financial request.

To maximize security and avoid duplicate ATCs, when multiple transactions are requested and a financial transaction follows a nonfinancial transaction, for example, a balance inquiry followed by a cash withdrawal, new chip data must be generated.

The terminal does not need to go through the full application selection process for subsequent requests but may restart the chip transaction using the last application selected.

For contactless transactions specifically, the contactless card or contactless payment device must be re-tapped to confirm the same cardholder is requesting this service.

R	All	CT CL	RA364.16	When a financial request follows a nonfinancial request, terminals must generate new chip data.
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Where the second transaction is a nonfinancial transaction, Mastercard recommends a new transaction with fresh chip data.

BP	All	CT CL	BA016.16	When a nonfinancial request follows a financial request, terminals should generate new chip data for each request.
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Terminal types

This section covers specific requirements for different types of terminals and operating environments.

ATMs

Requirements for ATMs are detailed in this section.

Terminal applications

All new contactless ATM readers submitted to Mastercard certification, Mastercard Terminal Integration Process (M-TIP) testing, as a new project and subsequently deployed must be compliant with Mastercard *Contactless Reader Specifications* v3.1 or later or *EMVCo Book C-2* v2.5 or later.

R	All	CL	RA394.22	All new contactless ATM readers submitted to M-TIP must be compliant with Mastercard <i>Contactless Reader Specifications</i> v3.1 or later or <i>EMVCo Book C-2</i> v2.5 or later.
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Technology selection

Mastercard Consumer-Presented QR must only be supported at POS, attended and unattended, and MPOS terminals. ATMs are not allowed to accept this type of transactions.

R	All	QR	RA445.25	ATMs must not support Mastercard Consumer-Presented QR transactions.
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Cardholder selection

ATMs must support cardholder selection. Refer to requirement RA078.19 in chapter "Acquirer requirements," section "Application selection."

Offline CAM requirements

ATMs compliant with Mastercard *Contactless Reader Specifications* v3.1.1 must support RRP and therefore must also support CDA.

Refer to requirement RA389.19 chapter "Acquirer requirements," section "Relay Resistance Protocol," and requirement RA398.16 in section "Card authentication."

CVM requirements

ATMs must support Online PIN. No other CVMs are permitted.

Refer to requirements RA054.20 and RA268.20. CDCVM is not an applicable CVM for ATMs that support contactless transactions even in cases where the reader is compliant with Mastercard Contactless Reader Specification v3.0 or later or EMVCo Book C-2.

Supporting mobile payment devices

Mobile payment devices support more security and protection by using the user interface, screen, and keypad of a mobile phone.

One of the security features is the option for an issuer to require cardholder confirmation through the user interface of the phone before the phone will allow an unattended cash transaction to be completed. This is achieved by the phone sending an Application Authentication Cryptogram (AAC) response to the GENERATE AC command until the cardholder has confirmed through the user interface they want to perform the transaction.

The process may require a minimum of two taps, as the initial tap will result in an AAC from the phone unless the cardholder has already confirmed the transaction.

For an ATM to support this security feature, Mastercard recommends the ATM respond immediately when receiving an AAC response to the GENERATE AC command, where an online authorization request was made. The ATM should decline the transaction without requesting authorization from the issuer or requesting PIN entry.

BP	All	CL	BA086.14	An ATM should immediately decline a transaction when an AAC is received in response to an Authorization Request Cryptogram (ARQC) request in the GENERATE AC command without requesting issuer authorization or asking the user to enter a PIN. The ATM should offer the user the option to tap their contactless device again or cancel the operation.
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Transaction processing

When a chip card is inserted into an ATM or any device with an integrated Magnetic Stripe and chip reader, it is common practice for the Magnetic Stripe to be read as the card enters the device.

Chip reading normally occurs when the card is fully inserted. This saves time because the Magnetic Stripe details have already been captured if no chip is present and it is not necessary to pass the card a second time over the Magnetic Stripe reading head.

If the Magnetic Stripe is not present or cannot be read, the device should attempt to read any chip that may be present on the card.

If the Magnetic Stripe data has been read and the service code indicates the presence of a chip, the device must attempt to read the chip. If the service code indicates that a chip is not present, the terminal does not need to attempt to read the chip.

BP	All	CT	BA081.14	Devices with an integrated Magnetic Stripe and chip reader should attempt to read any chip that may be present on the card even if the Magnetic Stripe is not present or cannot be read.
R	All	CT	RA186.13	If the Magnetic Stripe service code indicates the presence of a chip on the card, the device must attempt to read the chip.

Application identifier

ATMs must support the AID of each product that they can acquire. For example, an ATM acquiring Mastercard, Maestro, and Cirrus cards must support the Mastercard AID, the Maestro AID, and the Cirrus AID.

R	All	CT CL	RA169.12	ATMs must support the AID of each product that they can acquire.
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Processing Code

For cash disbursements, different brands use different values for the Processing Code, which is contained in DE 3 of the online request message.

The value used is normally identical to the Transaction Type (tag '9C') used by the terminal. However, a single value of the Transaction Type may be used by the terminal for all products for cash disbursements. In addition, the correct value of the Processing Code for the product may be added by the acquirer host. Therefore, the Processing Code in DE 3 and Transaction Type in DE 55 sent in network messages may differ.

Reader configuration

The integration of a contactless reader in the ATM implies certain constraints in its configuration to ensure the necessary behavior required for use at an ATM.

The contactless reader must be configured as online-only. This can be achieved by using an online only configuration with a value of '14' for the Terminal Type.

R	All	CL	RA310.14	The ATM contactless reader must not be configured to allow a Transaction Certificate (TC) to be requested in the GENERATE AC command of a chip contactless transaction.
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Contactless limits

In the contactless technical specifications, three limits are used by readers in processing a contactless transaction.

The Terminal Contactless Transaction Limit is a maximum transaction amount above which a contactless transaction must not be performed. For ATMs, this value is set to the maximum transaction amount possible for contact transactions.

Terminals do not require cardholder verification for transaction amounts equal to or below the Terminal CVM Required Limit and, unless specifically requested by the cardholder, do not require a printed receipt. For ATMs, this value must be set to zero.

The Terminal Contactless Floor Limit is a transaction amount above which online issuer authorization is required. For ATMs, this value is set to zero.

R	All	CL	RA318.16	The Terminal CVM Required Limit and the Terminal Contactless Floor Limit in the contactless ATM must be set to zero.
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ATM wake-up

Contact chip cards are physically held in the card reader and allow multiple interactions between the card and the ATM. While on the contactless interface, the interaction between the card and the ATM occurs only during the tap of the product on the contactless reader unless multiple taps are requested.

The actual operational process for moving the ATM from an idle to an active state and offering the required services to customers is determined by individual implementations and the services that will be offered. The process flows in the next four sections outline recommendations and specific requirements to achieve an acceptable cardholder experience when supporting a Mastercard contactless product for Fast Cash and Multiple Service offerings.

Fast Cash

During the idle state, the ATM can display a Fast Cash Only option, where the cardholder selects the fast cash option and the amount of cash required to wake up the ATM.

The ATM then requests the cardholder to swipe, insert, or tap, the contactless reader having been enabled by the selection of the fast cash option.

When the contactless product is tapped, the ATM performs an EMV contactless Cash Withdrawal transaction with the transaction amount equal to the requested cash amount and the transaction type equal to value '01' for Cash Withdrawal. On successful completion of the contactless exchange, the ATM requests Online PIN entry before creating and sending the online request. No further interaction with the card or device is required to complete the transaction successfully. The ATM immediately returns to an idle state after the process is completed. A suggested transaction flow is shown in appendix "Contactless ATM transaction flows."

Contactless transactions involving multiple requests

Most ATM implementations wake up from an idle state through the action of inserting or swiping a card in a card reader.

When this process is used for contactless, all interaction with the contactless card or device occurs before the required ATM service and the amount of any cash withdrawal are known. Therefore, a generic contactless ATM interaction will occur for use with all of the services offered by the ATM with a zero transaction amount and the transaction type (tag '9C') equal to value '93' (contactless ATM wakeup).

R	All	CL	RA345.15	If a contactless transaction is used to wake up the ATM before any ATM service is selected, the ATM must use the following data values in the GENERATE AC command: - Transaction Type (tag '9C') value of '93' (contactless ATM transaction). - Transaction Amount (tag '9F02') value of '000000000000'.
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Mastercard does not prescribe the transaction flow required to be followed. However, for examples, refer to appendix "Contactless ATM transaction flows."

Fallback to other technologies

There is no technical fallback to Magnetic Stripe mode for contactless transactions. The contactless reader automatically selects EMV mode if this is available on both the card and reader.

If a contactless ATM transaction fails to complete or is declined by the issuer, the ATM must allow a repeat attempt using contactless technology or another supported acceptance technology. Some contactless devices may allow ATM transactions only after user interaction with the device so a second attempt at the transaction may be successful. There must not be any automatic process to force fallback from contactless acceptance to any other form of acceptance technology before a second transaction is attempted.

R	All	CL	RA323.14	ATMs must allow a repeat attempt of a failed contactless transaction using any acceptance technology supported by the ATM, including trying contactless again.
R	All	CL	RA324.14	ATMs must not force fallback from contactless acceptance to any other form of acceptance technology before a second transaction is attempted.

Contactless processing restrictions

Exception File Checking by the ATM is not required. It may be done after the communication with the card is finished.

ATMs are not required to terminate contactless transactions if the Application Usage Control (AUC) read from the device does not indicate valid at ATMs or valid for domestic or international cash.

If these bits are not set in the AUC of the cardholder device, then, if the ATM performs Processing Restrictions checks, the TVR bit ([2][5]) "Requested service not allowed for card product" must be set. The ATM will follow normal Terminal Action Analysis and does not need to separately terminate transactions where the preceding AUC bits are not set.

Bank Branch Terminals

A Bank Branch Terminal (BBT) is an attended device located in a bank branch that is used to perform manual cash advance transactions.

Bank Branch Terminals may also support other administrative functions at the issuer's discretion, such as PIN unblocking.

Technology selection

Bank Branch Terminals should support technical fallback.

Refer to chapter 4 "Acquirer requirements," section "Technology selection," subsection "Technical fallback."

BP	All	CT	BA099.15	Bank Branch Terminals should support technical fallback.
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Bank Branch Terminals are not permitted to support Mastercard Consumer-Presented QR transactions.

R	MC MS	QR	RA443.19	Bank Branch Terminals must not support Mastercard Consumer-Presented QR transactions.
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Refer to the regional requirements for the Europe region in chapter "Acquirer requirements," section "Regional variations."

Cardholder selection

Bank Branch Terminals must support cardholder selection. Refer to requirement RA078.19 in section "Application selection" in chapter "Acquirer requirements."

Offline CAM requirements

Support for offline CAM is optional at Bank Branch Terminals.

CVM requirements

For CVM requirements, refer to section "Cardholder verification" in chapter "Acquirer requirements."

Cardholder Activated Terminals

A Cardholder Activated Terminal (CAT) is an unattended terminal, under the operational control of a merchant, that delivers goods and services.

Full details of CAT classifications are in the *Transaction Processing Rules*.

Online-capable CAT devices that support PIN, offline, online, or both, or CDCVM and No CVM are known as dual-capability devices. They may produce either CAT Level 1 or CAT Level 2 transactions, depending on which cardholder verification was completed.

A CAT that supports Offline PIN on the contact interface or CDCVM on the contactless interface, but not Online PIN, must have dual capability. Dual-capability devices produce CAT Level 1 transactions when PIN was used and produce CAT Level 2 transactions when No CVM was used.

R	MC	CT CL	RA493.22	A CAT not supporting Online PIN on either interface, but supporting Offline PIN on the contact interface or CDCVM on the contactless interface, must have dual capability.
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CAT EMV Terminal Type

EMV Terminal Type classifications are categorized according to environment, online capability, and operational control.

The following table maps the EMV Terminal Types that may be used for each CAT classification.

Online capability	EMV terminal type (in hex)	CAT classification
Offline with online capability	25	CAT Level 1
		CAT Level 2
		CAT Level 4
Online-only	24	CAT Level 1
		CAT Level 2
		CAT Level 4

Online capability	EMV terminal type (in hex)	CAT classification
Offline-only	26	CAT Level 3

Technology selection

If a hybrid card is swiped in the Magnetic Stripe reader of a hybrid terminal with separate readers, the terminal is normally required to prompt the cardholder or merchant to use the chip reader instead.

This could cause cardholder confusion at CATs where there is no attendant to help the cardholder. In these circumstances, the CAT is allowed to perform the transaction using the Magnetic Stripe as long as it is treated as technical fallback.

BP	MC	CT	BA100.15	Online-capable CATs should support technical fallback.
R	MC	CT	RA100.11	Offline-only CATs must not support technical fallback.

Refer to the regional requirements for the Europe region in chapter 4 "Acquirer requirements," section "Regional variations."

The maximum Transaction amount for a Magnetic Stripe transaction at offline-only CATs is zero.

R	MC	CT	RA356.15	Offline-only CATs must not support Magnetic Stripe transactions.
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Cardholder selection

Where a CAT terminal is able to support cardholder selection for contact transactions between alternative AIDs, it should do so.

It may not be possible to support cardholder selection due to one of the following reasons:

- The environment of use, such as a tollway, means that cardholder interaction is undesirable due to the need for fast transactions.
- The nature of the device, such as a vending machine, has a limited display or function keys.

Refer to requirement BA027.11 in chapter 4 "Acquirer requirements," section "Application selection."

CVM requirements

For CVM requirements, refer to chapter 4 "Acquirer requirements," section "Cardholder Verification."

IPOS terminals

Integrated Point-Of-Sale (IPOS) terminals are categorized as POS terminals and, therefore, follow the same requirements.

POS terminals

There are specific POS terminal requirements that must be followed for hybrid terminals.

Cardholder selection

POS must support cardholder selection.

Refer to requirement RA078.19 chapter "Acquirer requirements," section "Application selection."

Offline CAM requirement

For CAM requirements, refer to chapter "Acquirer requirements," section "Card authentication."

CVM requirements

For CVM requirements, refer to chapter "Acquirer requirements," section "Cardholder verification."

Automated Fuel Dispensers

Automated Fuel Dispensers (AFDs) are unattended terminals at gas and petrol stations.

Refer to the regional requirements for the United States region in chapter "Acquirer requirements," section "Regional variations."

Vending machines

Contactless-only payment acceptance is permitted on vending machines identified with MCC 5499.

Such vending machines can operate with one of two possible purchasing processes:

- "Select first," where the goods or service to be purchased are selected before payment is made.
- "Credit first," where the payment is made before selecting the goods or service.

Acquirers should note that only contactless readers installed on vending machines that use a "select first" ordering and payment process will be capable of supporting both online and offline authorization of contactless transactions. Such readers may also behave as offline-only or online-only.

Contactless readers installed on "credit first" vending machines cannot authorize contactless transactions offline and, therefore, must always request online authorization.

R	All	CL	RA280.14	Contactless-enabled vending machines that operate as "credit first" must always request online authorization.
R	All	CL	RA281.14	Offline-capable contactless-enabled vending machines must operate as "select first."

On-board terminals

On-board terminals are a particular type of POS used in environments where online authorizations are not normally available, such as aircraft, trains, and ferries.

In these environments, acquirers may, at their own risk, consider:

- configuring the terminal as online-capable, and sending deferred authorisations later, for example when arriving at the final destination.
- clearing chip-declined contact and contactless transactions.

For more information, refer to chapter "Acquirer requirements," section "Transaction types," subsection "Merchant approved and deferred authorization."

Technology selection

On-board terminals must not support technical fallback.

R	All	CT	RA086.11	On-board terminals must not support technical fallback.
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Cardholder selection

On-board terminals must support cardholder selection.

Refer to requirement RA078.19 in chapter "Acquirer requirements," section "Application selection."

Offline CAM requirements

For CAM requirements, refer to chapter "Acquirer requirements," section "Card authentication."

CVM requirements

This CVM requirement applies to on-board terminals.

R	All	CT	RA085.13	On-board terminals that accept Mastercard brand must support Offline PIN and Signature. On-board terminals that accept only Maestro cards must support Offline PIN only.
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Transaction processing

On-board terminals should be able to process transactions online.

BP	All	CT CL	BA094.15	On-board terminals should be online-capable.
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However, on-board terminals are permitted to be offline-only when using one of the MCCs listed in requirement RA087.22.

R	All	CT CL	RA087.22	On-board terminals must correctly identify the merchant for which offline-only processing is permitted using one of the following MCCs: • MCC 4111 (Transportation - Suburban and Local Commuter Passenger) • MCC 4112 (Passenger Railways) • MCC 4411 (Cruise Lines) • MCCs 3000-3350 and 4511 (Air Carriers, Airlines) • MCC 5811 (Caterers)
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Mobile Terminals

Mobile Terminal solutions allow acquirers to use Commercial Off The Shelf (CoTS) devices as terminals to process Mastercard and Maestro transactions. CoTS includes mobiles, tablets, and similar devices.

The two families of Mobile Terminals are:

- Mobile Terminal with accessory acceptance solutions - These use a POS card reader accessory that can either be plugged into the audio jack or USB port or connected to the mobile device through Bluetooth or any other interface to read Magnetic Stripe, chip contact, or contactless payment cards and may enable PIN entry. The mobile device provides communications and may perform other functions, such as PIN Entry or signature capture.
- Mobile Terminal without accessory acceptance solutions - These are where the mobile device is able to read the card and maybe perform PIN entry and other required functions without any hardware peripherals.

Depending on the implementation, card data may be collected through the contact, contactless, or Mastercard Consumer-Presented QR interfaces.

Mobile Terminals used by merchants at the Point-Of-Sale (POS) are known as MPOS terminals. Refer to chapter "Acquirer requirements," section "Terminal types," subsection "Mobile POS terminals." MPOS terminals must meet the requirements applicable to POS terminals, unless specific exceptions apply.

Mobile Terminals may be used for terminal types other than POS, such as Bank Branch Terminals, Cardholder Activated Terminals, Vending Machines, or On-board Terminals. The full requirements for these devices apply to Mobile Terminals unless specific exceptions apply. These exceptions are noted in the appropriate section for the terminal type.

The scope and capabilities of Mobile Terminals are evolving quickly and Mastercard may, from time to time, support new concepts on a trial basis.

For more information, contact mposprogram@mastercard.com.

Mobile POS terminals

Mobile Point-of-Sale (MPOS) acceptance solutions allow merchants to use mobile devices as POS terminals to process Mastercard and Maestro payment transactions.

Technology selection

MPOS terminals should support as many technologies and interfaces as possible to ensure the widest acceptance.

Particular attention must be paid to the penetration of different technologies in the local market and any local regulatory or Mastercard requirements.

The following table lists the MPOS solutions, with and without accessory, that may be deployed in accordance with the rules and requirements applicable in the country or region of deployment.

MPOS solutions with accessory	Capabilities
Chip-capable MPOS	• Supports contact chip and Magnetic Stripe, may acquire contactless transactions in the United States region for terminal accepting only Mastercard • Accessory used to read card • Does not support PIN
Chip-only MPOS	• Supports contact chip only • Accessory used to read card and enter PIN • PIN support, in the accessory, optional unless Maestro is supported
Hybrid MPOS	• Supports contact chip and Magnetic Stripe • May acquire contactless transactions • Supports PIN • Accessory used to read card and enter PIN
NOTE: Refer to the regional variation for the support of the Magnetic Stripe on MPOS terminals in the Europe region in section "Chip and contactless."	

MPOS solutions with accessory	Capabilities
SPoC (Software PIN on CoTS [Commercial Off The Shelf])	• Supports chip, may support Magnetic Stripe • May acquire contactless transactions • Accessory used to read card • Supports PIN, entered into the screen of the mobile device
MPOS solutions without accessory	Capabilities
CPoC (Contactless Payments on CoTS)	• Contactless only solution, uses mobile device NFC to read contactless cards, no accessory • Does not support PIN • Supports CDCVM and No CVM
MPoC (Mobile Payment on CoTS)	• Contactless only solution, uses mobile device NFC to read contactless cards, no accessory • Supports Online PIN, entered into the screen of the mobile device, CDCVM and No CVM

Chip and contactless acceptance

For MPOS terminals, the same requirements related to chip and contactless acceptance apply as for other POS terminals.

Refer to chip-related and contactless acceptance-related sections in chapter "Acquirer requirements."

Cardholder selection

For MPOS terminals, the same requirements related to cardholder selection apply as for other POS terminals.

Refer to requirement RA078.19 for contact and requirement RA470.19 for contactless. Both are in chapter "Acquirer requirements," section "Application selection."

Online capability

MPOS terminals operate as online only. All transactions must be authorized online.

R	All	CT CL QR	RA192.22	All transactions performed at an MPOS terminal must be authorized online.
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Refer to exceptions relating to merchant approved and deferred authorization in chapter "Acquirer requirements," section "Transaction types," subsection "Merchant approved and deferred authorization."

Offline CAM requirements

Card authentication requirements are as for POS terminals.

Refer to chapter "Acquirer requirements," section "Card authentication."

CVM requirements

At MPOS terminals, a transaction may be completed using any of the following Cardholder Verification Methods (CVMs):

- CDCVM
- Online PIN, either captured on a connected hardware device or on the mobile device
- Offline PIN, for contact transactions only
- Signature, either captured electronically or on paper
- No CVM

The determination of the applicable CVM for a particular MPOS terminal transaction is based on conditions that vary by product, technology, transaction amount, and other considerations. The same requirements apply for MPOS as for POS terminals. Refer to chapter "Acquirer requirements," section "Cardholder verification."

Signature support is optional for Mastercard hybrid MPOS terminals.

Chip-only MPOS and chip-capable MPOS terminals must support Signature, No CVM, or both.

R	MC	CT	RA191.13	Chip-only MPOS and chip-capable MPOS terminals must support Signature, No CVM, or both.
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Cardholder receipts

Receipts, where required, may be printed or provided electronically.

Mobile POS indicators

MPOS transactions must be identified in this authorization message field:

DE 61 (Point-of-Service Data), subfield 10 (Cardholder-Activated Terminal Level) must be set to "9" (MPOS Acceptance Device).

R	All	CT CL QR	RA514.22	MPOS transactions must be identified with the value "9" in DE 61, subfield 10 (Cardholder-Activated Terminal Level).
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Other fields must be configured for MPOS solutions relying on CoTS internal readers, known as software MPOS, to accept PIN entry, contactless transactions, or both:

- DE 22 (Point-of-Service Data Code), subfield 2 (POS Terminal PIN Entry Mode) must be set to one of the following:
 - "1" (Terminal has PIN entry capability)
 - "2" (Terminal does not have PIN entry capability)
 - "3" (MPOS Software-based PIN Entry Capability)
- DE 48 (Additional Data - Private Use), subelement 21 (Acceptance Data), subfield 01 (MPOS Acceptance Device Type) must be set to one of the following:
 - "0" (Dedicated MPOS Terminal with PCI-compliant dongle [with or without keypad]), used when the reader is external to the CoTS device
 - "1" (Off the Shelf Mobile Device), used when the reader is embedded in the CoTS device

The presence of these data elements in authorization and clearing messages is mandatory for all new and existing deployments of software-based MPOS terminals compliant with PCI SSC standards for mobile security. These are the SPoC, CPoC, and MPoC solutions.

R	All	CT CL QR	RA515.23	The presence of DE22, subfield 2 (POS Terminal PIN Entry Mode) and DE 48, subelement 21, subfield 01 (MPOS Acceptance Device Type) is mandatory in all authorization and clearing messages originating from SPoC, CPoC, and MPoC solutions.
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Mobile POS approval process

These requirements indicate the certifications that must be presented to initiate an M-TIP service for MPOS, with and without accessory.

R	All	CT CL	RA494.22	All MPOS with accessory submitted for M-TIP testing must have a valid: • EMVCo Level 1 and Level 2 Letter of Approval (LoA) • Terminal Quality Management (TQM) Label or a TQM Action Plan • Mastercard LoA for Contactless Product or EMVCo Contactless Product LoA for C2 kernel if contactless is supported • PCI PTS PED approval if the terminal supports PIN • PCI SPoC approval for the application and PCI PTS Secure Card Reader for PIN (SCRp) approval for the reader, for SPoC solutions
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R	All	CL	RA495.22	All MPOS without accessory submitted for M-TIP testing must have a valid: • EMVCo Proximity Coupling Device (PCD) LoA • Mastercard Terminal Test Assessment Summary (TTAS) and the Mastercard Component Conformity Statement (CCS) • PCI CPoC approval for the application, for CPoC solutions • Mastercard PIN security evaluation approval, for Mastercard MPoC pilots only
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Special acceptance environments

Different acceptance environments may require special processing procedures.

Payment of gratuity

When paying bills in restaurants, it is common for cardholders to add a gratuity or a tip to the amount they charge to their card. For chip transactions, the gratuity can either be added during the EMV transaction or after its completion.

Terminals that allow the gratuity to be added during the EMV transaction must allow the cardholder to enter it before CVM processing. This is especially important for PIN transactions, as the cardholder's PIN entry is implicitly approving the total amount.

Terminals that allow cardholders to add a tip to their card payment during the EMV transaction must use the procedure as detailed in requirement RA082.14.

If gratuities are to be included during contactless transactions, then the cardholder must be offered the opportunity to add the gratuity amount before the contactless transaction starts.

R	All	CT CL	RA082.14	Terminals that allow cardholders to add a tip to their card payment during the EMV transaction must use this procedure: 1. The amount of the bill, excluding tip, is displayed. 2. The terminal asks if the cardholder wants to add a tip. 3. If the cardholder chooses to add a tip, the terminal allows the cardholder to enter the gratuity amount. The design of the interface must make it clear that the tip is being entered so that the cardholder does not accidentally enter their PIN in this field. 4. The terminal displays the tip amount and the total amount to be charged and asks the cardholder to enter their PIN to confirm, if applicable. 5. The EMV transaction flow continues, and, if the transaction must be authorized online, the total transaction amount is used in the online request message.
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Visual card checks

For Magnetic Stripe Signature-based transactions, visual card checks should be performed to check the authenticity of the card being presented. With various types of chip transactions, the physical card is not handed over to the merchant and, therefore, visual checks cannot be completed.

There is no need to complete visual card checks for:

- Contactless transactions for amounts below the relevant CVM limit
- Contact or contactless transactions where the cardholder is verified with Online PIN, Offline PIN, or CDCVM

Therefore, the merchant does not have to:

- Check any visual security features, such as the presence of a Mastercard hologram
- Visually check the valid date and the expiration date on the face of the card
- Manually check Warning Bulletins
- Compare the four-digit truncated account number imprinted in the Signature panel with the last four digits of the embossed account number on the face of the card
- Compare the embossed account number on the face of the card with the number displayed or printed from the POS terminal
- Compare any photograph on the card with the person presenting the card
- Check that the card is signed

Any automation of these visual checks by the POS system, such as Swipe and Verify checks, must be capable of being overridden or disabled for the acceptance of relevant chip transactions.

Transit

In different transit implementations, a contactless transaction may be used to enable entrance or exit from the transit system or for checking authority to travel. These transactions will only be sent online periodically to be authorized by the issuer.

For more information on deferred authorization, refer to chapter 4 "Acquirer requirements", section "Transaction types," subsection "Merchant approved and deferred authorization."

To avoid that issuers are unaware that multiple transactions have been completed by the card or device, and thus the ATC has been incremented more than may be usual, which may in turn create unexpected declines, transit merchants that operate in this way must periodically send either a new authorization or an ATC Update to the issuer.

Transit merchants must, therefore, send an ATC update message at least once every 20 times a card is used since the last approved transaction from the issuer.

There is no requirement that the authorization or update message be completed in real time before the cardholder is granted access to the system. The message must contain the DE 55 information from the most recent card tap.

R	All	CL	RA333.14	Acquirers must generate a new authorization or an ATC Update message at least every 20 times the card is tapped in the transit merchant.
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For details of ATC Update messages, refer to the *Customer Interface Specification* and the *Single Message System Specifications* manuals.

Fleet acceptance

This section applies to Mastercard Fleet Card, MultiCard, and Government Commercial Card programs. The term Fleet Card is used here to refer to all the programs.

Based on local market considerations and subject to approval by Mastercard, terminals supporting fleet processing can use one of the following specifications:

- *Fleet Implementation Guide For the U.S. Region* (for US region fleet programs from 11 April 2025 onwards)
- *Fleet Implementation Guide - Mastercard Corporate Fleet card Program* (for Canada region and US region before 11 April 2025)
- *Fleet Implementation Guide Global* (for all regions except US and Canada)
- Or an alternative Mastercard agreed specification.

Detecting if Fleet Processing is required

Fleet terminals detect if fleet processing is required as per the appropriate Fleet Implementation Guide for the region or country.

The methods applicable may include:

- Presence of fleet-related data items on the chip
- BIN range
- A fleet-specific PIX extension such as 'FC xxxx' appended to a standard Mastercard brand AID

Fleet Processing

An important benefit of Fleet Cards is the flexibility to control what can be purchased and which prompts for information are displayed to the cardholder at the fuel merchant POS locations.

Fleet terminals should follow appropriate Fleet Implementation Guide instructions on fleet processing. This involves locating the driver prompt and processing restriction information found from the locations in the guide, and acting accordingly. It may also involve reading additional fixed fleet specific information from the chip.

In addition to the standard fleet processing defined in the Fleet Implementation Guide, terminals may optionally adopt additional special processing for particular subsets of fleet cards based on the AID used.

Hybrid terminals following the *Fleet Implementation Guide - Mastercard Corporate Fleet card Program* (for Canada region and US region before 11 April 2025) should continue to use legacy Magnetic Stripe-based product information, Product Restriction Code, and Product Type Code to support the proper processing of Magnetic Stripe Fleet Cards.

Data input by the driver into the terminal, in response to the prompts, and possibly additional fleet related data encoded on the card, is sent by the acquirer to Mastercard both online and in clearing. The terminal must be online capable.

BP	All	CT CL	BA076.24	Terminals that support special Fleet Card processing should drive the special processing requirements applicable to the product, by looking for the driver prompt and purchase restriction instructions from the fleet-related data on the card as instructed by the appropriate <i>Fleet Implementation Guide</i> . This includes the capture of fleet related data by the terminal, and sending of this data by the acquirer to Mastercard in online and clearing messages
R	All	CT CL	RA564.24	Fleet terminals must be online capable.

Refer to the regional variation for the United States region in chapter 4 "Acquirer requirements," section "Regional variations."

Alternative EMV transaction flows

Mastercard has rolled out optional EMV contact and contactless solutions using an alternative transaction flow currently used in the United States region, the Latin America and the Caribbean region, and Sweden to enhance the acceptance experience at the POS.

These solutions allow the cardholder to dip or tap their payment device before the final amount is known and, in the case of contact transactions, to withdraw the card from the reader without waiting for the authorization response to be received from the issuer. Such a flow has the following benefits:

- It may improve the speed of the transaction.
- It allows data to be read from the cardholder payment device and used by the merchant for proprietary purposes, such as a loyalty program, at an early stage in the payment process.

For more details about the alternative transaction flows in Sweden, refer to chapter "Acquirer requirements," section "Regional variation," subsection "Europe region," sub-subsection "Alternative contactless flow at terminals in Sweden."

For more details about the alternative transaction flows in the United States region and the Latin America and the Caribbean region, refer to chapter "Acquirer requirements," section "Regional variations," subsection "United States region," sub-subsection "M/Chip Fast."

Transaction types

This section covers specific requirements for different types of transactions. It covers both contact and contactless transactions.

Purchase

Purchase transactions are supported at POS terminals. Optionally, ATM can support purchase transactions, for example, mobile top-up transactions.

For contactless transactions at ATM, it is typically not known which transaction is to be performed at the moment of tapping the card on the reader. A generic contactless ATM interaction will, therefore, occur for use with all of the services offered by the ATM with a transaction amount (tag '9F02') of '000000000000' and the Transaction Type equal to value '93'. Refer to requirement RA345.15.

R	All	CT	RA297.19	An ATM or POS terminal performing a purchase transaction through the contact interface must set the Transaction Type (tag '9C') to '00'.
R	All	CL	RA433.19	A POS terminal performing a purchase transaction through the contactless interface must set the Transaction Type (tag '9C') to '00'.

Purchase with cash back

Purchase with cash back transactions may be supported for Maestro and Debit Mastercard cards on the contact and contactless interface.

Mastercard Consumer-Presented QR does not support purchase with cash back transactions.

Purchase with cash back transactions must be authorized online.

Cardholder verification must always be completed for purchase with cash back transactions. Refer to the regional requirements for the Asia/Pacific region, the Europe region, and the Middle East/Africa region in chapter 4 "Acquirer requirements," section "Regional variations."

The contactless CVM Limit does not apply to purchase with cash back transactions. Refer to requirement RA267.14.

R	All	CT CL	RA079.14	Purchase with cash back transactions must always be authorized online.
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For purchase with cash back, the terminal must populate the Amount, Other field with the amount of the cash back element. Refer to requirement RA126.22.

Acquirers may choose to support the use of Authorization Request Response/0110 or Financial Transaction Request Response/0210 message containing a code of "87" (Purchase Amount Only, No Cash Back Allowed).

The purchase with cash back offer can be made and accepted either before or after the cardholder dips or taps the card.

Offer is made and accepted before tap or dip

The recommended processing for this scenario is described here:

For contact transactions, the transaction is initiated using the Amount, Authorized (tag '9F02') set to the purchase amount plus the cash back amount, the Transaction Type (tag '9C') set to the Purchase with Cash Back value '09', and the Amount, Other (tag '9F03') set to the cash back amount. The cardholder dips the contact card, and the Application Usage Control on the card is consulted:

- If purchase with cash back is supported by the card, the terminal proceeds with all parameters already set for the purchase with cash back.
- If the contact card does not support purchase with cash back, the terminal should dynamically revert all parameters to the purchase-only transaction. If any of the Amount, Authorized, Amount, Other, or Transaction Type have already been communicated to the card in the GET PROCESSING OPTIONS command, a new transaction without cash back should be initiated. The new transaction may automatically select the same application that was used for the original transaction.

For contactless transactions, the transaction is initiated using the Amount, Authorized (tag '9F02') set to the purchase amount plus the cash back amount, the Transaction Type (tag '9C')

set to the Purchase with Cash Back value '09', and the Amount, Other set to the Cash Back amount (tag '9F03'). The contactless CVM Required Limit is set to zero. The cardholder taps the contactless card or the payment device, and the transaction follows normal transaction flow.

Offer is made and accepted after tap or dip

The offer may also be made and accepted after tap or dip because either the cardholder requesting cash back or the merchant proactively offering it. For contact transactions, this scenario could happen because the contact card is dipped and the Application Usage Control is consulted without any merchant or customer intervention. For contactless transactions, this scenario could happen because either the merchant offers cash back or the cardholder requests cash back after the tap.

The recommended processing for this scenario is described here:

For contact transactions, the transaction is initiated using the Amount, Authorized (tag '9F02') set to the purchase amount, the Transaction Type (tag '9C') set to the purchase value '00', and the Amount, Other (tag '9F03') set to zero. The cardholder dips the card and the Application Usage Control on the card is consulted:

- If purchase with cash back is supported by the contact card, the offer is made to the cardholder. If the cardholder accepts the offer, the cash back amount is chosen and the terminal should dynamically update the Amount, Authorized (tag '9F02') to the purchase amount plus cash back amount, the Transaction Type (tag '9C') to the Purchase with Cash Back value '09', and the Amount, Other (tag '9F03') to the cash back amount.

If any of the Amount, Authorized, Amount, Other, or Transaction Type have already been communicated to the card in the GET PROCESSING OPTIONS command, a new transaction with cash back should be initiated. The new transaction may automatically select the same application that was used for the original transaction.
- If the contact card does not support purchase with cash back, then the card proceeds with all parameters already set for purchase.

For contactless transactions, the original transaction is discarded and the transaction is reinitiated following the recommendation for the offer is made and accepted before tap or dip.

BP	All	CT CL	BA095.15	Terminals that support purchase with cash back should follow the guidance either for the offer made and accepted before tap or dip or the offer made and accepted after the cardholder dips or taps the card as described in the preceding scenarios.
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Manual cash advance

Cardholder verification is always required for manual cash advance.

R	All	CT CL	RA229.14	Cardholder verification must always be performed for manual cash advance transactions.
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The contactless CVM Limit does not apply to cash advance transactions. Refer to requirement RA231.17 in chapter "Acquirer requirements," section "Cardholder Verification," subsection "Contactless transactions."

Preauthorization transactions

Preauthorization transactions are used by certain merchants when providing goods or services that will not be paid for immediately, for example, in hotels and for car rentals. The preauthorization notifies the merchant that the cardholder has sufficient funds to pay for the goods or services when payment is due.

A chip card has no way of knowing whether it is being used for a preauthorization, so it performs a normal EMV transaction, including PIN entry if that is the chosen CVM. The transaction completes with a TC if the preauthorization is successful.

When a preauthorization is performed in a contactless environment, the CVM Limit should be set to zero to ensure that a CVM will always be used.

The sale may complete by the merchant submitting a clearing record with the final billing amount, which should include the chip data from the preauthorization, assuming the chip was not also used during the completion process.

BP	All	CT CL	BA020.11	In cases where the chip is not used for transaction completion, terminals should retain the chip data resulting from a preauthorization transaction for use in a subsequent clearing message.
BP	All	CL	BA103.16	The CVM Limit should be set to zero for preauthorization transactions.

In some circumstances, a preauthorization is used to take a deposit, for example, renting sports equipment. If the equipment is returned undamaged, the transaction is not used.

In some environments, for example, fuel dispensers, acquirers may send an advice message to inform the issuer of the final amount subsequent to a preauthorization. In that case, if the chip data used in the advice message is taken from the preauthorization message, it must not be modified. Changing any subelement may cause the cryptogram validation to fail.

R	All	CT CL	RA188.13	If an advice message is sent to inform the issuer of the final amount of a transaction, all DE 55 subelements, including the Transaction Amount (tag '9F02'), must remain unchanged between the authorization and the advice.
BP	All	CT CL	BA021.11	Terminals should produce a reversal for unused deposits to avoid blocking cardholder's funds unnecessarily.

Refunds

Refund transactions are made when a merchant credits the cardholder. This would occur when the cardholder returns an item that they purchased earlier. Refunds are always associated with a previous purchase transaction. A refund must be made to the same account as the original transaction.

Acquirers must be able to process refund transactions initiated through the contact and the contactless interface or through a Mastercard Consumer-Presented QR transaction.

If card refunds are supported by the merchant, refunds through both the contact and contactless interface must be supported. Merchant support for contactless refunds is recommended at a minimum of one contactless enabled terminal in a merchant location.

The requirement to support refunds using Mastercard Consumer-Presented QR payment is only applicable to attended terminals.

Cardholder verification is not required for refunds. The presence of chip data for EMV contact and contactless refund transactions is optional. However, if chip data is present, all mandatory tags in DE 55 must be present in the proper format and structure.

To initiate a refund, the terminal needs to read the PAN and expiration date from the card. This information can be read from the chip, contact or contactless, or the Magnetic Stripe, or can be entered manually, PAN Key Entry, when allowed by the product rules. If the Magnetic Stripe is used, the terminal should not flag the transaction as fallback to avoid a potential decline by the issuer.

A refund transaction is performed in the same way when using the chip as when using the Magnetic Stripe, except that the terminal reads the information necessary to perform the refund from the chip instead of the Magnetic Stripe.

Refunds over the contactless interface must be performed by reading the Track 2 details and then requesting an AAC. This prevents card risk management counters from being adversely impacted.

Refund transactions may be sent online to the issuer regardless of the card's generated cryptogram. This ensures that issuers are always aware in real time when a merchant gives a refund to the cardholder.

A cryptogram should be generated during online authorization of a refund transaction in the same manner as it occurs during the online authorization of a purchase transaction.

R	All	CL	RA220.14	Acquirers must be able to support refunds initiated through the contactless interface.
R	All	CL	RA223.14	If refunds initiated using the contact interface are supported at a POS, then the merchant must also allow refunds for transactions through the contactless interface.
R	MC MS	QR	RA439.19	Attended terminals that support Mastercard Consumer-Presented QR must support refund transactions.
BP	All	CT	BA022.11	Terminals that perform refunds by reading the contact chip should initiate a chip transaction and follow the normal EMV transaction flow to retrieve the Track 2 information from the chip.
R	All	CL	RA221.24	If EMV mode contactless refunds are supported by the merchant, an AAC must be requested from the card.
BP	All	CT	BA024.11	Terminals should be able to process the refund regardless of whether the card has produced a TC or an AAC.

The acquirer has proprietary rights regarding how refunds are processed by the merchant and terminal.

BP	All	CT CL	BA025.11	Acquirers that send refund transactions to the acquirer host for approval should do this after the dialogue with the chip card has been completed.
BP	All	CT CL	BA026.11	If an attempted chip-read refund fails, the merchant should reinitiate the refund transaction using PAN Key Entry when allowed by the product rules.

Merchant approved and deferred authorization

Some terminals support a Merchant Approved function, where the merchant overrides the transaction outcome to decline or go online and accepts the transaction anyway without communicating to the acquirer host.

Refer to chapter 4 "Acquirer requirements," section "Online request requirements," subsection "Authorization by acquirer host."

This function can be supported by both contact and contactless interfaces.

A merchant might, at their risk, override the transaction outcome to decline or go online and accept the transaction offline, possibly with a deferred authorization. In this case, the merchant should check:

- BIN validity

- Card support for deferred authorization transactions - according to tag '9F6E' (Third Party Data), "Product Extension for Transit", indicator "Deferred authorization not supported", if present
- Fraud risk impacts when the Terminal Verification Results (TVR) indicates any of the following:
 - ODA was not performed, SDA failed, DDA failed, CDA failed (TVR[1][8], TVR[1][7], TVR[1][4], TVR[1][3])
 - Card appears on terminal exception file (TVR[1][5])
 - Expired application (TVR[2][7])
 - Cardholder verification was not successful (TVR[3][8])
 - PIN Try Limit exceeded (TVR[3][6])
 - PIN entry required and PIN pad not present or not working (TVR[3][5])
 - PIN entry required, PIN pad present but PIN was not entered (TVR[3][4])

NOTE: In this context, successful DDA or CDA provides assurance that the terminal has authenticated the card with the claimed account number but not that the account can or will pay the merchant. Successful DDA or CDA is indicated when terminal and card both support DDA or both support CDA, for each Terminal Capabilities and card AIP, so that the "ODA was not performed" bit in the TVR is not set, but DDA or CDA has not failed, as indicated in the TVR.

For Maestro contact, CVM Results (tag '9F34') must indicate that the Offline PIN has been performed and validated successfully.

Maestro contact and contactless transactions that have been forced accepted are known as Merchant Approved Maestro POS transactions. Refer to *Transaction Processing Rules*.

For such transactions, acquirers must submit an online request to the issuer once their connection has resumed and they may only clear the transaction on authorization by the issuer.

BP	All	CT	BA121.20	Authorization should always be performed in real time.
		CL		

Balance inquiry

Acquirers that support balance inquiries for Magnetic Stripe cards must comply with the requirements listed in this section to support balance inquiries for chip cards.

To conduct a balance inquiry, the terminal performs a normal chip transaction and sends it online for authorization with the transaction type in DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code) set to "30". The CVM condition code "If unattended cash" is satisfied by a balance inquiry at an ATM.

For contactless transactions at ATM, it is typically not known which transaction is to be performed at the moment of tapping the card on the reader. A generic contactless ATM interaction will, therefore, occur for use with all of the services offered by the ATM, with a

transaction amount (tag '9F02') of '000000000000' and the transaction type (tag '9C') equal to value '93'. Refer to requirement RA345.15.

There are no requirements specific to CVM processing for balance inquiry transactions at POS.

R	All	CT	RA057.14	For the purposes of CVM Processing, a balance inquiry at an ATM must be treated as unattended cash.
R	All	CT	RA058.14	A terminal performing a balance inquiry must set the Transaction Type (tag '9C') to '30'.

Acquirers do not need to keep the ARQC produced by the card for a balance inquiry.

PIN Management

Acquirers may support PIN Management at ATMs. PIN Management transactions are used to manage the PIN used with a contact chip card.

There are two types of transactions:

- PIN Unblock (Transaction Type '91')
- PIN Change (Transaction Type '92')

For PIN change, cardholders normally enter their chosen new PIN twice and the encrypted values of these are compared before sending the encrypted value to the issuer.

PIN Management requests involve an EMV transaction for a zero amount. A PIN Management request is not a cash transaction, but the CVM condition code "If unattended cash" is satisfied by a PIN Management transaction.

Application Usage Control byte 1, bits 8 or 7 indicating support for cash transactions do not need to be set on the card.

If the outcome of the PIN Management transaction is unsuccessful, the ATM must display this to the cardholder. This may be following an issuer decline or if the delivery of any script to the card is unsuccessful. The ATM may need to display other messages, such as "contact issuer."

PIN Change requests must be reversed to the issuer if the script delivery is unsuccessful to enable the Online PIN value to be reversed to the old value.

R	All	CT	RA061.14	For the purposes of CVM Processing, PIN Management at an ATM must be treated as unattended cash.
BP	All	CT	BA042.13	An ATM performing PIN Management should set the Transaction Type (tag '9C') to '91' for PIN Unblock or '92' for PIN Change.
R	All	CT	RA065.11	The outcome of the PIN Management transaction must be displayed to the cardholder.

R	All	CT	RA066.11	PIN Change transactions must be reversed to the issuer if the script delivery is unsuccessful.
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Payment Transactions

A Payment Transaction is a transfer of funds to a card account.

Payment transactions enables a cardholder to use their card or access device to deposit cash into one of their accounts or to transfer funds between their accounts. For the following transactions:

- A cardholder verification different from No CVM must be performed.
- The account where the funds are being deposited or transferred must be associated with the card or access device used for authenticating the cardholder.

Payment Transactions may be supported for Mastercard, Maestro, and Cirrus products on the contact, contactless, and Mastercard Consumer-Presented QR interfaces. If Payment Transaction is supported by the terminal, it must be available to all products and interfaces supported by the terminal.

Payment Transactions conducted in a card-present environment must only be supported at Point-Of-Sale (POS) terminals, Automated Teller Machines (ATMs), Bank Branch Terminals (BBTs), and CAT Level 1 Cardholder Activated Terminals (CAT).

Payment Transactions are identified by all of the following:

- A value of "28" in DE 3 (Processing Code), subfield 1 (Cardholder Transaction Type Code)
- An applicable MCC in DE 18 (Merchant Type)
- A value of "P" in DE 48 Transaction Category Code (TCC)
- An applicable Transaction Type Identifier in DE 48 (Additional Data - Private Use), subelement 77 (Transaction Type Identifier [TTI])

For a list of MCCs, TTIs, and other required data applicable for Payment Transactions, refer to the *Customer Interface Specification*, *Single Message System Specification*, *Mastercard MoneySend and Funding Transactions Program Standards*, or any other applicable Mastercard program standards.

To conduct a Payment Transaction, the terminal performs a chip transaction and sends the transaction online.

R	All	CT CL QR	RA516.22	A terminal performing a Payment Transaction must set the Transaction Type (tag '9C') to '28', unless RA345.15 exception applies.
R	All	CT	RA517.22	For Payment Transactions, the Terminal Floor Limit must be set to zero.

R	All	CL	RA518.22	For Payment Transactions, the Terminal Contactless Floor Limit must be set to zero.
R	All	CL	RA519.22	For Payment Transactions, the Terminal CVM Required Limit must be set to zero.
R	All	CT CL QR	RA520.22	When a terminal supports Payment Transactions, it must be supported by all applications and interfaces supported by the terminal.

For more product-related information, refer to *Mastercard MoneySend and Funding Transactions Program Standards*, or any other applicable Mastercard program standards.

Account Status Inquiry

Account Status Inquiries (ASI) have DE 61.7="8" and either:

- DE 3.1 value "00" with DE 4 value 0.
- DE 3.1 value "28" and DE 4 value 0 or higher.

For more information on the content and use of ASI transactions, refer to the *Customer Interface Specification*, chapter 5, section "Account Status Inquiry Service."

For special programs approved and communicated by Mastercard, DE 55 is optional for Account Status Inquiry (ASI) transactions. If DE 55 is present, it must be TLV encoded with all the mandatory tags present and correct.

Online request requirements

This section relates both to the data carried between the terminal and the acquirer host and the data on the acquirer and network interface.

One of the major benefits of contactless transactions is fast transaction times. These benefits are maximized when used with high-speed authorization lines.

BP	All	CL	BA069.14	Connectivity of contactless terminals should be optimized where possible to allow online issuer authorization to be obtained within 4 seconds.
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Online request

This section relates both to Authorization Request/0100 and Financial Transaction Request/0200 messages.

All online-capable terminals must send the Application Cryptogram and all the data associated with it to the acquirer host.

Mastercard networks use the ISO 8583-defined data element DE 55 to transport chip-related data from acquirers to issuer host systems. The Mastercard networks transport DE 55 as a binary field without any conversion.

R	All	CT CL QR	RA114.19	Acquirers that support chip transactions must support the use of DE 55 in their interfaces to Mastercard authorization platforms for online requests, authorization response online authorization advice messages, and clearing messages.
R	All	CT CL	RA115.12	Acquirer and intermediary networks that support chip transactions must transport the minimum set of data elements defined by EMVCo in the EMV Book 4, Cardholder, Attendant, and Acquirer Interface Requirements for each chip transaction.

CVM Results (tag '9F34') must be supported by all acquirers.

R	All	CT CL	RA117.11	All acquirers must include the CVM Results (tag '9F34') in DE 55 in all Authorization Request/0100 and Financial Transaction/0200 messages.
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The data in DE 55 are used to create the cryptogram, and if they are altered in any way, it will not be possible to recreate or verify the cryptogram. Alteration or padding of data is, therefore, not allowed.

R	All	CT CL	RA118.11	The data elements in DE 55 must contain the exact values, bitwise, that were present on the card to terminal interface.
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Terminal Action Analysis

The outcome of a transaction must be dictated by the outcome of the Terminal Action Analysis, as determined by Terminal Action Codes and Issuer Action Codes settings.

Acquirers must not implement risk management measures that are not in line with Mastercard rules and requirements.

R	All	CT CL	RA468.19	Acquirers must not implement proprietary logic that overwrites the outcome of the EMV Terminal Action Analysis process.
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Data in the online request message

Chip-related data that is required in the online request message is given in this section. For full details of online request requirements, refer to the network interface specification documentation.

Data included in chip messages must be read from the chip interface being used. Data from either the Magnetic Stripe or an alternative chip interface must not be mixed in the message.

R	All	CT CL	RA121.22	In the Authorization Request/0100 and Financial Transaction Request/0200 message, for chip transactions, the following data elements must be populated with data read from the chip interface being used: • DE 2 (Primary Account Number) with the Application PAN (tag '5A') • DE 14 (Date, Expiration), if present, with the Application Expiration Date (tag '5F24') • DE 23 (PAN Sequence Number) with the Application PAN Sequence Number (tag '5F34') In the case that (tag '5F34') is missing, DE 23 must either not be populated or be populated with the default value "000". • DE 35 (Track 2 Data) with the Track 2 Equivalent Data (tag '57'). Acquirers must not use data from the individual component data elements from the chip to reconstruct a Track 2.
R	All	CT CL	RA123.14	Data included in online request messages must only be populated using data from the chip interface being used.
BP	All	CT CL	BA041.12	When an authorization is deferred for any reason, the chip data included in the authorization should be the data associated with the first GENERATE AC command.

R	MC MS	QR	RA444.22	For Mastercard Consumer-Presented QR transaction, in the online request and the clearing message, for EMV mode transactions the following data elements must be populated with data read from the QR Code: • DE 2 (Primary Account Number) with the Application PAN (tag '5A'). If tag '5A' is not present, the DE 2 must be populated with PAN from Track 2 Equivalent Data (tag '57'). • DE 14 (Date, Expiration), if present, with the Application Expiration Date (tag '5F24'). • DE 23 (PAN Sequence Number) with the Application PAN Sequence Number (tag '5F34'). If (tag '5F34') is missing, DE 23 must either not be populated or be populated with the default value "000". • DE 35 (Track 2 Data) with the Track 2 Equivalent Data (tag '57'), if present. Acquirers must not use data from the individual component data elements from the QR to reconstruct Track 2.
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When a purchase with cash back transaction is made, Amount, Other contains the cash back amount.

The Amount Authorized field is then the sum of the purchase amount and the cash back amount.

For a normal purchase transaction with no cash back, one of the following is true:

- Amount, Other (tag '9F03') is filled with binary zeros in DE 55.
- Amount, Other is not present in DE 55.

R	All	CT CL	RA126.22	For purchase with cash back transactions, the terminal must populate the Amount, Other field with the amount of the cash back element. If the transaction does not include cash back, Amount, Other (tag '9F03') must be either filled with binary zeroes or not be present in DE 55. Amount, Other may not contain any other value.
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The Transaction Sequence Counter is a counter maintained in the terminal that is incremented for each transaction performed.

BP	All	CT CL	BA030.11	DE 37 (Retrieval Reference Number) should contain the value of the Transaction Sequence Counter, right-justified and left-padded with zeroes.
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R	All	CL	RA282.14	Acquirers must process on their network interface and host system contactless transactions, as described in this section.
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To enable fraud monitoring of incorrect contactless kernels used, the terminal is recommended to always provide the Contactless Kernel Identifier-Terminal (tag '96') for contactless transactions.

BP	All	CL	BA127.25	Acquirers should transmit Kernel Identifier-Terminal (tag '96') in online authorization request messages for contactless transactions.
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Identifying contactless transactions

DE 61 and DE 22 in online request messages must identify the transaction as contactless. Other data elements contain the same data values as for existing transactions.

Contactless transactions from EMV mode-capable terminals are one of the following:

- EMV mode transactions with the same data elements as contact chip transactions
- Magnetic Stripe mode transactions with the same data elements as contact Magnetic Stripe transactions

Contactless transactions from Magnetic Stripe mode terminals are always Magnetic Stripe mode transactions with the same data elements as contact Magnetic Stripe transactions.

Acquirers who deploy contactless EMV mode terminals must be full grade acquirers and carry all the minimum data set required. DE 55 must be populated in the online request messages.

Device Type

Device Type describes the type of Card Access-device that the cardholder used to perform a transaction, for example Card, Mobile Phone, key fob, and so forth.

If present, Device Type is included in tag '9F6E' (Third Party Data). For more information, refer to chapter 5, "Data requirements" subsections "Payment scheme specific data objects" and "Third Party Data".

Acquirers should capture the Device Type, where present, and send this to the issuer:

- In Online messages (including Authorization Request/0100 and Financial Transaction Request/0200) in DE 48 SE 23 SF 1
- In GCMS Clearing messages (including First Presentment/1240): in IPM PDS 0198

BP	All	CT CL QR	BA071.24	Acquirers should include the Device Type, where present, in the online request message and clearing message.
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R	All	CL QR	RA557.24	Effective 1 January 2026, if Device Type (part of tag '9F6E' Third Party Data) is read from the card in a contactless or QR transaction, Acquirers must provide this in DE 48 SE 23 SF 1 in the Online message.
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Refer to the regional requirements for the Canada region and United States region in chapter 4 "Acquirer requirements," section "Regional variations."

Terminal Capabilities

Contactless readers support two sets of CVM. One is used when the transaction amount is equal to or below the CVM Limit and the other when the transaction amount is above the CVM Limit.

Similarly, some contact terminals may feature configurable kernels that are configured with several sets of supported CVM. In these cases, the Terminal Capabilities (tag '9F33') must reflect the CVM set that is actually used for processing the transaction.

R	All	CT CL	RA490.22	When Terminal Capabilities (tag '9F33') is present in DE 55, it must reflect the terminal capabilities set in the reader or kernel configuration actually used for the processing of the transaction.
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The mandate regarding the presence of the Terminal Capabilities (tag '9F33') in DE 55 for acquired transactions is described in these requirements.

R	All	CT CL	RA546.24	For all newly deployed (including replacement) terminals, acquirers must include Terminal Capabilities (tag '9F33') in DE 55 in all authorization and clearing messages.
R	All	CT CL	RA547.24	Effective 31 December 2024, for all terminal, acquirers must include Terminal Capabilities (tag '9F33') in DE 55 in all authorization and clearing messages.

Refer to the regional requirement RA350.20 in chapter 4 "Acquirer requirements," section "Regional variations," subsection "United States region."

Authorization response

For contact transactions, the Authorization Request Response/0110 or Financial Transaction Request Response/0210 message may contain DE 55, subelements '71', '72', and '91'.

When all of these subelements are received, they must be passed unaltered to the terminal.

If the Issuer Authentication Data is listed in the CDOL2 of the card, it must be sent to the card with the second GENERATE AC command. However, an online response may not contain Issuer

Authentication Data. All online-capable terminals must be able to send to the chip application the Issuer Authentication Data, if received, including the ARPC, received in the online response together with any script received.

R	All	CT	RA193.13	The DE 55 subelements of the Authorization Request Response/0110 or Financial Transaction Request Response/0210 message must be passed by the acquirer in full and unaltered to the terminal.
R	All	CT	RA129.11	Acquirers must not interpret or modify the contents of the Issuer Authentication Data.

If a response to an authorization is not received, transactions are approved at the acquirer's risk.

For EMV Mode contactless transactions, there is no second terminal or card risk management possible as there is for contact transactions.

Retaining the card at an attended terminal is optional, as it may be impractical for an attendant to retain a card that is not initially handed over to the merchant during payment.

Authorization Response Code

The Authorization Response Code (tag '8A') is either the ISO 8583 response code received by the terminal from the issuer or a Response Code generated by the terminal.

If the transaction was authorized online, the terminal normally sets the Authorization Response Code to the value in the Issuer Response Code, DE 39, in the authorization response.

The following Authorization Response Codes should be considered by the terminal as approved:

- "00" (Approved or completed successfully)
- "08" (Honor with ID)
- "10" (Partial Approval)
- "87" (Purchase Amount Only, No Cash Back Allowed)

On receiving one of these Response Codes, the terminal must request a TC at the second GENERATE AC.

An Authorization Response Code of "10" indicates a partial approval and an Authorization Response Code of "87" indicates an approval without cash back. The merchant and cardholder interface must make it clear exactly what has been approved.

R	All	CT	RA130.11	When an Authorization Response Code of "00" (Approval), "08" (Honor with ID), "10" (Partial Approval), or "87" (Purchase Amount Only, No Cash Back Allowed) is received, the terminal must request a TC at the second GENERATE AC.
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An Authorization Response Code of "01" indicates a referral, and the appropriate processing should be followed by the terminal.

Referral issuer responses are not required to be supported by acquirers for contactless transactions. Referral responses may be treated as a decline by the acquirer or merchant.

R	All	CT	RA132.11	If the issuer sends an authorization response with Issuer Response Code "01" (Refer to card issuer) and the terminal does not support referrals, the transaction must be treated as an online declined transaction.
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Authorization Response Code "85" (Not declined) should not be considered by the terminal as a decline. On receiving this Response Code, the terminal will regard the outcome as successful but must request an AAC at the second GENERATE AC. This Response Code might be used, for example, with balance inquiries or PIN Management requests.

R	All	CT	RA133.11	When an Issuer Response Code of "85" (Not declined) is received, the terminal must request an AAC at the second GENERATE AC.
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Other values of DE 39 indicate that the transaction is declined. On receiving one of these response codes, the terminal must request an AAC at the second GENERATE AC.

R	All	CT	RA134.11	When an Authorization Response Code other than "00" (Approval), "08" (Honor with ID), "10" (Partial Approval), or "87" (Purchase Amount Only, No Cash Back Allowed) is received, the terminal must request an AAC at the second GENERATE AC.
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If the terminal does not receive the exact value of DE 39 sent by the issuer for a transaction declined online from the acquirer, it may fill the Authorization Response Code with any value supported by Mastercard in DE 39 if this value results in the transaction being declined.

When transactions are not authorized online by either the issuer or its authorized agent, the terminal must be able to generate and send the following EMV-defined values of the Authorization Response Code to the card:

- "Y1" - Offline approved
- "Z1" - Offline declined
- "Y3" - Unable to go online-offline approved
- "Z3" - Unable to go online-offline declined

Refer to chapter 4 "Acquirer requirements," section "Unable to Go Online mode."

Dual message processing

In a dual message interface, the terminal may send an online message to inform the acquirer about the conclusion of the transaction.

If the transaction is approved, the terminal sends a second clearing message, either online or using a batch file, with the transaction details.

The TC generated after the successful conclusion of the transaction is normally sent through this clearing message.

If the transaction is declined by the card following receipt of an online approval, the acquirer should generate a Reversal Request/0400 message to inform the issuer earlier and keep the cardholder's open to buy updated.

BP	All	CT	BA092.15	In a dual message interface, if the transaction has been approved by the issuer but is not complete, the acquirer should generate a Reversal Request/0400 message for the issuer to keep the cardholder's open to buy updated.
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Single message processing

With the single message interface, the online message acts as both the online request message and the confirmation message.

Terminals working in a single message environment normally have a zero floor limit, thus all transactions go online to the acquirer.

The acquirer network interface may operate in single or dual message format.

Each Financial Transaction Request/0200 message contains the AC produced in the first GENERATE AC and no subsequent transaction data is sent. If any part of the transaction processes as a single message, a reversal message or a Financial Transaction Advice must be sent for declined transactions if there is a risk that the transaction could be considered approved by the issuer.

For example, if the issuer approves an authorization but the card generates an AAC in response to the second GENERATE AC, the authorization must be reversed to alert the issuer that the approved transaction did not complete.

R	All	CT	RA145.18	If the terminal to acquirer host or acquirer host-to-network interface operates with a single message protocol, the terminal must inform the acquirer host of any transaction that has been approved by the issuer but is not complete. For example, if the card produces an AAC in response to a request for a TC at the second GENERATE AC request. If the acquirer host-to-network interface operates with a single message protocol, the acquirer must generate an Acquirer Reversal Advice/0420 message or a Financial Transaction Advice/0220.
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Call referrals

Call referral is a fraud prevention tool that the issuer uses when it suspects or is attempting to prevent fraud at the point of interaction.

The issuer decides to ask for further confirmation of the cardholder identity before authorizing a transaction. The issuer asks the acquirer to contact them and the issuer can then ask the cardholder for more personal details, such as a password. After this procedure, the issuer may approve or decline the transaction.

Acquirers may use the same processing and keep the same merchant and cardholder interfaces for all call referrals, regardless of the technology used for the transaction.

If the issuer requests a call referral, the following steps apply:

- The online Authorization Response Code has value "01" response in DE 39.
- The terminal terminates the EMV transaction by requesting an AAC from the chip. The card may then be withdrawn from the chip reader.
- The merchant contacts the acquirer, who in turn contacts the issuer.
- The cardholder provides the required information to prove that they are the genuine cardholder, and the issuer gives an authorization approval code.
- The authorization approval code and the ARQC generated by the chip card for the online request must be kept by the acquirer.
- The approval code must be provided in the subsequent clearing message. If chip data is being cleared, the ARQC must be provided in the clearing message.
- For a normal EMV transaction, the acquirer must make the chip-related data and the ARQC available to the issuer on request.

R	All	CT	RA135.11	When a terminal receives a referral request from an issuer, it must terminate the EMV transaction by asking the card for an AAC.
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R	All	CT	RA136.11	After processing a referral, acquirers must retain the authorization approval code and the ARQC produced by the card.
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Authorization by acquirer host

A terminal may routinely go online to the acquirer host for all transactions regardless of whether the card generated an ARQC or TC.

A terminal may do this for various reasons, for example:

- The terminal operates with a single message interface to the acquirer, which means that all transactions must be sent online to the acquirer host.
- The acquirer host must check if the card is listed in the appropriate regional Warning Bulletin, Warning Notice, or other exception file.
- The acquirer may wish to perform other risk analyses, for example, on suspected fraudulent merchants.

If the card generated a TC, the acquirer host accepts or declines the transaction depending on the results of its additional checks. This decision is sent to the terminal, which then completes the transaction. In some situations, the acquirer host may request issuer authorization using the TC. Refer to chapter "Issuer requirements," section "Issuer host requirements," subsection "Online request," sub-subsection "TC received in online request." In all cases, if the card generated a TC, the transaction with the card is complete and the EMV processing is complete.

If the card generated an ARQC, normally the acquirer host follows EMV processing by seeking an issuer authorization. However, the acquirer may decide to proceed without issuer authorization. This may be for one of the following reasons:

- They already decide to decline the transaction because the card appears in a Warning Bulletin or has an invalid BIN for the transaction.
- They cannot get a timely valid response from the issuer due to network failure, so they decide that they will accept the transaction without issuer authorization.

In this case, the acquirer decision is sent to the terminal, which then completes the transaction accordingly. For contact transactions, the acquirer host must instruct the terminal to complete the transaction in the EMV Unable to Go Online mode. Refer to chapter "Acquirer requirements," section "Unable to Go Online mode." For contactless transactions, no further interaction with the card is possible and the terminal acts on the host instruction. Note that in this situation the acquirer may be liable for the transaction.

Note that EMV Unable to Go Online mode should not be used routinely by acquirers to accept a transaction but only when communication problems prevent valid timely issuer responses. This is because EMV Unable to Go Online mode may lead to a certain percentage of declines on contact transactions due to card risk management settings on the chip.

The acquirer host to terminal network must instruct the terminal to switch to the EMV Unable to Go Online mode and indicate that the online response did not originate from the issuer. If the acquiring path includes intermediate processors and associated networks, for example, retailer-

integrated Point-Of-Sale (POS) systems or when processing is outsourced to a processor, the acquirer must ensure that all the intermediate processors and associated networks between the acquirer host and the terminal also meet these requirements.

R	All	CT	RA146.14	If the transaction is authorized by the acquirer host and not by the issuer, the acquirer must ensure that the terminal has switched to the EMV Unable to Go Online mode and any intermediate entities are also aware of it.
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Voice authorizations

Voice authorizations allow merchants to call their acquirer to obtain approval for a transaction that is above the floor limit when their terminal cannot send transactions online.

Attended chip terminals should always be online-capable, meaning that they go online when the circumstances of the transaction make it necessary.

In some cases, there may be chip terminals that are not online-capable, for example, due to the acceptance environment or the hardware involved. Mastercard allows these devices to be used with voice authorization so that issuers get the benefits of chip technology at these acceptance locations.

R	MC	CT	RA138.14	Voice authorization must only be used for Mastercard transactions. It must not be used for Maestro.
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Mastercard allows voice authorization on attended online-capable POS terminals if the terminal is unable to establish an online communication with the issuer.

BP	MC	CT	BA031.11	Terminals that support voice authorization should set the terminal floor limit to published floor limit.
R	MC	CT	RA139.11	Terminals that support voice authorization must use a terminal type of 22, attended, online-capable device, even if they do not have an online connection so that voice authorization can be used correctly when required.

The chip transaction begins as usual. If the card responds with a TC or AAC at the first GENERATE AC, the transaction is completed, either approved or declined, as a normal EMV transaction and no voice authorization takes place.

If the card responds with an ARQC, the transaction should normally go online. If the terminal cannot go online, it enters Unable to Go Online mode and checks the TAC-Default and IAC-Default values.

When implementing voice authorization, acquirers must check that the Terminal Verification Results (TVR) does not indicate any of the following exception conditions:

- SDA failed, DDA failed, CDA failed (TVR[1][7], TVR[1][4], TVR[1][3])
- Card appears on terminal exception file (TVR[1][5])
- Expired application (TVR[2][7])
- Cardholder verification was not successful (TVR[3][8])
- PIN Try Limit exceeded (TVR[3][6])
- PIN entry required and PIN pad not Present or Not Working (TVR[3][5])
- PIN entry required, PIN pad present but PIN was not entered (TVR[3][4])

If the card responds with an AAC, the acquirer may accept the transaction at their own risk. The terminal risk management has not identified a reason to decline the transaction, and the acquirer may use voice authorization to seek an approval for the funds if they consider it is cost effective, to manage their risk.

If the transaction is below the floor limit and if the card replies TC, the transaction is approved as a normal EMV transaction and voice authorization is not needed.

If the transaction is above the floor limit and if the card replies with a TC, the acquirer should request a voice authorization even though the card produced a TC. A voice authorization provides an extra level of assurance for the acquirer concerning the risk of the transaction but may not protect against receiving authorization-related chargebacks. The approval by the card is conditional on the acquirer obtaining a voice authorization for the funds involved.

How the terminal manages voice authorization, the capture of the issuer authorization code, is proprietary and is usually the same for chip as for Magnetic Stripe. This is not in the scope of this document.

The acquirer should clear the TC, if available, or may replace it with the ARQC.

X-Code Processing

X-Code Processing is the term used to describe the process followed when an acquirer experiences a communications failure between its host system and the Mastercard Network.

X-Code Processing may be categorized as either Host X-Code or Mastercard interface processor (MIP) X-Code.

- Acquirer Host X-Code may take place if the communication failure is between the acquirer's host system and the acquirer's MIP.
- Acquirer MIP X-Code may take place if the communication failure is between the acquirer's MIP and the Mastercard Network.

R	All	CT	RA147.22	Acquirers that perform X-Code Processing in a chip environment must ensure that their infrastructure supports the EMV Unable to Go Online mode.
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X-Code Processing for contact chip transactions uses the EMV Unable to Go Online mode, and this should normally result in a decline if offline data authentication (ODA) has not been successfully performed.

BP	All	CT	BA033.22	An X-Code transaction should only be approved if the Unable to Go Online process results in a Transaction Certificate (TC) being produced by the card. Otherwise, the transaction should be declined.
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For more information on X-Code Processing, including acquirer responsibilities and liabilities, refer to the *Authorization Manual*.

Clearing requirements

This section provides requirements and best practices specific to the clearing of chip contact and EMV mode contactless transactions. For full details of clearing requirements, refer to the clearing specifications.

Inclusion of chip data in clearing messages for contact and contactless chip-read transactions is mandatory in all regions. Whenever chip data is submitted, it must include all mandatory subelements of DE 55. Chip data is not mandated for the clearing record of a refund transaction.

If the same data element appears several times on the card to terminal interface, DE 55 must contain the last value that was present on the card to terminal interface up to and including the GENERATE AC command and response from which the cryptogram (tag '9F26') in DE55 was obtained. Following are examples of how these requirements are interpreted in practice:

- In an online request message, the values of the ARQC and related data must be the same as the last values that were present on the card-terminal interface at the first GENERATE AC command.
- The Unpredictable Number may have different values when presented to the card in the INTERNAL AUTHENTICATE, first or second GENERATE AC commands, but the value used in clearing must be the same value that was used to generate the cryptogram that is being cleared. For example, if the ARQC is being cleared, the value of the Unpredictable Number presented to the card at the first GENERATE AC command must be the value that is used in clearing.
- The Issuer Application Data generated by the card may be different between the first and second GENERATE AC commands. The value provided in the clearing must be the same value as that which is used to generate the cryptogram that is being cleared.
- If an issuer responds with a partial approval containing a Response Code of "10" (Partial Approval) or "87" (Purchase Amount Only, No Cash Back Allowed) in DE 39, one of the following applies:

- If Amount Authorized and Amount, Other were requested in CDOL2 at second GENERATE AC, the terminal must provide the card with the values finally approved and populate DE 55 for clearing with these new values, assuming the TC is provided in the clearing message.
- If Amount Authorized and Amount, Other were not requested in CDOL2 second GENERATE AC, the terminal must populate DE 55 for clearing with the values last presented to the card, that is, the values used at the first GENERATE AC.

R	All	CT CL	RA149.11	Acquirers must submit DE 55 clearing data for all chip-read transactions except refund transactions.
R	All	CT CL	RA151.24	The data of the DE 55 subelements being cleared must correspond to the last values sent to or received from the card, up to and including the GENERATE AC command and its response from which the cryptogram (tag '9F26') being cleared is obtained.
BP	All	CT	BA034.15	The cryptogram being cleared should be the last generated cryptogram by the card.
R	All	CT CL	RA153.11	The integrity of the individual DE 55 subelements in the clearing message must be maintained without more padding or changes.
R	All	CT CL	RA152.14	In the clearing message, the following data elements must be populated with data read from the chip, if present: • DE 2 (Primary Account Number) with the Application PAN (tag '5A') • DE 14 (Date, Expiration) with the Application Expiration Date (tag '5F24') • DE 23 (PAN Sequence Number) with the Application PAN Sequence Number (tag '5F34') • DE 35 (Track 2 Data) with the Track 2 Equivalent Data (tag '57'). Acquirers must not use data from the individual component data elements from the chip to reconstruct a Track 2.
R	All	CT CL	RA155.14	Data in clearing messages must only be populated using data from the chip interface being used.
R	All	CL QR	RA558.24	Effective 1 January 2026, if Device Type (see tag '9F6E' Third Party Data) is read from the card in a contactless or QR transaction, Acquirers must provide this in GCMS IPM PDS 0198 in the Clearing message

Details on the values of DE 22 in the clearing can be found in the *IPM Clearing Formats* manual.

No aggregation or truncation of contactless transactions is permitted, except in certain transit situations, as documented in the *Transaction Processing Rules*.

R	All	CL	RA286.14	Acquirers must process on their clearing interface and host system contactless transactions as described in this section "Clearing Requirements."
BP	All	CL	BA072.14	If the Device Type is read from the card during a transaction, acquirers should include it in Private Data Subelement (PDS) 0198 of the First Presentment/1240 message.
BP	All	CL	BA128.25	Acquirers should transmit Kernel Identifier-Terminal (tag '96') in the First Presentment/1240 clearing message for contactless transactions.

Refer to the regional requirements for the Canada region and United States region in chapter 4 "Acquirer requirements," section "Regional variations."

Acquirers must keep the TC and the transaction data that were used to calculate it to resolve any disputes that might arise in the future.

In some circumstances, the TC is not available and an AAC or an ARQC may be retained instead. For example:

- When the acquirer operates a single message interface between the terminal and the acquirer host system.
- Contactless transactions where an ARQC was sent in response to the first GENERATE AC command.
- When a card returns an AAC in response to a GENERATE AC command but the terminal forces acceptance anyway, for example, call referrals and voice authorizations.

R	All	CT CL	RA158.11	Acquirers must retain the TC or ARQC, as appropriate, and the data that was used to create it and provide this data to the issuer on request even if the chip data was included in the clearing message.
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For contact transactions only, the acquirer may also want to retain the Issuer Authentication Data so they can verify the issuer's decision in the event of a dispute. If retaining them, the data items should be retained for the maximum dispute time frame.

Acquirers do not need to keep the AAC and the related data elements for transactions that were declined offline and were not Merchant Approved.

Single message requirements

Offline chip transactions may take place in single message environments, for example, non-European Maestro transactions, whenever the floor limit configured in the terminal is greater than the transaction amount. Specific requirements apply for offline-approved transactions.

These offline-approved transactions are processed using a Financial Transaction Advice/0220 message containing DE 55 and DE 23. The DE 55 data requirements for these transactions are the same as for those in dual message environments.

In addition:

- DE 15 (Date, Settlement) must contain the current processing day.
- DE 39 must contain a value of "00" (Approved or completed successfully).

R	All	CT CL	RA160.11	Offline-authorized chip transactions in single message environments must be processed with a Financial Transaction Advice/0220 message containing both DE 55 and DE 23.
R	All	CT CL	RA161.11	In a Financial Transaction Advice/0220 message: • DE 15 must contain the current processing day. • DE 39 must contain a value of "00" (Approved or completed successfully).

DE 63 (Network Data) and DE 90 (Original Data Elements) are generated by the Single Message System and do not need to be populated.

Acquirer logging and archiving

In addition to requirements that apply to Magnetic Stripe transactions, this section describes logging and archiving requirements for acquirers.

R	All	CT CL	RA157.11	If any potentially sensitive data, for example, Track 2 data, needs to be stored, it must be stored in a way that complies with any applicable data storage standards.
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Regional variations

This section describes regional requirements and best practices that vary from the global requirements.

Asia/Pacific region

The additional requirements in this section apply to acquirers in the Asia/Pacific region.

For terminals in American Samoa, Guam, and North Mariana Islands, also refer to requirements relating to the Durbin Regulations in section "United States region."

Chip and contactless acceptance

For Australia and New Zealand, EMV and contactless mandates are already in place. All terminals in Australia and New Zealand must support EMV contact transactions. All POS terminals deployed in Australia and New Zealand must be contactless-enabled.

R	All	CT	RA180.20	All POS and CAT terminals and ATMs in Australia and New Zealand must be EMV certified and process EMV cards.
R	All	CT CL	RA330.14	All POS terminals in Australia and New Zealand must be contactless-enabled.

All newly deployed (excluding replacement) POS terminals in the Asia/Pacific region, must be dual interface hybrid terminals that support and enable both EMV contact and EMV mode contactless transactions.

R	All	CT CL	RA414.24	All newly deployed (excluding replacement) POS and MPOS terminals in the Asia/Pacific region must support and enable both EMV contact and EMV mode contactless technology. Contactless-only MPOS terminals are excluded from this requirement.
R	All	CT CL	RA545.25	All newly deployed (including replacement) POS terminals in Japan, including MPOS terminals and unattended terminals, must be dual interface hybrid terminals that support and enable both EMV contact and EMV mode contactless payment functionality. Contactless-only MPOS terminals, AFDs (MCC 5542) and iPOS terminals at fuel locations (MCCs 5541, 5983) are all excluded from this requirement.
R	All	CT CL	RA565.25	Effective 1 July 2026, in Japan, all newly deployed (including replacement) AFDs (MCC 5542) and iPOS terminals at fuel locations (MCCs 5541, 5983), must be dual interface hybrid terminals that support and enable both EMV contact and EMV mode contactless payment functionality.
R	All	CL	RA450.20	Any new contactless-enabled terminals in the Asia/Pacific region submitted for M-TIP testing must only support EMV mode contactless transactions. Magnetic Stripe mode contactless must not be supported.

CVM requirements

In the Asia/Pacific region, Online PIN support is progressively mandated on the contact and contactless interfaces of dual interface POS terminals and on contactless-only POS terminals.

In China, all attended POS terminals, including MPOS, must support Online PIN for domestic transactions. For cross-border transactions, Online PIN must be supported in accordance with requirements RA504.25 and RA523.24.

R	All	CT	RA502.24	In China, all attended POS terminals, including MPOS terminals, that accept China domestic transactions must support Online PIN.
R	All	CL	RA521.24	In China, all attended POS terminals, including MPOS terminals, that accept China domestic transactions must support Online PIN for contactless transaction amounts above the CVM Limit.
R	All	CT	RA504.25	All newly deployed (including replacement) (or submitted to Mastercard for M-TIP testing) attended dual interface POS terminals in the Asia/Pacific region, excluding Japan, Republic Korea, and Taiwan, must support Online PIN on the contact interface. The support of Online PIN is optional for MPOS terminals outside of China.
R	All	CL	RA533.24	All newly deployed (including replacement) (or submitted to Mastercard for M-TIP testing) contactless attended POS terminals in the Asia/Pacific region, excluding Japan, Republic Korea, and Taiwan, must support Online PIN on the contactless interface for transaction amounts above the CVM Limit. The support of Online PIN is optional for MPOS terminals outside of China.
R	All	CT	RA523.24	All newly deployed (including replacement) (or submitted to Mastercard for M-TIP testing) MPOS terminals in China must support Online PIN on the contact interface.
R	All	CL	RA534.24	All newly deployed (including replacement) (or submitted to Mastercard for M-TIP testing) MPOS terminals in China must support Online PIN on the contactless interface for transaction amounts above the CVM Limit.

Single Tap and PIN Request

Mastercard has introduced the Single Tap and PIN Request functionality.

Single Tap and PIN Request provides a seamless cardholder experience when low value contactless transactions require an SCA. Cardholders can enter their PIN for the same contactless transaction instead of having to dip their card.

BP	All	CL	BA122.24	Newly deployed (including replacement) terminals in the Asia/Pacific region should support Single Tap and PIN Request functionality.
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Technology selection

Attended, online-capable POS terminals should support technical fallback. Refer to chapter "Acquirer requirements," section "Technology selection," subsection "Technical Fallback."

BP	All	CT	BA117.19	Attended, online-capable POS terminals in the Asia/Pacific region should support technical fallback.
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Application selection on a multi-application card

The mandate to support cardholder selection on multi-application cards (refer to requirement RA078.19) does not apply to Japan. However, terminals are recommended to allow cardholders to choose their preferred application.

BP	All	CT	BA119.19	POS terminals, ATMs, Bank Branch Terminals, and On-board terminals in Japan should support cardholder selection to allow cardholders to choose their preferred application on a multi-application card.
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Purchase with cash back

These requirements for the support of purchase with cash back transactions are applicable in Australia.

R	MC	CT	RA484.24	Terminals in Australia supporting PWCB must support Online PIN and Offline PIN. Domestic purchase with cash back transactions in Australia must be online authorized and verified with a PIN.
R	MC	CL	RA553.24	Terminals in Australia supporting PWCB must support Online PIN and CDCVM. Domestic purchase with cash back transactions in Australia must be online authorized and verified with a PIN or CDCVM

Canada region

The additional requirements in this section apply to acquirers in the Canada region.

Chip and contactless acceptance

In the Canada region, contactless-enabled terminals must only support EMV mode contactless technology.

R	All	CL	RA449.23	All contactless-enabled terminals in the Canada region must only support EMV mode contactless transactions. Magnetic Stripe mode contactless must not be supported.
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Authorization and clearing

In the Canada region, these requirements related to the Device Type apply.

R	All	CL	RA285.15	If the Device Type is read from the card during a contactless, EMV and Magnetic Stripe mode, transaction, Canada region acquirers must transmit it in DE 48 of online request messages.
R	All	CL	RA288.15	If the Device Type is read from the card during a contactless, EMV and Magnetic Stripe mode, transaction, Canada region acquirers must transmit it in private data subelement (PDS) 0198 (Device Type Indicator) of First Presentment/1240 messages.

Europe region

The additional requirements in this section apply to acquirers in the Europe region.

Chip and contactless acceptance

This requirement applies to all acquirers residing in any of the countries or territories designated as the geographical scope of SEPA, as listed in the *Mastercard Rules* and in selective non-SEPA countries in Europe.

R	All	CT	RA001.25	All terminals except for MPOS, in the SEPA countries, and also in Albania, Bosnia and Herzegovina, Kosovo, Macedonia, Moldova, Montenegro, and Serbia, must support both Magnetic Stripe and EMV chip technology.
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NOTE: This mandate does not apply to MPOS terminals. MPOS terminals deployed in the Europe region are not required to have a Magnetic Stripe reader.

The EMV mandate is expanded to the remaining countries in the Europe region.

R	All	CT	RA419.23	All POS and MPOS terminals in the Europe region must support EMV technology.
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In addition to EMV contact transactions, POS and MPOS, in Europe must support contactless transactions except for AFDs deployed in Estonia, Finland, France, Iceland, Ireland, Latvia, Lithuania, and the United Kingdom.

Mastercard recommends AFDs deployed in Finland, France, Iceland, Ireland, and the United Kingdom to support contactless transactions.

In Estonia, Latvia, and Lithuania, newly deployed AFDs must be contactless-enabled. In Ireland and the United Kingdom, all AFDs must be contactless-enabled by 1 April 2025.

R	All	CL	RA312.22	All POS and MPOS terminals in the Europe region, except AFDs in Estonia, Finland, France, Iceland, Ireland, Latvia, Lithuania, and the United Kingdom, must be contactless-enabled.
BP	All	CL	BA116.20	AFDs deployed in Finland, France, Iceland, Ireland, and the United Kingdom should be contactless-enabled.
R	All	CL	RA506.24	Newly deployed (including replacement) AFDs in Estonia, Latvia, and Lithuania must be contactless-enabled.
R	All	CL	RA544.23	Effective 1 April 2025, all AFDs in Ireland and the United Kingdom must be contactless-enabled.

The contactless mandate is expanded to ATMs in some European countries.

All ATMs in the Czech Republic and Poland must support EMV contactless functionality.

ATMs in Bosnia and Herzegovina, Serbia, and Montenegro that are deployed after 21 January 2022 must be contactless-enabled.

R	All	CL	RA472.24	All ATMs in the Czech Republic and Poland must be contactless-enabled.
R	All	CL	RA491.25	All newly deployed (including replacement) ATMs in Bosnia and Herzegovina, Montenegro, and Serbia must be contactless-enabled.

For ATMs in Albania, Austria, Bulgaria, Croatia, Cyprus, Germany, Greece, Hungary, Kosovo, Liechtenstein, Malta, North Macedonia, Romania, Slovakia, Slovenia and Switzerland, the effective dates are described in these requirements.

R	All	CL	RA541.24	All newly deployed (including replacement) ATM terminals in Albania, Austria, Bulgaria, Croatia, Cyprus, Germany, Greece, Hungary, Kosovo, Liechtenstein, Malta, North Macedonia, Romania, Slovakia, Slovenia, and Switzerland must support and enable EMV contactless functionality.
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R	All	CL	RA542.24	All ATM terminals in Albania, Austria, Bulgaria, Croatia, Cyprus, Germany, Greece, Hungary, Kosovo, Liechtenstein, Malta, North Macedonia, Romania, Slovakia, Slovenia, and Switzerland supporting EMV contactless functionality for other acceptance brands must also support and enable EMV contactless functionality for the Mastercard, Maestro, and Cirrus acceptance brands.
R	All	CL	RA543.23	Effective 19 July 2028, all ATM terminals in Albania, Austria, Bulgaria, Croatia, Cyprus, Germany, Greece, Hungary, Kosovo, Liechtenstein, Malta, North Macedonia, Romania, Slovakia, Slovenia, and Switzerland must support and enable EMV contactless functionality.

About the technology mode supported on the contactless interface, all contactless-enabled terminals must support EMV mode contactless transactions, and all newly deployed contactless-enabled terminals must support EMV mode contactless transactions only and must not support Magnetic Stripe mode contactless transactions.

R	All	CL	RA447.24	All contactless-enabled terminals in the Europe region must support EMV mode contactless transactions. Magnetic Stripe mode contactless must not be supported.
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PSD2

Payment Service Directive 2 (PSD2) and the related Regulatory Technical Standards (RTS) define how Strong Customer Authentication (SCA) must be applied for payments occurring within the EEA countries, the United Kingdom, and Gibraltar.

These requirements maintain high security standards that protect cardholders and their accounts.

These requirements under the RTS directly impact CAT Level 2, CAT Level 3, and CAT Level 4 terminals deployed in the EEA countries, the United Kingdom, and Gibraltar that do not have a PIN pad and that are not covered by Article 12 of the RTS, as these terminals will be generating noncompliant transactions.

R	All	CT	RA457.22	CAT Level 2 terminals in the EEA countries, the United Kingdom, and Gibraltar supporting contact transactions that do not operate in a transport or parking environment (MCCs 4111, 4112, 4131, 4784, 4789 and 7523) must be updated in one of the following ways or be removed from deployment: • Be upgraded to have dual capability by the addition of an Offline PIN-capable PIN pad • Be upgraded to become a CAT Level 1 terminal by the addition of an Online PIN-capable PIN pad • Have contact chip functionality removed, resulting in contactless-only acceptance
R	All	CT	RA458.22	CAT Level 3 terminals in the EEA countries, the United Kingdom, and Gibraltar supporting contact transactions that do not operate in a transport or parking environment (MCCs 4784 and 7523) must be updated in one of the following ways or be removed from deployment: • Be upgraded by the addition of an Offline PIN-capable PIN pad • Be upgraded to become a CAT Level 1 terminal by the addition of an Online PIN-capable PIN pad • Have contact chip functionality removed, resulting in contactless-only acceptance
R	All	CT	RA459.22	CAT Level 4 terminals in the EEA countries, the United Kingdom, and Gibraltar supporting contact transactions must be updated in one of the following ways or be removed from deployment: • Be upgraded to have dual capability by the addition of an Offline PIN-capable PIN pad • Have contact chip functionality removed, resulting in contactless-only acceptance

Single Tap and PIN Request

Mastercard has introduced a mechanism to enable an issuer to request an online PIN verification on low-value contactless transactions without the need to use the contact chip reader.

This mechanism is currently mandated in Andorra, Austria, Belgium, Bulgaria, Croatia, Cyprus, Czech Republic, Denmark, Estonia, Georgia, Germany, Greece, Hungary, Iceland, Kazakhstan, Latvia, Lithuania, Luxembourg, Malta, Netherlands, Norway, Poland, Portugal, Romania, Slovakia, Slovenia, Spain, and Sweden.

R	All	CL	RA462.20	Acquirers in the Europe region supporting Single Tap and PIN Request processing must support sending DE 48, subelement 22, subfields 02 (Single Tap Indicator) and 03 (Response to PIN Request), and receiving DE 48, subelement 22, subfield 04 (Issuer PIN Request in a Single Tap Mode), indicating the issuer requests a PIN in single tap mode.
R	All	CL	RA497.22	All terminals in Andorra, Austria, Belgium, Bulgaria, Croatia, Cyprus, Czech Republic, Denmark, Estonia, Georgia, Germany, Greece, Hungary, Iceland, Kazakhstan, Latvia, Lithuania, Luxembourg, Malta, Netherlands, Norway, Poland, Portugal, Romania, Slovakia, Slovenia, Spain, and Sweden must support Single Tap and PIN Request functionality.

The mandate for the support of Single Tap and PIN Request has been extended to other European countries with different effective dates.

R	All	CL	RA478.24	All terminals in Finland, France, Italy, Monaco, San Marino, and Vatican City must support Single Tap and PIN Request functionality.
R	All	CL	RA551.24	All contactless-enabled terminals in Armenia, Azerbaijan, Belarus, Kyrgyzstan, Moldova, Tajikistan, Turkey, Turkmenistan, Ukraine, and Uzbekistan submitted to M-TIP must support Single Tap and PIN Request functionality.
R	All	CL	RA552.23	Effective 31 December 2024, all contactless-enabled terminals in Armenia, Azerbaijan, Belarus, Kyrgyzstan, Moldova, Tajikistan, Turkey, Turkmenistan, Ukraine, and Uzbekistan must support Single Tap and PIN Request functionality.

Support of Single Tap and PIN Request is recommended in all other markets in the Europe region that support Online PIN at the POS. Any acquirer in the Europe region that wants to support Single Tap and PIN Request functionality must comply with requirement RA462.20.

BP	All	CL	BA125.22	Acquirers in the Europe region supporting Online PIN should support Single Tap and PIN Request functionality.
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PIN Entry Bypass

PIN Entry Bypass must not be allowed for contactless transactions (refer to requirement RA264.14) and is not permitted at unattended terminals (refer to requirement RA103.11).

For the Europe region, the support of PIN Entry Bypass for contact transactions at attended POS terminals must also be withdrawn.

R	All	CT	RA508.23	Attended POS terminals in the Europe region must not support PIN Entry Bypass.
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Terminal Risk Management Data

Mastercard recommends contactless unattended payment terminals located in EEA countries, the United Kingdom, and Gibraltar that offer high throughput transportation or parking services to provide an indication about their Cardholder Verification Method (CVM) exemption to the device that is performing the transaction.

The valid transport and parking MCCs are:

- 4111 (Local Commuter Transport)
- 4112 (Passenger Railways)
- 4131 (Bus Lines/Charters Tours)
- 4784 (Tolls/Bridges Fees)
- 4789 (Transportation Services not Elsewhere Classified)
- 7523 (Parking Lots/Garages)

This indication is provided by a minor configuration update to the Terminal Risk Management Data (TRMD) (tag '9F1D').

This update will improve the current means of detecting such terminals based on MCCs, allowing CVM exemption while transacting at contactless unattended terminals for transaction amounts below the CVM Limit.

BP	All	CL	BA120.22	Contactless-enabled unattended terminals within the EEA countries, the United Kingdom, and Gibraltar supporting MCC 4111, 4112, 4131, 4784, 4789, or 7523 should have the Terminal Risk Management Data (tag '9F1D') element, byte 4, bit 7 (CVM exempt) set to indicate that cardholder verification is not required for the transaction.
R	All	CL	RA463.22	Byte 4, bit 7 (CVM Exempt) of the Terminal Risk Management Data (tag '9F1D') must not be set to 1b for terminals that have a terminal type other than unattended or MCC other than 4111, 4112, 4131, 4784, 4789, or 7523.

Terminal applications

In the Europe region, all attended contactless readers that comply with *Mastercard Contactless Reader Specification v3.0* or later or *EMVCo Book C-2* must be configured to support transactions above the CVM Limit.

R	All	CL	RA298.14	Attended contactless readers that comply with <i>Mastercard Contactless Reader Specification v3.0</i> or later or <i>EMVCo Book C-2</i> must be configured to support transactions above the CVM Limit.
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Application selection

Terminals deployed in the European Union (EU) or in Norway, Iceland, and Liechtenstein acquiring cards personalized with tag '9F0A' with ID='0001' may support new alternative methods of performing the Application Selection process.

- Merchants can decide to remove the support for specific Mastercard products from their terminals. In this case, the same policy must apply to all competitive schemes acquired by the terminal.
- Merchants may ignore the value of the application priority indicator and change the order of priority in which the applications are displayed to the cardholder.
- Merchants can preselect their preferred application but they must always enable cardholder's choice. The merchant must not override cardholder's decision.
- If a contactless card supports more than one application, the terminal may support cardholder selection as it is supported for contact cards.

R	All	CT CL	RA368.16	Merchants that decide not to support specific Mastercard products in their terminals must not support equivalent products from competitive schemes.
R	All	CT CL	RA370.16	Merchants that preselect their preferred application must not override cardholder's decision.
R	All	CT	RA369.16	Terminals must support application selection with cardholder assistance over the contact interface.

More details on the new alternative methods of performing the Application Selection Process at POS terminals have been published. Acquirers should update their terminals to follow the new process. Mastercard recommends that acquirers consult the *SEPA Cards Standardization Volume v7.1, Bulletin 01 - 20160229 - Book 2*.

R	All	CL	RA469.19	Terminals that follow <i>SEPA Cards Standardization Volume v7.1, Bulletin 01 - 20160229 - Book 2</i> must allow the cardholder to perform the transaction with the merchant's preferred application through a single tap with no cardholder interaction or selection. If supported by the terminal, the possibility to override the merchant's preferred application must be clearly indicated to the cardholder and instructions must be provided to guide the cardholder through the selection process.
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The cardholder selection may be performed through but not limited to:

- A "Corr"/yellow button with function keys
- Other keys like softkeys or touchpad keys next to the POI screen
- A virtual button on the touchscreen of the POI

Online Processing

In the United Kingdom, the Terminal Contactless Floor Limit must be set to zero, except at some specific merchants.

R	All	CL	RA456.19	All contactless transactions in the United Kingdom, except for parking, tolls, transit, vending, charities, and event transactions, must be authorized online.
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Maestro acceptance at No CVM locations

Mastercard permits Maestro No CVM contact card acceptance in the Europe region in specific acceptance environments up to the maximum transaction amount set out in this table.

The same solution for contactless terminals can also be deployed in this specific acceptance environments without any limitation in the maximum transaction amount set out in this table.

Acceptance environment	Maximum transaction amount
Tollways (MCC 4784)	EUR 100 or local currency equivalent
Parking Lots and Garages (MCC 7523)	EUR 50 or local currency equivalent
Transit Vending Machines (MCCs 4111, 4112, 4131, 4789)	EUR 25 or local currency equivalent

This No CVM acceptance may take place at both attended and unattended environments. The terminal may support PIN for other brands or other amounts.

The permitted MCCs listed in the table may be configured with No CVM as the only supported CVM for transactions equal to or below the applicable limit. Maestro transactions that take place without PIN in these environments must not exceed the stated maximum transaction amount. Otherwise, the terminal must abort the transaction unless standard Maestro PIN processing takes place.

A transit vending machine, a parking garage, or a tollway terminal that only supports No CVM must process all Maestro cards presented. The terminal must not routinely decline transactions because cardholder verification has failed due to the lack of a common cardholder verification method between the terminal and the card. The terminal must otherwise follow normal EMV processing.

If a card declines, the terminal must continue the transaction by reading the Magnetic Stripe and submitting an online request based on its data. An alternative method is to submit an online authorization using the chip data and include the AAC as the cryptogram.

Terminals that only support Maestro No CVM transactions do not need to support Offline PIN.

R	MS	CT	RA106.14	A Maestro No CVM terminal must support only No CVM for transactions up to or equal the applicable limit.
R	MS	CT CL	RA109.13	Terminals must correctly classify the merchant using the applicable permitted MCC.
R	MS	CT	RA110.13	A Maestro No CVM transaction must not exceed the stated maximum transaction amount for each MCC.
R	MS	CT	RA111.13	If the card declines and an AAC is provided in response to the terminal request for an ARQC or TC, the terminal must perform one of the following options: • Submit an Authorization Request/0100 message using the card data read from the Magnetic Stripe and identify the transactions as Magnetic Stripe in the online request message with DE 22 (POS Entry Mode) value of "90" • Submit an Authorization Request/0100 message using the chip data with DE 22 (POS Entry Mode) value of "05" and including the AAC as the cryptogram

These are the requirements regarding the terminal certification for Maestro No CVM.

R	MS	CT	RA112.13	For Maestro transactions equal to or below the applicable limit, the terminal kernel must have been EMVCo Level 2 approved based on a configuration with No CVM as the only CVM.
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R	MS	CT CL	RA113.19	The terminal must have successfully completed Mastercard chip certification for Maestro transit vending, parking lots, and garages and tollways No CVM transactions.
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NOTE: POS terminals that support chip and Magnetic Stripe No CVM transit vending machine transactions must not be deployed in the Netherlands.

CVM requirements

This mandate applies to all POS terminals in SEPA countries and in Albania, Bosnia and Herzegovina, Kosovo, Macedonia, Moldova, Montenegro, and Serbia.

R	All	CT	RA002.18	All POS terminals in the SEPA countries and in Albania, Bosnia and Herzegovina, Kosovo, Macedonia, Moldova, Montenegro, and Serbia must support PIN.
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NOTE: Requirement RA002.18 does not apply to chip-capable MPOS without PIN support or terminals only accepting Maestro No CVM transactions.

Estonia, Ireland, Latvia, Lithuania, and the United Kingdom are also countries that previously did not support Online PIN, but these countries are now in a transition period to support Online PIN.

All new POS terminals deployed in Estonia, Latvia, and Lithuania must support Online PIN on the contactless interface.

Acquirers and merchants residing in the remaining countries in the Europe Region must support Online PIN as a CVM on the contactless interface of POS terminals.

Also, acquirers and merchants that currently do not support Online PIN on the contactless interface of POS terminals must not allow Signature to be used as a CVM on the contactless interface for Mastercard contactless transactions.

R	All	CL	RA299.25	In the Europe region, except for Estonia, Ireland, Israel, Latvia, Lithuania, and the United Kingdom, dual interface attended POS terminals must support Online PIN on the contactless interface for transaction amounts above the CVM Limit.
R	All	CL	RA528.23	All new dual interface terminals in Ireland and the United Kingdom undergoing Mastercard certification must support Online PIN for contactless transaction amounts above the CVM Limit if Online PIN is also supported on the contact interface.

R	All	CL	RA372.25	All newly deployed (including replacement) dual interface attended POS terminals in Estonia, Latvia, and Lithuania must support Online PIN on the contactless interface for transaction amounts above the CVM Limit.
R	All	CL	RA236.14	In the Europe region, if a dual interface POS terminal does not support Online PIN on the contactless interface, it must not support Signature on the contactless interface for EMV mode transactions.

Alternative contactless flow at terminals in Sweden

Terminals in Sweden are permitted to follow an alternative contactless flow, which allows the cardholder payment device to be presented to the contactless reader before the final transaction amount is known.

This enables data to be read from the cardholder payment device and used by the merchant for proprietary purposes, such as a loyalty program, within the context of the transaction session.

Before the final transaction amount is known

The cardholder is invited to present their contactless payment device to the reader before the final transaction amount has been determined. Interaction between the reader and the payment device is completed using a configurable placeholder amount. Transaction data, including a one-time online cryptogram, is retrieved from the payment device. Once the final amount is known, an online authorization request is sent to the issuer.

This solution may result in a difference between the value of the placeholder amount that is submitted to the payment device and the value of the final transaction amount that is acknowledged by the cardholder and submitted in field DE 4 of the resulting authorization message. Cardholder verification may occur before the online cryptogram is generated by the payment device, for example, if CDCVM is used, or, if required, after the payment device is withdrawn using Online PIN or Signature.

When CDCVM is supported by the payment device

If supported on the payment device, the cardholder may be authenticated using CDCVM prior to tapping the payment device. Because the final transaction amount is not yet known, during the interaction with the payment device the contactless reader executes the standard contactless transaction flow based on a placeholder amount. This includes application selection, transaction initiation, reading of data records, and the retrieval of the unique onetime cryptogram. The placeholder amount may be set to a value that is above or below the CVM Limit configured in the reader, which must be set to the value published in Appendix C of the *Chargeback Guide*.

During the interaction with the payment device, the reader determines the CVM for the transaction and indicates this in the CVM Results. The reader is configured to request an online cryptogram from the payment device. On receipt of the online cryptogram, the reader indicates that the payment device may be withdrawn. As with all contactless transactions, the reader interacts only for a short period with the payment device, typically less than half a second. After

the withdrawal of the payment device, if an online cryptogram has not been returned, the transaction is declined. Otherwise, processing continues and, if required, offline card authentication is completed.

After the final transaction amount is known

Once the final transaction amount is known and if it exceeds the CVM limit and if Online PIN is the selected CVM, the transaction amount is displayed and the cardholder is invited to enter their PIN. The PIN is encrypted into a PIN block and temporarily stored according to PCI PED requirements. If CDCVM has already been performed or if Signature is the chosen CVM for the transaction or if the final transaction amount does not exceed the CVM limit, the Terminal displays the final transaction amount and requests confirmation from the cardholder for the transaction before continuing.

If Online PIN is the chosen CVM but the final transaction amount does not exceed the CVM Limit, PIN entry is not required and may be skipped. Similarly, if Signature is the chosen CVM but the final transaction amount does not exceed the CVM Limit, Signature is not required and may be skipped and the transaction receipt need not contain a placeholder for the Signature. Depending on the setting of the placeholder amount and the value of the final Transaction Amount, the CVM Results may need to be updated to accurately reflect the final outcome of the transaction. This may result in a mismatch between the TVR and the CVM Results. However, the contents of the TVR must not be amended at this stage. All transactions using the alternative contactless flow must be sent online to the issuer.

R	MC	CL	RA400.17	Alternative Contactless Flow transactions must be authorized online by the issuer.
	MS			

The authorization request sent to the issuer includes the final transaction amount in DE 4 and the one-time cryptogram in the chip data either in DE 55, including the ARQC and placeholder amount, or in the track data in DE 35/45 (CVC3). Note that the difference between the amounts in DE 4 and DE 55 is not new to Mastercard issuers, as this can occur in various acceptance environments, for example, transit. If the authorization response indicates that the transaction has been approved, a receipt is printed, if required. If Signature is required, the cardholder is invited to sign for the final transaction amount.

Acquirers should be aware that transactions submitted for which the Terminal Verification Results indicates Online PIN entry but for which there is no PIN block present may be declined by the issuer's authorization host. Payment devices from issuers in Sweden that have adapted their authorization hosts to accommodate this solution will not encounter problems in this respect. However, merchants with a sizeable proportion of international customers should be aware of the risk of declines for this reason.

Terminal types

This section covers specific requirements for different terminal types in Europe.

ATMs

Technical fallback support at ATMs in Europe is prohibited.

R	All	CT	RA053.11	All ATMs within the Europe region must not allow technical fallback.
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Bank Branch Terminals

Second-level technical fallback to PAN Key Entry or paper-based solutions is not permitted at Bank Branch Terminals (BBTs) in Europe.

R	All	CT	RA170.12	Bank Branch Terminals located in the Europe region must not support technical fallback from chip cards to PAN Key Entry or paper-based technical fallback.
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CATs

Technical fallback is not permitted at online-capable CATs in Europe.

R	All	CT	RA099.11	Online-capable CATs within the Europe region must not support technical fallback.
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Technology selection

Second-level technical fallback to PAN Key Entry or paper-based solutions is not permitted in Europe.

R	All	CT	RA171.12	Attended online-capable POS terminals located in the Europe region must not support technical fallback from chip cards to PAN Key Entry or paper-based technical fallback.
R	All	CT	RA072.11	Attended online-capable POS terminals that support fallback must complete fallback transactions using voice authorization when the online connection is not available.

Transaction types

This section covers specific requirements for different types of transactions in Europe.

Purchase with cash back

Acquirers in the Europe region may support purchase with cash back (PWCB) transactions. For acquirers in Romania, the support of purchase with cash back is mandatory for all their POS terminals. The support of purchase with cash back is mandatory for all POS terminals in Serbia and Montenegro.

Also, all terminals in Albania, Austria, Bulgaria, Cyprus, the Czech Republic, Greece, Hungary, Kosovo, Macedonia, Malta, Poland, Slovakia, and Slovenia that support PWCB on the contact interface must also support the service on the contactless interface.

This mandate will apply to all deployed terminals in these countries from 30 June 2023.

R	All	CT CL	RA420.22	Attended POS terminals in Romania must support purchase with cash back transactions for both domestic and international transactions.
R	All	CT CL	RA492.22	Attended POS terminals in Serbia and Montenegro must support purchase with cash back transactions for both domestic and international transactions.
R	All	CT CL	RA510.23	All attended POS terminals in Albania, Austria, Bulgaria, Cyprus, the Czech Republic, Greece, Hungary, Kosovo, Macedonia, Malta, Poland, Slovakia, and Slovenia supporting PWCB transactions on the contact interface must also support it on the contactless interface for both domestic and international transactions.

These requirements apply to acquirers in Moldova supporting purchase with cash back transaction.

R	All	CT CL	RA548.23	Acquirers in Moldova supporting purchase with cash back transactions must support purchase-only approval.
R	All	CT CL	RA549.23	Acquirers in Moldova must not support POI Currency Conversion for cross-border purchase with cash back transactions.

These requirements list the allowed CVMs to be used for purchase with cash back transactions in different European countries.

R	All	CT CL	RA464.20	All domestic purchase with cash back transactions in Russia must be authorized with an Online PIN for the full amount, including both the purchase and cash back amounts.
R	All	CT CL	RA461.22	All domestic purchase with cash back transactions in Switzerland must be authorized with an Online PIN or CDCVM for the full amount, including both the purchase and cash back amounts.
R	All	CT CL	RA467.19	All domestic purchase with cash back transactions in Germany must be authorized with an Online PIN or Offline PIN on contact transactions and with Online PIN or CDCVM on contactless.
R	All	CT CL	RA485.20	All domestic purchase with cash back transactions in Ukraine must be authorized with an Online PIN or CDCVM for the full amount, including both the purchase and cash back amounts.
R	All	CT CL	RA511.22	All domestic and international purchase with cash back transactions in Albania, Austria, Bulgaria, Cyprus, the Czech Republic, Greece, Hungary, Kosovo, Macedonia, Malta, Montenegro, Poland, Romania, Serbia, Slovakia, and Slovenia must be authorized with PIN, Offline or Online, or CDCVM for the full amount, including both the purchase and the cash back amount.
R	All	CT CL	RA550.23	All domestic purchase with cash back transactions in Moldova must be authorized with an Online PIN or CDCVM for the full amount, including both the purchase and cash back amounts.

Latin America and the Caribbean region

The additional requirements in this section apply to acquirers in the Latin America and the Caribbean region.

For terminals in Puerto Rico and American Virgin Islands, also refer to requirements relating to the Durbin Regulations in section "United States region."

Chip and contactless acceptance

New ATMs deployed in the Latin America and Caribbean region must be EMV.

R	All	CT	RA003.24	Newly deployed (including replacement) ATM terminals in the Latin America and the Caribbean region must be EMV-compliant.
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All POS, MPOS, and IPOS terminals must be dual interface hybrid terminals that support and enable both EMV contact and EMV mode contactless transactions.

R	All	CT CL	RA425.23	All POS terminals, including MPOS and IPOS, in the Latin America and the Caribbean region must support and enable both EMV contact and EMV mode contactless technology. Contactless-only MPOS are excluded from this requirement.
R	All	CL	RA440.20	All contactless-enabled terminals in the Latin America and the Caribbean region must only support EMV mode contactless technology. Magnetic Stripe mode contactless must not be supported.

CVM requirements

In the Latin America and the Caribbean region, all POS terminals (except MPOS terminals) must support Online PIN. For MPOS terminals, support for Online PIN is optional.

NOTE: For Mexico, the effective dates are different, as detailed in these requirements.

R	All	CT	RA500.25	All dual interface attended POS terminals, except MPOS, in the Latin America and the Caribbean region, except in Mexico, must support Online PIN on the contact interface.
R	All	CL	RA536.24	All attended contactless POS terminals, except MPOS, in the Latin America and the Caribbean region, except in Mexico, must support Online PIN on the contactless interface for transaction amounts above the CVM Limit.
R	All	CT	RA562.24	Effective 1 January 2025, all newly deployed (including replacement) attended dual interface POS terminals, except iPOS and MPOS, in Mexico must support Online PIN on the contact interface.
R	All	CL	RA563.24	Effective 1 January 2025, all newly deployed (including replacement) attended contactless POS terminals, except iPOS and MPOS, in Mexico must support Online PIN on the contactless interface for transaction amounts above the CVM Limit.
R	All	CT	RA525.24	Effective 1 June 2024, all attended dual interface POS terminals submitted to Mastercard for M-TIP testing, except IPOS and MPOS, in Mexico must support Online PIN on the contact interface.
R	All	CL	RA537.24	Effective 1 June 2024, all attended contactless POS terminals submitted to Mastercard for M-TIP testing, except IPOS and MPOS, in Mexico must support Online PIN on the contactless interface for transaction amounts above the CVM Limit.

R	All	CT	RA526.24	Effective 1 March 2025, all newly deployed (including replacement) attended dual interface IPOS terminals, except MPOS, in Mexico must support Online PIN on the contact interface.
R	All	CL	RA538.24	Effective 1 March 2025, all newly deployed (including replacement) attended contactless IPOS terminals, except MPOS, in Mexico must support Online PIN on the contactless interface for transaction amounts above the CVM Limit.
R	All	CT	RA527.24	Effective 1 December 2025, all attended dual interface POS terminals, except MPOS, in Mexico must support Online PIN on the contact interface.
R	All	CL	RA539.24	Effective 1 December 2025, all attended contactless POS terminals, except MPOS, in Mexico must support Online PIN on the contactless interface for transaction amounts above the CVM Limit.

In Peru, POS terminals deployed before 1 May 2025 may be configured so that No CVM is the only CVM supported for contact chip transactions conducted with a Mastercard card issued in Peru, provided the transaction amount is equal to or less than the value specified in the *Transaction Processing Rules* document. In this terminal configuration, No CVM replaces both signature and PIN in the terminal's list of supported CVMs. The acquirer must only use this terminal configuration for domestic transactions.

Refer to Mastercard Announcement LAC 10669.1.

R	MC	CT	RA529.25	Attended POS terminals, except MPOS terminals, that accept Mastercard cards must support Signature. In Peru, contact chip terminals deployed before 1 May 2025 may be configured with only No CVM for domestic transactions below an amount as specified in the <i>Transaction Processing Rules</i> document.
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Application selection in Brazil

Cardholder selection at POS terminals and ATMs in Brazil may be different from the process defined in EMV Book 1, Section 12.4, Step 4.

POS terminals and ATMs in Brazil may support an application preselection step in the beginning of the transaction where the cardholder is asked to choose the preferred type of the application to conduct the payment. In fact, the cardholder may be asked to choose between Credit or Debit in the beginning of every transaction even when the card supports only one application, preselection. Credit represents all Mastercard applications, and Debit represents Maestro and Debit Mastercard applications.

After the cardholder's preselection on the type of application, the terminal checks which applications are supported by the chip and compares them with the preselected application type. The supported applications that have the same application type to the one selected by the cardholder, Credit or Debit, are placed into the Candidate List of applications.

Then, the cardholder is asked to choose an application from the Candidate List. If there is only one application in the Candidate List, that application is automatically selected by the terminal. In case the preselected application type does not match any of the supported card applications, the terminal aborts the transaction with the message of "invalid application."

R	All	CT CL	RA406.22	POS terminals and ATMs in Brazil must support cardholder selection as defined either in this section or in EMV Book 1, Section 12.4, Step 4.
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Single Tap and PIN Request

Mastercard has introduced the Single Tap and PIN Request functionality.

Single Tap and PIN Request provides a seamless cardholder experience when low value contactless transactions require an SCA. Cardholders can enter their PIN for the same contactless transaction instead of having to dip their card.

BP	All	CL	BA123.24	Newly deployed (including replacement) terminals in the Latin America and the Caribbean region should support Single Tap and PIN Request functionality.
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M/Chip Fast

M/Chip Fast is the Mastercard implementation of the EMV transaction flow specifically designed for environments where faster transaction times are particularly important. It allows an EMV transaction to be performed before the final transaction amount has been determined.

Full details of the M/Chip Fast solution, including acquirer implementation requirements, are described in chapter "Acquirer requirements," section "Regional variations," subsection "United States region."

Middle East/Africa region

The additional requirements in this section apply to acquirers in the Middle East/Africa region.

Chip and contactless acceptance

New ATMs deployed in the Middle East/Africa region must be EMV.

R	All	CT	RA006.24	Newly deployed (including replacement) ATM terminals in the Middle East/Africa region must be EMV-compliant.
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All POS and MPOS terminals deployed in the Middle East/Africa region must be dual interface hybrid terminals that support and enable both EMV contact and EMV mode contactless transactions.

Magnetic Stripe mode contactless must not be supported on any contactless-enabled terminals.

R	All	CT CL	RA428.23	All POS terminals, including MPOS and IPOS, in the Middle East/Africa region must support and enable both EMV contact and EMV mode contactless technology. Contactless-only MPOS terminals are excluded from this requirement.
R	All	CL	RA454.20	All contactless-enabled terminals in the Middle East/Africa region must only support EMV mode contactless technology. Magnetic Stripe mode contactless must not be supported.

CVM requirements

Terminals in the Middle East/Africa region are required to support Online PIN on the contact and contactless interfaces of a dual interface POS terminal and on contactless-only POS terminals.

MPOS terminals and unattended POS terminals are excluded from this requirement, as support for Online PIN is optional.

R	All	CT	RA501.24	All dual interface attended POS terminals, except MPOS, in the Middle East/Africa region must support Online PIN on the contact interface.
R	All	CL	RA540.24	All attended contactless POS terminals, except MPOS, in the Middle East/Africa region must support Online PIN on the contactless interface for transaction amounts with amount above the CVM Limit.

Single Tap and PIN Request

Mastercard has introduced the Single Tap and PIN Request functionality.

Single Tap and PIN Request provides a seamless cardholder experience when low value contactless transactions require an SCA. Cardholders can enter their PIN for the same contactless transaction instead of having to dip their card.

R	All	CL	RA124.24	All contactless-enabled terminals in the Middle East/Africa region submitted to M-TIP must support Single Tap and PIN Request functionality.
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R	All	CL	RA530.23	Effective 31 December 2024, all contactless-enabled terminals in the Middle East/Africa region must support Single Tap and PIN Request functionality.
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Purchase with cash back and cash back without purchase

Merchants in South Africa and in Kenya that have received prior approval from their acquirers may offer a purchase with cash back (PWCB) transaction or cash back without purchase (CBWP) transaction to any cardholder for intra-country transactions.

In South Africa, PIN or CDCVM verification must be obtained for either PWCB, CBWP or both transactions.

In Kenya, all PWCB and CBWP transactions must be authorized with Online PIN.

R	All	CT CL	RA303.14	Terminals in South Africa supporting PWCB and/or CBWP must support Online PIN, Offline PIN (and CDCVM for contactless). A PIN (or CDCVM for contactless) CVM must be used for each PWCB or CBWP transaction conducted in South Africa.
R	MC	CT CL	RA460.19	All PWCB and CBWP intra-country transactions in Kenya must be authorized with an Online PIN for the full amount, including both the purchase and cash back amounts.

United States region

The additional requirements in this section apply to acquirers in the United States region.

As Durbin Regulations also apply to U.S. Territories, all the requirements in the "Application selection," "Editing the Candidate List," "CVM requirements," and "PIN Entry Bypass" sections, plus any other requirements relating to the U.S. Maestro AID, also apply to terminals in U.S. overseas territories.

Card reader requirements

When dip readers are used at ATMs in the United States region:

- The Magnetic Stripe data should be read as the card is inserted so that if the service code indicates the presence of a chip, it is not necessary to reinsert the card.
- Once the chip is identified, the card should be locked to prevent withdrawal before the chip transaction, including delivering any script commands, is completed.

BP	All	CT	BA105.16	Dip readers in the United States region should read the Magnetic Stripe data as the card is inserted.
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BP	All	CT	BA106.16	Dip readers in the United States region should lock the chip card until the chip transaction is completed.
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Together with the issuing mandate that imposes the progressive removal of the physical Magnetic Stripe from the back of the cards (refer to requirements RI427.22 and RI428.22 in chapter "Issuer requirements"), terminals must be able to process cards that do not support the physical Magnetic Stripe by 1 April 2027 in the United States region.

R	All	CT	RA512.22	Effective 1 April 2027, all chip-capable ATMs and POS terminals in deployment in the United States region must be able to correctly process chip cards with no physical Magnetic Stripe.
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Chip and contactless acceptance

Mastercard has put in place a timetable to remove Magnetic Stripe mode contactless technology.

R	MC MS	CL	RA436.23	All contactless-enabled POS terminals in the United States region must only support EMV mode contactless transactions. Magnetic Stripe mode contactless must not be supported.
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The support of Magnetic Stripe mode for U.S. Maestro AID is not allowed.

R	USM	CL	RA375.16	A contactless terminal accepting the U.S. Maestro AID must not support Magnetic Stripe mode for this AID.
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Technology selection

Attended online-capable POS terminals should support technical fallback. Refer to chapter "Acquirer requirements," section "Technology selection," subsection "Technical Fallback."

BP	ALL	CT	BA114.19	Attended online-capable POS terminals in the United States region should support technical fallback.
BP	ALL	CT	BA115.19	Technical fallback in the United States region should always be attempted in any situation where it is not forbidden.

Cardholder receipts

Requirements RA274.18 and RA275.18 apply to approved transactions. For declined transactions, receipt printing support is optional.

Online processing

Acquirers in the United States region must ensure that all terminals be either online-only or online-preferring with respect to authorization processing, except where the Standards specifically permit offline processing.

R	All	CT CL	RA397.16	Terminals in the United States region must be either online-only or online-preferring, except where the Standards specifically permit offline processing.
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Application selection

For both contact and contactless transactions, terminals use the standard application selection process described in EMV. The terminal may filter the Candidate List to remove multiple AIDs that relate to debit or prepaid programs that access the same underlying funding account.

For more details, refer to section "Editing the Candidate List."

To meet the requirements of the Durbin amendment to the Dodd-Frank Wall Street Reform and Consumer Protection Act (Durbin Regulations) to enable merchants or acquirers to choose to route transactions from debit and prepaid card programs through unaffiliated networks, Mastercard has introduced the U.S. Maestro AID 'A0000000042203'.

Mastercard chip cards issued in the U.S. and overseas territories under debit and prepaid card programs will support both an international AID, Mastercard or Maestro, and the U.S. Maestro AID. If the Mastercard or global Maestro AID is used to initiate the chip transaction, the resulting transaction must be routed through a Mastercard Network.

If the U.S. Maestro AID is used to initiate the chip transaction, the resulting transaction may also be routed through unaffiliated debit networks that have executed the necessary access agreement with Mastercard. Eligibility for transactions to be routed through different networks is beyond the scope of this document but is typically identified in BIN tables and may be contingent on other aspects of the transaction, such as the cardholder verification method used.

Terminals must not assume that if the U.S. Maestro AID is present, this can be automatically selected to the exclusion of all other AIDs on the card, as these AIDs may relate to alternative products that the cardholder wishes to use.

A merchant that supports the U.S. Maestro AID should also support both the Mastercard and Maestro AIDs.

R	USM	CT CL	RA376.16	Terminals must not assume that if the U.S. Maestro AID is present, this can be automatically selected to the exclusion of all other AIDs on the card that are supported by the terminal.
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BP	All	CT CL	BA107.16	A merchant that supports the U.S. Maestro AID should also support both the Mastercard and Maestro AIDs.
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Although terminals are required to follow EMV defined cardholder selection process, non-consumer facing POS terminals (that is: no cardholder facing screen + keyboard to permit application selection or PIN entry (or PIN bypass) by the cardholder) are allowed to automatically select the highest priority AID from the card instead of displaying the options to the cardholder for selection.

R	All	CT	RA410.17	ATMs, Bank Branch Terminals, On-board terminals, and POS terminals, except for non-consumer facing POS terminals, must support cardholder selection as defined in EMV Book 1, Section 12.4, Step 4, to allow cardholders to choose their preferred payment application on a multi-account card.
R	MC MS	CT	RA411.17	Non-consumer facing POS that does not support cardholder selection as defined in EMV Book 1, Section 12.4, Step 4, must not support PIN.

Domestic AIDs can only be used for transactions performed in the issuing country, domestically. Cross border transaction processing must always be processed using Mastercard AIDs through Mastercard Network based on transaction processing rules. Therefore, all terminals in the United States that process EMV transactions with EMV cards, branded, co-branded or co-badged, from other countries, cross border EMV transactions, must perform these transactions using Mastercard AIDs and route transactions through Mastercard networks regardless of the existence of another AID on the card, such as a domestic network AID of the issuing country. For cards issued by non-U.S. issuers, a U.S. terminal must omit any other, domestic, network AIDs and initiate the transaction using Mastercard AID and process the transaction on Mastercard Network.

R	All	CT CL	RA429.18	Terminals in the United States must remove from the candidate list any domestic AID of the issuing country when they acquire cross border transactions.
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Editing the Candidate List

All AIDs that relate to debit or prepaid programs will be identified in the data read by the terminal at application selection.

The presence of both of the following data elements identifies the AID as relating to a U.S. debit or prepaid program:

- Issuer Country Code (two-digit alpha) (tag '5F55') with a value of '5553' ("US")
- Issuer Identification Number (IIN) (tag '42')

When a card has just two AIDs where these data elements are present and the values of the IIN in each AID is the same, the terminal may assume that the two AIDs access the same underlying funding account and can eliminate one of the AIDs from the Candidate List. The terminal may automatically select the remaining AID in the Candidate List, scenario 1 in the the "United States Region: Guidance for editing the Candidate List during Application Selection" table.

When there are other AIDs present on the card, for contact transactions the terminal must present the choice to the cardholder in accordance with EMV, scenario 2 in the "United States Region: Guidance for editing the Candidate List during Application Selection" table. For contactless transactions, the terminal automatically selects the AID in the Candidate List with the highest Application Priority Indicator, as described in EMV.

When a card has multiple AIDs where these data elements are present and the values of the IIN in each pair of AIDs is different, the terminal may assume that any two AIDs with the same IIN access the same underlying funding account and can eliminate one of the AIDs from the Candidate List, scenario 3 in the "United States Region: Guidance for editing the Candidate List during Application Selection" table.

When a card has multiple AIDs where these data elements are present and the values of the IINs are the same, if there are multiple U.S. Maestro AIDs present, the terminal may assume that each U.S. Maestro AID relates to a different underlying funding account. The terminal may either eliminate all the U.S. Maestro AIDs or all the international AIDs identified as part of a debit or prepaid program from the Candidate List, scenario 4 in the "United States Region: Guidance for editing the Candidate List during Application Selection" table.

In both scenarios 3 and 4, there may be other AIDs on the card that are not part of a debit or prepaid program, and the choice between these products must be offered to the cardholder in accordance with EMV for contact transactions or based on the value of the Application Priority Indicator for contactless transactions. When the choice is presented to the cardholder, AIDs should be presented according to the Application Priority Indicator as designated by EMV.

Table 6: United States Region: Guidance for editing the Candidate List during Application Selection

Product	AID (in hex)	Country Code tag		Application label	Customer free choice	Mastercard restricted choice
		'5F55' in ASCII	IIN tag '42' (in hex)			
					Candidate List	Candidate List
Scenario 1: Card accesses single debit funding account						

Product	AID (in hex)	Country Code tag '5F55' in ASCII	IIN tag '42' (in hex)	Application label	Customer free choice	Mastercard restricted choice
					Candidate List	Candidate List
Debit Mastercard®	A0000000 041010	US	550000	Mastercard Debit	Mastercard Debit	Merchant may automatically select either application.
U.S. Maestro® AID	A0000000 042203	US	550000	US Debit	US Debit	
Scenario 2: Card accessing credit account and single debit funding account						
Mastercard Credit	A0000000 041010			MastercardCredit	MastercardCredit	MastercardCredit
Mastercard Debit	A0000000 041010	US	550000	Mastercard Debit	Mastercard Debit	"Mastercard Debit" or "US Debit"
U.S. Maestro AID	A0000000 042203	US	550000	US Debit	US Debit	
Scenario 3: Card accessing two debit funding accounts (accounts have different IINs)						
Mastercard Debit 1	A0000000 041010AA 01	US	550000	Mastercard Debit 1	Mastercard Debit 1	"Mastercard Debit 1" or "US Debit 1"
U.S. Maestro AID 1	A0000000 042203AA 01	US	550000	US Debit 1	US Debit 1	
Mastercard Debit 2	A0000000 041010AA 02	US	529999	Mastercard Debit 2	Mastercard Debit 2	"Mastercard Debit 2" or "US Debit 2"
U.S. Maestro AID 2	A0000000 042203AA 02	US	529999	US Debit 2	US Debit 2	
Scenario 4: Card accessing two debit funding accounts (accounts have the same IINs)						

Product	AID (in hex)	Country Code tag '5F55' in ASCII	IIN tag '42' (in hex)	Application label	Customer free choice	Mastercard restricted choice
					Candidate List	Candidate List
Mastercard Debit	A0000000041010AA01	US	550000	Mastercard Debit 1	Mastercard Debit 1	"Mastercard Debit 1" and "Mastercard Debit 2"
U.S. Maestro AID	A0000000042203AA01	US	550000	US Debit 1	US Debit 1	or "US Debit 1" and "US Debit 2"
Mastercard Debit	A0000000041010AA02	US	550000	Mastercard Debit 2	Mastercard Debit 2	
U.S. Maestro AID	A0000000042203AA02	US	550000	US Debit 2	US Debit 2	

NOTE: For Scenario 2, refer to requirement RI379.18.

In each of the scenarios described in the table, how the terminal identifies the AID to eliminate from the Candidate List is beyond the scope of this document. Mastercard recommends that terminals are developed to enable this identification method configurable to support changes in merchant and acquirer preferences.

Legacy processes, such as debit or credit option buttons, must not be used.

CVM requirements

POS terminals that accept Mastercard cards must support Signature. They may support PIN. When PIN is supported, the terminal must support Online PIN and Offline PIN. In certain merchant categories, terminals may accept transactions with No CVM.

Terminals that accept Maestro cards must support Online PIN and Offline PIN. Refer to requirements RA075.11 and RA036.24.

Terminals that accept the U.S. Maestro AID must support Online PIN. Offline PIN may also be supported. In certain merchant categories, terminals may accept transactions with no cardholder verification and thus support No CVM for PIN bypass. But US-Maestro accepting terminals cannot support only No CVM without supporting Online PIN– they must permit cardholder ability to: enter PIN or choose to bypass it.

When the U.S. Maestro AID is used, the merchant can process a No CVM chip transaction and then treat it as a Signature-based transaction based on direction from its acquirer.

Terminals that support Offline PIN should support offline CAM.

For terminals that support Offline PIN:

- Offline enciphered PIN must be supported.
- Refer to chapter 4 "Acquirer Requirements," section "CVM requirements," subsection "Contact transactions" for requirements on supporting Offline Plaintext PIN.

R	MC	CT	RA173.24	Hybrid POS terminals located in the United States region that support PIN as a CVM must support both Online PIN and Offline PIN processing.
R	USM	CT CL	RA377.16	Attended POS terminals that accept the U.S. Maestro AID must support Online PIN.
BP	All	CT	BA108.16	Terminals that support Offline PIN should support offline CAM.

Although PINs will normally be four digits, the full EMV requirement should be supported. This means that the customer will need to press Enter after completing their PIN entry.

For contactless transaction amounts equal to or below the CVM Required Limit, the terminal must support No CVM only. For transaction amounts above the CVM Limit, terminals that support U.S. Maestro AID must support Online PIN and must not support CDCVM.

R	USM	CL	RA378.16	Contactless-enabled terminals must support Online PIN for contactless transaction amounts above the CVM Limit for U.S. Maestro AID.
R	USM	CL	RA379.16	Contactless-enabled terminals must support No CVM for contactless transaction amounts equal to or below the CVM Limit for U.S. Maestro AID.
R	USM	CL	RA380.16	Contactless-enabled terminals must not support CDCVM for U.S. Maestro AID.

PIN Entry Bypass

PIN Entry Bypass must only be used when requested by the cardholder and must not be used systematically to avoid PIN for certain transactions.

Support for alternative solutions outside of the scope of EMV, such as debit or credit button choices, should not be used to bypass the PIN.

PIN Entry Bypass is permitted for contactless transactions on the US-Maestro AID.

If Online PIN entry is bypassed then:

- Acquirer must send the TVR unchanged to the Issuer, exactly as it was sent to the chip, otherwise the issuer will find ARQC is incorrect; refer to RA554.24.
- If this is a contactless transaction, CVM results must be set to '3F0001'.

R	All	CT CL	RA513.22	PIN Entry Bypass must only be used when requested by the cardholder.
BP	USM	CT CL	BA110.16	Support for PIN Entry Bypass for terminals supporting U.S. Maestro AID is strongly recommended.
R	USM	CL	RA572.25	For all terminals submitted for M-TIP testing, if Online PIN entry is bypassed in a contactless transaction, CVM results sent in the Authorization Request in DE55 tag '9F34' must be set to '3F0001'.
R	USM	CL	RA573.25	Effective 1 January 2026, for all newly deployed (including replacement) terminals, if Online PIN entry is bypassed in a contactless transaction, CVM results sent in the Authorization Request in DE 55 tag '9F34' must be set to '3F0001'.

Requirement RA103.11 does not apply to AFDs in the United States region.

Subsequent contact transactions

This best practice applies to contactless terminals with PIN pad that also support contact transactions.

BP	All	CL	BA111.16	If a contact interface with PIN pad is available on the terminal and the card has a contact interface, contactless terminals should display an appropriate message to prompt for a contact transaction to be attempted when a contactless transaction is declined by the issuer using an Authorization Response Code "65" (Exceeds withdrawal count limit).
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Automated Fuel Dispensers

The cardholder experience at Automated Fuel Dispensers (AFDs) is described in this section.

Transaction flow when card is inserted

This procedure describes the transaction flow for when a card is inserted.

1. The terminal may prompt the cardholder with prepayment questions, which may include but not be limited to other services, payment options, payment card type selection, and so on.
2. The cardholder may be prompted to select the appropriate payment application from choices available on the card if the card is used is a multi-account card.
3. The AFD terminal processes the cardholder verification according to the CVM List on the card.
 - a. Cardholders with PIN above No CVM in the CVM List will be prompted for a PIN.

4. The cardholder may opt to bypass PIN, as available.
5. The AFD may optionally prompt for the Zip Code (AVS) or other Fleet Card prompts for Fleet Card processing.
6. The AFD is activated after preauthorization. The pump automatically stops when the maximum dispense amount is reached unless cardholder has stopped sooner.
7. The cardholder is offered an optional receipt.
8. Authorization is complete when the final amount is processed by the AFD.

Transaction flow when contactless card or access device is tapped

This procedure describes the transaction flow for when a contactless card or access device is tapped.

1. The terminal may prompt the cardholder with prepayment questions, which may include but not be limited to other services, payment options, payment card type selection, and so on.
2. If the AFD is set to process a preauthorization amount below the CVM Limit, the AFD will process the transaction using the No CVM option.
 - a. For U.S. Debit cards, the AFD may initiate a transaction using U.S. Maestro AID and process the transaction accordingly.
3. If the AFD is set to process a preauthorization amount over the CVM Limit, the AFD will process the cardholder verification according to the preferred CVM supported by the card or device.
 - a. The cardholder may be prompted to complete CDCVM in case on device cardholder verification is needed and not completed before the initial tap. A second tap might be needed in such a case.
4. No further prompts are recommended for contactless transaction processing to improve the cardholder experience. However, the AFD may still optionally prompt for the Zip Code (AVS) or more Fleet Card prompts for Fleet Card processing.
5. The AFD is activated after preauthorization. The pump automatically stops when the maximum dispense amount is reached unless the cardholder has stopped sooner.
6. The cardholder is offered an optional receipt.
7. Authorization is complete when the final amount is processed by the AFD.

Requirements for AFD implementations

The following requirements apply to AFD implementations:

- The terminal must operate as a dual capability device, that is, it can perform both CAT Level 1 and CAT Level 2 transactions.
- The contactless CVM Limit must be set according to the CVM Limit published by Mastercard. Where CDCVM is supported by the mobile device being used, this will always be requested.
- The acquirer and merchant are protected against lost and stolen fraud chargebacks if they make a PIN pad available for chip transactions (Chip/PIN liability shift).
- The maximum dispense amount (AFD Limit) is the same for contact and contactless transactions. That amount is published in the Mastercard *Chargeback Guide* in section "Automated Fuel Dispenser (MCC 5542) Transactions."

Standard EMV transaction flow for contact or contactless interface

The standard EMV transaction flow is followed for either contact or contactless interface, as appropriate. This includes application selection, card authentication, and cardholder verification. Either a transaction amount of USD 1 or the maximum transaction amount may be used.

- If CDCVM is not supported by the device, the appropriate CVM is selected by the terminal according to normal EMV processing. The sequence of the CVM List set by the issuer will determine which CVM is requested. If PIN is identified as the required CVM, this is input by the cardholder for either contact or contactless transactions.
- PIN Entry Bypass may be supported for contact transactions. If supported, it may only be actioned if selected by the cardholder. PIN Entry Bypass must not be actioned to systematically avoid PIN for certain transactions.
- Transactions must be authorized before the fuel is dispensed. The preauthorization amount is either USD 1 or the maximum transaction amount.
- Once the transaction is completed, if processed through the dual message system, the acquirer must send an authorization advice request message (0120) for the full dispense amount. The message must be sent within 60 minutes of the original authorization request and must include in DE 48, subfield 63 the Trace ID value of DE 63 from the authorization response message.

The authorization advice request message must contain the chip data from the original authorization. Refer to requirement RA188.13 in chapter "Acquirer requirements," section "Transaction types," subsection "Preauthorization transactions." For single message processing, a 0420 message must be sent. Refer to *Transaction Processing Rules*.

R	All	CT CL	RA382.16	AFDs in the United States region must operate as dual capability devices.
R	All	CT CL	RA384.16	Transactions completed at AFDs in the United States region must be completed with an Authorization Advice Request/0120 message within 60 minutes of the original authorization request. The message must include in DE 48, subfield 63 the value of DE 63 from the authorization response message. For single message processing, an Acquirer Reversal Advice/0420 message must be used.

AFDs follow this CVM requirement on the contact interface and the contactless interface for transactions above the CVM Limit.

AFDs in the United States region may support Signature CVM on both interfaces.

R	All	CT	RA385.16	AFDs in the United States region must support Online PIN, Offline PIN, and No CVM on the contact interface.
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R	All	CL	RA386.16	AFDs in the United States region must support CDCVM, Online PIN, and No CVM on the contactless interface for transactions above the CVM Limit.
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Fleet processing

Hybrid terminals in the United States region that support special Fleet Card processing must use one of following Fleet Implementation Guides:

- *Fleet Implementation Guide For the U.S. Region* (for fleet programs from 11 April 2025 onwards)
- *Fleet Implementation Guide - Mastercard Corporate Fleet card Program* (for earlier fleet programs)

The newer *Fleet Implementation Guide For the U.S. Region* document explains how terminals should support fleet processing.

If terminals use the older *Fleet Implementation Guide - Mastercard Corporate Fleet card Program* document, they must drive the special processing requirements applicable to the product by:

- Using the Product Control Information in the Third Party Data (tag '9F6E'), if present
- Otherwise, use legacy Magnetic Stripe-based product information, Product Restriction Code and Product Type Code. These are present on the chip in the following locations:
 - Track 2 Equivalent Data (tag '57')
 - Track 1 Discretionary Data (tag '9F1F')
 - Track 2 Discretionary Data (tag '9F20')

Refer to chapter "Acquirer requirements," section "Special acceptance environments," subsection "Fleet acceptance."

R	All	CT CL	RA387.24	Terminals deployed in the United States region that support special Fleet Card processing according to the <i>Fleet Implementation Guide - Mastercard Corporate Fleet card Program</i> document, must use the Product Control Information in the Third Party Data (tag '9F6E') to drive the special processing requirements applicable to the product.
R	All	CL CT	RA388.24	Terminals deployed in the United States region that support special Fleet Card processing according to the <i>Fleet Implementation Guide - Mastercard Corporate Fleet card Program</i> document, must continue to use legacy Magnetic Stripe-based product information to support the proper processing of chip cards where the Product Control Information is not present.

M/Chip Fast

M/Chip Fast is the Mastercard implementation of the EMV transaction flow specifically designed for environments where faster transaction times are particularly important. It allows an EMV transaction to be performed before the scanning of goods has been completed and the final transaction amount has been determined.

An M/Chip Fast transaction is conducted based on a placeholder amount. The transaction is then authorized online once the final transaction amount is known. M/Chip Fast may be used for both contact and contactless transactions.

M/Chip Fast is most beneficial in environments where it may take some time before the final transaction amount is known, for example, due to the scanning of goods in a multi-lane retail environment. It may also be used for merchants that wish to optimize the transaction in other environments, for example, Quick Service restaurants.

M/Chip Fast supports all Mastercard brands and all Application Identifiers (AIDs) associated with the Mastercard brands, including the U.S. Maestro AID. Mastercard also allows the use of M/Chip Fast for cards carrying other brands.

The solution is currently available only to terminals deployed in the United States region and the Latin America and the Caribbean region. For merchants where speed does not have a significant business impact, standard and optimized EMV processing is still recommended.

The *M/Chip Fast Technical Requirements U.S. Market* manual contains the technical requirements for developers and a detailed description of the transaction flow.

R	All	CT CL	RA404.20	M/Chip Fast must not be implemented at ATMs.
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The M/Chip Fast flow

An M/Chip Fast implementation performs a standard EMV transaction up to and including the processing of the first GENERATE AC command, where terminals must be configured to request an Authorization Request Cryptogram (ARQC) from the card.

This can be achieved through the configuration of a zero floor limit. Application selection, offline card authentication, and CVM processing are all part of an M/Chip Fast transaction.

The EMV flow is completed by the terminal through a second GENERATE AC command with an Authorization Response Code "Unable to go online, offline declined".

Once the final amount is known, an online authorization request must be sent to the issuer containing the final transaction amount in DE 4 together with DE 55 containing the chip data that was generated based on the placeholder amount.

R	All	CT CL	RA405.17	M/Chip Fast transactions must be authorized online by requesting an ARQC from the card.
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Configuring the placeholder amount

For contactless transactions, care must be taken to ensure that the placeholder amount is configured with a suitable value relative to the CVM Limit.

If the merchant environment is one in which the transaction total is highly unlikely to exceed the CVM Limit, then the merchant may wish to configure a placeholder amount that is lower than the CVM Limit. This will result in no cardholder verification being requested by the terminal for any contactless transaction regardless of the amount. Note that for mobile solutions where the mobile device is configured to always request Consumer Device CVM (CDCVM), this will still take place. If the final amount exceeds the CVM Limit, the merchant may choose either to accept the potential risk and continue with the transaction or restart the transaction, that is, perform a second transaction using the final amount.

For mixed value environments where it is likely that a proportion of transactions will be for an amount below the CVM Limit and a proportion above it, different behaviors will be required depending on the final transaction amount, which is not known at the time of interaction with the chip. In these environments, the placeholder amount should be set to an amount that is above the CVM Limit.

If CVM processing requires a PIN to be entered, the PIN should only be requested from the cardholder if the final amount exceeds the CVM Limit. During PIN entry, the final amount must be displayed to the cardholder. If necessary, the CVM Results should be amended to accurately reflect the final outcome of the transaction. The contents of the TVR must not be amended at this stage.

In environments where the final amount is available earlier in the process, this can be used instead of the placeholder amount. For these environments, the benefit of M/Chip Fast, for contact transactions, is being able to withdraw the card from the reader while the authorization message is sent to the issuer.

Where a placeholder amount is used, its value must be greater than zero.

R	All	CT CL	RA409.17	Where a placeholder amount is used, it must be greater than zero.
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Purchase with cash back

There are no special requirements associated with implementing purchase with cash back using M/Chip Fast. It can be implemented in the same way as for a standard EMV environment

provided the cash back amount is known once the AID has been selected at the beginning of the EMV transaction.

The Amount, Authorized field (tag '9F02') should be populated with the sum of the placeholder amount and the cash back amount. All other amount fields should be populated as for a standard EMV purchase with cash back transaction.

Authorization and clearing

In the United States region, these requirements related to the Device Type apply.

R	All	CL	RA284.22	If the Device Type is read from the card during a contactless EMV mode transaction, United States region acquirers must transmit it in DE 48, subelement 23, subfield 1 (Device Type) of online request messages.
R	All	CL	RA287.17	If the Device Type is read from the card during a contactless EMV mode transaction, United States region acquirers must transmit it in PDS 0198 (Device Type Indicator) of First Presentment/1240 messages.

The presence of the Terminal Capabilities (tag '9F33') is mandatory for acquired transactions.

R	All	CT CL	RA350.20	Acquirers in the United States must include the Terminal Capabilities (tag '9F33') in the DE 55 of all acquired transactions.
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Chapter 5 Data requirements

This chapter contains more detailed information about the underlying data objects that drive the required behavior of the chip card and chip terminal. The chapter provides an explanation of the usual settings.

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Overview

For full details of card personalization settings, refer to the *M/Chip Personalization Data Specifications for Contact and Contactless* manual.

Application identification

EMV uses the AID, Application Label, and Application Preferred Name to help enable terminals to identify the applications present on a card and support the selection of the appropriate application by the cardholder or terminal, when required.

These data objects may also be used within wallets to enable users to select their preferred application to perform a transaction before a contactless transaction is commenced.

Application Identifiers

As defined by ISO 7816, an *AID* is a variable length data object with a minimum length of 5 bytes and a maximum length of 16 bytes.

The AID consists of the Registered Application Provider Identifier (RID) of 5 bytes and a Proprietary Application Identifier Extension (PIX). The PIX can be any length from 0 to 11 bytes.

The AID is the mechanism by which the terminal recognizes the cards that can be accepted.

R	All	CT CL	RI067.11	Cards and terminals must support the AIDs corresponding to the brands being issued or acquired.
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To clearly understand the Application Selection process described in *EMV Book 1, Application Independent ICC to Terminal Interface Requirements*, it is necessary to distinguish between the AID held in the terminal, supported AID list, and the AID held in the card (tag '4F'). As explained in *EMV Book 1*, these might not be identical even for matching applications. In the Application Selection procedure, the term *AID* is used for the application identifier held in the terminal, while the application identifier held in the card is referred to as the "Dedicated File (DF) Name."

On successful final selection, the terminal must set the value of the terminal data element Application Identifier (AID) (tag '9F06'), to the same value as the DF Name (tag '84') returned in the File Control Information (FCI).

A Mastercard® AID consists of at least the Mastercard RID ('A0000000004') and a 2-byte PIX that identifies the payment brand associated with the application, for example, the PIX associated with Maestro® is '3060', so the shortest Maestro AID is 'A00000000043060'.

To meet the business needs associated with the different Mastercard card products, the AID can be longer than 7 bytes. This helps enable issuers to differentiate between applications that may

have the same acceptance brand. An example of when this might happen is if a card contained both a Mastercard credit and a Debit Mastercard® application.

Issuers using DF Names that are longer than 7 bytes for EMV application selection must ensure that:

- All DF Names for the same product are longer than 7 bytes
- The card supports partial AID selection, meaning the card supports the SELECT command with the parameter P2 set to '02', indicating SELECT NEXT

R	All	CT	RI069.11	Card applications must support partial selection if they contain DF Names longer than 7 bytes.
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All terminals must select applications based on the 7-byte AID for the payment product, for example, Mastercard.

R	All	CT CL	RA294.14	All terminals must support partial selection with at least 7 bytes.
----------	------------	----------	----------	---

If the card responds with a DF Name that is longer than the 7 bytes sent in the SELECT command, the terminal uses the SELECT NEXT command to find all the DF Names on the card that partially match the AID held in the terminal. All the partially matching DF Names are included in the Candidate List.

R	All	CT	RA166.24	To ensure that all applications are entered into the Candidate List, the terminal Application Selection Indicator must be set to "partial AID match supported" for each of the Mastercard applications that it supports.
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Terminals can also select applications using an AID longer than 7 bytes, up to the maximum allowed 16 bytes. The procedure to add DF Names to the Candidate List is the same as for terminals that use a 7-byte AID.

Mobile payment devices may use extended AIDs within the wallet to identify different payment products. The user chooses the payment product they wish to use before the contactless transaction starts. Once the payment product has been chosen by the cardholder, the AID that appears in the PPSE will only use the 7-byte AID for the product. The mobile application will know to which underlying payment application this refers.

Extended AIDs with special acceptance functionality

This section deals with the use of AIDs in the context of application selection by the terminal during a contact or contactless EMV transaction. It does not cover the use of the AID within wallets.

When an extended AID is known to the terminal, it can be used to drive specific functionality in that environment.

Domestic AID extensions

When specific functionality is required by a group of customers to support a product feature unique to a domestic market, Mastercard will allocate an AID in the following form:

'A0 00 00 00 04 pp pp Dc cc xx'

Where:

- pp pp is the PIX for the primary acceptance branch '1010' (for Mastercard, '3060' for Maestro, or '6000' for Cirrus®)
- D is 'D' (4 bits coded as 1101b)
- c cc is the three-digit ISO numeric Country Code
- xx.. is a unique number

This can be used by issuers on their branded cards that will be accepted at terminals throughout the world due to the partial 'A0 00 00 00 04 pp pp' AID and will be identified specifically to terminals in their home market by the 'Dc cc xx..' to drive special processing.

The xx extension will be managed by Mastercard to avoid duplication. Mastercard may delegate the management to a local registration authority. Only terminals that provide the special functionality need to be aware of the AID extension.

Club AID extensions

When specific functionality is required by a group of customers other than a domestic market, for example, customers associated with a specific co-branding scheme, Mastercard will allocate an AID in the following form:

'A0 00 00 00 04 pp pp Cn nn nn yy'

Where:

- pp pp is the PIX for the primary acceptance brand ('1010' for Mastercard, '3060' for Maestro, or '6000' for Cirrus)
- C is 'C' (4 bits coded as 1100b)
- n nn nn is a unique club identifier issued by Mastercard
- yy yy.. is a unique number managed by the club

This can be used by issuers on their branded cards that will be accepted at terminals throughout the world due to the partial 'A0 00 00 00 04 pp pp' AID and will be identified specifically to terminals in their club by the 'Cn nn nn yy yy..' to drive special processing.

The 'n nn nn' extension will be issued by Mastercard to avoid duplication. The 'yy yy' part of the extension is managed by the club. Only terminals that provide the special functionality need to be aware of the AID extension.

Co-branded AID extensions

When a Mastercard card is co-branded with some other payment functionality, customers may request Mastercard to issue an AID extension for this specific purpose in one of the following forms:

'A0 00 00 00 04 99 99 Cn nn nn yy..'

'A0 00 00 00 04 99 99 Dc cc xx..'

These applications will not be selected by any terminals other than those that participate in the special functionality. A card using these AIDs usually also supports another AID from the Mastercard range.

Private Label issuers that have permission to use the Mastercard RID should also follow this convention.

The 'n nn nn' and 'c cc xx..' extensions will be issued by Mastercard to avoid duplication.

Multi-issuer platforms

This section applies to the use of the AID within wallets.

Where multiple issuers may have applications supported on a single device, such as a mobile phone, issuers should use an AID extension in one of the following forms:

- Extension based on the Bank Identification Code
- Extension based on the Interbank Card Association (ICA) number
- Extension based on the Bank Identification Number (BIN)

Extension based on the Bank Identification Code

'A0 00 00 00 04 pp pp BB bb bb bb bb zz'

Where:

- pp pp is the PIX for the primary acceptance brand ('1010' for Mastercard or '3060' for Maestro)
- BB is 'BB' (8 bits coded as 10111011b)
- bb bb bb bb is the first four characters of the Bank Identification Code (BIC) of the issuing institution (ASCII encoded), that is, the Bank Code according to ISO 9362
- zz is a value chosen by the issuer to make each application on the device unique

Issuers with presence in more than one country should use zz accordingly to identify the country. Issuers are responsible for the management of the assigned zz values.

Extension based on the ICA number

'A0 00 00 00 04 pp pp BC CA ii ii ii zz'

Where:

- pp pp and zz are as described in the description of an extension based on the BIC
- BC is 'BC' (8 bits coded as 10111011b)
- CA is 'CA' (8 bits coded as 11001010b)
- ii ii ii is the ICA number allocated to the customer by Mastercard. The field is always 3 bytes long, with the ICA number expressed in hexadecimal form, right-justified with leading zeroes (format n).

Extension based on the Bank Identification Number (BIN)

'A0 00 00 00 04 pp pp BC Bn bb bb bb zz'

Where:

- pp pp, BC, and zz are as in the description of an extension based on the BIC
- B is 'B' (4 bits coded as 1011b)
- n is the length (in bytes) of the BIN value that follows (how many characters/2)
- bb bb bb is the BIN number of the issuer. The field is n bytes long, defined in the description of an extension based on the BIC, with the BIN value coded expressed in hexadecimal form, left-justified with a single trailing 'F', if required (format cn).

Fleet Cards

For cards issued under a Fleet Card program, requiring a program-specific AID, issuers should use an AID extension in the following form:

'A0 00 00 00 04 pp pp FC uu uu'

Where:

- pp pp is the PIX for the primary acceptance brand ('1010' for Mastercard or '3060' for Maestro)
- FC is 'FC' (8 bits coded as 11111100b)
- uu uu is the fleet scheme identifier issued by Mastercard

This can be used by issuers on their Fleet Cards that will be accepted at terminals due to the partial 'A0 00 00 00 04 pp pp' AID and will be identified specifically to terminals in their fleet program by the Mastercard allocated fleet scheme identifier uu uu to drive special processing, if required.

Extended AIDs with no special acceptance functionality

When no specific functionality is required at the terminal beyond the core brand processing but the card supports multiple applications within the same payment brand, cardholders must be given the choice of which application to use for a transaction.

To support this requirement, issuers should use an AID extension in the following form:

'A0 00 00 00 04 pp pp AA zz'

Where:

- pp pp is the PIX for the primary acceptance branch ('1010' for Mastercard, '3060' for Maestro, or '6000' for Cirrus)
- AA is 'AA' (8 bits coded as 10101010b)
- zz is a value chosen by the issuer to make each application on the card unique

This can be used by issuers on their branded cards that will be accepted at terminals throughout the world due to the partial 'A0 00 00 00 04 pp pp' AID, and cardholders will be offered a choice between the two or more applications supported by the card.

The zz extension will not be issued by Mastercard and may be selected by the issuer. The value will not be unique across all Mastercard cards and will drive no unique terminal functionality.

Terminals must take no special action for cards with AID extensions in the AA range beyond supporting cardholder selection.

Summary of Application Identifiers

For the table "Application Identifiers and related default application labels" the following legend applies to the "AID extension" column.

Table 7: Components used in Application Identifier (AID) extensions

AID component	Definition
AA	Hex 'AA' (8 bits coded as 10101010b)
B	Hex 'B' (4 bits coded as 1011b)
BB	Hex 'B' (8 bits coded as 10111011b)
BC	Hex 'BC' (8 bits coded as 10111100b)
C	Hex 'C' (4 bits coded as 1100b)
CA	Hex 'CA' (8 bits coded as 11001010b)
D	Hex 'D' (4 bits coded as 1101b)
FC	Hex 'FC' (8 bits coded as 11111100b)
bb bb bb bb	First four digits of Bank Identification Code (BIC) of the issuing institution, ASCII encoded, the Bank Code according to ISO 9362
ccc	ISO three-digit numeric Country Code
ii ii ii	Interbank Card Association (ICA)
nnnnn	Club identification defined by Mastercard
uuuu	Fleet scheme identifier allocated by Mastercard
xx..	Managed by Mastercard or by a local registration authority
yy..	Managed by the club

AID component	Definition
zz..	Chosen by the issuer

The following table provides the AIDs that must be associated with the Mastercard products for payment applications and the values for the Application Labels in the card. Labels may use uppercase and lowercase.

Refer to Chapter 3 in the "M/Chip Personalization Data Specification" for Application Label text formatting rules.

Table 8: Application Identifiers and related default application labels

Product	RID (in hex)	PIX (in hex)	AID extension	Application label
Mastercard Credit	A000000004	1010		"Mastercard" In the United States region and U.S. Territories, when a Mastercard Credit application is on the same card as a debit application, the application label for the Credit Mastercard application must be "MastercardCredit".
Debit Mastercard	A000000004	1010		"Debit Mastercard" In the United States region and U.S. Territories, the application label for the Debit Mastercard application must be "Mastercard Debit".
Maestro	A000000004	3060		"Maestro"
Cirrus	A000000004	6000		"Cirrus"
U.S. Maestro AID	A000000004	2203		"US Debit"
Mastercard Cash	A000000004	1208		"Mastercard Cash"
Cardholder Authentication Application	A000000004	8002		As for underlying product
Mastercard Credit on multi-issuer platform	A000000004	1010	BB bb bb bb bb zz	"Mastercard"
Debit Mastercard on multi-issuer platform	A000000004	1010	BB bb bb bb bb zz	"Debit Mastercard"

Product	RID (in hex)	PIX (in hex)	AID extension	Application label
Maestro on multi-issuer platform	A000000004	3060	BB bb bb bb bb zz	"Maestro"
An application that is not a Mastercard product on multi-issuer platform	A000000004	9999	BB bb bb bb bb zz	Issuer defined
Mastercard Credit on card with multiple instances of the same brand	A000000004	1010	AA zz	Each application must contain an Application Label starting with "Mastercard" and contain other text that makes it distinguishable from all other applications.
Debit Mastercard on card with multiple instances of the same brand	A000000004	1010	AA zz	Each application must contain an Application Label starting with "Mastercard" and contain other text that makes it distinguishable from all other applications.
Maestro on card with multiple instances of the same brand	A000000004	3060	AA zz	Each application must contain an Application Label starting with "Maestro" and contain other text that makes it distinguishable from all other applications.
Cirrus on card with multiple instances of the same brand	A000000004	6000	AA zz	Each application must contain an Application Label starting with "Cirrus" and contain other text that makes it distinguishable from all other applications.
Mastercard Credit with special functionality in domestic environment	A000000004	1010	Dc cc xx	"Mastercard"
Debit Mastercard with special functionality in domestic environment	A000000004	1010	Dc cc xx	"Debit Mastercard"
Maestro with special functionality in domestic environment	A000000004	3060	Dc cc xx	"Maestro"

Product	RID (in hex)	PIX (in hex)	AID extension	Application label
Cirrus with special functionality in domestic environment	A000000004	6000	Dc cc xx..	"Cirrus"
An application that is not a Mastercard product with special functionality in a domestic environment only	A000000004	9999	Dc cc xx..	Issuer defined
Mastercard Credit with special functionality at certain club terminals	A000000004	1010	Cn nn nn yy	"Mastercard"
Debit Mastercard with special functionality at certain club terminals	A000000004	1010	Cn nn nn yy	"Debit Mastercard"
Maestro with special functionality at certain club terminals	A000000004	3060	Cn nn nn yy	"Maestro"
Cirrus with special functionality at certain club terminals	A000000004	6000	Cn nn nn yy	"Cirrus"
An application that is not a Mastercard product with special functionality at certain club terminals	A000000004	9999	Cn nn nn yy	Issuer defined
Mastercard Credit on multi-issuers platforms based on the BIC	A000000004	1010	BB bb bb bb bb zz..	"Mastercard"
Mastercard Debit on multi-issuers platforms based on the BIC	A000000004	1010	BB bb bb bb bb zz..	"Debit Mastercard"

Product	RID (in hex)	PIX (in hex)	AID extension	Application label
Maestro on multi-issuers platforms based on the BIC	A000000004	3060	BB bb bb bb bb zz..	"Maestro"
Cirrus on multi-issuers platforms based on the BIC	A000000004	6000	BB bb bb bb bb zz..	"Cirrus"
Mastercard Credit on multi-issuers platforms based on the ICA	A000000004	1010	BC CA ii ii ii zz..	"Mastercard"
Mastercard Debit on multi-issuers platforms based on the ICA	A000000004	1010	BC CA ii ii ii zz..	"Debit Mastercard"
Maestro on multi-issuers platforms based on the ICA	A000000004	3060	BC CA ii ii ii zz..	"Maestro"
Cirrus on multi-issuers platforms based on the ICA	A000000004	6000	BC CA ii ii ii zz..	"Cirrus"
Mastercard Credit on multi-issuers platforms based on the BIN	A000000004	1010	BC Bn bb bb bb zz ..	"Mastercard"
Mastercard Debit on multi-issuers platforms based on the BIN	A000000004	1010	BC Bn bb bb bb zz ..	"Debit Mastercard"
Maestro on multi-issuers platforms based on the BIN	A000000004	3060	BC Bn bb bb bb zz ..	"Maestro"
Cirrus on multi-issuers platforms based on the BIN	A000000004	6000	BC Bn bb bb bb zz ..	"Cirrus"
Mastercard Credit Fleet Card Program	A000000004	1010	FCuuuu	"Mastercard"
Debit Mastercard Fleet Card Program	A000000004	1010	FCuuuu	"Debit Mastercard"
Maestro Fleet Card Program	A000000004	3060	FCuuuu	"Maestro"

Product	RID (in hex)	PIX (in hex)	AID extension	Application label
Co-branded Fleet Card Program	A000000004	9999	FCuuuu	Issuer defined

Application Selection Registered Proprietary Data

The Application Selection Registered Proprietary Data (tag '9F0A') is a variable length data element with binary format.

The value field of the Application Selection Registered Proprietary Data follows the format $ID_1L_1V_1ID_2L_2V_2, \dots$ where for each occurrence of an $ID_nL_nV_n$ item:

- ID is a 2-byte Proprietary Data Identifier assigned by EMVCo.
- L is the length of the value field coded in 1 byte (0 to 255).
- V is the value field.

The value '0001' for ID (Product Identification) has been allocated by EMVCo and indicates a card issued in EEA (and previously the United Kingdom and Gibraltar.) The Value Field for ID '0001' has a variable length of 1 to 5 bytes. The first byte describes the application's product type, according to the EU's Interchange Fee Regulation (IFR.) The values are defined in the following table.

Value (in hex)	IFR product type
01	Debit
02	Credit
03	Commercial
04	Prepaid

Bytes 2 to 5 are reserved for future use, and if present, must be filled with zeroes for the current version.

The following table provides examples of valid coding of tag '9F0A' for each product type.

IFR product type	ID1 (in hex)	L1 (in hex)	V1 (in hex)
Debit	0001	05	0100000000
	0001	04	01000000
	0001	03	010000
	0001	02	0100
	0001	01	01

IFR product type	ID1 (in hex)	L1 (in hex)	V1 (in hex)
Credit	0001	05	0200000000
	0001	04	02000000
	0001	03	020000
	0001	02	0200
	0001	01	02
Commercial	0001	05	0300000000
	0001	04	03000000
	0001	03	030000
	0001	02	0300
	0001	01	03
Prepaid	0001	05	0400000000
	0001	04	04000000
	0001	03	040000
	0001	02	0400
	0001	01	04

Application Interchange Profile

The Application Interchange Profile (AIP) indicates the capability of the card to support specific functions in the application. The AIP is returned by the card in response to the GET PROCESSING OPTIONS command.

The values of the AIP for different card profiles are provided in the *M/Chip Personalization Data Specifications for Contact and Contactless* manual.

Application Usage Control

The Application Usage Control (AUC) contains various bits that are used to indicate to the terminal the kinds of transaction that the card allows, for example, international or domestic cash withdrawals.

R	All	CT	RI072.15	On hybrid cards, the services allowed and forbidden by the AUC must be consistent with the services allowed and forbidden by the service code in the Magnetic Stripe to ensure consistency between chip and Magnetic Stripe transactions.
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With prior approval, Maestro and Mastercard cards that are part of a prepaid program may restrict card acceptance to certain environments or merchants. For such programs, the AUC may be varied to restrict acceptance, as appropriate.

The values of the AUC for different card profiles are provided in the *M/Chip Personalization Data Specifications for Contact and Contactless* manual.

Application Version Number

The Application Version Number is used to ensure compatibility between the card and the terminal and is assigned by the payment system.

The current terminal application version number for EMV transactions is '0002'.

R	All	CT CL	RA176.13	The Application Version Number must be set to '0002' for Mastercard products.
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Cardholder Verification Method List

The Cardholder Verification Method (CVM) List identifies the cardholder verification methods that are supported by the card or issuer and the sequence and conditions under which each should be executed.

Each entry in the list also identifies what action the terminal should take if this option is not completed successfully, either to stop CVM processing or try the next available option.

CVM rules condition codes '06', '07', '08', and '09' use the amount of the transaction to decide on the CVM. These codes will only apply when the transaction currency is the same as the

Application Currency Code of the card. The codes may be used for domestic or currency zone requirements.

All cards that can be accepted at ATMs should include Online PIN at the beginning of the list, with an appropriate condition code for ATMs. This prevents an ATM transaction from being completed with any other CVM.

Maestro cards must be accepted at Bank Branch Terminals with Online PIN. Therefore, the cards must include Online PIN at the beginning of the list, with an appropriate condition code for Bank Branch Terminals. This prevents the cash advance transaction from being completed with any other CVM.

Any card that supports No CVM must include it as the last entry in the CVM List.

This includes on the contact interface for Maestro cards issued in the Europe region.

R	MS	CL	RI417.20	Maestro cards must include No CVM as the last entry in the CVM List.
R	MC	CT CL	RI386.19	Mastercard cards must include No CVM as the last entry in the CVM List.

Requirement RI386.19 does not apply to the United States region.

Refer to the regional requirement RI163.20 for the Europe region in chapter "Issuer requirements," section "Europe region," subsection "CVM requirements for Maestro cards."

Cards that support Offline PIN may support Offline enciphered PIN, Offline plaintext PIN, or both. Cards that support Offline enciphered PIN are recommended to also support Offline plaintext PIN.

BP	MC MS	CT	BI231.14	Enciphered Offline PIN should precede Offline plaintext PIN if both are present in the CVM List.
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For Mastercard cards, when PIN, offline or online, is not successful at POS terminals, then the transaction should fall back to another method.

BP	MC	CT	BI232.14	If Offline PIN, enciphered, plaintext, or both, or Online PIN is applied in POS terminals, the CVM List should be configured to apply the next CVM method, if unsuccessful.
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The personalization values of the CVM List are described in the *M/Chip Personalization Data Specifications for Contact and Contactless* manual.

FCI Issuer Discretionary Data

The FCI Issuer Discretionary Data (tag 'BF0C') may contain extra optional tags not defined by Mastercard at the issuer's discretion. The issuer defined tags are context-specific class data objects in the ranges 'BF01' to 'BF0B', 'BF0D' to 'BF0F', and 'BF70' to 'BF7F'.

The use of these tags is specific to an issuer, so different issuers might use the same tag for a different purpose. Issuers should be aware that information carried in these tags will only be understood by terminals that have been specially developed to recognize a specific issuer's use of these tags. Acquirers should remember that usage of the tags is issuer-specific and so should identify the issuer before acting on information in a specific tag. A different issuer might use the same tag with a different purpose or meaning.

BP	All	CT	BI113.15	Issuers should be aware that information carried in tags in the issuer-specific range will only be understood correctly by specially developed terminals.
		CL		
BP	All	CT	BA096.15	Acquirers should identify the issuer before acting on information carried in tags that are from the issuer-specific range.
		CL		

Issuer Action Codes

Issuer Action Codes (IACs) are used by the terminal during Terminal Action Analysis phase of the transaction to decide how to process transactions in the context of the results of the various tests performed.

TAC and IAC settings are always considered together. Therefore, if either the TAC or IAC decline bit is set for a particular condition, the transaction will be declined if this condition is met.

The conditions under which the terminal sets the corresponding bits in the TVR are defined by EMV. For full details, refer to the EMV specifications.

Issuers will require different values for IACs depending on:

- The product supported, for example, Mastercard, Maestro
- The capabilities of the underlying application, for example, M/Chip Advance
- The product options supported, for example, whether the card supports Offline PIN

IAC options are defined by Mastercard in the *M/Chip Personalization Data Specifications for Contact and Contactless* manual.

This section explains the reasons for some of the IAC settings defined by Mastercard.

Offline Data Authentication Was Not Performed

The "Offline Data Authentication Was Not Performed" bit (byte 1, bit 8) in the TVR may be set when processing a genuine card if any of the following apply:

- The card does not support offline CAM.
- The terminal is online-only and does not support offline CAM.
- The transaction has been subjected to certain wedge device attacks.

SDA, DDA, or CDA Failed

The "SDA," "DDA," and "CDA Failed" bits (byte 1, bits 7, 4, and 3) in the TVR may be set because one of the following is true:

- The terminal does not possess the correct payment system key.
- There is an error in the terminal software.
- The card being used is counterfeit.

If CDA fails because a wedge device has manipulated data on the card or terminal interface or because the card is counterfeit, this may occur after the TVR bit is set. The terminal will decline the transaction.

SDA Selected

The "SDA Selected" bit (byte 1, bit 2) is set in the TVR when the card and the terminal support SDA but they do not both support DDA or CDA.

New Card

The "New Card" bit (byte 2, bit 4) is only set if the card requests the terminal to perform risk management, which M/Chip applications do not.

Cardholder Verification Was Not Successful

Transactions where the "Cardholder Verification Was Not Successful" bit (byte 3, bit 8) is set should never be approved offline.

Unrecognized CVM

When the "Unrecognized CVM" bit (byte 3, bit 7) is set, it means that the terminal did not recognize a proprietary CVM. Issuers can make whatever decision they consider appropriate.

If the "Unrecognized CVM" is a proprietary method used in addition to the other methods, Mastercard recommends that the IACs corresponding to this bit are set to zero.

PIN Try Limit Exceeded, PIN Pad Not Working, and PIN Entry Bypass

The "PIN Try Limit Exceeded," "PIN Pad Not Working," and "PIN Entry Bypass" conditions are noted in these bits in the TVR.

Meaning	Byte	Bit
PIN Try Limit Exceeded	3	6
PIN Entry Required and PIN Pad not present or not working	3	5
PIN Entry Required, PIN Pad present but PIN was not entered	3	4

When the TVR indicates that the PIN has not been successful, the issuer has several options for IAC settings.

The settings (0,0,0), (0,1,0), or (0,1,1) indicate that the issuer does not want to automatically decline if PIN fails and they may want to try another CVM. Therefore, the issuer must set byte 1, bit 7 of the Cardholder Verification Rule to 1b to signify that if there is failure, the next CVM rule must be applied.

Transaction Exceeds Floor Limit

When the "Transaction Exceeds Floor Limit" bit (byte 4, bit 8) in the TVR is set and online authorization is not possible, the terminal may either decline or accept the transaction offline with a voice authorization. The issuer allows voice authorization through setting a value of 0b for the IAC Default.

Static Data to be Authenticated

The Static Data to be Authenticated is data signed by the issuer Private Key in the Signed Static Application Data (tag '93').

This static data is also used to produce the Integrated Circuit Card (ICC) Public Key Certificate (tag '9F46').

Data elements that are included in the Static Data to be Authenticated cannot be altered without detection because any change to this data causes offline CAM to fail.

Cards normally only support a single value for the Signed Static Application Data (tag '93') or of the ICC Public Key Certificate (tag '9F46'), except if the data that is used to calculate the certificate has different values for different transactions. For example, if the CVM List is different for a domestic transaction and an international transaction and the CVM List is included in the static data to be authenticated, the card must contain two certificates produced from each CVM List.

R	All	CT CL	RI074.13	The Static Data to be Authenticated must be composed of the data elements listed in the next table.
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Table 9: Elements specified by the AFL to be authenticated

Value	Format	Length	Tag (in hex)
Application Effective Date (If present)	n 6	3	5F25
Application Expiration Date	n 6	3	5F24
Application Usage Control	b	2	9F07
Application Primary Account Number (PAN)	cnvar. up to 19	Var. up to 10	5A
Application PAN Sequence number	n 2	1	5F34
CVM List	b	Var. up to 252	8E
Issuer Action Code - Default	b	5	9F0D
Issuer Action Code - Denial	b	5	9F0E
Issuer Action Code - Online	b	5	9F0F
Static Data Authentication Tag List	- -	2	9F4A
		The Static Data Authentication Tag List, (tag '9F4A'), must contain only the Application Interchange Profile (tag '82').	
Issuer Country Code	n 3	2	5F28
CDOL1 (If the card supports CDA)	b	Var. up to 252	8C
CDOL2 (If the card supports CDA) (Not required on the contactless interface)	b	Var. up to 252	8D

Value	Format	Length	Tag (in hex)
Application Currency Code (If the CVM List contains CVM condition codes '06', '07', '08', or '09')	n 3	2	9F42

Terminal Action Codes

Settings for Terminal Action Codes (TACs) for different types of contact terminal are provided. The settings are intended to ensure interoperability for cards personalized according to Mastercard requirements.

For contactless terminals, the applicable TAC settings can be found in the *Mastercard Contactless Kernel Configuration* manual. Therefore, this section does not apply to the contactless interface.

For terminals in a transit environment, the applicable TAC settings can be found in the *Mastercard Transit Terminal Requirements* manual.

R	All	CT	RA164.11	Chip terminals must use the TAC settings defined in this document.
R	All	CT CL	RA273.14	The terminal must maintain TACs for use in contact and contactless transactions independent of each other.
R	All	CL	RA487.20	Contactless terminals must use the TAC settings as defined in the <i>Mastercard Contactless Kernel Configuration</i> manual.
R	All	CL	RA488.20	Contactless Transit Terminals must use the TAC as defined in the <i>Mastercard Transit Terminal Requirements</i> manual.

The following assumptions are made for all profiles in this section:

- The terminal supports SDA. If the terminal does not support SDA, byte 1, bit 7 "SDA Failed" is set to (0,0,0).
- The terminal supports setting the "SDA Selected" bit in TVR. If the terminal does not support this, byte 1, bit 2 "SDA Selected" is set to (0,0,0).
- The terminal supports DDA. If the terminal does not support DDA, byte 1, bit 4 "DDA Failed" is set to (0,0,0).
- The terminal supports CDA. If the terminal does not support CDA generation, byte 1, bit 3 "CDA Failed" is set to (0,0,0).

NOTE: CATs support for No CVM does not alter the TACs used. Similarly, support for Signature is always assumed for POS terminals supporting Mastercard credit cards.

Terminal Action Codes for contact ATMs

These are the TACs for contact ATMs.

Byte	Bit	Meaning	Denial	Online	Default
1	8	Offline data authentication was not performed	0	1	1
	7	SDA failed	0	1	1
	6	ICC Data missing	0	1	1
	5	Card appears on terminal exception file	0	1	1
	4	DDA failed	0	1	1
	3	CDA failed	0	1	1
	2	SDA selected	0	1	1
	1	RFU	0	0	0
2	8	ICC and terminal have different application versions	0	0	0
	7	Expired application	0	1	1
	6	Application not yet effective	0	0	0
	5	Requested service not allowed for card product	0	1	1
	4	New Card	0	0	0
	3-1	RFU	0	0	0
3	8	Cardholder verification was not successful	0	1	1
	7	Unrecognized CVM	0	0	0
	6	PIN Try Limit Exceeded	0	0	0
	5	PIN entry required and PIN pad not present or not working	0	1	1
	4	PIN entry required, PIN pad present but PIN was not entered	0	1	1
	3	Online PIN entered	0	1	1
	2-1	RFU	0	0	0
4	8	Transaction exceeds floor limit	0	1	1
	7	Lower consecutive offline limit exceeded	0	0	0
	6	Upper consecutive offline limit exceeded	0	0	0
	5	Transaction selected randomly for online processing	0	0	0

Byte	Bit	Meaning	Denial	Online	Default
	4	Merchant forced transaction online	0	0	0
	3-1	RFU	0	0	0
5	8	Default TDOL used	0	0	0
	7	Issuer authentication was unsuccessful	0	0	0
	6	Script processing failed before final GENERATE AC	0	0	0
	5	Script processing failed after final GENERATE AC	0	0	0
	4-1	RFU	0	0	0

Terminal Action Codes for contact Branch Terminals (online-only)

These are the TACs for contact Branch Terminals, online-only.

Byte	Bit	Meaning	Denial	Online	Default
1	8	Offline data authentication was not performed	0	1	1
	7	SDA failed	0	1	1
	6	ICC Data missing	0	1	1
	5	Card appears on terminal exception file	0	1	1
	4	DDA failed	0	1	1
	3	CDA failed	0	1	1
	2	SDA selected	0	1	1
	1	RFU	0	0	0
2	8	ICC and terminal have different application versions	0	0	0
	7	Expired application	0	1	1
	6	Application not yet effective	0	1	1
	5	Requested service not allowed for card product	0	1	1
	4	New Card	0	1	1
	3-1	RFU	0	0	0
3	8-1	Use Byte 3 settings according to supported CVMs.			
4	8	Transaction exceeds floor limit	0	1	1

Data requirements

Terminal Action Codes Byte 3 for contact Branch Terminals (online-only) with no PIN entry (Mastercard only)

Byte	Bit	Meaning	Denial	Online	Default
	7	Lower consecutive offline limit exceeded	0	0	0
	6	Upper consecutive offline limit exceeded	0	0	0
	5	Transaction selected randomly for online processing	0	0	0
	4	Merchant forced transaction online	0	0	0
	3-1	RFU	0	0	0
5	8	Default TDOL used	0	0	0
	7	Issuer authentication was unsuccessful	0	0	0
	6	Script processing failed before final GENERATE AC	0	0	0
	5	Script processing failed after final GENERATE AC	0	0	0
	4-1	RFU	0	0	0

NOTE:

Byte 3 settings are in the tables in the following sections:

- Terminal Action Codes Byte 3 for contact Branch Terminals (online-only) with no PIN entry (Mastercard only)
- Terminal Action Codes Byte 3 for contact Branch Terminals (online-only) supporting Online PIN
- Terminal Action Codes Byte 3 for contact Branch Terminals (online-only) supporting Offline PIN (Mastercard only)
- Terminal Action Codes Byte 3 for contact Branch Terminals (online-only) supporting Online PIN and Offline PIN

Terminal Action Codes Byte 3 for contact Branch Terminals (online-only) with no PIN entry (Mastercard only)

These are the TACs Byte 3 for contact Branch Terminals, online-only, with no PIN entry, Mastercard only.

Byte	Bit	Meaning	Denial	Online	Default
3	8	Cardholder verification was not successful	1	0	0
	7	Unrecognized CVM	0	0	0
	6	PIN Try Limit Exceeded	0	0	0

Byte	Bit	Meaning	Denial	Online	Default
	5	PIN entry required and PIN pad not present or not working	0	0	0
	4	PIN entry required, PIN pad present but PIN was not entered	0	0	0
	3	Online PIN entered	0	0	0
	2-1	RFU	0	0	0

Terminal Action Codes Byte 3 for contact Branch Terminals (online-only) supporting Online PIN

These are the TACs for Byte 3 contact Branch Terminals, online-only, supporting Online PIN.

Byte	Bit	Meaning	Denial	Online	Default
3	8	Cardholder verification was not successful	1	0	0
	7	Unrecognized CVM	0	0	0
	6	PIN Try Limit Exceeded	0	0	0
	5	PIN entry required and PIN pad not present or not working - for Maestro	1	0	0
	5	PIN entry required and PIN pad not present or not working - for Mastercard	0	1	1
	4	PIN entry required, PIN pad present but PIN was not entered	1	0	0
	3	Online PIN entered	0	1	1
	2-1	RFU	0	0	0

Terminal Action Codes Byte 3 for contact Branch Terminals (online-only) supporting Offline PIN (Mastercard only)

These are the TACs Byte 3 for contact Branch Terminals, online-only, supporting Offline PIN, Mastercard only.

Byte	Bit	Meaning	Denial	Online	Default
3	8	Cardholder verification was not successful	1	0	0
	7	Unrecognized CVM	0	0	0
	6	PIN Try Limit Exceeded	0	1	1

Byte	Bit	Meaning	Denial	Online	Default
	5	PIN entry required and PIN pad not present or not working	0	1	1
	4	PIN entry required, PIN pad present but PIN was not entered	1	0	0
	3	Online PIN entered	0	0	0
	2-1	RFU	0	0	0

Terminal Action Codes Byte 3 for contact Branch Terminals (online-only) supporting Online PIN and Offline PIN

These are the TACs Byte 3 for contact Branch Terminals, online-only, supporting Online and Offline PIN.

Byte	Bit	Meaning	Denial	Online	Default
3	8	Cardholder verification was not successful	1	0	0
	7	Unrecognized CVM	0	0	0
	6	PIN Try Limit Exceeded	0	1	1
	5	PIN entry required and PIN pad not present or not working - for Cirrus and Maestro BBT	1	0	0
	5	PIN entry required and PIN pad not present or not working - for Mastercard BBT	0	1	1
	4	PIN entry required, PIN pad present but PIN was not entered	1	0	0
	3	Online PIN entered	0	1	1
	2-1	RFU	0	0	0

Terminal Action Codes for contact ATMs

These are the TACs for contact ATMs.

Offline CAM supported			CVM supported		Terminal Action Codes
SDA	DDA	CDA	Online PIN	Offline PIN	
No	No	No	Yes	No	TAC Denial '00 00 00 00 00' TAC Online 'B0 50 9C 80 00' TAC Default 'B0 50 9C 80 00'
Yes	Yes	No	Yes	No	TAC Denial '00 00 00 00 00' TAC Online 'FA 50 9C 80 00' TAC Default 'FA 50 9C 80 00' 'F8' should be used instead of 'FA' if the terminal does not support the "SDA selected" setting.
Yes	Yes	Yes	Yes	No	TAC Denial '00 00 00 00 00' TAC Online 'FE 50 9C 80 00' TAC Default 'FE 50 9C 80 00' 'FC' should be used instead of 'FE' if the terminal does not support the "SDA selected" setting.

Terminal Action Codes for Branch Terminals (Cirrus and Maestro)

These are the TACs for Branch Terminals for Cirrus and Maestro.

Offline CAM supported			CVM supported		Terminal Action Codes
SDA	DDA	CDA	Online PIN	Offline PIN	
No	No	No	Yes	No	TAC Denial '00 00 98 00 00' TAC Online 'B0 78 04 80 00' TAC Default 'B0 78 04 80 00'
Yes	Yes	No	Yes	No	TAC Denial '00 00 98 00 00' TAC Online 'FA 78 04 80 00' TAC Default 'FA 78 04 80 00' 'F8' should be used instead of 'FA' if the terminal does not support the "SDA selected" setting.

Offline CAM supported			CVM supported		Terminal Action Codes
SDA	DDA	CDA	Online PIN	Offline PIN	
Yes	Yes	Yes	Yes	No	TAC Denial '00 00 98 00 00' TAC Online 'FE 78 04 80 00' TAC Default 'FE 78 04 80 00' 'FC' should be used instead of 'FE' if the terminal does not support the "SDA selected" setting.

Terminal Action Codes for contact Branch Terminals (Mastercard)

These are the TACs for contact Branch Terminals for Mastercard.

Offline CAM supported			CVM supported		Terminal Action Codes
SDA	DDA	CDA	Online PIN	Offline PIN	
No	No	No	No	No	TAC Denial '00 00 80 00 00' TAC Online 'B0 78 00 80 00' TAC Default 'B0 78 00 80 00'
			Yes	No	TAC Denial '00 00 88 00 00' TAC Online 'B0 78 14 80 00' TAC Default 'B0 78 14 80 00'
			No	Yes	TAC Denial '00 00 88 00 00' TAC Online 'B0 78 30 80 00' TAC Default 'B0 78 30 80 00'
			Yes	Yes	TAC Denial '00 00 88 00 00' TAC Online 'B0 78 34 80 00' TAC Default 'B0 78 34 80 00'
Yes	Yes	No	No	No	TAC Denial '00 00 80 00 00' TAC Online 'FA 78 00 80 00' TAC Default 'FA 78 00 80 00' 'F8' should be used instead of 'FA' if the terminal does not support the "SDA selected" setting.

Data requirements

Terminal Action Codes for contact Branch Terminals (Mastercard)

Offline CAM supported			CVM supported		Terminal Action Codes
SDA	DDA	CDA	Online PIN	Offline PIN	
			Yes	No	TAC Denial '00 00 88 00 00' TAC Online 'FA 78 14 80 00' TAC Default 'FA 78 14 80 00' 'F8' should be used instead of 'FA' if the terminal does not support the "SDA selected" setting.
			No	Yes	TAC Denial '00 00 88 00 00' TAC Online 'FA 78 30 80 00' TAC Default 'FA 78 30 80 00' 'F8' should be used instead of 'FA' if the terminal does not support the "SDA selected" setting.
			Yes	Yes	TAC Denial '00 00 88 00 00' TAC Online 'FA 78 34 80 00' TAC Default 'FA 78 34 80 00' 'F8' should be used instead of 'FA' if the terminal does not support the "SDA selected" setting.
Yes	Yes	Yes	No	No	TAC Denial '00 00 80 00 00' TAC Online 'FE 78 00 80 00' TAC Default 'FE 78 00 80 00' 'FC' should be used instead of 'FE' if the terminal does not support the "SDA selected" setting.
			Yes	No	TAC Denial '00 00 88 00 00' TAC Online 'FE 78 14 80 00' TAC Default 'FE 78 14 80 00' 'FC' should be used instead of 'FE' if the terminal does not support the "SDA selected" setting.

Offline CAM supported			CVM supported		Terminal Action Codes
SDA	DDA	CDA	Online PIN	Offline PIN	
			No	Yes	TAC Denial '00 00 88 00 00' TAC Online 'FE 78 30 80 00' TAC Default 'FE 78 30 80 00' 'FC' should be used instead of 'FE' if the terminal does not support the "SDA selected" setting.
			Yes	Yes	TAC Denial '00 00 88 00 00' TAC Online 'FE 78 34 80 00' TAC Default 'FE 78 34 80 00' 'FC' should be used instead of 'FE' if the terminal does not support the "SDA selected" setting.

Terminal Action Codes for contact online-capable POS and CAT

These are the TACs for contact online-capable POS terminals and CATs.

Byte	Bit	Meaning	Denial	Online	Default
1	8	Offline data authentication was not performed	0	1	1
	7	SDA failed	0	1	1
	6	ICC Data missing	0	1	1
	5	Card appears on terminal exception file	0	1	1
	4	DDA failed	0	1	1
	3	CDA failed	0	1	1
	2	SDA selected	0	1	1
	1	RFU	0	0	0
2	8	ICC and terminal have different application versions	0	0	0
	7	Expired application	0	1	1
	6	Application not yet effective	0	0	0
	5	Requested service not allowed for card product	0	1	1
	4	New Card	0	0	0

Byte	Bit	Meaning	Denial	Online	Default
	3-1	RFU	0	0	0
3	8-1	Use Byte 3 settings according to supported CVMs.			
4	8	Transaction exceeds floor limit	0	1	0/1 ¹
	7	Lower consecutive offline limit exceeded	0	1	0
	6	Upper consecutive offline limit exceeded	0	1	1
	5	Transaction selected randomly for online processing	0	1	0
	4	Merchant forced transaction online	0	1	0
	3-1	RFU	0	0	0
5	8	Default TDOL used	0	0	0
	7	Issuer authentication was unsuccessful	0	0	0
	6	Script processing failed before final GENERATE AC	0	0	0
	5	Script processing failed after final GENERATE AC	0	0	0
	4-1	RFU	0	0	0

Online-capable terminals should use the TAC values that correspond to online-capable terminals regardless of the value of the floor limit. Online-capable terminals configured with a zero floor limit cannot use the TAC values defined for online-only terminals. The value of the floor limit is configurable, and if the acquirer decides to change it, the TACs do not need to change as well.

NOTE:

Byte 3 settings are in the tables in the following sections:

- **Terminal Action Codes for contact offline-only CAT**
- **Terminal Action Codes Byte 3 for contact offline-only CAT with no PIN entry**
- **Terminal Action Codes Byte 3 for contact offline-only CAT supporting Offline PIN**
- **Terminal Action Codes for contact offline-only CAT**

Terminal Action Codes Byte 3 for contact online-capable POS and CAT with no PIN entry

These are the TACs Byte 3 for contact online-capable POS terminals and CATs with no PIN entry.

¹ An attended POS must have setting 1b for Maestro transactions. The setting must be 0b for Mastercard transactions if voice authorizations are supported otherwise 1b. CATs must set to 1b.

Byte	Bit	Meaning	Denial	Online	Default
3	8	Cardholder verification was not successful	0	1	1
	7	Unrecognized CVM	0	0	0
	6	PIN Try Limit Exceeded	0	0	0
	5	PIN entry required and PIN pad not present or not working	0	0	0
	4	PIN entry required, PIN pad present but PIN was not entered	0	0	0
	3	Online PIN entered	0	0	0
	2-1	RFU	0	0	0

Terminal Action Codes Byte 3 for contact online-capable POS and CAT supporting Online PIN

These are the TACs Byte 3 for contact online-capable POS terminals and CATs supporting Online PIN.

Byte	Bit	Meaning	Denial	Online	Default
3	8	Cardholder verification was not successful	0	1	1
	7	Unrecognized CVM	0	0	0
	6	PIN Try Limit Exceeded	0	0	0
	5	PIN entry required and PIN pad not present or not working	0	1	1
	4	PIN entry required, PIN pad present but PIN was not entered	0	1	1
	3	Online PIN entered	0	1	1
	2-1	RFU	0	0	0

Terminal Action Codes Byte 3 for contact online-capable POS and CAT supporting Offline PIN

These are the TACs Byte 3 for contact online-capable POS terminals and CATs supporting Offline PIN.

Byte	Bit	Meaning	Denial	Online	Default
3	8	Cardholder verification was not successful	0	1	1
	7	Unrecognized CVM	0	0	0
	6	PIN Try Limit Exceeded	0	1	1
	5	PIN entry required and PIN pad not present or not working	0	1	1
	4	PIN entry required, PIN pad present but PIN was not entered	0	1	1
	3	Online PIN entered	0	0	0
	2-1	RFU	0	0	0

Terminal Action Codes Byte 3 for contact online-capable POS and CAT supporting Online PIN and Offline PIN

These are the TACs Byte 3 for contact online-capable POS terminals and CATs supporting Online and Offline PIN.

Byte	Bit	Meaning	Denial	Online	Default
3	8	Cardholder verification was not successful	0	1	1
	7	Unrecognized CVM	0	0	0
	6	PIN Try Limit Exceeded	0	1	1
	5	PIN entry required and PIN pad not present or not working	0	1	1
	4	PIN entry required, PIN pad present but PIN was not entered	0	1	1
	3	Online PIN entered	0	1	1
	2-1	RFU	0	0	0

Terminal Action Codes for contact online-capable POS and CAT

These are the TACs for contact online-capable POS terminals and CATs.

Offline CAM supported			CVM supported		Terminal Action Codes
SDA	DDA	CDA	Online PIN	Offline PIN	
Yes	Yes	No	No	No	TAC Denial '00 00 00 00 00' TAC Online 'FA 50 80 F8 00' TAC Default 'FA 50 80 A0 00' 'F8' should be used instead of 'FA' if the terminal does not support the "SDA selected" setting. '20' should be used instead of 'A0' if voice authorization supported for Mastercard transactions in "Unable to go Online" mode.
			Yes	No	TAC Denial '00 00 00 00 00' TAC Online 'FA 50 9C F8 00' TAC Default 'FA 50 9C A0 00' 'F8' should be used instead of 'FA' if the terminal does not support the "SDA selected" setting. '20' should be used instead of 'A0' if voice authorization supported for Mastercard transactions in "Unable to go Online" mode.
			No	Yes	TAC Denial '00 00 00 00 00' TAC Online 'FA 50 B8 F8 00' TAC Default 'FA 50 B8 A0 00' 'F8' should be used instead of 'FA' if the terminal does not support the "SDA selected" setting. '20' should be used instead of 'A0' if voice authorization supported for Mastercard transactions in "Unable to go Online" mode.

Data requirements

Terminal Action Codes for contact online-capable POS and CAT

Offline CAM supported			CVM supported		Terminal Action Codes
SDA	DDA	CDA	Online PIN	Offline PIN	
			Yes	Yes	TAC Denial '00 00 00 00 00' TAC Online 'FA 50 BC F8 00' TAC Default 'FA 50 BC A0 00' 'F8' should be used instead of 'FA' if the terminal does not support the "SDA selected" setting. '20' should be used instead of 'A0' if voice authorization supported for Mastercard transactions in "Unable to go Online" mode.
Yes	Yes	Yes	No	No	TAC Denial '00 00 00 00 00' TAC Online 'FE 50 80 F8 00' TAC Default 'FE 50 80 A0 00' 'FC' should be used instead of 'FE' if the terminal does not support the "SDA selected" setting. '20' should be used instead of 'A0' if voice authorization supported for Mastercard transactions in "Unable to go Online" mode.
			Yes	No	TAC Denial '00 00 00 00 00' TAC Online 'FE 50 9C F8 00' TAC Default 'FE 50 9C A0 00' '20' should be used instead of 'A0' if voice authorization supported for Mastercard transactions in "Unable to go Online" mode. 'FC' should be used instead of 'FE' if the terminal does not support the "SDA selected" setting.

Offline CAM supported			CVM supported		Terminal Action Codes
SDA	DDA	CDA	Online PIN	Offline PIN	
			No	Yes	TAC Denial '00 00 00 00 00' TAC Online 'FE 50 B8 F8 00' TAC Default 'FE 50 B8 A0 00' 'FC' should be used instead of 'FE' if the terminal does not support the "SDA selected" setting. '20' should be used instead of 'A0' if voice authorization supported for Mastercard transactions in "Unable to go Online" mode.
			Yes	Yes	TAC Denial '00 00 00 00 00' TAC Online 'FE 50 BC F8 00' TAC Default 'FE 50 BC A0 00' 'FC' should be used instead of 'FE' if the terminal does not support the "SDA selected" setting. '20' should be used instead of 'A0' if voice authorization supported for Mastercard transactions in "Unable to go Online" mode.

Terminal Action Codes for contact offline-only CAT

These are the TACs for contact offline-only CAT.

Byte	Bit	Meaning	Denial	Online	Default
1	8	Offline data authentication was not performed	1	0	0
	7	SDA failed	1	0	0
	6	ICC Data missing	1	0	0
	5	Card appears on terminal exception file	1	0	0
	4	DDA failed	1	0	0
	3	CDA failed	1	0	0
	2	SDA selected	1	0	0
	1	RFU	0	0	0

Byte	Bit	Meaning	Denial	Online	Default
2	8	ICC and terminal have different application versions	0	0	0
	7	Expired application	1	0	0
	6	Application not yet effective	0	0	0
	5	Requested service not allowed for card product	1	0	0
	4	New Card	0	0	0
	3-1	RFU	0	0	0
3	8-1	Use Byte 3 settings according to supported CVMs.			
4	8	Transaction exceeds floor limit	1	0	0
	7	Lower consecutive offline limit exceeded	0	0	0
	6	Upper consecutive offline limit exceeded	1	0	0
	5	Transaction selected randomly for online processing	0	0	0
	4	Merchant forced transaction online	0	0	0
	3-1	RFU	0	0	0
5	8	Default TDOL used	0	0	0
	7	Issuer authentication was unsuccessful	0	0	0
	6	Script processing failed before final GENERATE AC	0	0	0
	5	Script processing failed after final GENERATE AC	0	0	0
	4-1	RFU	0	0	0

NOTE:

Byte 3 settings are in the tables in the following sections:

- Terminal Action Codes Byte 3 for contact offline-only CAT with no PIN entry
- Terminal Action Codes Byte 3 for contact offline-only CAT supporting Offline PIN

Terminal Action Codes Byte 3 for contact offline-only CAT with no PIN entry

These are the TACs Byte 3 for contact offline-only CATs with no PIN entry.

Byte	Bit	Meaning	Denial	Online	Default
3	8	Cardholder verification was not successful	1	0	0
	7	Unrecognized CVM	0	0	0
	6	PIN Try Limit Exceeded	0	0	0
	5	PIN entry required and PIN pad not present or not working	0	0	0
	4	PIN entry required, PIN pad present but PIN was not entered	0	0	0
	3	Online PIN entered	0	0	0
	2-1	RFU	0	0	0

Terminal Action Codes Byte 3 for contact offline-only CAT supporting Offline PIN

These are the TACs Byte 3 for contact offline-only CATs supporting Offline PIN.

Byte	Bit	Meaning	Denial	Online	Default
3	8	Cardholder verification was not successful	1	0	0
	7	Unrecognized CVM	0	0	0
	6	PIN Try Limit Exceeded	0	0	0
	5	PIN entry required and PIN pad not present or not working	0	0	0
	4	PIN entry required, PIN pad present but PIN was not entered	0	0	0
	3	Online PIN entered	0	0	0
	2-1	RFU	0	0	0

Terminal Action Codes for contact offline-only CAT

These are the TACs for contact offline-only CATs.

Offline CAM supported			CVM supported		Terminal Action Codes
SDA	DDA	CDA	Online PIN	Offline PIN	
Yes	Yes	No	No	No	TAC Denial 'F8 50 80 A0 00' TAC Online '00 00 00 00 00' TAC Default '00 00 00 00 00'
			No	Yes	TAC Denial 'F8 50 80 A0 00' TAC Online '00 00 00 00 00' TAC Default '00 00 00 00 00'
Yes	Yes	Yes	No	No	TAC Denial 'FC 50 80 A0 00' TAC Online '00 00 00 00 00' TAC Default '00 00 00 00 00'
			No	Yes	TAC Denial 'FC 50 80 A0 00' TAC Online '00 00 00 00 00' TAC Default '00 00 00 00 00'

Terminal Action Codes for contact online-only POS and CAT

These are the TACs for contact online-only POS terminals and CATs.

Byte	Bit	Meaning	Denial	Online	Default
1	8	Offline data authentication was not performed	0	1	1
	7	SDA failed	0	1	1
	6	ICC Data missing	0	1	1
	5	Card appears on terminal exception file	0	1	1
	4	DDA failed	0	1	1
	3	CDA failed	0	1	1
	2	SDA selected	0	1	1
	1	RFU	0	0	0
2	8	ICC and terminal have different application versions	0	0	0
	7	Expired application	0	1	1
	6	Application not yet effective	0	0	0

Byte	Bit	Meaning	Denial	Online	Default
	5	Requested service not allowed for card product	0	1	1
	4	New Card	0	0	0
	3-1	RFU	0	0	0
3	8-1	Use Byte 3 settings according to supported CVMs.			
4	8	Transaction exceeds floor limit	0	1	1
	7	Lower consecutive offline limit exceeded	0	0	0
	6	Upper consecutive offline limit exceeded	0	0	0
	5	Transaction selected randomly for online processing	0	0	0
	4	Merchant forced transaction online	0	0	0
	3-1	RFU	0	0	0
5	8	Default TDOL used	0	0	0
	7	Issuer authentication was unsuccessful	0	0	0
	6	Script processing failed before final GENERATE AC	0	0	0
	5	Script processing failed after final GENERATE AC	0	0	0
	4-1	RFU	0	0	0

NOTE:

Byte 3 settings are in the tables in the following sections:

- Terminal Action Codes Byte 3 for contact online-only POS and CAT with no PIN entry
- Terminal Action Codes Byte 3 for contact online-only POS and CAT supporting Online PIN
- Terminal Action Codes Byte 3 for contact online-only POS and CAT supporting Offline PIN
- Terminal Action Codes Byte 3 for contact online-only POS and CAT supporting Online PIN and Offline PIN

Terminal Action Codes Byte 3 for contact online-only POS and CAT with no PIN entry

These are the TACs Byte 3 for contact online-only POS terminals and CATs with no PIN entry.

Byte	Bit	Meaning	Denial	Online	Default
3	8	Cardholder verification was not successful	0	1	1

Byte	Bit	Meaning	Denial	Online	Default
	7	Unrecognized CVM	0	0	0
	6	PIN Try Limit Exceeded	0	0	0
	5	PIN entry required and PIN pad not present or not working	0	0	0
	4	PIN entry required, PIN pad present but PIN was not entered	0	0	0
	3	Online PIN entered	0	0	0
	2-1	RFU	0	0	0

Terminal Action Codes Byte 3 for contact online-only POS and CAT supporting Online PIN

These are the TACs Byte 3 for contact online-only POS terminals and CATS supporting Online PIN.

Byte	Bit	Meaning	Denial	Online	Default
3	8	Cardholder verification was not successful	0	1	1
	7	Unrecognized CVM	0	0	0
	6	PIN Try Limit Exceeded	0	0	0
	5	PIN entry required and PIN pad not present or not working	0	1	1
	4	PIN entry required, PIN pad present but PIN was not entered	0	1	1
	3	Online PIN entered	0	1	1
	2-1	RFU	0	0	0

Terminal Action Codes Byte 3 for contact online-only POS and CAT supporting Offline PIN

These are the TACs Byte 3 for contact online-only POS terminals and CATs supporting Offline PIN.

Byte	Bit	Meaning	Denial	Online	Default
3	8	Cardholder verification was not successful	0	1	1
	7	Unrecognized CVM	0	0	0
	6	PIN Try Limit Exceeded	0	1	1

Byte	Bit	Meaning	Denial	Online	Default
	5	PIN entry required and PIN pad not present or not working	0	1	1
	4	PIN entry required, PIN pad present but PIN was not entered	0	1	1
	3	Online PIN entered	0	0	0
	2-1	RFU	0	0	0

Terminal Action Codes Byte 3 for contact online-only POS and CAT supporting Online PIN and Offline PIN

These are the TACs Byte 3 for contact online-only POS terminals and CATS supporting Online and Offline PIN.

Byte	Bit	Meaning	Denial	Online	Default
3	8	Cardholder verification was not successful	0	1	1
	7	Unrecognized CVM	0	0	0
	6	PIN Try Limit Exceeded	0	1	1
	5	PIN entry required and PIN pad not present or not working	0	1	1
	4	PIN entry required, PIN pad present but PIN was not entered	0	1	1
	3	Online PIN entered	0	1	1
	2-1	RFU	0	0	0

Terminal Action Codes for contact online-only POS and CAT

These are the TACs for contact online-only POS terminals and CATs.

Offline CAM supported			CVM supported		Terminal Action Codes
SDA	DDA	CDA	Online PIN	Offline PIN	
No	No	No	No	No	TAC Denial '00 00 00 00 00'
					TAC Online 'B0 50 80 80 00'
					TAC Default 'B0 50 80 80 00'

Data requirements
Terminal Action Codes for contact online-only POS and CAT

Offline CAM supported			CVM supported		Terminal Action Codes
SDA	DDA	CDA	Online PIN	Offline PIN	
			Yes	No	TAC Denial '00 00 00 00 00' TAC Online 'B0 50 9C 80 00' TAC Default 'B0 50 9C 80 00'
			No	Yes	TAC Denial '00 00 00 00 00' TAC Online 'B0 50 B8 80 00' TAC Default 'B0 50 B8 80 00'
			Yes	Yes	TAC Denial '00 00 00 00 00' TAC Online 'B0 50 BC 80 00' TAC Default 'B0 50 BC 80 00'
Yes	Yes	No	No	No	TAC Denial '00 00 00 00 00' TAC Online 'FA 50 80 80 00' TAC Default 'FA 50 80 80 00' 'F8' should be used instead of 'FA' if the terminal does not support the "SDA selected" setting.
			Yes	No	TAC Denial '00 00 00 00 00' TAC Online 'FA 50 9C 80 00' TAC Default 'FA 50 9C 80 00' 'F8' should be used instead of 'FA' if the terminal does not support the "SDA selected" setting.
			No	Yes	TAC Denial '00 00 00 00 00' TAC Online 'FA 50 B8 80 00' TAC Default 'FA 50 B8 80 00' 'F8' should be used instead of 'FA' if the terminal does not support the "SDA selected" setting.
			Yes	Yes	TAC Denial '00 00 00 00 00' TAC Online 'FA 50 BC 80 00' TAC Default 'FA 50 BC 80 00' 'F8' should be used instead of 'FA' if the terminal does not support the "SDA selected" setting.

Offline CAM supported			CVM supported		Terminal Action Codes
SDA	DDA	CDA	Online PIN	Offline PIN	
Yes	Yes	Yes	No	No	TAC Denial '00 00 00 00 00' TAC Online 'FE 50 80 80 00' TAC Default 'FE 50 80 80 00' 'FC' should be used instead of 'FE' if the terminal does not support the "SDA selected" setting.
			Yes	No	TAC Denial '00 00 00 00 00' TAC Online 'FE 50 9C 80 00' TAC Default 'FE 50 9C 80 00' 'FC' should be used instead of 'FE' if the terminal does not support the "SDA selected" setting.
			No	Yes	TAC Denial '00 00 00 00 00' TAC Online 'FE 50 B8 80 00' TAC Default 'FE 50 B8 80 00' 'FC' should be used instead of 'FE' if the terminal does not support the "SDA selected" setting.
			Yes	Yes	TAC Denial '00 00 00 00 00' TAC Online 'FE 50 BC 80 00' TAC Default 'FE 50 BC 80 00' 'FC' should be used instead of 'FE' if the terminal does not support the "SDA selected" setting.

Terminal Action Codes for contact offline-only on-board merchant supporting Offline PIN

These are the TACs for contact offline-only on-board merchants supporting Offline PIN.

Byte	Bit	Meaning	Denial	Online	Default
1	8	Offline data authentication was not performed	1	0	0
	7	SDA failed	1	0	0
	6	ICC Data missing	1	0	0

Data requirements

Terminal Action Codes for contact offline-only on-board merchant supporting Offline PIN

Byte	Bit	Meaning	Denial	Online	Default
	5	Card appears on terminal exception file	1	0	0
	4	DDA failed	1	0	0
	3	CDA failed	1	0	0
	2	SDA selected	1	0	0
	1	RFU	0	0	0
2	8	ICC and terminal have different application versions	0	0	0
	7	Expired application	1	0	0
	6	Application not yet effective	0	0	0
	5	Requested service not allowed for card product	1	0	0
	4	New Card	0	0	0
	3-1	RFU	0	0	0
3	8	Cardholder verification was not successful	1	0	0
	7	Unrecognized CVM	0	0	0
	6	PIN Try Limit Exceeded	1	0	0
	5	PIN entry required and PIN pad not present or not working	1	0	0
	4	PIN entry required, PIN pad present but PIN was not entered	1	0	0
	3	Online PIN entered	0	0	0
	2-1	RFU	0	0	0
4	8	Transaction exceeds floor limit	1	0	0
	7	Lower consecutive offline limit exceeded	0	0	0
	6	Upper consecutive offline limit exceeded	1	0	0
	5	Transaction selected randomly for online processing	0	0	0
	4	Merchant forced transaction online	0	0	0
	3-1	RFU	0	0	0
5	8	Default TDOL used	0	0	0
	7	Issuer authentication was unsuccessful	0	0	0
	6	Script processing failed before final GENERATE AC	0	0	0

Byte	Bit	Meaning	Denial	Online	Default
	5	Script processing failed after final GENERATE AC	0	0	0
	4-1	RFU	0	0	0

Terminal Action Codes for contact offline-only on-board merchant supporting Offline PIN

These are the TACs for contact offline-only on-board merchants supporting Offline PIN.

Offline CAM supported			CVM supported		Terminal Action Codes
SDA	DDA	CDA	Online PIN	Offline PIN	
Yes	Yes	No	No	Yes	TAC Denial 'F8 50 B8 A0 00' TAC Online '00 00 00 00 00' TAC Default '00 00 00 00 00'
Yes	Yes	Yes	No	Yes	TAC Denial 'FC 50 B8 A0 00' TAC Online '00 00 00 00 00' TAC Default '00 00 00 00 00'

Network data

This section provides a description of the chip data contained in authorization and clearing messages in the Mastercard Network. Formats of messages in national networks are likely to be different.

Network data legend

Data elements impacted by chip adhere to a specific format.

Table 10: Legend

M	Mandatory
C	Conditional. The data element must be present in the message only if the conditions described in the Presence column are met.
O	Optional

LLVAR	The format of a variable-length data element is defined as LLVAR or LLLVAR, where LL or LLL is a two-digit or three-digit length specifier that indicates the actual number of digits or characters of data contained in the data element, excluding the length specifier. VAR is defined as <attribute(s)> .. <maximum number of digits or characters allowed>. For example, b..255;LLVAR means that the data element is of variable length and may contain up to a maximum of 255 bytes with binary value.
PDS	Private Data Subelement
SE	Subelement
LTV	Length, Tag, Value
IPM	Integrated Product Messages. Format used by Mastercard for clearing and interchange activity
b	Binary value
VAR	Variable length data

DE 55 data for messages from Acquirer

Data in DE 55 for Authorization Request/0100, Financial Transaction Request/0200, Authorization Advice Request/0120-Acquirer generated, Financial Transaction Advice/0220, and First Presentment/1240 messages.

Table 11: Data in DE 55 for messages from Acquirer

Data element	Condition	Data element name	Format	Presence
DE 55	C	ICC System Related Data	b..255; LLLVAR	If subfield 1 of DE 22 (POS Entry Mode) in the request message contains "05" or "07", DE 55 must be present. (The presence of DE 55 is optional for refund and Account Status Inquiry (ASI) transactions.) If present, DE 55 must be TLV encoded and must contain the mandatory, recommended, and conditional subelements, as described in appendix B.
SE 9F26	M	Application Cryptogram	b8	
SE 9F27	M	Cryptogram Information Data	b1	
SE 9F37	M	Unpredictable Number	b4	

Data element	Condition	Data element name	Format	Presence
SE 9F36	M	Application Transaction Counter	b2	
SE 95	M	Terminal Verification Results	b5	
SE 9A	M	Transaction Date	b3 (n6)	
SE 9C	M	Transaction Type	b1 (n2)	
SE 9F02	M	Amount, Authorized (Numeric)	b6 (n12)	
SE 5F2A	M	Transaction Currency Code	b2 (n3)	
SE 82	M	Application Interchange Profile	b2	
SE 9F1A	M	Terminal Country Code	b2 (n3)	
SE 9F03	C	Amount, Other (Numeric)	b6 (n12)	Must be provided if a cash back amount is part of the transaction. If there is no cash back, the value of Amount, Other may be zero or Amount, Other may be absent.
SE 9F10	C	Issuer Application Data	b..32 VAR	Must be provided if supplied by the card to terminal.
SE 9F34	M	Cardholder Verification Method Results	b3	Mandatory
SE 9F33	M	Terminal Capabilities	b3	Mandatory for the United States region. Refer to requirements RA546.24 and RA547.24 for all other regions.
SE 9F35	O	Terminal Type	b1 (n2)	
SE 9F1E	O	Interface Device Serial Number	b8 (an8)	
SE 9F53	O		b1 (an1)	
SE 84	M	Dedicated File Name	b5..16 VAR	Mandatory
SE 9F09	O	Terminal Application Version Number	b2	

Data element	Condition	Data element name	Format	Presence
SE 9F41	O	Transaction Sequence Counter	b2..4 (n4..8) VAR	
SE 91	O	Issuer Authentication Data	b8..16 VAR	
SE 9F0A	O	Application Selection Registered Proprietary Data	b VAR up to 255	Tag '9F0A' with ID='0001' may be used only for transactions performed with cards or devices issued in the European Union and in Gibraltar, Iceland, Liechtenstein, Norway, and the United Kingdom.
SE 5F34	O	Application Primary account Number (PAN) Sequence Number	n 2	
SE 9F6E	O	Third Party Data	b5..32	
SE 9F60	O	Authenticated Application Data	b VAR up to 127	
SE 96	O	Kernel Identifier-Terminal	b8	

DE 55 data for 0110 and 0210 response messages

Data in DE55 of an Authorization Request Response/0110 and Financial Transaction Request Response/0210 message.

Table 12: Data in DE55 for 0110 and 0210 response messages from Issuer

Data element	Condition	Data element name	Format	Presence
DE 55	C	ICC System Related Data	b..255; LLLVAR	May be present if subfield 1 of DE 22 (POS Entry Mode) contains a value of "05" and DE 55 is present in the request. Otherwise not present.
SE 91	O	Issuer Authentication Data	b8..16 VAR	
SE 71	O	Issuer Script Template 1	b..127 VAR	

Data element	Condition	Data element name	Format	Presence
SE 72	O	Issuer Script Template 2	b..127 VAR ²	

Other data in messages from Acquirer

The following table provides a summary of the impacts of chip on fields other than DE 55 in the Authorization Request/0100, Financial Transaction Request/0200, Authorization Advice/0120-Acquirer generated, and Financial Transaction Advice/0220 messages. Advice messages generated by Stand-In are filled with the same data as in the original Authorization Request/0100 or Financial Transaction Request/0200 message.

For a full description of all data elements and their subfields, refer to the *Customer Interface Specification* manual.

Table 13: Other data in 0100, 0200, 0120 and 0220 messages

Data element	Requirement
DE 22	For a chip contact transaction, acquirers must provide the value "05x". For a chip contactless transaction, acquirers must provide the value "07x". For a Mastercard Consumer-Presented QR, acquirers must provide the value "03x". For a contact transaction where the Magnetic Stripe was used as a technical fallback, acquirers must provide the value "80x". In this case, acquirers must provide the full, unaltered track data. For a transaction where voice authorization is used as a technical fallback, acquirers must provide the value "79x".
DE 23	If DE 22 = "05x" or "07x" and if the Application PAN Sequence Number tag '5F34' is present on the chip, acquirers must send DE 23 containing the Application PAN sequence number tag '5F34'. If DE 22 = "05x" or "07x" and the Application PAN sequence number tag '5F34' is not present on the chip, acquirers must either not send DE 23 or send DE 23 with the default value "000".
DE 35	For a chip transaction, acquirers must provide Track 2 Equivalent Data (tag '57') if the data object was present on the ICC. If this data object is not present in the chip, DE 35 must not be sent. When the Magnetic Stripe was used as technical fallback, DE 35 if present, must contain the actual Track 2 data from the Magnetic Stripe.

² EMV states that terminals must be able to support scripts of 128 bytes, including the tag '71' or '72' and the length. As a result, the maximum length of the value of the script is 126 bytes. Therefore, issuers must send scripts of no more than 126 bytes in length, even though the networks support 127 bytes.

Data element	Requirement
DE 37	Mastercard recommends that the value of the Transaction Sequence Counter (EMV tag '9F41') is included in subfield 2 of DE 37, right-justified, left-padded, with zeroes.
DE 52 and DE 53	These data elements contain PIN-related data. Both DE 52 and DE 53 must be provided if the CVM was Online PIN, where the security format of the PIN Block is double encryption, the encrypted PIN Block Key is present in a subfield of DE 48. They must not be provided if any other CVM was used. DE 53 is not supported on all Mastercard Network interfaces.

IPM Clearing messages from Acquirer requirements

The following table provides a summary of the impacts of chip on the clearing messages for acquirers.

Table 14: Acquirer requirements for First Presentment/1240 Messages

Data element	Requirement
DE 22	Refer to the DE 22 Point-of-Service Data Code in section "DE 22 Point of Service Data code in the clearing."
DE 23	If DE 22 = "05x" or "07x" and if the Application PAN sequence number tag '5F34' is present on the chip, the acquirer must send DE 23 containing the Application PAN sequence number tag '5F34'. If DE 22 = "05x" or "07x" and the Application PAN sequence number tag '5F34' is not present on the chip, the acquirer must either not send DE 23 or send DE 23 with the default value "000".
DE 40	Service code on the card. Acquirers are strongly recommended to populate this field. This field is needed if the values of the Interchange Rate Designator (IRD) are "33" or "34".
DE 55	DE 55 (ICC System Related Data) must be provided. If present, DE 55 must be TLV encoded and must contain the information, mandatory and recommended.
PDS 0023	Indicates the CAT level.

DE 22 Point of Service Data code in the clearing

DE 22 (Point of Service Data) indicates to the issuer the capabilities of the terminal and the circumstances of the transaction, for example, whether the terminal was PIN-capable and whether the chip or the Magnetic Stripe was read.

Refer to the *IPM Clearing Formats* manual for a full description of this data element and its subfields. This section gives additional information relevant to chip transactions.

DE 22, subfield 8 (Cardholder Authentication Method) and subfield 9 (Cardholder Authentication Entity) are used in combination to indicate which method of CVM was used.

Possible values for DE 22, subfield 8 are shown in the following table.

Table 15: DE 22 Subfield 8 values

Name	Value	Value description
Cardholder Authentication Method	0	Not authenticated
	1	PIN
	2	Electronic Signature analysis
	5	Manual Signature verification
	6	Other manual verification, such as driver's license number
	9	Unknown, data not available
	S	Other systematic verification

Possible values for DE 22 subfield 9 are shown in the following table.

Table 16: DE 22 Subfield 9 values

Name	Value	Value description
Cardholder Authentication Entity	0	Not authenticated
	1	ICC - Offline PIN
	2	Card acceptance device
	3	Authorizing agent - Online PIN
	4	Merchant/card acceptor Signature
	5	Other

Name	Value	Value description
	9	Unknown, data not available

When PIN is used, subfield 8 is set to "1". Subfield 9 then indicates whether the PIN check was performed online or offline. The following table summarizes these settings.

Table 17: Cardholder Verification Method (CVM)

DE 22 subfield	Offline PIN	Online PIN	Signature	No CVM
Subfield 8	1	1	5	0
Subfield 9	1	3	4	0

Digital Secure Remote Payments

Digital Secure Remote Payment transactions contain dynamic data, also referred to as cryptograms, which are generated by the payment application on the mobile device to secure the transaction.

Cryptograms may be received in the following three forms, depending on the capabilities of the merchant and acquiring systems that have captured the transaction:

- Full EMV data - In this case, the full EMV data, carried in DE 55, is sent in the transaction messages. Mastercard has introduced a new Point-of-Service (POS) Entry Mode for these transactions (POS Entry Mode of "09").
- UCAF data - In this case, the EMV cryptogram and a subset of the EMV data are carried in the Universal Cardholder Authentication Field (UCAF) in DE 48, SE 43. These transactions are coded as e-commerce transactions (POS entry mode of "81").
- DPC - DE 104, subelement 001 (Digital Payment Cryptogram) is where the DSRP Cryptogram will be sent to be processed and verified.

The issuer must be prepared to receive authorization or financial transaction request messages containing the dynamic data and be prepared to verify the cryptograms.

For a detailed description of the system impacts of Digital Secure Remote Payment, on acquirers, refer to the *Digital Secure Remote Payment, Acquirer Implementation Guide*.

Payment scheme specific data objects

The coding of primitive context-specific class data objects in the range '9F50' to '9F7F' is reserved for the payment systems. This section lists Mastercard defined data objects used from this range that are between the card and the terminal for contactless transactions.

Acquirers must ensure that data carried in tags from the payment-specific range is interpreted for Mastercard transactions according to the definitions given in this section.

All length indications are given in bytes. Data object formats are binary (b) or alphanumeric (an).

R	All	CT CL	RA355.15	For Mastercard transactions, acquirers must use the Mastercard format definition for data objects carried in tags from the payment-specific range.
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Application capabilities information

Tag	'9F5D'
Length	3
Format	b
Description	The Application Capabilities Information is an optional data object included in the File Control Information Template of the contactless card. It lists various card features beyond regular payment and is coded as defined in the next table.

Byte	Bit	Description
Byte 1	b8-5	Version Number
		0000: VERSION 0
		Other values: RFU
	b4-1	Data Storage Version Number
		0000: DATA STORAGE NOT SUPPORT
		0001: VERSION 1
		0010: VERSION 2
		Other values: RFU

Byte	Bit	Description
Byte 2	b8-4	RFU
	b3	Support for field off detection
	b2	Support for balance reading
	b1	CDA Indicator
		0: CDA SUPPORTED AS IN EMV
		1: CDA SUPPORTED OVER TC, ARQC AND AAC
Byte 3	b8-1	SDS Scheme Indicator
		00000000: Undefined SDS configuration
		00000001: All 10 tags 32 bytes
		00000010: All 10 tags 48 bytes
		00000011: All 10 tags 64 bytes
		00000100: All 10 tags 96 bytes
		00000101: All 10 tags 128 bytes
		00000110: All 10 tags 160 bytes
		00000111: All 10 tags 192 bytes
		00001000: All SDS tags 32 bytes except '9F78', which is 64 bytes
		Other values: RFU

Terminal Risk Management Data

Terminal Risk Management Data is an EMV data object with tag '9F1D', whose format is application-specific. It is defined as shown in this section for the Mastercard contactless payment application specified in *Mastercard Contactless Reader Specification v3.0* or later or *EMVCo Book C-2*.

It must be stored in the TLV database of the reader when the contactless Kernel is instantiated, with the value as indicated in the following table. The mechanism for doing this is implementation-specific.

R	All	CL	RA207.14	Contactless readers that comply with <i>Mastercard Contactless Reader Specification v3.0</i> or later or <i>EMVCo Book C-2</i> must support the Terminal Risk Management Data data object.
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Terminal Risk Management Data is defined in the following table, and its value is the same for all transactions below and above the CVM Limit.

Tag	'9F1D'
Length	8
Format	b
Description	Application-specific value used by the contactless card or payment device for risk management purposes. All RFU bits must be set to zero.

Byte	Bit	Meaning	Value
1	8	Restart supported	0
	7	Enciphered PIN verified online (Contactless)	0/1
	6	Signature (Contactless)	0/1
	5	Enciphered PIN verification performed by ICC (Contactless)	0
	4	No CVM required (Contactless)	0/1
	3	CDCVM (Contactless)	0/1
	2	Plaintext PIN verification performed by ICC (Contactless)	0
	1	Present and Hold supported	0
2	8	CVM Limit Exceeded	0
	7	Enciphered PIN verified online (Contact)	0/1
	6	Signature (Contact)	0/1
	5	Enciphered PIN verification performed by ICC (Contact)	0/1
	4	No CVM required (Contact)	0/1
	3	CDCVM (Contact)	0
	2	Plaintext PIN verification performed by ICC (Contact)	0/1

Byte	Bit	Meaning	Value
	1	RFU	0
3	8	Magnetic Stripe mode contactless transactions not supported	0/1
	7	EMV mode contactless transactions not supported	0/1
	6	CDCVM without CDA supported	0/1
	5-1	RFU	00000
4	8	CDCVM bypass requested	0/1
	7	CVM exempt	0/1
	6-1	RFU	000000
5	8-1	RFU	00000000
6	8-1	RFU	00000000
7	8-1	RFU	00000000
8	8-1	RFU	00000000

Third Party Data

The Third Party Data contains proprietary information from a third party and is coded as shown in the following table.

Tag	'9F6E'
Length	5-32
Format	b
Description	If present in the card, the Third Party Data is returned in the File Control Information Template. The Device Type subfield is present when the most significant bit of byte 1 of the Unique Identifier is set to 0b. In this case, the maximum length of the Proprietary Data field is 26 bytes. Third Party Data may be used to communicate the Device Type to the terminal, even when there is no Proprietary Data being used. In this case, a static default value of '0000' for the Unique Identifier is used.

Table 18: Third Party Data format

Data field	Length	Format	Value
Country Code	2	n 3	Country Code according to [ISO 3166-1]
Unique Identifier	2	b	Value assigned by Mastercard or '0000' if Proprietary Data not used
Device Type	0 or 2	an	For a list of values, refer to the <i>Customer Interface Specification</i> manual.
Proprietary Data	1-26 or 28	b	Determined by issuer or third party Default value of '00' if field not used

Product Extension

The Third Party data object may be used to carry specific product information to be optionally used by the terminal in processing transactions.

The Proprietary Data field can be used to store data that is not specific to a third party but instead to a type of product. This extra data formatted as defined in the table is called a Product Extension.

The value '0003' or '8003' of the Unique Identifier within the Third Party Data indicate that a Mastercard defined Product Extension is present in the Proprietary Data in the "Use of Proprietary Data with Product Extension" table.

The value '0003' is used when the Device Type indicator is present in bytes 5 and 6 of the Third Party Data and the value '8003' is used when the Device Type indicator is not present.

The first 2 bytes of the Product Extension contain the Product Identifier. The remaining bytes contain the Product Control Information. The structure and use of this Product Control Information is defined for each product.

The Product Identifier value indicates to the terminal if the subsequent information in the Product Control Information is relevant to the environment.

The Proprietary Data, when a Product Extension is used, is structured as in the following table.

Table 19: Use of Proprietary Data with Product Extension

Data field	Length	Format
Product Identifier	2	RFU
Product Control Information	0-24 or 26	Defined for each product

Fleet Cards

When using the Product Extension for Fleet, a Product Identifier has the value '0001' and the Proprietary Data is structured as in the following table.

Table 20: Product Extension for Fleet Cards

Byte	Bit	Meaning
1-2		Product Identifier ('0001')
3		Product Restriction Code
	8-3	RFU
	2	Good for fuel only
	1	Good for fuel and other products
4		Product Type Code
	8-5	RFU
	4	Prompt for Odometer
	3	Prompt for Driver Number
	2	Prompt for Vehicle Number
	1	Prompt for ID Number
5		Card Type
	8	Driver card
	7	Vehicle card
	6-1	RFU
6-8		RFU

Transit

When using the Product Extension for Transit, a Product Identifier has the value '0002' and the Proprietary Data is structured as in the following table.

Table 21: Product Extension for Transit

Byte	Bit	Meaning
1-2		Product Identifier ('0002')
3	8	Deferred Authorization Not Supported
	7-1	RFU
4	8-1	Concession Code: '01' - Senior Citizen, potentially eligible for senior citizen discounts '02' - Student, potentially eligible for student-based discounts '03' - Active military and veterans, potentially eligible for service member discounts '04' - Low Income household, potentially eligible for means-based discounts '05' - Disability, eligible for paratransit and other disability discounts '06' - Minor Child, potentially enables free travel for under 16 and young kids '07' - Transit Staff, potentially enables free travel for transit staff '08' - City, Government, and Preferred Employees, potentially enables discounted travel for federal and preferred employees Other values RFU
5	8-1	RFU

Issuers not wanting their cards to be used on transit systems should set "Deferred Authorization Not supported" (byte 3, bit 8) of the Product Extension for Transit.

BP	MC	CL	BI234.22	Issuers should set "Deferred Authorization Not supported" (byte 3, bit 8) of the Product Extension for Transit on cards that are not allowed to be used on transit systems.
	MS			

Appendix A Data dictionary

This appendix describes the format and contents of data elements defined by EMV and Mastercard.

Data dictionary.....	333
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Data dictionary

The table in this section is based on the equivalent table in *EMV Book 3, Application Specification*.

In addition to the information given in EMV®, the following table contains several Mastercard® specific data elements and also indicates whether the presence of each data item is optional, mandatory, or conditional.

Issuers, acquirers, or networks must not use tags defined by EMV as reserved for payment systems (in the range '9F50' to '9F7F') for proprietary purposes.

R	The coding of primitive context-specific class data objects in the range '9F50' to '9F7F' is reserved for Mastercard and must not be used by issuers, acquirers, or networks.
RA175.11	

Table 22: Legend

Code	Description
-	Data Object that does not require configuration.
M	If the source of this data element is the ICC, this data element is mandatory. If the source of this data element is the terminal, the terminal must supply a correct value in the format fixed by EMV if the card requests the data element in a Data Object List (DOL).
O	This data element is optional. If the source of this data element is the terminal, the terminal is allowed to send a value containing binary zeros, which means that the data is not supported by the terminal if the card requests the data element in a DOL.
M1	This data element is mandatory for terminals that support offline CAM.
M2	This data element is mandatory for ICCs that implement SDA.
M3	This data element is mandatory for ICCs that implement DDA.
M4	This data element is mandatory for ICCs that implement CDA.
M(CL)	Mandatory on contactless interface
M(CT)	Mandatory on the contact interface
M(CMAG)	Mandatory if Magnetic Stripe mode transactions supported. Otherwise not present.
M(MAS4C)	Mandatory for Mastercard Authentication Solutions for Chip (MAS4C).
M(PPSE)	Mandatory in PPSE entry for contactless interface.
M(v3,C-2)	Mandatory for Mastercard Contactless Reader Specification v3.0 or EMV Book C-2. Otherwise not present.
C(CT)	Conditional on the contact interface

Code	Description
C(PSE)	Must be present if PSE is present on the contact interface.
C(DS)	Must be present if Data Storage is enabled. M/Chip Advance only.
C(TORN)	Must be present if support for torn transaction recovery is enabled. M/Chip Advance only.
C	Conditional data elements are not present unless specific conditions are met, in which case the data element becomes mandatory.
a	Alphabetic data elements contain a single character for each byte. The permitted characters are alphabetic only (a to z and A to Z, uppercase and lowercase).
an	This data element is mandatory for terminals that Alphanumeric data elements contain a single character for each byte. The permitted characters are alphabetic (a to z and A to Z, uppercase and lowercase) and numeric (0 to 9).
ans	Alphanumeric Special data elements contain a single character for each byte. The permitted characters and their coding are shown in the Common Character Set table in <i>EMV Version 4.3 Book 4 Annex B</i> , except for Application Preferred Name, where the characters supported are the non-control characters defined in the ISO 8859 part designated in the Issuer Code Table Index associated with the Application Preferred Name.
b	These data elements consist of either unsigned binary numbers or bit combinations that are defined elsewhere in the specification. Binary example: The Application Transaction Counter (ATC) is defined as "b" with a length of 2 bytes. An ATC value of 19 is stored as Hex '00 13'. Bit combination example: Terminal Capabilities is defined as "b" with the format shown in Section 1.4 of <i>EMV2000 Version 4.3 Book 3</i>.
cn	Compressed Numeric data elements consist of 2 numeric digits (having values in the range Hex '0' to '9') for each byte. These data elements are left-justified and padded with trailing hexadecimal 'F's.
n	Numeric data elements consist of 2 numeric digits (having values in the range Hex '0' to '9') for each byte. These digits are right-justified and padded with leading hexadecimal zeros. Other specifications sometimes refer to this data format as Binary Coded Decimal (BCD) or unsigned packed. Example: Amount Authorized (Numeric) is defined as "n 12" with a length of 6 bytes. A value of "12345" is stored in Amount Authorized (Numeric) as Hex '00 00 00 01 23 45'.
var.	Variable data elements are variable length and may contain any bit combination. Additional information on the formats of specific variable data elements is available elsewhere.

When data is moved from one entity to another, for example, card to terminal, it must always be passed in the order from high order to low order, regardless of how it is stored internally. The same rule applies when concatenating data. The length of data in the following table is expressed in bytes.

Terminals and cards must support the mandatory data. They must support the optional and conditional data, where applicable. The values of the Terminal and ICC data elements are the responsibility of the acquirer and issuer, respectively.

Table 23: Data dictionary

Data object	Description	Format	Length	Source	Tag (in hex)	Presence
Account Type	Indicates the Type of Account selected on the terminal.	n	2	Terminal	5F57	O
Acquirer Identifier	Uniquely identifies the acquirer within each payment system.	n 6-11	6	Terminal	9F01	O
Additional Terminal Capabilities	Indicates the data input and output capabilities of the terminal.	b	5	Terminal	9F40	M
Amount, Authorized (Binary)	Authorized amount of the transaction, excluding adjustments.	b	4	Terminal	81	M
Amount, Authorized (Numeric)	Authorized amount of the transaction, excluding adjustments.	n 12	6	Terminal	9F02	M
Amount, Other (Binary)	Secondary amount associated with the transaction representing a cash back amount.	b	4	Terminal	9F04	O
Amount, Other (Numeric)	Secondary amount associated with the transaction representing a cash back amount.	n 12	6	Terminal	9F03	O
Amount, Reference Currency	Authorized amount expressed in the reference currency.	b	4	Terminal	9F3A	O

Data object	Description	Format	Length	Source	Tag (in hex)	Presence
Application Capabilities Information	Lists a number of card features beyond regular payment.	b	3	ICC (template 'BF0C')	9F5D	O
Application Cryptogram	Cryptogram returned by the ICC in response to the GENERATE AC command.	b	8	ICC (template '77' or '80')	9F26	M
Application Currency Code	Indicates the currency in which the account is managed in accordance with ISO 4217. Used by the terminal if CVM Condition Codes '06', '07', '08', or '09' is used.	n 3	2	ICC (template '70' or '77')	9F42	C
Application Currency Exponent	Indicates the implied position of the decimal point from the right of the amount represented in accordance with ISO 4217.	n 1	1	ICC (template '70' or '77')	9F44	O
Application Dedicated File (ADF) Name	Identifies the application as described in ISO/IEC 7816-5.	b	5-16	ICC (template '61')	4F	M(PPSE) C(PSE)
Application Effective Date	Date from which the application may be used. The date is expressed in the YYMMDD format. For Mastercard applications, the date always reads 20YYMMDD.	n 6 YYMMDD	3	ICC (template '70' or '77')	5F25	O

Data object	Description	Format	Length	Source	Tag (in hex)	Presence
Application Expiration Date	Date after which application expires. The date is expressed in the YYMMDD format. For Mastercard applications, the date always reads 20YYMMDD.	n 6 YYMMDD	3	ICC (template '70' or '77')	5F24	M
Application File Locator (AFL)	Indicates the location (Short File Identifier [SFI] range of records) of the Application Elementary Files (AEFs) associated with a particular AID on the chip and read by the terminal during a transaction.	var.	var. up to 252	ICC (template '77' or '80')	94	M

Data object	Description	Format	Length	Source	Tag (in hex)	Presence
Application Identifier (AID)	Identifies the application as described in ISO/IEC 7816-5. Refer to chapter "Data Requirements" for possible AIDs for Mastercard applications. Any Application Definition File (ADF) or Directory Definition File (DDF) in the ICC is referenced by its DF name. Each DF name must be unique within a given card. EMV allows a DF name to be longer than, but begin with, the corresponding AID. The AID of a Mastercard application must match the DF name of this application on the ICC exactly. For this reason, the DF Name is used to identify the AID in DE 55.	b	5-16	Terminal		M
Application Interchange Profile	Indicates the capabilities of the card to support specific functions in the application.	b	2	ICC (template '77' or '80')	82	M

Data object	Description	Format	Length	Source	Tag (in hex)	Presence
Application Label	Name associated with the AID in accordance with ISO/IEC 7816-5.	ans with the special character limited to space	1-16	ICC (template '61' or 'A5') Terminal	50	M
Application Life Cycle Data	This data is mandatory for M/Chip Advance applications and must follow the requirements of the card application specification. On all other card platforms, it is optional.	b	48	ICC (template - none)	9F7E	C
Application Preferred Name	Preferred name associated with the AID, for example, a domestic debit brand name.	ans	1-16	ICC (template '61' or 'A5')	9F12	O
Application Primary Account Number (PAN)	Valid cardholder account number.	cn, var. up to 19	var. up to 10	ICC (template '70' or '77')	5A	M
Application Primary Account Number (PAN) Sequence Number	Identifies and differentiates cards with the same PAN.	n 2	1	ICC (template '70' or '77')	5F34	M
Application Priority Indicator	Indicates the priority of a given application or group of applications in a directory. It is mandated on all contactless cards and multi-application contact cards.	b	1	ICC (template '61' or 'A5')	87	M (CL) C (CT)

Data object	Description	Format	Length	Source	Tag (in hex)	Presence
Application Reference Currency	1-4 currency codes used between the terminal and the ICC when the Transaction Currency Code is different from the Application Currency Code. Each code is three digits in accordance with ISO 4217.	n 3	2-8	ICC (template '70' or '77')	9F3B	O
Application Currency Reference Exponent	Indicates the implied position of the decimal point from the right of the amount, for each of the 1-4 reference currencies represented in accordance with ISO 4217.	n 1	1-4	ICC (template '70' or '77')	9F43	O
Application Selection Registered Proprietary Data	A variable length data object with binary format, intended for proprietary use. The value field is a series of IDLV blocks in the form ID1L1V1ID2L2V2 where each ID is a unique identifier governed and allocated by EMVCo.		var. up to 255	ICC Contact (template '73' and 'BF0C') Contactless (template '61' and 'BF0C')	9F0A	O M ³

³ For cards issued in the EU and Norway.

Data object	Description	Format	Length	Source	Tag (in hex)	Presence
Application Selection Indicator	For an application in the ICC to be supported by an application in the terminal, the Application Selection Indicator indicates whether the associated AID in the terminal must match the AID in the card exactly, including the length of the AID, or only up to the length of the AID in the terminal. There is only one Application Selection Indicator for each AID supported by the terminal.	Terminal dependent	Terminal dependent	Terminal		M
Application Template	Contains one or more data objects relevant to an application directory entry in accordance with ISO/IEC 7816-5.	b	var. up to 252	ICC (template '70')	61	C
Application Transaction Counter (ATC)	Counter maintained by the application in the ICC, incrementing the ATC managed by the ICC.	b	2	ICC (template '77' or '80')	9F36	M
Application Usage Control	Indicates issuer's specified restrictions on the geographic use and services allowed for the application.	b	2	ICC (template '77' or '80')	9F07	M

Data object	Description	Format	Length	Source	Tag (in hex)	Presence
Application Version Number	Version number assigned by the payment system for the application.	b	2	ICC (template '70' or '77')	9F08	M
Application Version Number	Version number assigned by the payment system for the application.	b	2	Terminal	9F09	M
Authenticated Application Data	Allows issuer proprietary data provided by the card to the terminal to be communicated securely to the issuer host.	b	Var-127	ICC	9F60	O
Authorization Code	Value generated by the issuer for an approved transaction.	ans	6	Issuer	89	O
Authorization Response Code	Code that defines the disposition of a message.	an	2	Issuer/ Terminal	8A	M
Bank Identifier Code (BIC)	Uniquely identifies a bank as defined in ISO 9362.	var.	8 or 11	ICC (template '73' and 'BF0C')	5F54	O
Card Risk Management Data Object List 1 (CDOL1)	The Card Risk Management Data Object List 1 (CDOL1) is a data object in the ICC that provides the terminal with a list of data objects that must be passed from the terminal to the ICC in the first GENERATE AC command.	b	var. up to 252	ICC (template '70' or '77')	8C	M

Data object	Description	Format	Length	Source	Tag (in hex)	Presence
Card Risk Management Data Object List 2 (CDOL2)	The Card Risk Management Data Object List 2 (CDOL2) is a data object in the ICC that provides the terminal with a list of data objects that must be passed from the terminal to the ICC in the second GENERATE AC command.	b	var. up to 252	ICC (template '70' or '77')	8D	M
Card Status Update (CSU)	Contains data sent to the ICC to indicate whether the issuer approves or declines the transaction and to indicate actions specified by the issuer. Transmitted to the card in Issuer Authentication Data.	b	4	Issuer		O
Card Verification Results (CVR)	Informs the issuer about the transaction context as part of the Issuer Application Data. Used as part of decision-making process in card risk management.	b	6	ICC (part of Issuer Application Data (tag '9F10'))	9F52	-

Data object	Description	Format	Length	Source	Tag (in hex)	Presence
Cardholder Name	Indicates cardholder name in accordance with ISO 7813. The cardholder name as encoded in Track 1 of the Magnetic Stripe if there is a Track 1 on the Magnetic Stripe. This data element is conditional on the presence of the Magnetic Stripe's Track 1. If it is present in the Magnetic Stripe's Track 1, the value in tag '5F20' is normally identical to that. However, for card programs where the cardholder name is not known, for example, anonymous prepaid cards, on the Magnetic Stripe and in the chip, a dummy value can be used, for example, "{space}"/" in Track 1 and "MASTERCARD/CARD" in the chip.	ans 2-26	2-26	ICC (template '70' or '77')	5F20	C
Cardholder Name Extended	Indicates the whole cardholder name when greater than 26 characters using the same coding convention as in ISO 7813.	ans 27-45	27-45	ICC (template '70' or '77')	9F0B	O

Data object	Description	Format	Length	Source	Tag (in hex)	Presence
Cardholder Verification Method (CVM) List	Identifies the methods of verification of the cardholder supported by the application.	b	10-252	ICC (template '70' or '77')	8E	M
Cardholder Verification Method (CVM) Results	Indicates the results of the last CVM performed.	b	3	Terminal	9F34	M
Certification Authority Public Key Check Sum (Certification Authority is the Payment System.)	A check value calculated on the concatenation of all parts of the Certification Authority Public Key (RID, Certification Authority Public Key Index, Certification Authority Public Key Modulus, Certification Authority Public Key Exponent) using SHA-1.	b	20	Terminal		O
Certification Authority Public Key Exponent	Value of the exponent part of the Certification Authority Public Key.	b	1 or 3	Terminal		M1
Certification Authority Public Key Index	Identifies the certification authority's public key in conjunction with the RID.	b	1	ICC (template '70' or '77')	8F	M1
Certification Authority Public Key Modulus	Value of the modulus part of the Certification Authority Public Key.	b	N _{CA} (up to 248)	Terminal		M1

Data object	Description	Format	Length	Source	Tag (in hex)	Presence
Command Template	Identifies the data fields of a command message.	b	var.	Terminal	83	M
Cryptogram Information Data	Indicates the type of cryptogram and the actions to be performed by the terminal.	b	1	ICC (template '77' or '80')	9F27	M
CVC 3 (Track 1)	A 2-byte cryptogram returned by the Card in the response to the COMPUTE CRYPTOGRAPHIC CHECKSUM command generated based on Track 1 Data during a Magnetic Stripe mode contactless transaction.	b	2	ICC (template '77')	9F60	-
CVC 3 (Track 2)	A 2-byte cryptogram returned by the Card in the response to the COMPUTE CRYPTOGRAPHIC CHECKSUM command generated based on Track 2 Data during a Magnetic Stripe mode contactless transaction.	b	2	ICC (template '77')	9F61	-

Data object	Description	Format	Length	Source	Tag (in hex)	Presence
Data Authentication Code (DAC)	An issuer-assigned value that is retained by the terminal during the verification process of the Signed Static Application Data.	b	2	ICC	9F45	M2 ⁴
Data Recovery Data Object List (DRDOL)	List of data objects (tag and length) to be passed to the ICC in the RECOVER AC command data.	b	3	ICC (template '70' or '77')	9F51	C(TORN)
Data Storage Data Object List (DSDOL)	For M/Chip Advance cards using Data Storage, it contains the list of tags and lengths of data storage related data in the GENERATE AC command data.	b	12	ICC	9F5B (template '70' or '77')	C(DS)
Dedicated File (DF) Name	Identifies the name of the DF as described in ISO/IEC 7816-4.	b	5-16	ICC (template '6F')	84	M

⁴ Not applicable for MCBP implementations.

Data object	Description	Format	Length	Source	Tag (in hex)	Presence
Default Dynamic Data Authentication Data Object List (DDOL)	DDOL to be used for constructing the INTERNAL AUTHENTICATE Terminals contain a Default Dynamic Data Authentication Data Object List (DDOL), which is used if the DDOL is not present in the card. The Default DDOL for a Mastercard application must only contain the Unpredictable Number (tag '9F37').	b	var.	Terminal		M3
Default Transaction Certificate Data Object List (TDOL)	List of data objects (tag and length) to be used by the terminal in generating the TC Hash Value, if the TDOL in the card is not present. Not present for Mastercard applications.	b	var.	Terminal		
Directory Definition File (DDF) Name	Identifies the name of a DF associated with a directory.	b	5-16	ICC (template '61')	9D	C
Directory Discretionary Template	Issuer discretionary part of the directory in accordance with ISO/IEC 7816-5.	var.	var. up to 252	ICC (template '61')	73	O

Data object	Description	Format	Length	Source	Tag (in hex)	Presence
Dynamic Data Authentication Data Object List (DDOL)	List of data objects (tag and length) to be passed to the ICC in the INTERNAL AUTHENTICATE command.	b	var. up to 252	ICC (template '70' or '77')	9F49	M3
Enciphered Personal Identification Number (PIN) Data	Transaction PIN enciphered at the PIN pad for online verification or for offline verification if the PIN pad and Interface Device (IFD) are not a single integrated device.	b	8	Terminal		O
File Control Information (FCI) Issuer Discretionary Data	Issuer discretionary part of the FCI.	var.	var. up to 222	ICC (template 'A5')	BF0C	O
File Control Information (FCI) Proprietary Template	Identifies the data object proprietary to this specification in the FCI template in accordance with ISO/IEC 7816-4.	var.	var.	ICC (template '6F')	A5	M
File Control Information (FCI) Template	Identifies the FCI template, in accordance with ISO/IEC 7816-4.	var.	var. up to 252	ICC (template - none)	6F	M
Integrated Circuit Card (ICC) Dynamic Number	Time-variant number generated by the ICC, to be captured by the terminal.	b	2-8	ICC	9F4C	M3 M4

Data object	Description	Format	Length	Source	Tag (in hex)	Presence
ICC PIN Encipherment Public Key Certificate	ICC PIN Encipherment Public Key certified by the issuer. This data item is mandatory if the chip supports Offline PIN encipherment with a specific ICC PIN encipherment Public Key.	b	N _I	ICC (template '70' or '77')	9F2D	C
ICC PIN Encipherment Public Key Exponent	ICC PIN Encipherment Public Key Exponent used for PIN encipherment. This data item is mandatory if the chip supports Offline PIN encipherment with a specific ICC PIN encipherment Public Key.	b	1 or 3	ICC (template '70' or '77')	9F2E	C

Data object	Description	Format	Length	Source	Tag (in hex)	Presence
ICC PIN Encipherment Public Key Remainder	Remaining digits of the ICC PIN Encipherment Public Key Modulus. This data item is mandatory if the chip supports Offline PIN encipherment with a specific ICC PIN encipherment Public Key and if the length of the ICC PIN Encipherment Public Key Modulus (N_{PE}) plus 42 exceeds the length of the Issuer Public Key Modulus (N_I), that is, if $N_{PE}+42 > N_I$.	b	$N_{PE} - N_I + 42$	ICC (template '70' or '77')	9F2F	C
Integrated Circuit Card (ICC) Public Key Certificate	ICC Public Key certified by the issuer.	b	N_I	ICC (template '70' or '77')	9F46	M3 M4
Integrated Circuit Card (ICC) Public Key Exponent	ICC Public Key Exponent used for the verification of the Signed Dynamic Application Data.	b	1 to 3	ICC (template '70' or '77')	9F47	M3 M4

Data object	Description	Format	Length	Source	Tag (in hex)	Presence
Integrated Circuit Card (ICC) Public Key Remainder	Remaining digits of the ICC Public Key Modulus. This data item is mandatory if the chip supports DDA or CDA or PIN encipherment with the ICC Public key, and if the length of the ICC Public Key Modulus (N_{IC}) plus 42 exceeds the length of the Issuer Public Key Modulus (N_I), that is, if $N_{IC}+42 > N_I$.	b	$N_{IC} - N_I + 42$	ICC (template '70' or '77')	9F48	C
Interface Device (IFD) Serial Number	Unique and permanent serial number assigned to the IFD by the manufacturer.	an 8	8	Terminal	9F1E	O
International Bank Account Number (IBAN)	Uniquely identifies the account of a customer at a financial institution as defined in ISO 13616.	var.	var. up to 34	ICC (template '73' and 'BF0C')	5F53	O
Issuer Action Code - Default	Specifies the issue's conditions that cause a transaction to be rejected if it might have been approved online but the terminal was unable to process the transaction online.	b	5	ICC (template '70' or '77')	9F0D	M

Data object	Description	Format	Length	Source	Tag (in hex)	Presence
Issuer Action Code - Denial	Specifies the issuer's conditions that cause the denial of a transaction without attempt to go online.	b	5	ICC (template '70' or '77')	9F0E	M
Issuer Action Code - Online	Specifies the issuer's conditions that cause a transaction to be transmitted online.	b	5	ICC (template '70' or '77')	9F0F	M
Issuer Application Data	Contains proprietary application data for transmission to the issuer in an online transaction.	b	var. up to 32	ICC (template '77' or '80')	9F10	O
Issuer Authentication Data	Data sent to the ICC for online issuer authentication and action. Not generated by Magnetic Stripe grade issuer hosts.	b	8-16	Issuer	91	C
Issuer Authentication Flags	Payment System Specific data used in the Mastercard Authentication Solutions for Chip (MAS4C) to specify the CAP token generation options using either CAP technology or PLA technology.	b	1	ICC (template '70')	9F55	O M(MAS4C)

Data object	Description	Format	Length	Source	Tag (in hex)	Presence
Issuer Code Table Index	Indicates the code table, in accordance with ISO 8859, for displaying the application Preferred Name. If the Issuer Code Table Index is not present, the Application Preferred Name will not be displayed.	n 2	1	ICC (template 'A5')	9F11	O
Issuer Country Code (For the complete list, refer to the <i>Quick Reference Booklet</i> .)	Indicates the country of the issuer in accordance with ISO 3166.	n 3	2	ICC (template '70' or '77')	5F28	M
Issuer Country Code (alpha 2 format)	Indicates the country of the issuer as defined in ISO 3166 (using a 2-character alphabetic code).	a 2	2	ICC (template '73' and 'BF0C')	5F55	O
Issuer Country Code (alpha 3 format)	Indicates the country of the issuer as defined in ISO 3166 (using a 3-character alphabetic code).	a 3	3	ICC (template '73' and 'BF0C')	5F56	O
Issuer Identification Number (IIN)	The number that identifies the major industry and the card issuer and that forms the first part of the Primary Account Number (PAN).	n 6	3	ICC (template '73' and 'BF0C')	42	O

Data object	Description	Format	Length	Source	Tag (in hex)	Presence
Issuer Proprietary Bitmap	Payment System Specific data used in the Mastercard Authentication Solutions for Chip (MAS4C) to determine the compression of the assembled CAP token. Each bit in the IPB determines whether the corresponding bit in the assembled token is included in the CAP token.	b	11-44	ICC (template '70')	9F56	O M(MAS4C)
Issuer Public Key Certificate	Issuer public key certified by a certification authority.	b	N _{CA}	ICC (template '70' or '77')	90	M2 M3 M4
Issuer Public Key Exponent	Issuer public key exponent used for the verification of the Signed Static Application Data.	b	1-3	ICC (template '70' or '77')	9F32	M2 M3 M4

Data object	Description	Format	Length	Source	Tag (in hex)	Presence
Issuer Public Key Remainder	Remaining digits of the issuer Public Key Modulus. This data item is mandatory if the chip supports SDA or DDA or CDA or the enciphered PIN, and if the length of the Issuer Public Key Modulus (N_I) plus 36 exceeds the length of the Certification Authority Public Key Modulus (N_{CA}), that is, if $N_{CA}+36 > N_I$.	b	$N_I - N_{CA} + 36$	ICC (template '70' or '77')	92	C
Issuer Script Command	Contains a command for transmission to the ICC.	b	var. up to 261	Issuer (template '71' or '72')	86	O
Issuer Script Identifier	Identification of the issuer Script.	b	4	Issuer (template '71' or '72')	9F18	O
Issuer Script Results	Indicates the result of the terminal script processing.	b	var.	Terminal		M
Issuer Script Template 1	Contains proprietary issuer data for transmission to the ICC before the second GENERATE AC command.	b	var.	Issuer	71	O
Issuer Script Template 2	Contains proprietary issuer data for transmission to the ICC after the second GENERATE AC command.	b	var.	Issuer	72	O

Data object	Description	Format	Length	Source	Tag (in hex)	Presence
Issuer URL	The URL provides the location of the ICC issuer's library server on the Internet. The Issuer Uniform Resource Locator (URL) points to the Internet location of a brand or financial institution.	ans	var.	ICC (template '73' and 'BF0C')	5F50	O
Kernel Identifier-Terminal	Identifies the kernel used for the transaction.	b	8	Terminal	96	O
Language Preference	1-4 languages stored in order of preference, each represented by two alphabetical characters, in accordance with ISO 639. NOTE: EMVCo strongly recommends that cards be personalized with data object 5F2D coded in lowercase, but that terminals accept the data object whether it is coded in uppercase or lowercase with ISO 639.	an 2	2-8	ICC (template 'A5')	5F2D	O
Last Online Application Transaction Counter (ATC) Register	ATC value of the last transaction that went online.	b	2	ICC (template - none)	9F13	O

Data object	Description	Format	Length	Source	Tag (in hex)	Presence
Log Entry	Provides the SFI of the Transaction Log File and its number of records.	b	2	ICC (template '73' and 'BF0C')	9F4D	O
Log Format	List (in tag and length format) of data objects representing the logged data objects that are passed to the terminal when a transaction log record is read.	b	var.	ICC (template - none)	9F4F	O
Lower Consecutive Offline Limit	Issuer-specified preference for the maximum number of consecutive offline transactions for this ICC application allowed in a terminal with online capability. When Velocity Checking is performed by the terminal during the Terminal Risk Management, the Lower Consecutive Offline Limit is a mandatory data object.	b	1	ICC (template '70' or '77')	9F14	C
Magstripe Application Version Number (Reader)	Version number assigned by Mastercard for the specific Magnetic Stripe mode contactless functionality of the terminal.	b	2	Terminal	9F6D	M

Data object	Description	Format	Length	Source	Tag (in hex)	Presence
Magstripe Application Version Number (Card)	Version number assigned by Mastercard for the specific Magnetic Stripe mode contactless functionality of the card.	b	2	ICC (template '70')	9F6C	O
Maximum Target Percentage to be used for Biased Random Selection	Value used in Terminal Risk Management for random transaction selection.			Terminal		O
Merchant Category Code	Classifies the type of business being done by the merchant, represented in accordance with ISO 8583:1993 for Card Acceptor Business Code.	n 4	2	Terminal	9F15	O
Merchant Custom Data	Proprietary merchant data to be stored in the Transaction Log File.	ans	20	Terminal	9F7C	O
Merchant Identifier	When concatenated with the Acquirer Identifier, uniquely identifies a given merchant.	ans 15	15	Terminal	9F16	O
Merchant Name and Location	Indicates the name and location of the merchant.	ans	var.	Terminal	9F4E	O
Message Type	Indicates whether the batch data capture record is a financial record or advice.	n 2	1	Terminal		M

Data object	Description	Format	Length	Source	Tag (in hex)	Presence
Mobile Support Indicator	Informs the card that the contactless terminal supports extensions for mobile and requires CDCVM.	b	1	Terminal	9F7E	M(v3,C-2)
NATC (Track 1)	Represents the number of digits of the Application Transaction Counter to be included in the discretionary data field of Track 1 Data during a Magnetic Stripe mode contactless transaction.	b	1	(template '70')	9F64	M(CMAG)
NATC (Track 2)	Represents the number of digits of the Application Transaction Counter to be included in the discretionary data field of Track 2 Data during a Magnetic Stripe mode contactless transaction.	b	1	(template '70')	9F67	M
Offline Balance/ Offline Accumulator Balance 1	Represents the amount of offline spending available on the card. If allowed by the application, it is retrieved by the GET DATA command.	n 12	6	ICC (template - none)	9F50	- ⁵

⁵ As supported by the card application.

Data object	Description	Format	Length	Source	Tag (in hex)	Presence
Offline Accumulator Balance 2	Represents the amount of offline spending available on the card. If allowed by the application, it is retrieved by the GET DATA command.	n 12	6	ICC (template - none)	9F58	-
Offline Counter Balance 1	Represents the number of remaining offline transactions as determined by Counter 1. If allowed by the application, it is retrieved by the GET DATA command.	b	1	ICC (template - none)	9F7A	-
Offline Counter Balance 2	Represents the number of remaining offline transactions as determined by Counter 2. If allowed by the application, it is retrieved by the GET DATA command.	b	1	ICC (template - none)	9F59	-
Payment Account Reference (PAR)	Contains the data object associated with an Application PAN. NOTE: Lowercase alphabetic characters are not permitted for the PAR.	an	29	ICC (template '70' or '77')	9F24	O

Data object	Description	Format	Length	Source	Tag (in hex)	Presence
PCVC3 (Track 1)	Indicates to the terminal the positions in the discretionary data field of the Track 1 Data where the CVC 3 (Track1) digits must be copied during a Magnetic Stripe mode contactless transaction.	b	6	ICC (template '70')	9F62	M
PCVC3 (Track 2)	Indicates to the terminal the positions in the discretionary data field of the Track 2 Data where the CVC 3 (Track 2) digits must be copied during a Magnetic Stripe mode contactless transaction.	b	2	ICC (template '70')	9F65	M
Personal Identification Number (PIN) PAD Secret Key	Secret key of a symmetric algorithm used by the PIN pad to encipher the PIN and by the card reader to decipher the PIN if the PIN pad and card reader are not integrated.			Terminal		
Personal Identification Number (PIN) Try Counter	Number of PIN tries remaining. Required if Offline PIN is supported.	b	1	ICC (template - none)	9F17	C

Data object	Description	Format	Length	Source	Tag (in hex)	Presence
POI information	Contactless terminals can optionally send this to the contactless card, to aid appropriate application selection. Applies to contactless interface only. See EMV Contactless Book B Entry Point Specification and EMV specification bulletin No. 299.	b	var. up to 64	Terminal	8B	O
Point-of-Service (POS) Entry Mode	Indicates the method used for entering the PAN in accordance with the first two digits of the ISO 8583:1987.	n 2	1	Terminal	9F39	M
Processing Options Data Object List (PDOL)	Contains a list of terminal-resident data objects (tags and lengths) needed by the ICC in processing the GET PROCESSING command.	b	var.	ICC (template 'A5')	9F38	O

Data object	Description	Format	Length	Source	Tag (in hex)	Presence
Proprietary Authentication Data	Contains issuer data for transmission to the card in the Issuer Authentication Data of an online transaction. For a cryptogram defined by the Common Core Definitions with a Cryptogram Version of 4, the Proprietary Authentication Data object shall be 0 bytes long. The only Cryptogram Version currently defined for the Common Core Definitions is 4.	b	var. up to 8	Issuer		O
Protected Data Envelope 1	The Protected Data Envelopes contain proprietary information from the issuer, payment system or third party. The Protected Data Envelope can be retrieved with the GET DATA command. Updating the Protected Data Envelope with the PUT DATA command requires secure messaging.	b	var. up to 192	ICC (template - none)	9F70	O

Data object	Description	Format	Length	Source	Tag (in hex)	Presence
Protected Data Envelope 2	Same as Protected Data Envelope 1.	b	var. up to 192	ICC (template - none)	9F71	O
Protected Data Envelope 3	Same as Protected Data Envelope 1.	b	var. up to 192	ICC (template - none)	9F72	O
Protected Data Envelope 4	Same as Protected Data Envelope 1.	b	var. up to 192	ICC (template - none)	9F73	O
Protected Data Envelope 5	Same as Protected Data Envelope 1.	b	var. up to 192	ICC (template - none)	9F74	O
PUNATC (Track 1)	Indicates to the contactless reader the positions in the discretionary data field of Track 1 Data where the Unpredictable Number (Numeric) digits and Application Transaction Counter digits have to be copied.	b	6	ICC (template '70')	9F63	M
PUNATC (Track 2)	Indicates to the contactless reader the positions in the discretionary data field of Track 2 Data where the Unpredictable Number (Numeric) digits and Application Transaction Counter digits have to be copied.	b	2	ICC (template '70')	9F66	M

Data object	Description	Format	Length	Source	Tag (in hex)	Presence
READ RECORD Response Message Template	Contains the contents of the record read. (Mandatory for SFIs 1-10. Response messages for SFIs 11-30 are outside the scope of EMV, but may use template '70'.)	var.	var. up to 252	ICC (template - none)	70	O
Response Message Template Format 1	Contains the data objects (without tags and lengths) returned by the ICC in response to a command.	var.	var.	ICC (template - none)	80	M
Response Message Template Format 2	Contains the data objects (with tags and lengths) returned by the ICC in response to a command.	var.	var.	ICC (template - none)	77	M
Service Code	Service code as defined on Tracks 1 and 2.	n 3	2	ICC (template '70' or '77')	5F30	O
Short File Identifier (SFI)	Identifies the SFI to be used in the commands related to a given AEF or DDF. The SFI data object is a binary field with the three high-order bits set to zero.	b	1	ICC (template 'A5')	88	M
Signed Dynamic Application Data	Digital Signature on critical application parameters for DDA or CDA.	b	N _{IC}	ICC (template '77' or '80')	9F4B	M3 M4

Data object	Description	Format	Length	Source	Tag (in hex)	Presence
Signed Static Application Data	Digital Signature on critical application parameters for SDA.	b	N ₁	ICC (template '70' or '77')	93	M2
Static Data Authentication Tag List	List of tags of primitive data objects defined in this specification for which the value fields must be included in the Signed Static or Dynamic Application Data.		var.	ICC (template '70' or '77')	9F4A	M2 M3 M4
Target Percentage to be used for Random Selection	Value used in terminal risk management for random transaction selection.			Terminal		O
Terminal Action Code - Default	Specifies the acquirer's conditions that cause a transaction to be rejected if it might have been approved online, but the terminal is unable to process the transaction online.	b	5	Terminal		M
Terminal Action Code - Denial	Specifies the acquirer's conditions that cause the denial of a transaction without attempt to go online.	b	5	Terminal		M

Data object	Description	Format	Length	Source	Tag (in hex)	Presence
Terminal Action Code - Online	Specifies the acquirer's conditions that cause a transaction to be transmitted online.	b	5	Terminal		M
Terminal Capabilities	Indicates the card data input, CVM, and security capabilities of the terminal.	b	3	Terminal	9F33	M
Terminal Country Code	Indicates the country of the terminal, represented in accordance with ISO 3166.	n 3	2	Terminal	9F1A	M
Terminal Floor Limit	Indicates the floor limit in the terminal in conjunction with the AID.	b	4	Terminal	9F1B	M
Terminal Identification	Designates the unique location of a terminal at a merchant.	an 8	8	Terminal	9F1C	M
Terminal Risk Management Data	Application-specific value used by the card for risk management purposes.	b	1-8	Terminal	9F1D	O
Terminal Type	Indicates the environment of the terminal, its communications capability, and its operational control.	n 2	1	Terminal	9F35	M
Terminal Verification Results	Status of the different functions from the terminal perspective.	b	5	Terminal	95	M

Data object	Description	Format	Length	Source	Tag (in hex)	Presence
Third Party Data	Contains proprietary information from a third party, issuer country code and device type. Refer to Chapter 5 (Data requirements), section "Payment scheme specific data objects", section "Third Party Data" for complete definition.	b	5-32	ICC (template 'BF0C' or '70' or '61' or '73')	9F6E	O
Threshold Value for Biased Random Selection	Value used in terminal risk management for random transaction selection. If present, the Threshold Value for Biased Random Selection may be set to zero to indicate that transactions below the floor limit must not be selected for random online processing.			Terminal		O
Token Requestor ID	Uniquely identifies the pairing of the Token Requestor with the Token Domain, as defined in the <i>EMV Payment Tokenisation Specification - Technical Framework</i> .	n11	6	ICC (template '70' or '77')	9F19	O

Data object	Description	Format	Length	Source	Tag (in hex)	Presence
Track 1 Data	Contains the data objects of the Track 1 according to (ISO/IEC 7813), excluding start sentinel, end sentinel and LRC.	ans	var. up to 76	ICC (template '70')	56	M(CMAG)
Track 1 Discretionary Data	Discretionary part of Track 1, in accordance with ISO/IEC 7813.	ans	var.	ICC (template '70' or '77')	9F1F	O
Track 2 Data	Contains the data objects of the Track 2 according to ISO/IEC 7813, excluding start sentinel, end sentinel, and LRC.	b	var. up to 19	ICC (template '70')	9F6B	M(CMAG)
Track 2 Discretionary Data	Discretionary part of Track 2 in accordance with ISO/IEC 7813.	cn	var.	ICC (template '70' or '77')	9F20	O
Track 2 Equivalent Data	Contains the data objects of the Track 2.	b	var. up to 19	ICC (template '70' or '77')	57	M
Transaction Amount	Clearing amount of the transaction, including tips and other adjustments.	n 12	6	Terminal		M
Transaction Category Code	This is a data object defined by Mastercard, which indicates the type of transaction being performed.	an	1	Terminal	9F53	O
Transaction Certificate Data (TDOL) Object List	List of data objects (tag and length) to be used by the terminal in generating the TC Hash Value.	b	var. up to 252	ICC (template '70' or '77')	97	O

Data object	Description	Format	Length	Source	Tag (in hex)	Presence
Transaction Certificate (TC) Hash Value	Result of a SHA-1 hash function, specified by ISO/IEC 10118-3. The TC Hash Value puts the burden of hashing transaction data that is signed by the card but not needed by the card for its Card Risk Management on the terminal, and not on the card.	b	20	Terminal	98	M
Transaction Currency Code	Indicates the currency code of the transaction in accordance with ISO 4217.	n 3	2	Terminal	5F2A	M
Transaction Currency Exponent	Indicates the implied position of the decimal point from the right of the transaction amount represented in accordance with ISO 4217.	n 1	1	Terminal	5F36	M
Transaction Date	Local date that the transaction was authorized.	n 6 YYMMDD	3	Terminal	9A	M
Transaction Personal Identification Number (PIN) Data	Data entered by the cardholder to support PIN verification.	b	var.	Terminal	99	O

Data object	Description	Format	Length	Source	Tag (in hex)	Presence
Transaction Reference Currency Code	Code defining the common currency used by the terminal when the Transaction Currency Code is different from the Application Currency Code. This data is present only if the terminal supports the EMV currency conversion by the terminal. Mastercard recommends that terminals do not support this option.	n 3	2	Terminal	9F3C	O
Transaction Reference Currency Conversion	Factor used in the conversion from the Transaction Currency Code to the Transaction Reference Currency Code.	n 8	4	Terminal		O

Data object	Description	Format	Length	Source	Tag (in hex)	Presence
Transaction Reference Currency Exponent	Indicates the implied position of the decimal point from the right of the transaction amount, with the Transaction Reference Currency Code represented in accordance with ISO 4217. This data is only present if the terminal supports the EMV currency conversion by the terminal. Mastercard recommends that terminals do not support this option.	n 1	1	Terminal	9F3D	O
Transaction Sequence Counter	Counter maintained by the terminal that is incremented by one for each transaction.	n 4-8	2-4	Terminal	9F41	M (CT)
Transaction Status Information	Indicates the functions performed in a transaction.	b	2	Terminal	9B	M (CT)
Transaction Time	Local time that the transaction was authorized.	n 6 hhmmss	3	Terminal	9F21	M

Data object	Description	Format	Length	Source	Tag (in hex)	Presence
Transaction Type	Indicates the type of financial transaction, represented by the first two digits of ISO 8583:1987 Processing Code. For the full list of values, refer to the <i>Customer Interface Specification</i> manual.	n 2	1	Terminal	9C	M
Unpredictable Number Data Object List (UDOL)	Specifies the data objects to be included in the data field of the COMPUTE CRYPTOGRAPHIC CHECKSUM command during a contactless transaction.	b	var.	ICC (template '70')	9F69	O
Unpredictable Number	32-bit data object that is unpredictable even with knowledge of previous values.	b	4	Terminal	9F37	M
Unpredictable Number (Numeric)	Unpredictable number generated by the Kernel during a Magnetic Stripe mode contactless transaction. It is passed to the card in the data field of the COMPUTE CRYPTOGRAPHIC CHECKSUM command.	n 8	4	Terminal	9F6A	-

Data object	Description	Format	Length	Source	Tag (in hex)	Presence
Upper Consecutive Offline Limit	Issuer-specified preference for the maximum number of consecutive offline transactions for this ICC application allowed in a terminal without online capability. When Velocity Checking is performed by the terminal during Terminal Risk Management, the Upper Consecutive Offline Limit is a mandatory data object.	b	1	ICC (template '70' or '77')	9F23	C

Appendix B Analysis of DE 55

This appendix provides a bit-level analysis of the mandatory subelements that must be present and the recommended subelements that may be present in DE 55.

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About DE 55 tags

For each tag of DE 55 there is a short description and a definition table containing (a) the name of the tag, (b) the source of the tag, ICC or terminal, (c) the format of the data element, (d) the name of the tag, and (e) the length of the value in bytes.

In addition, there is an analysis of each tag, where needed, and a few examples from M/Chip Advance for contact and contactless. However, this does not replace the published specifications for these card applications. Issuers and acquirers that are interested in the tag analysis of other applications supported by Mastercard should refer to the appropriate documentation.

Mandatory subelements in online request

When DE 55 is present in the Authorization Request/0100 or Financial Transaction Request/0200 message, specific subelements must be included.

Application Cryptogram, '9F26'

The Application Cryptogram contains the cryptogram returned by the ICC in response to the GENERATE AC command.

Name	Source	Format	Tag	Length
Application Cryptogram	ICC	b	'9F26'	8

Cryptogram Information Data, '9F27'

The Cryptogram Information Data indicates the type of cryptogram and the actions to be performed by the terminal.

Name	Source	Format	Tag	Length
Cryptogram Information Data (CID)	ICC	b	'9F27'	1

Table 24: Analysis of Cryptogram Information Data

Byte	Bit	Meaning
1	8	Type of Cryptogram
	7	00: AAC
		01: TC
		10: ARQC
		11: RFU
	6	Payment System-specific Cryptogram
	5	These bits are always set to zero.
	4	Advice Required This bit is always set to zero, with the exception of M/Chip Advance when it operates in two host backward compatibility modes.
	3	Reason/Advice Code
	2	000: No Information Given
	1	

Issuer Application Data, '9F10'

The Issuer Application Data contains the proprietary application data for transmission to the issuer in an online transaction.

Name	Source	Format	Tag	Length
Issuer Application Data (IAD)	ICC	b	'9F10'	Var up to 32

For the M/Chip Advance application, the Issuer Application Data is the concatenation of the data elements identified in the following table.

Data Element	Length	Description
Key Derivation Index	1	The value of the Key Derivation Index enables the issuer to identify which issuer master key to use to verify the Application Cryptogram. The value is proprietary to the issuer.

Data Element	Length	Description
Cryptogram Version Number	1	The value of the Cryptogram Version Number enables the issuer to identify which session key derivation method to use and other data input to the Application Cryptogram.
Card Verification Results	6	Contains data sent to the issuer indicating exception conditions that occurred during the current and previous transactions.
Data Authentication Code (DAC) / ICC Dynamic Number (DN)	2	The DAC/DN helps the issuer to determine whether SDA or DDA offline card authentication was attempted in the event of a dispute. • If the terminal performed SDA successfully, the DAC obtained by the terminal from the issuer key certificate is included. • If the terminal performed DDA successfully, the two left-most bytes of the ICC Dynamic Number are included. • If the terminal performed CDA, '0000' is included, whether successful or not. This is because if CDA is successful, the DN is not obtained by the terminal prior to GENERATE AC, thus the DN is not sent to the card, so cannot be included in '9F10'. • If the terminal did not perform SDA or DDA successfully, '0000' is included. NOTE: If CDA was successful, the terminal will have obtained the Dynamic Number, but it is not present in this '9F10' field because it was not obtained by the terminal prior to GENERATE AC, therefore could not be sent in the GENERATE AC request.
Counters (Plaintext or Encrypted)/Accumulators	8 or 16	The value of the accumulators or counters may have an impact on the current authorization decision, especially for preauthorized cards where the balance is managed offline by the card. The value may also impact the issuer decision on how the accumulators or counters should be reset or the limits updated. For MCBP implementations, the MD cryptogram is contained within the counters.

Data Element	Length	Description
Last Online ATC	2 (optional)	Last Online ATC is the value of the Application Transaction Counter for the last transaction where issuer authentication was completed, or where an online response has been received while the application is configured to accept transactions without issuer authentication and the outcome of the transaction is a TC, and can be optionally included in the Issuer Application Data.

Therefore, the length of the Issuer Application Data for M/Chip Advance may be 18, 20, 26, or 28.

Cryptogram Version Number

The Cryptogram Version Number provides information about the algorithm and data used for the Application Cryptogram computation during online transactions in the online request and after transaction completion in the clearing record.

Name	Source	Format	Tag	Length
Cryptogram Version Number	ICC	b		1

Table 25: Definition of Cryptogram Version Number

Byte	Bit	Meaning
1	8	0001
	7	
	6	
	5	
	4	0 RFU
	3	Session key used for AC computation:
2	0 0	Mastercard Proprietary SKD session key
	1 0	EMV CSK session key
1		Counters included in AC computation:
	0	Counters not included in AC data
	1	Counter included in AC data

Card Verification Results

Contains data sent to the issuer indicating exception conditions that occurred during the current and previous transactions.

Name	Source	Format	Tag	Length
Card Verification Results	ICC	b		5

Definition of Card Verification Results

The coding of Card Verification Results for M/Chip Advance is displayed in the following table.

Byte	Bit	Meaning
1	8	AC Returned in Second GENERATE AC
	7	0 0 AAC
		0 1 TC
		1 0 Not requested
		1 1 RFU
	6	AC Returned in First GENERATE AC
	5	0 0 AAC
		0 1 TC
		1 0 ARQC
		1 1 RFU
	4	Date Check Failed
		0 Date Check Not Failed
		1 Date Check Failed
	3	Offline PIN Verification Performed
		0 Offline PIN Verification Not Performed
		1 Offline PIN Verification Performed
	2	Offline Encrypted PIN Verification Performed
		0 Offline Encrypted PIN Verification Not Performed
		1 Offline Encrypted PIN Verification Performed

Byte	Bit	Meaning
2	1	Offline PIN Verification Successful 0 Offline PIN Verification Not Successful 1 Offline PIN Verification Successful
	8	DDA Returned 0 DDA Not Returned 1 DDA Returned
	7	Combined DDA/AC Generation Returned in First GENERATE AC 0 Combined DDA/AC Generation Not Returned in First GENERATE AC 1 Combined DDA/AC Generation Returned in First GENERATE AC
	6	Combined DDA/AC Generation Returned in Second GENERATE AC 0 Combined DDA/AC Generation Not Returned in Second GENERATE AC 1 Combined DDA/AC Generation Returned in Second GENERATE AC
	5	Issuer Authentication Performed 0 Issuer Authentication Not Performed 1 Issuer Authentication Performed
	4	CIAC Default Skipped on CAT3 0 No CIAC Default Skipped on CAT3 1 CIAC Default Skipped on CAT3
	3	Offline Change PIN Result 0 Offline Change PIN Result Unsuccessful 1 Offline Change PIN Result Successful
	2	Issuer Discretionary
	1	
3	8	Right nibble of Script Counter
	7	
	6	
	5	

Byte	Bit	Meaning
	4	Right nibble of PIN Try Counter
	3	
	2	
	1	
4	8	Last Online Transaction Not Completed 0 Last Online Transaction Completed 1 Last Online Transaction Not Completed
	7	Unable To Go Online Indicated 0 Unable To Go Online Not Indicated 1 Unable To Go Online Indicated
	6	Offline PIN Verification Not Performed 0 Offline PIN Verification Performed 1 Offline PIN Verification Not Performed
	5	Offline PIN Verification Failed 0 No Failure of Offline PIN Verification 1 Offline PIN Verification Failed
	4	PTL Exceeded 0 PTL Not Exceeded 1 PTL Exceeded
	3	International Transaction 0 Domestic Transaction 1 International Transactions
	2	Domestic Transaction 0 International Transaction 1 Domestic Transaction
	1	Terminal Erroneously Considers Offline PIN OK 0 Terminal Does Not Erroneously Consider Offline PIN OK 1 Terminal Erroneously Considers Offline PIN OK

Byte	Bit	Meaning
5	8	Lower Consecutive Counter 1 Limit Exceeded 0 Lower Consecutive Counter 1 Limit Not Exceeded 1 Lower Consecutive Counter 1 Limit Exceeded
	7	Upper Consecutive Counter 1 Limit Exceeded 0 Upper Consecutive Counter 1 Limit Not Exceeded 1 Upper Consecutive Counter 1 Limit Exceeded
	6	Lower Cumulative Accumulator 1 Limit Exceeded 0 Lower Cumulative Accumulator Limit Not Exceeded 1 Lower Cumulative Accumulator Limit Exceeded
	5	Upper Cumulative Accumulator 1 Limit Exceeded 0 Upper Cumulative Accumulator Limit Not Exceeded 1 Upper Cumulative Accumulator Limit Exceeded
	4	Go Online On Next Transaction Was Set 0 Go Online On Next Transaction Was Not Set 1 Go Online On Next Transaction Was Set
	3	Issuer Authentication Failed 0 No Issuer Authentication Failed 1 Issuer Authentication Failed
	2	Script Received 0 No Script Received 1 Script Received
	1	Script Failed 0 No Script Failed 1 Script Failed
6	8	Lower Consecutive Counter 2 Limit Exceeded 0 Lower Consecutive Counter 2 Limit Not Exceeded 1 Lower Consecutive Counter 2 Limit Exceeded
	7	Upper Consecutive Counter 2 Limit Exceeded 0 Upper Consecutive Counter 2 Limit Not Exceeded 1 Upper Consecutive Counter 2 Limit Exceeded

Byte	Bit	Meaning
	6	Lower Cumulative Accumulator 2 Limit Exceeded 0 Lower Cumulative Accumulator 2 Limit Not Exceeded 1 Lower Cumulative Accumulator 2 Limit Exceeded
	5	Upper Cumulative Accumulator 2 Limit Exceeded 0 Upper Cumulative Accumulator 2 Limit Not Exceeded 1 Upper Cumulative Accumulator 2 Limit Exceeded
	4	MTA Limit Exceeded 0 MTA Limit Not Exceeded 1 MTA Limit Exceeded
	3	Number Of Days Offline Limit Exceeded
	2	Match Found In Additional Check Table 0 No Match Found In Additional Check Table 1 Match Found In Additional Check Table
	1	No Match Found In Additional Check Table 0 Match Found In Additional Check Table 1 No Match Found In Additional Check Table

Counters (plaintext or encrypted)/Accumulators

In M/Chip Advance, the reporting of the counters and accumulators and in the Issuer Application Data is configurable for each interface through the Accumulator X Control and Counter X Control data objects. The values may appear in plain text or encrypted form and may or may not be signed by the Application Cryptogram. It should be noted that if the counters/accumulators are sent encrypted, the padding is also encrypted and it does not show up as 'FF' or padding.

In MCBP, the Plaintext/Encrypted Counters contains a truncation of the Application Cryptogram (MD) for the device to be authenticated.

The first 8-byte part is always present and can contain:

- Only padding (No Accumulators and Counters)
- One Accumulator plus padding
- One Accumulator and one Counter plus padding
- Both Counters plus padding
- If MCBP, the MD cryptogram truncated plus padding

The following table shows the exact format of the first bytes for MCBP products.

Byte 1	'00'
Byte 2	Application Cryptogram (MD), Byte 1
Byte 3	Application Cryptogram (MD), Byte 2
Byte 4	Application Cryptogram (MD), Byte 3
Byte 5	Application Cryptogram (MD), Byte 4
Byte 6	Application Cryptogram (MD), Byte 5
Byte 7	'00'
Byte 8	'FF'

The second 8-byte part is optional and can contain:

- The second Accumulator plus padding
- The second Accumulator and second Counter plus padding

Example of Issuer Application Data analysis

If the value of tag '9F10' (Issuer Application Data) is '0110608003220000B6A400000000000000FF', this indicates:

'01'	'10'	'608003220000'	'B6A4'	'00000000000000FF'
byte 1	byte 2	byte 3-8	bytes 9-10	bytes 11-18
Key Derivation Index	Cryptogram Version Number	Card Verification Results	ICC Dynamic Number	Clear Text Accumulators/Counters

The conversion of the value of CVR in binary is shown in the following table.

'608003220000'	0110 0000	1000 0000	0000 0011	0010 0010	0000 0000	0000 0000
	byte 1	byte 2	byte 3	byte 4	byte 5	byte 6

The following table outlines the meaning of the bits that are set.

Byte	Bit	Meaning
1	8	AC Returned in Second GENERATE AC
	7	0 1 TC

Byte	Bit	Meaning
	6	AC Returned in First GENERATE AC
	5	1 0 ARQC
2	8	DDA Returned
		1 DDA Returned
3	4	Right nibble of PIN Try Counter
	3	
	2	
	1	
4	6	Offline PIN Verification Not Performed
		1 Offline PIN Verification Not Performed
	2	Domestic Transaction
		1 Domestic Transaction

The analysis of the accumulators/counters, which appear in plaintext, are given in the following table:

'00000000000000FF'	'00 00 00 00 00 00'	'00'	'FF'
	byte 1-6	byte 7	byte 8

byte 1-6: '000000000000' → Cumulative value of offline transactions expressed in the chosen currency

byte 7: '00' → Number of consecutive offline transactions

byte 8: 'FF' → Padding

Unpredictable Number, '9F37'

The Unpredictable Number is a value that provides variability and uniqueness to the generation of a cryptogram.

Name	Source	Format	Tag	Length
Unpredictable Number	Terminal	b	'9F37'	4

Application Transaction Counter, '9F36'

The Application Transaction Counter is a value maintained by the application in the ICC and increments for each transaction.

Name	Source	Format	Tag	Length
Application Transaction Counter	ICC	b	'9F36'	2

Terminal Verification Results, '95'

The Terminal Verification Results (TVR) is a series of bits set by the terminal reading an EMV card based on logical tests, for example, an expired card. This data object is used in the terminal's decision to accept, decline, or go online for a transaction.

Name	Source	Format	Tag	Length
Terminal Verification Results (TVR)	Terminal	b	'95'	5

Table 26: Analysis of Terminal Verification Results

Byte	Bit	Meaning
1	8	Offline data authentication was not performed
	7	SDA failed
	6	ICC Data missing
	5	Card appears on terminal exception file
	4	DDA failed
	3	CDA failed
	2	SDA selected
	1	RFU
2	8	ICC and terminal have different application
	7	Expired application
	6	Application not yet effective
	5	Requested service not allowed for card product
	4	New Card
	3	RFU
	2	RFU

Byte	Bit	Meaning
3	1	RFU
	8	Cardholder verification was not successful
	7	Unrecognized CVM
	6	PIN Try Limit exceeded
	5	PIN entry required and PIN pad not present or not working
	4	PIN entry required, PIN pad present but PIN was not entered
	3	Online PIN entered
	2	RFU
4	1	RFU
	8	Transaction exceeds floor limit
	7	Lower consecutive offline limit exceeded
	6	Upper consecutive offline limit exceeded
	5	Transaction selected randomly for online processing
	4	Merchant forced transaction online
	3	RFU
	2	RFU
5	1	RFU
	8	Default TDOL used
	7	Issuer authentication was unsuccessful
	6	Script processing failed before final GENERATE AC
	5	Script processing failed after final GENERATE AC
	4	Relay resistance threshold exceeded (only for contactless)
	3	Relay resistance time limits exceeded (only for contactless)
	2-1	Relay resistance performed (only for contactless) 00: Relay resistance not supported 01: Relay resistance not performed 10: Relay resistance performed 11: RFU

Example of Terminal Verification Results analysis

If the value of TVR is '4000008000' in hexadecimal mode, this value should be converted into binary mode. The conversion is shown in the following table.

Table 27: Terminal verification results

	byte 1	byte 2	byte 3	byte 4	byte 5
'4000008000'	(01000000	00000000	00000000	10000000	00000000) _{bin}
4000008000	(01000000	00000000	00000000	10000000	00000000) _{bin}
	byte 1	byte 2	byte 3	byte 4	byte 5

This means that bit 7 from Byte 1 and bit 8 from Byte 4 are set. Considering the analysis of TVR, the result indicates that the SDA was unsuccessful during the transaction and the transaction amount was over the terminal's floor limit.

Transaction Date, '9A'

The Transaction Date represents the local date (YYMMDD) that the transaction was authorized.

Name	Source	Format	Tag	Length
Transaction Date	Terminal	n 6	'9A'	3

Amount Authorized, '9F02'

The amount normally reflects the actual transaction amount in the transaction currency, but there are exceptions to this rule which include:

- If transaction type (tag '9C') is '93' (Contactless ATM transaction) where '9F02' may be zero or placeholder amount. This only occurs for contactless transactions.
- If this is a contactless transit transaction, where '9F02' may be zero or a placeholder amount.
- If POI Currency Conversion occurred on a contactless transaction, where '9F02' may reflect the amount in the original currency before conversion.
For more information, refer to chapter 4 (Acquirer requirements), section "Terminal requirements," subsection "Currency conversion."
- If the "Alternative contactless flow at terminals in Sweden" applies, '9F02' contains a placeholder or estimated amount. This only occurs for contactless transactions.

For more information, refer to chapter 4 (Acquirer requirements), section "Regional variations," subsection "Europe region," subsection "Alternative contactless flow at terminals in Sweden."

- If the "M/Chip Fast flow at terminals in the United States region" applies, '9F02' contains a placeholder or estimated amount. This can occur for both contact and contactless transactions.

For more information, refer to chapter 4 (Acquirer requirements), section "Regional variations," subsection "United States region," subsection "M/Chip Fast."

- If any other alternate EMV transaction flow is implemented.
For more information, refer to chapter 4 (Acquirer requirements), section "Special acceptance environments," subsection "Alternate EMV transaction flows."

Name	Source	Format	Tag	Length
Amount, Authorized (Numeric)	Terminal	n 12	'9F02'	6

Transaction Type, '9C'

The Transaction Type indicates the type of financial transaction represented by the first two digits of the ISO 8583:1987 Processing Code. The actual values to be used for the Transaction Type data element are defined by the relevant payment system.

Name	Source	Format	Tag	Length
Transaction Type	Terminal	n 2	'9C'	1

Table 28: Analysis of Transaction Type

Values (in hex)	Meaning
00	Purchase
01	Withdrawal
02	Debit Adjustment
09	Purchase with Cash Back
17	Cash Disbursement
18	Script Issue
20	Purchase Return/Refund
21	Deposit
22	Credit Adjustment
23	Check Deposit Guarantee

Values (in hex)	Meaning
24	Check Deposit
28	Payment Transaction
30	Balance Inquiry
40	Account Transfer
90	Reserved For Future Use
91	PIN Unblock
92	PIN Change
93	Contactless ATM transaction

Transaction Currency Code, '5F2A'

The Transaction Currency Code indicates the currency code of the transaction according to ISO 4217.

Name	Source	Format	Tag	Length
Transaction Currency Code	Terminal	n 3	'5F2A'	2

Application Interchange Profile, '82'

The Application Interchange Profile indicates the capabilities of the card to support specific functions in the application.

Name	Source	Format	Tag	Length
Application Interchange Profile (AIP)	ICC	b	'82'	2

Table 29: Analysis of Application Interchange Profile

Byte	Bit	Meaning
1	8	RFU
	7	SDA is supported
	6	DDA is supported
	5	Cardholder verification is supported
	4	Terminal risk management is to be performed

Byte	Bit	Meaning
	3	Issuer authentication is supported When this bit is set to 1, Issuer Authentication using the EXTERNAL AUTHENTICATE command is supported.
	2	CDCVM is supported (only for contactless)
	1	CDA supported
2	8	EMV mode is supported (only for contactless)
	7	RFU
	6	RFU
	5	RFU
	4	RFU
	3	RFU
	2	RFU
	1	Relay resistance protocol is supported (only for contactless)

Terminal Country Code, '9F1A'

The Terminal Country Code indicates the country of the terminal represented according to the *Quick Reference Booklet*.

Name	Source	Format	Tag	Length
Terminal Country Code	Terminal	n 3	'9F1A'	2

Cardholder Verification Method Results, '9F34'

The Cardholder Verification Method Results indicate the results of the last CVM performed.

The presence of tag '9F34' is mandatory for all online request messages containing DE 55.

Name	Source	Format	Tag	Length
Cardholder Verification Method (CVM) Results	Terminal	b	'9F34'	3

Table 30: Analysis of CVM Results: Byte 1 - CVM performed, Last CVM of the CVM List actually performed by the terminal

CVM performed (Byte 1)			
Value (in hex)	Value (in binary)	Meaning	
00	00000000	Fail CVM processing	Fail cardholder verification if this CVM is unsuccessful
01	00000001	Plaintext PIN verification performed by ICC	Fail cardholder verification if this CVM is unsuccessful
02	00000010	Enciphered PIN verified online	Fail cardholder verification if this CVM is unsuccessful
03	00000011	Plaintext PIN verification performed by ICC and Signature	Fail cardholder verification if this CVM is unsuccessful
04	00000100	Enciphered PIN verification performed by ICC	Fail cardholder verification if this CVM is unsuccessful
05	00000101	Enciphered PIN verification performed by ICC and Signature	Fail cardholder verification if this CVM is unsuccessful
1E	00011110	Signature	Fail cardholder verification if this CVM is unsuccessful
1F	00011111	No CVM Required	Fail cardholder verification if this CVM is unsuccessful
3F	00111111	No CVM Performed	Fail cardholder verification if this CVM is unsuccessful
40	01000000	Fail CVM Processing	Apply succeeding CVM Rule if this CVM is unsuccessful
41	01000001	Plaintext PIN verification performed by ICC	Apply succeeding CVM Rule if this CVM is unsuccessful
42	01000010	Enciphered PIN verified online	Apply succeeding CVM Rule if this CVM is unsuccessful
43	01000011	Plaintext PIN verification performed by ICC and Signature	Apply succeeding CVM Rule if this CVM is unsuccessful
44	01000100	Enciphered PIN verification performed by ICC	Apply succeeding CVM Rule if this CVM is unsuccessful
45	01000101	Enciphered PIN verification performed by ICC and Signature	Apply succeeding CVM Rule if this CVM is unsuccessful
5E	01011110	Signature	Apply succeeding CVM Rule if this CVM is unsuccessful

CVM performed (Byte 1)			
Value (in hex)	Value (in binary)	Meaning	
5F	01011111	No CVM Required	Apply succeeding CVM Rule if this CVM is unsuccessful

Table 31: Analysis of CVM Results: Byte 2 - One-byte CVM Condition Code of the CVM List

CVM Condition Codes value (Byte 2) (in hex)	Meaning
00	Always
01	If unattended cash
02	If not unattended cash and not manual cash and not purchase with cash back
03	If terminal supports the CVM
04	If manual cash
05	If purchase with cash back
06	If transaction is in the application currency and is under X value
07	If transaction is in the application currency and is over X value
08	If transaction is in the application currency and is under Y
09	If transaction is in the application currency and is over Y value
0A-7F	RFU
80-FF	Reserved for use by individual payment systems

Table 32: Analysis of CVM Results: Byte 3 - This indicates the result of the last CVM performed as known to the terminal

Result of the last CVM performed (Byte 3) (in hex)	Meaning
00	Unknown (for example, for Signature)
01	Failed (for example, for Offline PIN)
02	Successful (for example, for Offline PIN)

Example of CVM Results Analysis

If the value of CVM results in the online request message is '410302', this means:

Table 33: CVM results examples

byte 1	byte 2	byte 3
'41'	'03'	'02'

The first byte indicates "Offline plaintext PIN" was the last CVM method performed during the transaction.

The second byte is translated into "If terminal supports the CVM," which represents the used CVM condition during the transaction.

The third byte is translated into "Successful," which represents the result of the last performed CVM. Ultimately, this means the last CVM performed was "Offline plaintext PIN" and the result was "Successful."

Dedicated File Name, '84'

The Dedicated File (DF) Name identifies the name of the DF as described in ISO/IEC 7816-4.

Name	Source	Format	Tag	Length
Dedicated File (DF) Name	ICC	b	'84'	5-16

The DF Name contains the value of the Application Identifier (AID) from the card used to generate the current transaction. The following table shows the most common AIDs. For additional information, refer to chapter "Data requirements," section "Application identification," subsection "Application Identifiers."

Table 34: Values of Dedicated File Name

Product	AID (in hex)	Recommended Application Label
Mastercard Credit or Debit Mastercard	A0000000041010	"Mastercard" or "Debit Mastercard"
Maestro	A0000000043060	"Maestro"
Cirrus	A0000000046000	"Cirrus"

Terminal Capabilities, '9F33'

The Terminal Capabilities indicates the card data input, CVM, and security capabilities of the terminal. The presence of tag '9F33' is mandatory for all transactions acquired in the United States region. For all other regions, refer to requirements RA546.24 and RA547.24.

Name	Source	Format	Tag	Length
Terminal Capabilities	Terminal	b	'9F33'	3

The Terminal Capabilities contains:

- Card data input capability - Indicates all the methods supported by the terminal for entering the information from the card into the terminal (first byte).
- Cardholder Verification Method (CVM) capability - Indicates all the methods supported by the terminal during the processing of the current transaction for verifying the identity of the cardholder at the terminal (second byte).
- Security capability - Indicates all the methods supported by the terminal for authenticating the card at the terminal and whether the terminal has the ability to capture a card (third byte).

Table 35: Analysis of Terminal Capabilities

Byte	Bit	Meaning
1	8	Manual key entry
	7	Magnetic Stripe
	6	IC with contacts
	5	RFU
	4	RFU
	3	RFU
	2	RFU
2	1	RFU
	8	Plaintext PIN for ICC verification
	7	Enciphered PIN for online verification
	6	Signature
	5	Enciphered PIN for offline verification, using RSA
	4	No CVM Required
	3	Online Biometric

Byte	Bit	Meaning
3	2	Offline Biometric
	1	Enciphered PIN for offline verification, using ECC
	8	SDA
	7	DDA
	6	Card capture
	5	RFU
	4	CDA
	3	XDA
	2	RFU
	1	RFU

Example of Terminal Capabilities analysis

If the value of terminal capabilities is 'E0F0C8' in hexadecimal mode, this value should be converted into binary mode. The conversion is shown in the following table.

Table 36: Terminal capabilities

	byte 1	byte 2	byte 3
'E0F0C8'	(111100000	111110000	(11001000) _{bin}

This means that the following units are set:

- bit 8, bit 7, and bit 6 from Byte 1
- bit 8, bit 7, bit 6, and bit 5 from Byte 2
- bit 8, bit 7, and bit 4 from Byte 3

When considering the analysis of terminal capabilities, the terminal supports:

- Manual key entry
- Magnetic Stripe
- IC with contacts
- Plaintext PIN for ICC verification
- Enciphered PIN for online verification
- Signature (either captured electronically or on paper)
- Enciphered PIN for offline verification
- SDA

- DDA
- CDA

Recommended subelements in online request

When DE 55 is present in the Authorization Request/0100 or Financial Transaction Request/0200 message, the subelements described in this section are recommended in the DE 55 field.

Terminal Type, '9F35'

The Terminal Type indicates the environment of the terminal, its communications capability, and operational control.

Name	Source	Format	Tag	Length
Terminal Type	Terminal	n 2	'9F35'	1

Table 37: Analysis of values of Terminal Type

Terminal Type	Value (Operational control provided by)		
Environment	Financial Institution	Merchant	Cardholder
Attended			
Online only	'11'	'21'	-
Offline with online capability	'12'	'22'	-
Offline only	'13'	'23'	-
Examples of attended terminals	Branch terminal	Electronic cash register, portable POS terminal, stand-alone POS terminal, host concentrating POS terminal	-
Unattended			
Online only	'14'	'24'	'34'
Offline with online capability	'15'	'25'	'35'
Offline only	'16'	'26'	'36'

Terminal Type		Value (Operational control provided by)	
Environment	Financial Institution	Merchant	Cardholder
Examples of unattended terminals	ATM	Automated fuel dispenser, pay telephone, ticket dispenser, vending machine	Home terminal, personal computer, screen telephone, pay phones, digital interactive television, and set top boxes

Kernel Identifier-Terminal, '96'

This identifies which contactless kernel was used in a contactless EMV mode transaction. It provided it helps the issuer detect if a kernel different to what was configured on the access device was used.

Name	Source	Format	Tag	Length
Kernel Identifier-Terminal	Terminal	b	'96'	8

Transaction Sequence Counter, '9F41'

The Transaction Sequence Counter represents a counter maintained by the terminal that is incremented by one for each transaction.

Name	Source	Format	Tag	Length
Transaction Sequence Counter	Terminal	n 4-8	'9F41'	2-4

Conditional subelements in online request

When DE 55 is present in the Authorization Request/0100 or Financial Transaction Request/0200 message, the subelement described in this section is required in DE 55 only if specific conditions are fulfilled.

Amount, Other, '9F03'

The Amount, Other is a secondary amount associated with the transaction representing a cash back amount. The exponent is defined by tag '9F44' (Application Currency Exponent), which indicates the implied position of the decimal point from the right of the amount represented according to ISO 4217.

Name	Source	Format	Tag	Length
Amount, Other (Numeric)	Terminal	n 12	'9F03'	6

Appendix C Contactless ATM transaction flows

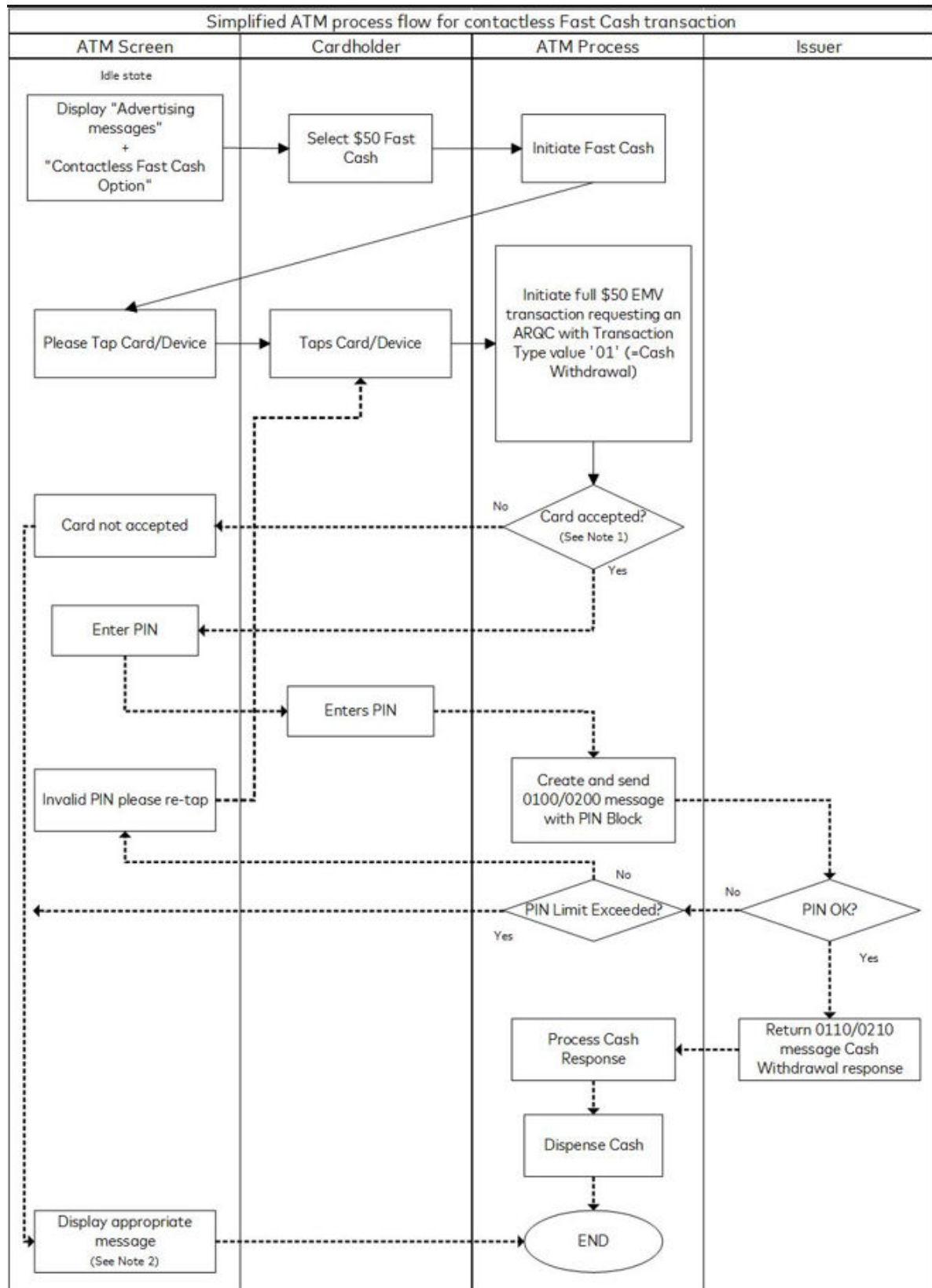
The diagrams in this section provide examples of ATM transaction process flows.

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Contactless ATM Transaction Flows

The diagrams in this section provide examples of ATM transaction process flows.

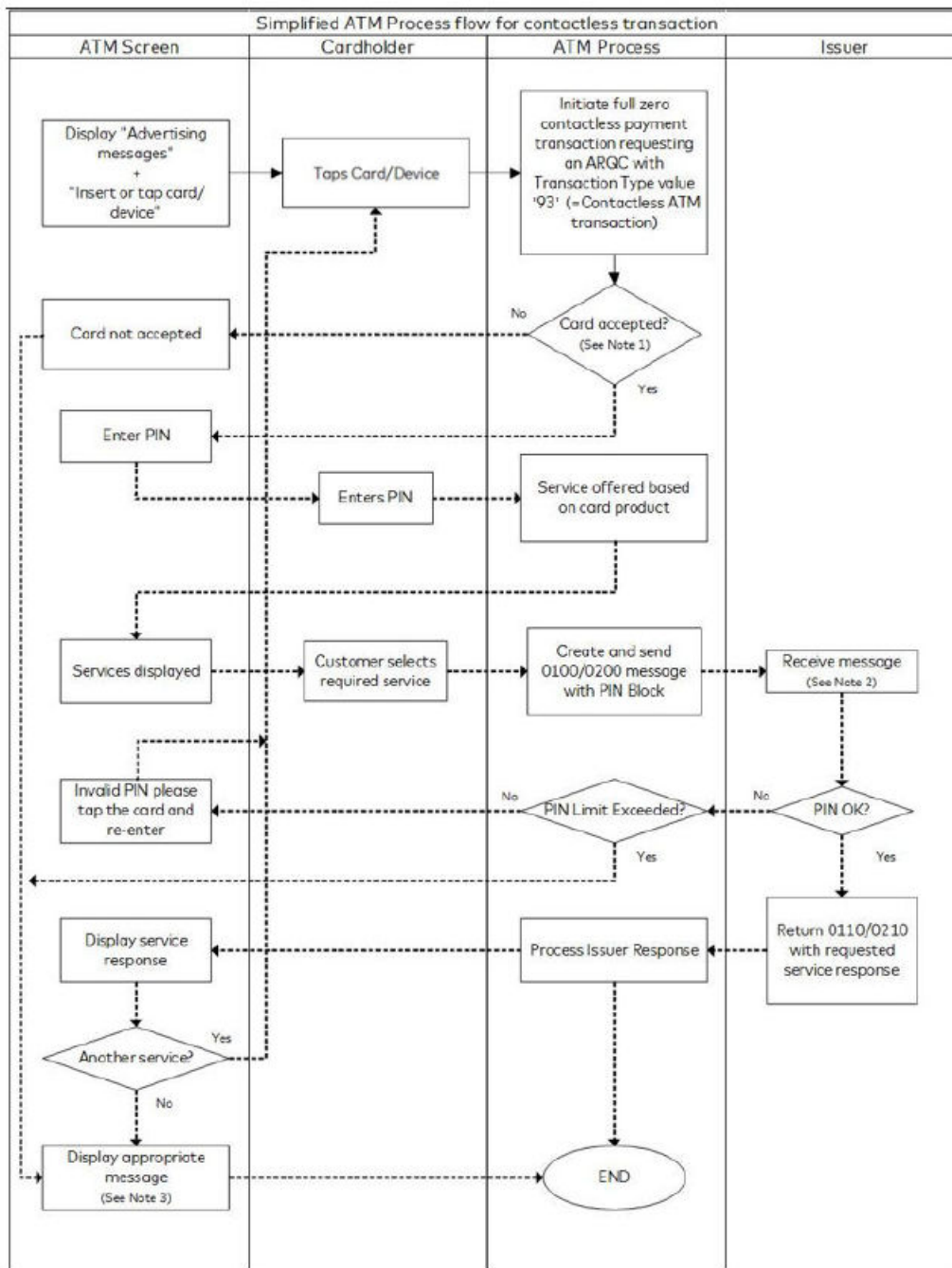
Figure 2: Simplified ATM process flow for Contactless Fast Cash Transaction



NOTE: In addition to any existing validation performed by the ATM, to support additional security features available in Mastercard® mobile products, if a card/device responds to the **GENERATE AC** command that has requested an Online Authorization (ARQC), with an AAC (Offline Decline), the ATM should terminate the transaction without requesting issuer authorization or PIN entry.

NOTE: If the ATM/Contactless reader supports version 3 or later of the *Mastercard Contactless Reader Specification*, then it must support display messages associated with the POS Cardholder Interaction Information (PCII).

Figure 3: Simplified Example ATM process flow for Contactless Transactions



NOTE: In addition to any existing validation performed by the ATM, to support additional security features available in Mastercard mobile products, if a card/device responds to the GENERATE AC command that has requested an Online Authorization (ARQC), with an AAC (Offline Decline), the ATM should terminate the transaction without requesting issuer authorization or PIN entry. Refer to requirement BA086.14.

NOTE: A single-tap flow can be supported for multiple ATM services when a nonfinancial request (any transaction that does not cause a transfer of funds) follows a financial or a nonfinancial request. While Mastercard recommends that terminals generate new chip data for each request (refer to requirement BA016.16), issuers must still be able to receive the same contactless chip cryptogram data, for example, same ATC and ARQC data (refer to requirement RI445.23).

A re-tap of the card or device along with new chip data is required if a financial request follows a nonfinancial request, for example, a balance inquiry followed by a cash withdrawal, to ensure that the correct cardholder is still at the ATM and new chip data is sent to the issuer. Refer to RA364.16.

NOTE: If the ATM Contactless reader supports version three or later of the Mastercard Contactless Reader Specification, then it must support display messages associated with the PCII.

Appendix D Abbreviations

This appendix contains a list of abbreviations used throughout the manual.

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Abbreviations

Abbreviations used throughout the manual are listed in the following table.

Abbreviation	Definition
AAC	Application Authentication Cryptogram
AC	Application Cryptogram
ADF	Application Definition File
AEF	Application Elementary File
AFD	Automated Fuel Dispenser
AFL	Application File Locator
AID	Application Identifier
AIP	Application Interchange Profile
an	alphanumeric
APDU	Application Protocol Data Unit
API	Application Priority Indicator
ARPC	Authorization Response Cryptogram
ARQC	Authorization Request Cryptogram
ASCII	American Standard Code for Information Interchange
ATC	Application Transaction Counter
ATM	Automated Teller Machine
AUC	Application Usage Control
AVS	Address Verification Service
b	binary
BBT	Bank Branch Terminal
BCD	Binary Code Decimal
BIC	Bank Identification Code
BIN	Bank Identification Number
BP	Best Practice
C	Conditional
CAM	Card Authentication Method
CAST	Compliance Assessment and Security Testing

Abbreviation	Definition
CAT	Cardholder Activated Terminal
CAT1/2	Dual capability CAT Level 1 and CAT Level 2
CBWP	cash back without purchase
CCN	Card Certification Number
CCS	Component Conformity Statement
CDA	Combined DDA/AC Generation
CDCVM	Consumer Device Cardholder Verification Method
CDOL	Card Risk Management Data Object List
CEE	Central and Eastern Europe
CIAC	Card Issuer Action Code
CL	Contactless
CMS	Credentials Management System
CNS	Change Notification Statement
CoTS	Commercial Off The Shelf
CPA	Common Payment Application
CPoC	Contactless Payments on CoTS
CPV	Card Personalization Validation
CQM	Card Quality Management
CRM	Card Risk Management
CSU	Card Status Update
CT	Contact
CVC	Card Validation Code
CVM	Cardholder Verification Method
CVR	Card Verification Results
DAC	Data Authentication Code
DDA	Dynamic Data Authentication
DDF	Directory Definition File
DDOL	Dynamic Data Authentication Data Object List
DE	Data Element
DF	Dedicated File
DN	Dynamic Number

Abbreviation	Definition
DOL	Data Object List
DPAN	Device Primary Account Number
DRDOL	Data Recovery Data Object List
DSDOL	Data Storage Data Object List
DSRP	Digital Secure Remote Payment
ECR	Emergency Card Replacement
EEA	European Economic Area
EMV CSK	EMV Common Session Key derivation
EU	European Union
FCI	File Control Information
FESE	Fully Encapsulated Secure Element
FPAN	Funding Primary Account Number
GCMS	Global Clearing Management System
HCE	Host Card Emulation
HEX	hexadecimal
IAC	Issuer Action Code
IBAN	International Bank Account Number
IC	Integrated Circuit
ICA	Interbank Card Association
ICC	Integrated Circuit Card
IEC	International Electrotechnical Commission
IFD	Interface Device
IFR	Interchange Fee Regulation
IFSF	International Forecourt Standards Forum
IIN	Issuer Identification Number
IPM	Integrated Product Message
IPOS	Integrated Point-Of-Sale
IRD	Interchange Rate Designator
ISO	International Standards Organization
KDI	Key Derivation Index
LDA	Local Dynamic Authentication

Abbreviation	Definition
LoA	Letter of Approval
LVT	low-value transactions
M	Mandatory
MAC	Message Authentication Code
MCBP	Mastercard Cloud-Based Payments
MCC	card acceptor business code
MD	Mobile Device
MDES	Mastercard Digital Enablement Service
MIP	Mastercard interface processor
MPA	Mobile Payment Application
MPCN	Mobile Payment Certification Number
MPoC	Mobile Payment on CoTS
MPOS	Mobile Point-Of-Sale
MTBP	Mastercard TEE Based Payments
M-TIP	Mastercard Terminal Integration Process
NATC _{TRACK1}	Track 1 Number of ATC Digits
NATC _{TRACK2}	Track 2 Number of ATC Digits
NFC	Near Field Communications
nUN	Number of positions in the discretionary data of Track 1 Data and Track 2 Data for transporting UN
O	Optional
ODA	offline data authentication
OEM	Original Equipment Manufacturer
OTA	over-the-air
PAN	Primary Account Number
PAR	Payment Account Reference
PCC	POI Currency Conversion
PCD	Proximity Coupling Device
PCI	Payment Card Industry
PCI DSS	Payment Card Industry Data Security Rules
PCII	POS cardholder Interaction Information

Abbreviation	Definition
PCI-PTS	Payment Card Industry-PIN Transaction Security
PCN	Payment Certificate Number
PCVC3 _{TRACK1}	Track 1 Bit Map for CVC 3
PCVC3 _{TRACK2}	Track 2 Bit Map for CVC 3
PDOL	Processing Options Data Object List
PDS	Private Data Subelement
PED	PIN entry device
PIN	Personal Identification Number
PIX	Proprietary Application Identifier Extension
PKI	Public Key Infrastructure
POI	point-of-interaction
POS	Point-Of-Sale
PPSE	Proximity Payment System Environment
PSD2	Payment Services Directive 2
PSE	Payment System Environment
PSPK	Payment System Public Key
PTL	PIN Try Limit
PTS	PIN Transaction Security
PUNATC _{TRACK1}	Track 1 Bit Map for UN and ATC
PUNATC _{TRACK2}	Track 2 Bit Map for UN and ATC
PVV	PIN Verification Value
PWCB	purchase with cash back
RCCN	Restricted Card Certification Number
RCCS	Restricted Component Conformity Statement
RFU	Reserved for Future Use
RID	Registered Application Provider Identifier
RMPCN	Restricted Mobile Payment Certification Number
RRP	Relay Resistance Protocol
RSA	Rivest, Shamir, Adleman [Algorithm]
RSTAS	Restricted Security Test Assessment Summary
RTS	Regulatory Technical Standards

Abbreviation	Definition
SBMP	Software-Based Mobile Payments
SCA	Strong Customer Authentication
SCRP	Secure Card Reader for PIN
SDA	Static Data Authentication
SE	Secure Element
SEI TSM	Secure Element Issuer Trusted Service Manager
SEPA	Single European Payments Area
SFI	Short File Identifier
STAS	Security Test Assessment Summary
TAC	Terminal Action Code
TC	Transaction Certificate
TDOL	Default Transaction Certificate Object List
TDS	Transaction Detail Service
TEE	Trusted Execution Environment
TLV	Tag, Length, Value
TQM	Terminal Quality Management
TRM	Terminal Risk Management
TRMD	Terminal Risk Management Data
TTAS	Terminal Test Assessment Summary
TTI	Transaction Type Identifier
TVR	Terminal Verification Results
UCAF	Universal Cardholder Authentication Field
UDOL	Unpredictable Number Data Object List
UI	User Interface
UMD	User and Mobile Device Authentication
UN	Unpredictable Number
URL	Uniform Resource Locator
USM	U.S. Maestro AID
VAR	variable length data

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